

PUBLIC HEALTH & MICROBIOLOGY

NOTES

FOURTH EDITION

PRE-SUMMARIZED
READY-TO-STUDY
HIGH-YIELD NOTES

FOR THE TIME-POOR
MEDICAL, PRE-MED,
USMLE OR PA STUDENT



MED
STUDENT
NOTES
.COM

199 PAGES





A Message From Our Team

Studying medicine or any health-related degree can be stressful; believe us, we know from experience! The human body is an incredibly complex organism, and finding a way to streamline your learning is crucial to succeeding in your exams and future profession. Our goal from the outset has been to create the greatest educational resource for the next generation of medical students, and to make them as affordable as possible.

In this fourth edition of our notes we have made a number of text corrections, formatting updates, and figure updates which we feel will enhance your study experience. We have also endeavoured to use only open-source images and/or provide attribution where possible.

If you are new to us, here are a few things to help get the most out of your notes:

1. **Once saved, the notes are yours for life!** However, we strongly advise that you download and save the files immediately upon purchasing for permanent offline access.
2. **Sharing notes is prohibited.** All files are share-protected and our system will automatically revoke access to and lock files if it detects a customer attempting to share or distribute our notes.
3. **Your license permits you to do the following:**
 - a. You may download/save/view files on up to 2 simultaneous devices.
 - b. You may save the files to an external hard drive for backup purposes only.
 - c. You may print your notes to hard copy on any home printer/photocopier.
4. **Once saved, you do not need to download the notes again.** You can simply transfer your file/s to your second personal device (eg: your iPad/tablet) without the need to re-download your files. If you wish to retire an old device, simply transfer your files to your new device, then delete the files from your old device.
5. **Each order has an allocated download allowance.** Exceeding your download quota will require you to request an extension by emailing us at admin@medstudentnotes.com and quoting your order number.
6. **If you have any further queries,** feel free to check out our FAQ page: <https://www.medstudentnotes.com/pages/faqs> or email us at admin@medstudentnotes.com
7. **Please ensure you use the SAME Email address for any future purchases.** That way, all of your files will be visible in the one place on your account page.
8. **Feel free to use the discount code "TAKE20OFF"** for 20% off any future purchases. (Valid for a limited time)
9. **If you find these notes valuable but you haven't yet got our [Bundle Deal](#), (which includes ALL of our subjects at an 80% discount), then NOW is the time! To upgrade, simply visit this link for instructions: <https://www.medstudentnotes.com/pages/bundle-upgrade-process> .**



Table Of Contents:

What's included: Ready-to-study notes of various population health & infectious disease topics presented in succinct, intuitive and richly illustrated downloadable PDF documents. Once downloaded, you may choose to either print and bind them, or make annotations digitally on your iPad or tablet PC.

Clickable Hyperlinks Below:

- [PUBLIC HEALTH OVERVIEW](#)
- [MEASURING HEALTH CONCEPTS](#)
- [HEALTH BEHAVIOUR](#)
- [CHANGING BEHAVIOUR](#)
- [CHRONIC DISEASE & RISK FACTORS](#)
- [PANDEMICS](#)
- [VACCINATION/IMMUNIZATION](#)
- [BASIC CONCEPTS OF INFECTIOUS DISEASES](#)
- [MICROBIOLOGY: PRIONS](#)
- [MICROBIOLOGY: VIRUSES](#)
- [MICROBIOLOGY: PARASITES](#)
- [MICROBIOLOGY: BACTERIA](#)
- [NOTABLE INFECTIVE DISEASES](#)
 - [INFECTIVE ENDOCARDITIS](#)
 - [LYMPHANGITIS](#)
 - [MYOCARDITIS – VIRAL & TOXIC](#)
 - [PERICARDITIS](#)
 - [IMPETIGO \(SCHOOL SORES\)](#)
 - [ERYSIPELAS](#)
 - [CELLULITIS](#)
 - [SCABIES](#)
 - [LICE \(PEDICULOSIS\)](#)
 - [HERPES SIMPLEX](#)
 - [CHICKEN POX \(VARICELLA ZOSTER\)](#)
 - [HERPES ZOSTER \(SHINGLES\)](#)
 - [MEASLES VIRUS](#)
 - [RUBELLA VIRUS \(“GERMAN MEASLES”\)](#)
 - [HUMAN PARVOVIRUS B19 \(“5TH DISEASE”\)](#)
 - [SCARLET FEVER](#)
 - [DERMATOPHYTOSIS - “RINGWORM”/“TINEA”](#)
 - [PARASITIC GUT INFECTIONS](#)
 - [POLIOMYELITIS](#)
 - [MENINGITIS](#)
 - [ENCEPHALITIS](#)
 - [COMMON COLD \(ACUTE RHINITIS\)](#)
 - [PHARYNGITIS \(SORE THROAT\)](#)
 - [ACUTE LARYNGOTRACHEOBRONCHITIS \(CROUP\)](#)
 - [ACUTE EPIGLOTTITIS](#)
 - [PERTUSSIS - WHOOPING COUGH](#)
 - [Q-FEVER](#)
 - [LEPTOSPIROSIS](#)
 - [MELIOIDOSIS](#)
 - [PNEUMONIAS \(“Infections of the Lung”\)](#)
 - [BRONCHIOLITIS](#)
 - [SEASONAL FLU \(INFLUENZA A & B\)](#)
 - [BIRD FLU \(H5N1\)](#)

- SWINE FLU (H1N1)
- SARS & COVID – SEVERE ACUTE RESPIRATORY SYNDROME
- GENITAL HERPES SIMPLEX
- HUMAN PAPILLOMA VIRUS
- SYPHILIS
- CHLAMYDIA
- GONORRHOEA
- DONOVANOSIS
- HEPATITIS C
- HUMAN IMMUNODEFICIENCY VIRUS
- TRACHOMA
- PULMONARY TUBERCULOSIS
- INTESTINAL TUBERCULOSIS
- LEPROSY
- WHIPPLES DISEASE
- METAZOAN PARASITES
- LYMPHATIC FILARIASIS
- MALARIA
- LEISHMANIASIS
- ARBOVIRUSES
- ROSS RIVER VIRUS (RRV)
- DENGUE VIRUS
- YELLOW FEVER
- MURRAY VALLEY ENCEPHALITIS

PUBLIC HEALTH OVERVIEW:

PUBLIC HEALTH OVERVIEW:

Definitions:

- **Population Health:**
 - Relates the health of certain groups of people to their health-determinants, health-trends, and health-inequalities
 - May be dependent upon:
 - Physical factors
 - Biological factors
 - Social factors
 - Environmental factors
 - Economic factors
 - Personal health behaviours
 - Available health services
- **Public Health:**
 - The programs/policies instituted by a society to protect, promote, and restore individual's health and prevent morbidity and mortality
 - Includes:
 - Practices
 - Programs
 - Policies
 - Institutions
- **Epidemiology:**
 - The study of the distribution and determinants of disease in a population
- **Preventative Medicine:**
 - The arm of medicine devoted to addressing health problems at the risk-factor level in order to minimise the manifestation of disease in a population
- **Aetiology (Etiology):**
 - The Cause of disease or study of factors involved in development of disease
- **Risk factor:**
 - Something associated with an increased risk of developing a particular disease or condition
 - Demographic
 - Behavioural
 - Biomedical
 - Genetic
 - Environmental
 - Social
 - Other factors which may interact to increase or reduce effect

Common Goals of Public Health Services:

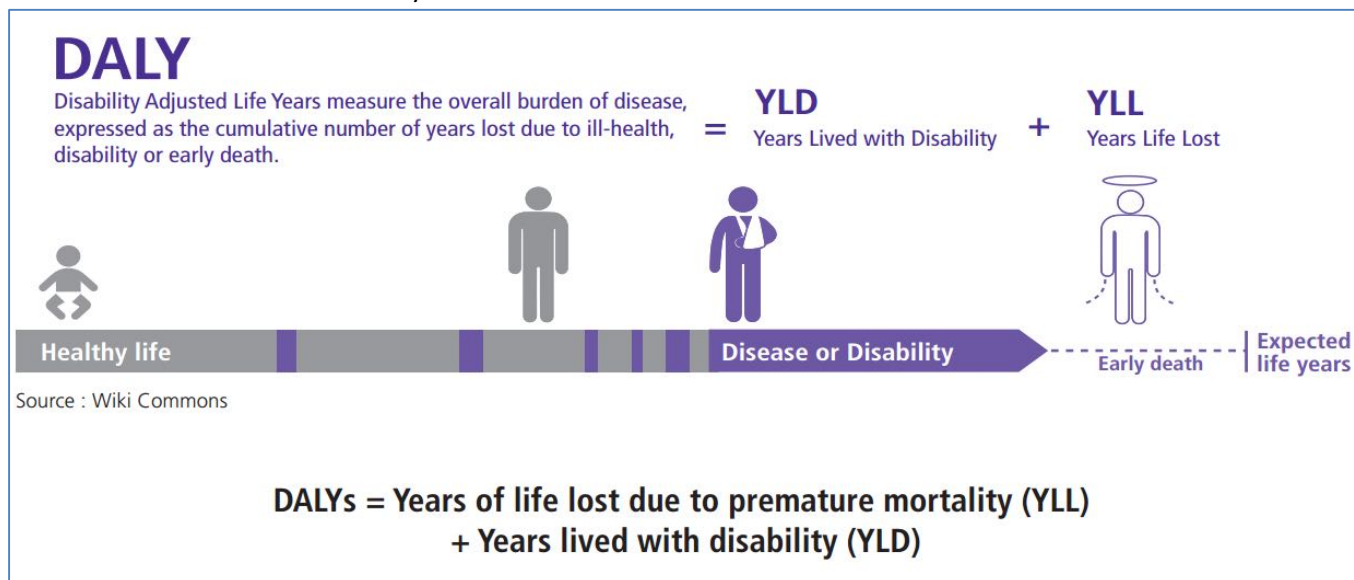
- **Health Protection:**
 - Addressing potential health risks at the population level (Eg: Safe water / food regulation)
- **Surveillance:**
 - Monitoring and early identification of epidemics or events/behaviours likely to cause negative health outcomes
- **Disease/Injury Prevention:**
 - Eg: Vaccination
 - Eg: Mandating PPE in certain workplaces
- **Population assessment:**
 - Studying & engaging with a community to better understand their needs
- **Promoting health:**
 - Promote improved health via policy, interventions, community organizing etc
- **Preparedness & Response Planning**
 - Eg: For natural disasters
 - Eg: For pandemics
 - Eg: For man-made disasters

People Have Different Concepts of Health:

- **Wellness:**
 - o State of dynamic physical, mental, social, and spiritual well-being that enables a person to achieve full potential and have an enjoyable life
- **Disease:**
 - o Abnormal, medically-defined changes in the structure or function of the human body
- **Illness:**
 - o An individual's experience or subjective perception of a lack of physical or mental well-being and consequent inability to function normally in social roles
- **Sickness:**
 - o Views the individual and their society hold towards a health condition, affecting their thoughts and actions
- **Impairment:**
 - o Any loss or abnormality of psychological, physiological, or anatomical structure or function
- **Disability:**
 - o Any restriction or lack of ability to perform an activity within the range considered normal for a human being

Disease Prevention Measurements:

- **DALY** = Disability Adjusted Life Years:
 - o An indicator of the time lived with a disability and the time lost due to premature mortality
- **YLL** = (years of life lost):
 - o Years Lost due to premature death
- **YLD** = Years Lost to Disability



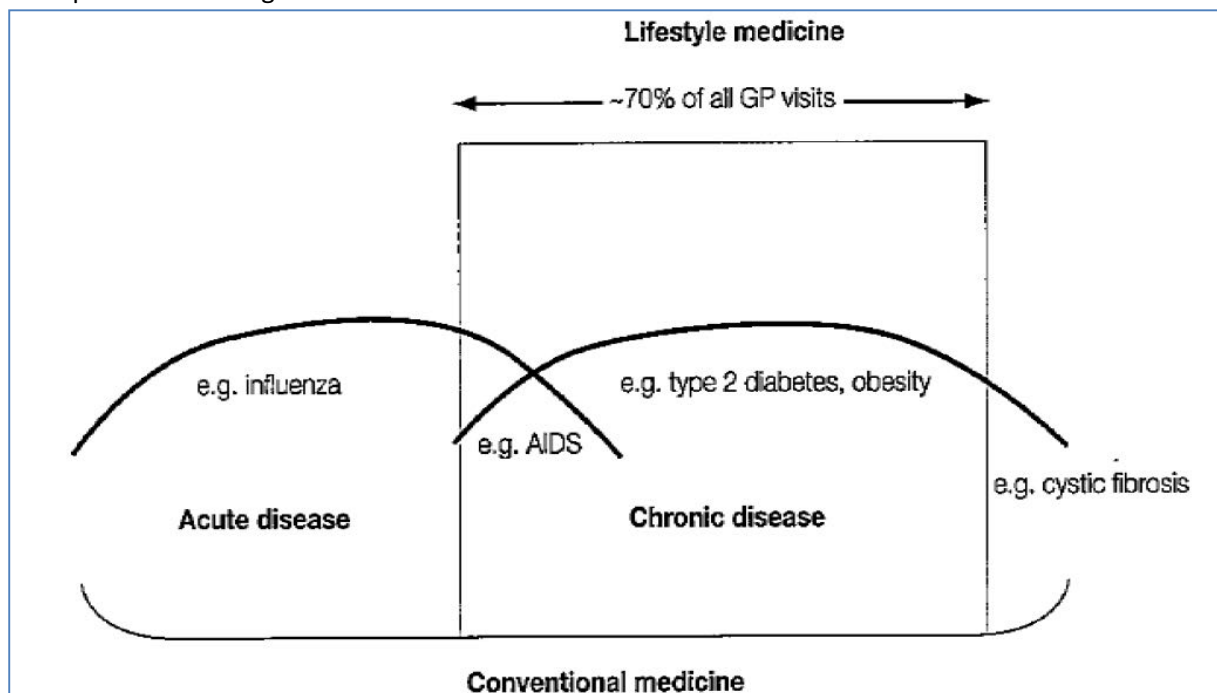
<https://nccid.ca/publications/understanding-summary-measures-used-to-estimate-the-burden-of-disease/>

Risk Factor Contribution to DALYs:

Risk factor	Percentage
Overweight	8.6
Tobacco smoking	7.9
High blood pressure	7.3
Physical inactivity	6.7
High cholesterol	6.1
Alcohol harm	3.8
Alcohol benefit	-1.8
Occupational exposure	2.0
Illicit drugs	1.9
Lack of fruit and vegetables	1.0

Why Prevent Disease?:

- Beneficial for patient
- Prevents disability/mortality
- Some diseases aren't curable (Eg: AIDs), but are preventable
- Cheaper than treating chronic disease – Some 70% of all GP visits are due to Chronic Disease:



Types of Prevention:

- **Primary Prevention:**
 - o Preventing the disease from developing in the first place by modifying removing risk factors
 - Eg: Changing eating habits to prevent obesity
 - Eg: Immunisation
 - Eg: Fitting vehicles with seat-belts
- **Secondary Prevention (Screening):**
 - o Prevent disease progression by early detection of disease & Early Intervention
 - Eg: Identifying someone with hypertension → early treatment to prevent CVD
 - Eg: Mammography
 - Eg: Routine pap smears
- **Tertiary Prevention:**
 - o Interventions to prevent or minimise complications with an Established disease
 - Eg: Bariatric surgery for morbidly obese people with poor diabetic control to avoid needing insulin therapy
 - Eg: Monitoring diabetes with HbA1c, eye exams, foot exams
 - Eg: Medications

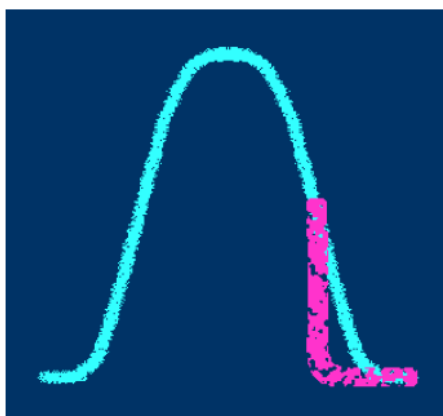
Screening:

- "Identifying individuals who are *More Likely To Be HELPED THAN HARMED* by further tests/treatment"
- **Criteria for Selecting Diseases to Screen for:**
 - o **1:** It should be an obvious burden for the Individual/Community
 - Deaths
 - Suffering
 - Economic/Social Costs
 - o **2:** It should have an initial Latent Stage, or be determined by risk factors, which can be detected by tests
 - o **3:** The Tests should be simple, safe, precise, socially-acceptable & validated
 - o **4:** Treatment/Intervention is crucial to prognosis
 - o **5:** Early intervention must provide a BETTER prognosis (Mortality/Morbidity/QOL)

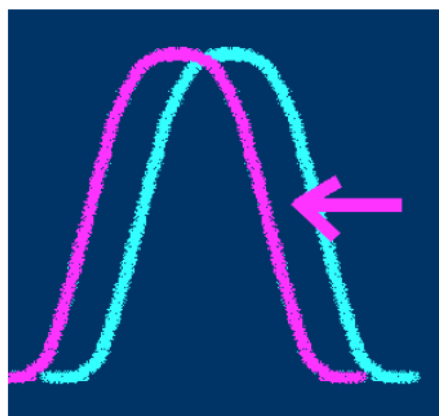
Prevention Strategies:

- **1: “High Risk” Prevention Strategies:**
 - Selecting individuals at high risk of a disease → Medical Intervention
 - Opportunistic Screening
 - **Advantages:**
 - Appropriate for the individual
 - Cost-Effective
 - Good Risk-Benefit Ratio
 - **Disadvantages:**
 - Problems with screening
 - Who
 - How
 - When
 - Borderline Cases
 - Behaviourally Inappropriate (Eg: Pap smears)
 - \$Costs\$
 - Difficult to predict the *Absolute Risk* of disease in an individual:
 - Some people with risk factors don’t get the disease
 - Many people with the disease, don’t have the risk factors
- **2: “Population” Prevention Strategies:**
 - Where you attempt shift the whole *distribution* of an exposure in a favourable direction by controlling the determinants of the disease (Environmental/Behavioural/etc)
 - Ie: Trying to reduce the underlying causes of a disease across an entire population
 - **Advantages:**
 - A small change can make a huge difference when it occurs across an entire population
 - **Disadvantages:**
 - Low Benefit-Risk ratio:
 - Limited benefit to the individual (Eg: Immunisation – even @ low risk of disease)
 - Poor motivation

High Risk & Population Approaches to Prevention



Truncate high risk end of exposure distribution (e.g. organise an obesity clinic).
Clinical approach to disease prevention.



Reduce a small amount of risk in a large number of people (e.g. reduce fat a little in fast-food outlets).
Lifestyle change plus environmental approach.

Commonplace Disease Prevention Practices:

- Pregnancy:

- **Folic Acid (Folate) Supplementation** – Prevents Neural Tube Defects (Eg: Spina Bifida)
- **Get Genetic Testing for Fragile X**
- **Check Rubella Immunity**
- **Stop Smoking**
- **Stop Drinking**
- **Prevent Listeriosis** – A bacterial infection typically contracted from 'Deli-foods'
 - Unpasteurised Dairy Products
 - Soft Cheeses
 - Cold Meats
 - Raw Seafood
 - Maintain good Personal/Food-Hygiene

TIPS FOR HEALTHY PREGNANCY



REGULAR
PRENATAL
CHECKS



MONITOR
BLOOD
PRESSURE



PROTEIN
& SUGAR
URINE TEST



SUGAR
BLOOD TEST



TRACK
WEIGHT GAIN



EARLY TREATMENT
REDUCES RISK OF
COMPLICATIONS
DURING DELIVERY



TAKE
VITAMINS



TAKE
NUTRITIOUS
FOOD



REGULAR
EXERCISE

HAVE A POSITIVE
STATE OF MIND



AND ENJOY
YOUR PREGNANCY!

VACCINE

WHY?



Mother's immunity is passed to baby during pregnancy and protects a newborn during the first few months of life before he/she is vaccinated.

Certain vaccines that contain live strains of viruses and may pose health risks for women who are pregnant.



WHO?



FATHER



MOTHER



Individuals who will have
CLOSE CONTACT
with infant

WHICH?



DEPENDS ON FACTORS

- Lifestyle
- High-risk conditions (if any)
- Travel type or location
- Previous vaccination

WHEN?



**BEFORE
YOU PLAN TO
HAVE A BABY**



Consult your doctor for further information.

<https://www.mymumnbaby.com/pregnancy-your-age/>

Breast Cancer:

- Screening not necessary until 50yrs
- If 50+, screen every 2 years – Mammogram & Breast Examination

LET'S BEAT BREAST CANCER

1 in 7
IDAHO WOMEN ARE AT RISK OF BEING DIAGNOSED WITH BREAST CANCER

EARLY DETECTION
IS KEY TO SURVIVING BREAST CANCER

IDAHO RANKS **50TH** IN THE NATION FOR SCREENING

SCHEDULE A MAMMOGRAM

BEGINNING NO LATER THAN AGE **50*** EVERY **1-2** YEARS MAKE AN APPOINTMENT TODAY

*If you have a family history of breast cancer, talk to your healthcare provider about when you should start screening.

BREAST CANCER SCREENINGS ARE COVERED IN FULL BY MOST HEALTH INSURANCE PLANS.
If you're uninsured, help is available.

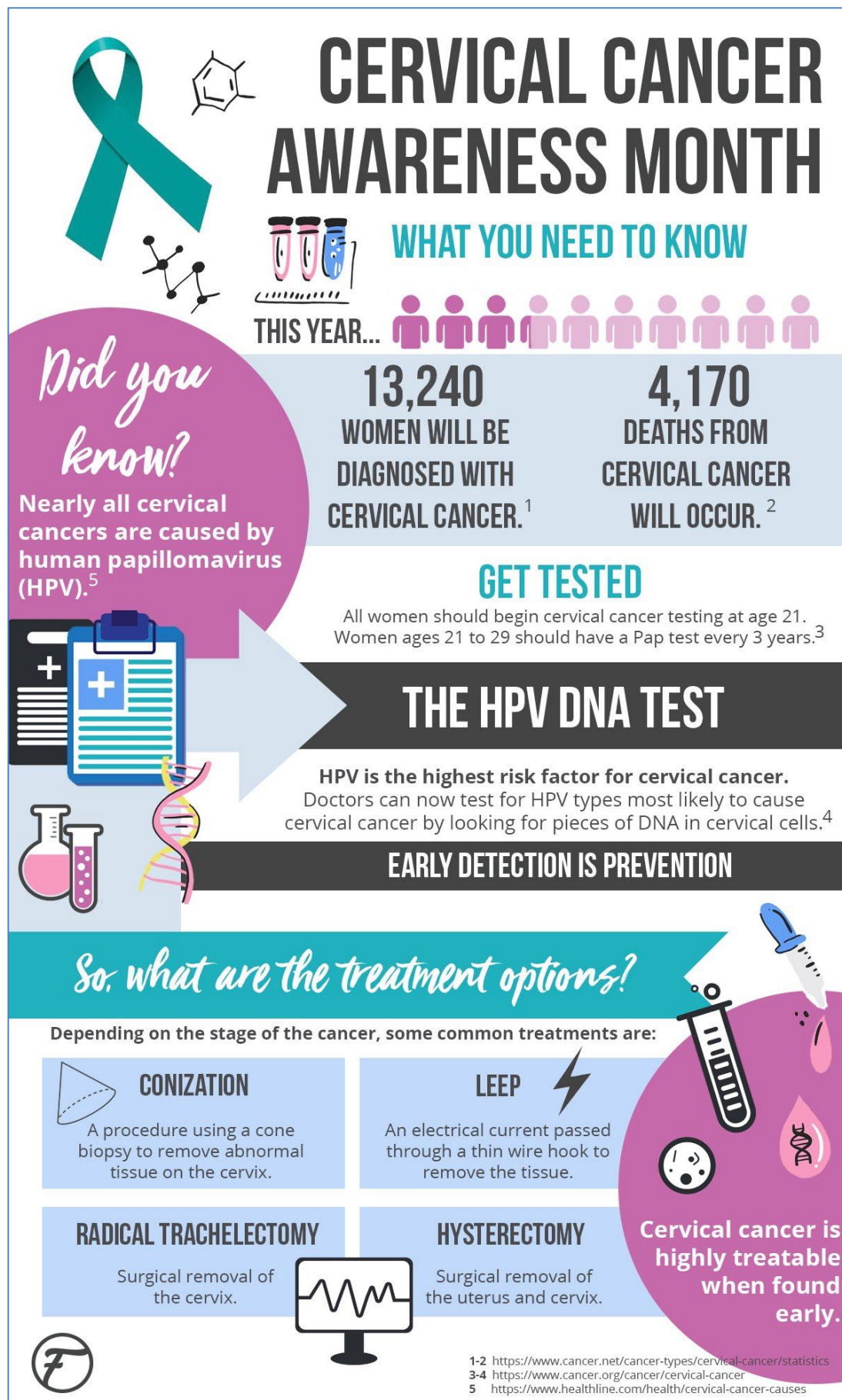
"I really didn't want to do a mammogram, but I scheduled one anyway thanks to my doctor's recommendation. I'm so glad I did! My cancer was in Stage 1 when we found it. Now I'm cancer-free." — **Cynthia**

FOR MORE INFORMATION ABOUT CANCER IN IDAHO AND HELP PAYING FOR SCREENINGS, VISIT HEALTHANDWELFARE.IDAHO.GOV/CANCER.

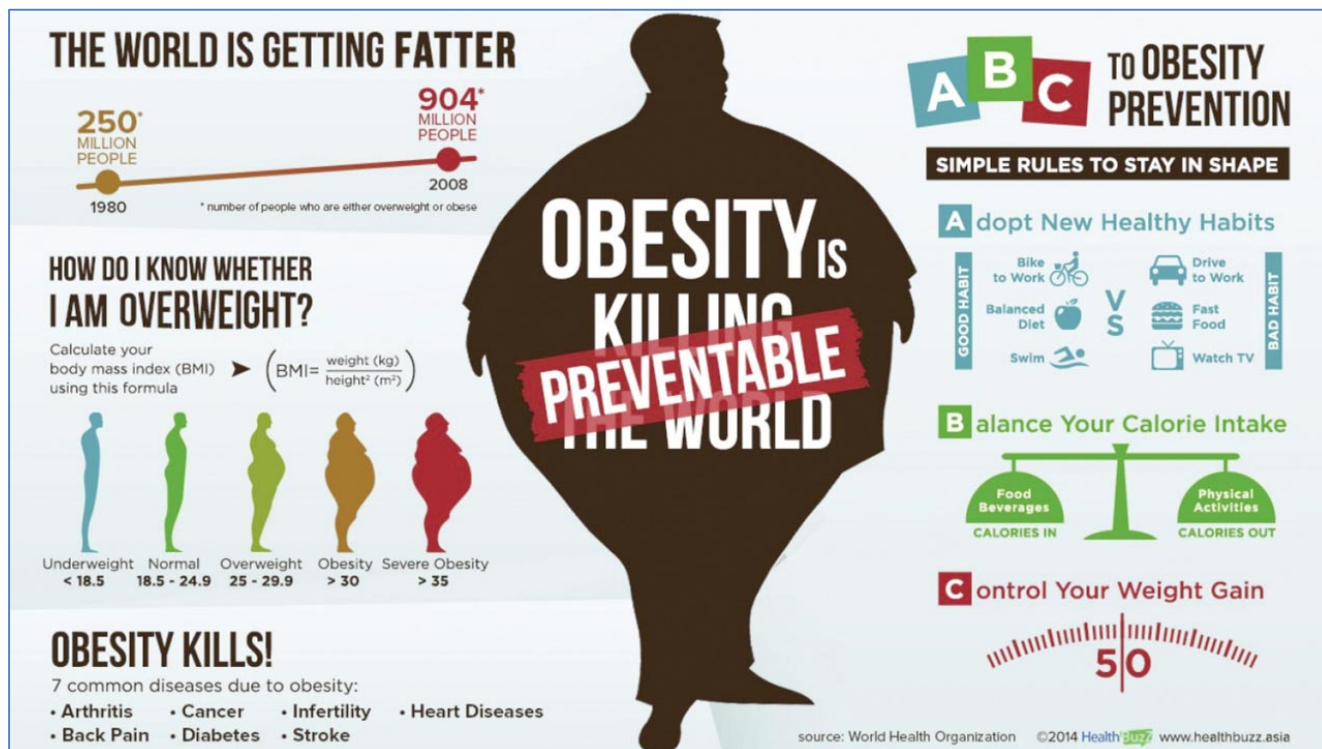
<https://healthtools.dhw.idaho.gov/products/copy-of-cancer-infographic-poster-breast-cancer-2-max-10-per-order>

Cervical Cancer:

- Screen 2yrly
- Pap-smear
- Immunisation (Gardasil)

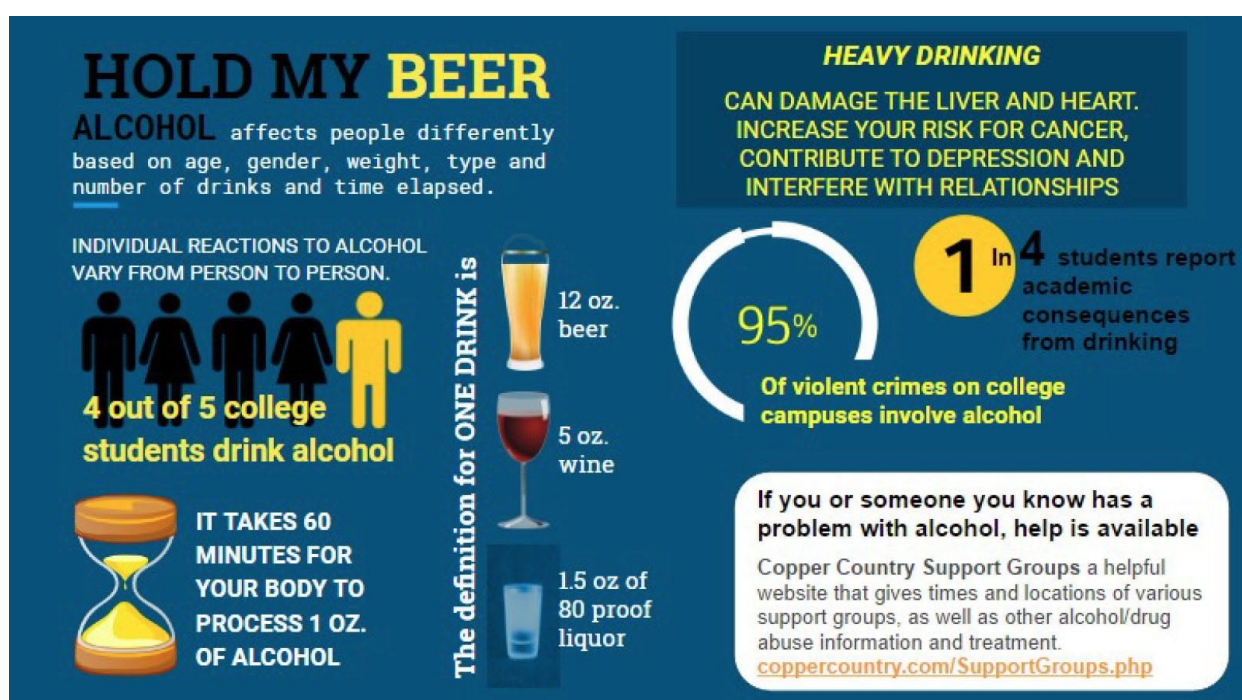


- **Overweight & Obesity:**
 - Screen 12mthly for:
 - Blood Pressure
 - Cholesterol & Lipids
 - Diabetes
 - Screen 6mthly for:
 - Nutritional Advice
 - Ideal Waist Circumference = <94cm



Source: WHO via www.healthbuzz.asia

- **Alcohol:**
 - Reduce consumption as much as possible
 - Ensure 2x 'Alcohol-Free Days' per week



<https://www.finlandia.edu/news/sophomore-nursing-class-hosting-events-alcohol-awareness-month/>

- **Falls:**

- **Common in elderly due to:**
 - Vision Problems (Eg: Glaucoma – screen @ 55+ yrs)
 - Inner Ear Problems → ↓ Balance
 - Multiple Meds → Nauseating
 - Gait
- **Screening Procedures:**
 - Check all of above
 - Suggest Installation of handles/non-slip surfaces in their home
 - Suggest having a carer



Fall Prevention

Help Prevent and Reduce Falls



1 in 4 Americans
over the age of 65 fall each year



Every 11 seconds
an older adult is in the emergency room
being treated for a fall



Every 19 minutes
an elderly patient dies
from a fall

In 2015, medical costs for falls totaled more than \$50 billion

Common Causes for Falls



- Low blood pressure
- Poor balance and impaired mobility
- Limited physical-activity endurance and muscle weakness
- Foot problems that cause pain
- Impaired vision

5 Simple Steps to Help Prevent Falls



Stay Active

A good exercise program includes activities to improve balance, strength, and flexibility



Speak up

Talk to your doctor about taking supplements to improve bone, muscle, and nerve health



Eye Exam

Get your vision checked annually as poor vision can increase the likelihood of falling



Keep Your Home Safe

- Install handrails and grab bars
- Use non-slip mats
- Wear socks with a non-slip tread
- Wear slip resistant soles in the shower

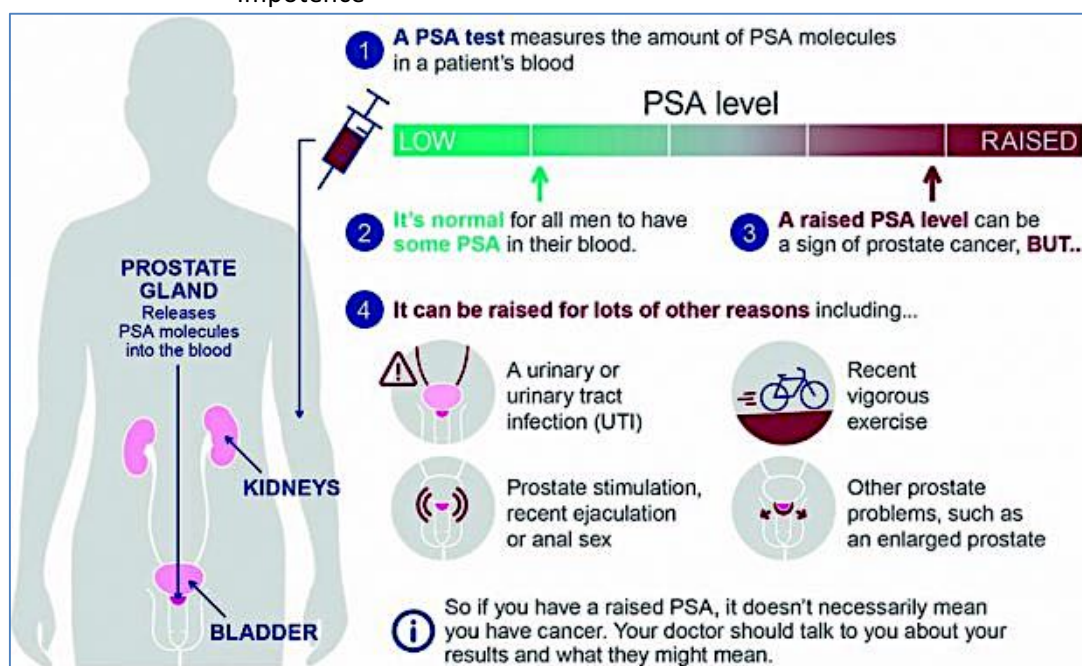


Learn the Facts

In reality most falls can be prevented. Learn more about fall prevention and debunking the myths of falls among the elderly

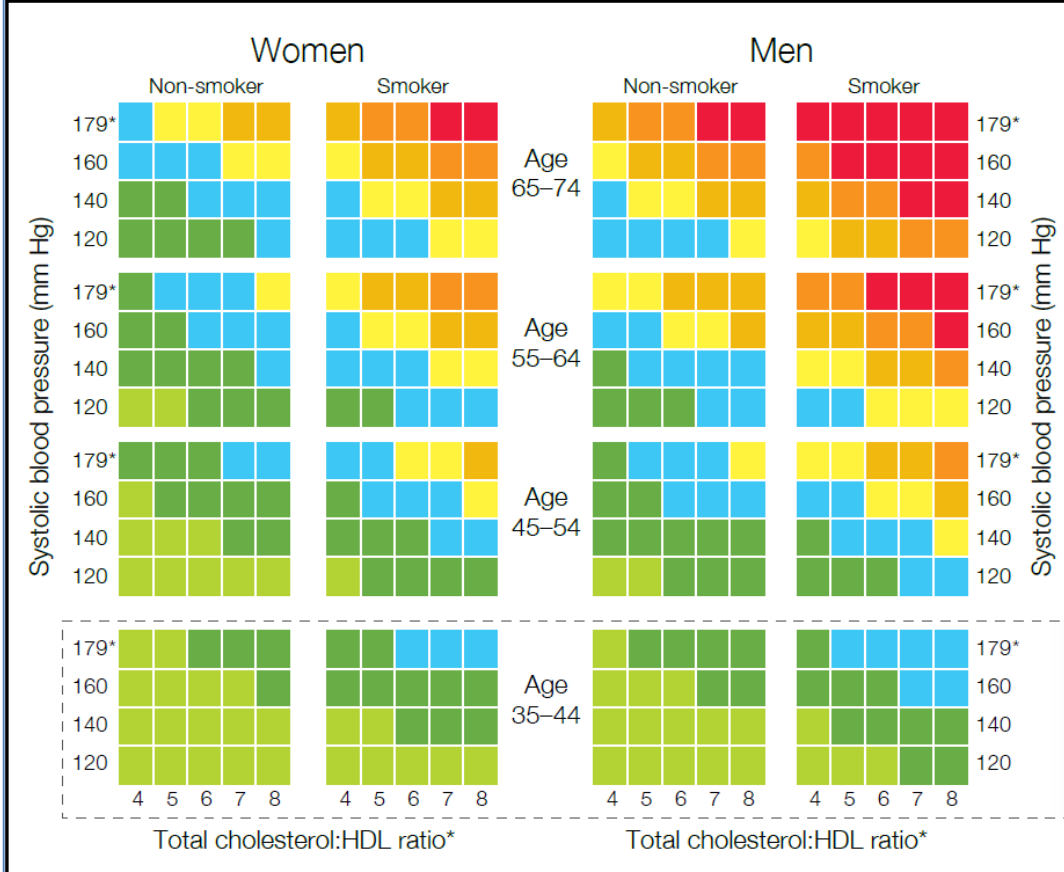
- **Prostate Cancer:**

- **Risk Factors:**
 - #1 – Family History (The closer the affected relatives, the more likely one is to be affected)
 - Age – Typically seen in men over 50yrs (40% of men over 50yrs have prostate cancer)
 - Race: Highest = African American; Lowest = Chinese
- **Screen 2yrlly for 50⁺yrs**
- **Note:** 85% of cases have a 20yr survival rate with no treatment → Most die with it, not of it
- **Note:** Early surgery only saves 1:12 (NNT=12)
- **Screening Procedures:**
 - Digital Rectal Exam (DRE)
 - Prostate Specific Antigen (PSA) blood test
- **PSA Screening:**
 - **↑PSA occurs with:**
 - Carcinoma – (The purpose of the test)
 - However, also with:
 - Benign prostatic hypertrophy
 - Prostatitis/UTI
 - Recent Ejaculation
 - Bike Riding
 - **Sensitivity** = Relatively Sensitive (A Few false negatives)
 - Ie: ≈99% of *Normal* PSAs are Not Cancer
 - **Specificity** = Poorly Specific (Many false positives)
 - Ie: ≈33% of *Abnormal* PSAs Are Cancer
 - Note: False positives → Anxiety, further tests & possible treatment → ↓QOL
- **Best Treatment:**
 - Uncertain; Can't predict who will benefit from early treatment (Ie: No way of knowing which cancers are fatal)
 - **Options:**
 - Wait & Watch
 - Radical Prostatectomy
 - Radiation Therapy
 - Hormone Therapy
 - **Side Effects:**
 - Infection
 - Urinary Incontinence (Very Common)
 - Chronic Diarrhoea & Rectal Bleeding (From radiation)
 - Impotence



Cardiovascular Risk Calculators:

People without diabetes

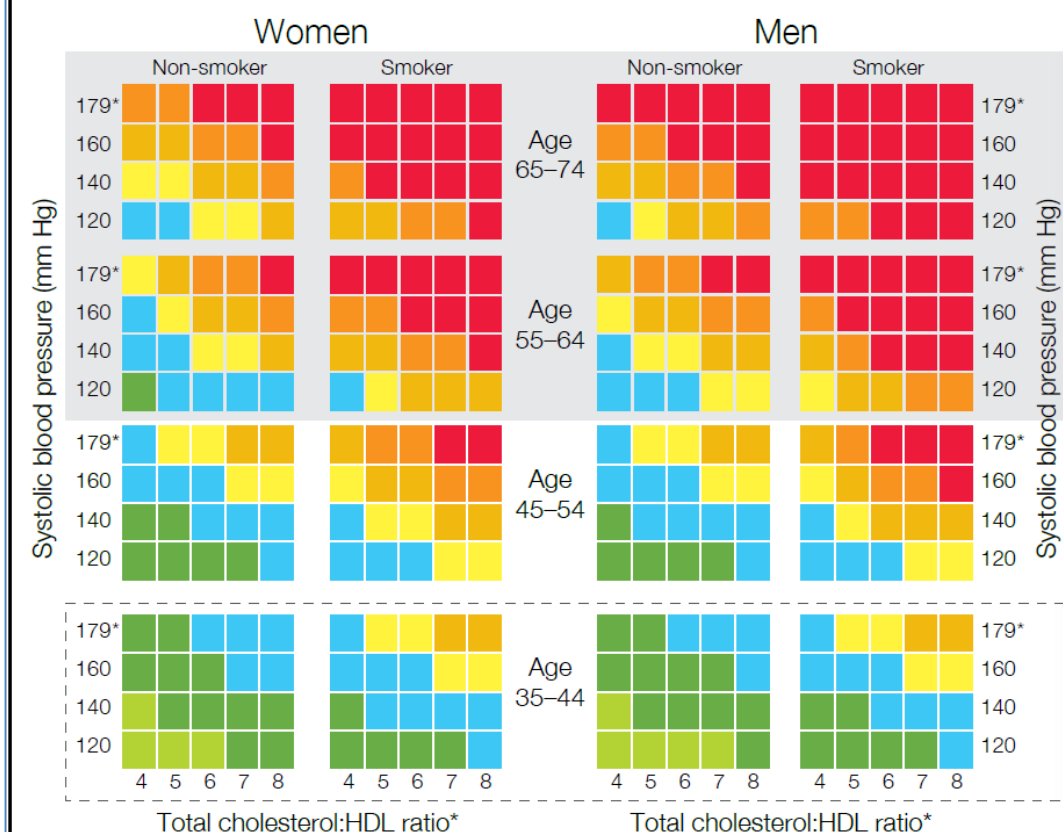


* In accordance with Australian guidelines, patients with systolic blood pressure ≥ 180 mm Hg, or a total cholesterol of > 7.5 mmol/L, should be considered at increased absolute risk of CVD.

Risk level for 5-year cardiovascular (CVD) risk

High risk	Moderate risk	Low risk
≥ 30% 25-29% 20-24% 16-19%	10-15%	5-9% < 5%

People with diabetes



Adults over the age of 60 with diabetes are equivalent to high risk (> 15%), regardless of their calculated risk level. Nevertheless, reductions in risk factors in this age group can still lower overall absolute risk.

Charts in this age bracket are for use in Aboriginal and Torres Strait Islander populations only.

*In accordance with Australian guidelines, patients with systolic blood pressure ≥ 180 mm Hg, or a total cholesterol of > 7.5 mmol/L, should be considered at increased absolute risk of CVD.

Risk level for 5-year cardiovascular (CVD) risk

High risk	Moderate risk	Low risk
$\geq 30\%$ 25–29% 20–24% 16–19%	10–15 %	5–9% < 5%

<https://www.summithealth.org.au/wp-content/uploads/2013/03/chronic-disease-referral-pathways.pdf>

Disease Prevention Success Stories:

- **Vaccination →**
 - Eradication of Smallpox/Polio
 - Control of Measles/Rubella/Tetanus/HiB
- **Car Safety →**
 - Personal Behaviour Change (Seat-belts/Helmets/Drink-Driving)
 - ↑Engineering of Roads & Vehicles
 - → Large Reduction in Deaths
- **Occupational Hazards →**
 - Injury reductions due to legislation (Health & Safety at all sites/Smoking Ban)
 - → ↓ “Black Lung”/Asbestosis/Workplace Deaths/etc
- **Communicable Disease Control →**
 - Clean Water & sanitation
 - Antibiotics
 - Vector control
- **Cardiovascular Disease →**
 - Risk factor reduction
 - BP Control
 - Smoking Cessation
 - Earlier Detection
 - Safer, more-effective treatment
- **Food Safety →**
 - ↓Microbial Content (Eg: Pasteurisation)
 - ↑Nutritional Content (Eg: Food fortification – Eg: Iodised Table Salt)
 - Food safety legislation for handlers
 - Elimination of major nutritional deficiency diseases (Rickets, Goitre, Pellagra)
- **Mothers’ & Babies’ Health →**
 - Hygiene & Nutrition
 - Antibiotics
 - Access to healthcare
 - Technology
 - → Infant & maternal mortality decreased by 90%⁺
- **Fluoridation of Water →**
 - Entire population benefits
 - 40% Reduction in adult tooth-loss
 - 60% Reduction in Child Tooth Decay
- **Antismoking Campaigns →**
 - Recognition of tobacco as a health hazard
 - Legislation – Sales to minors, Advertising banned, No Smoking in Public/Work-Places
 - Smoking reduced from 40% → 20%

MEASURING HEALTH CONCEPTS:

MEASURING HEALTH CONCEPTS:

Sensitivity Vs Specificity:

- **Sensitivity:**

- The ability of a test to pick up people who truly have the disease of interest
- Ie: Few/No False Negatives
- **Calculating Sensitivity:**

$$\text{Sensitivity} = \frac{\text{Number of True Test Positives}}{\text{Actual Positives}}$$

Ie: The % of the diseased people that the test recognised as diseased

- **Specificity:**

- The ability of a test to weed out people who are truly Free of the disease of interest
- Ie: No False Positives
- **Calculating Specificity:**

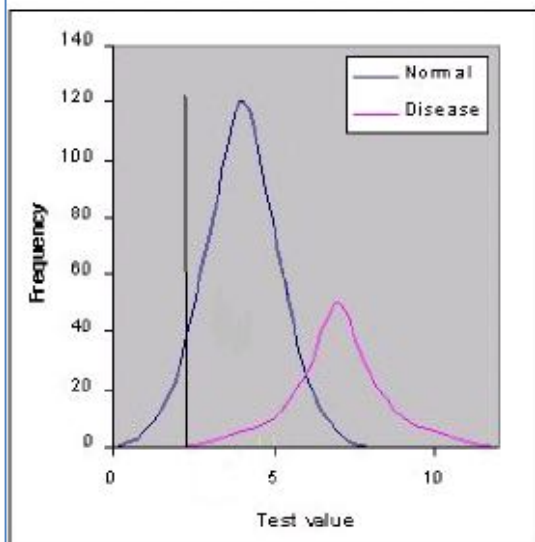
$$\text{Specificity} = \frac{\text{Number of True Test Negatives}}{\text{Actual Negatives}}$$

Ie: The % of the healthy people that the test recognised as healthy

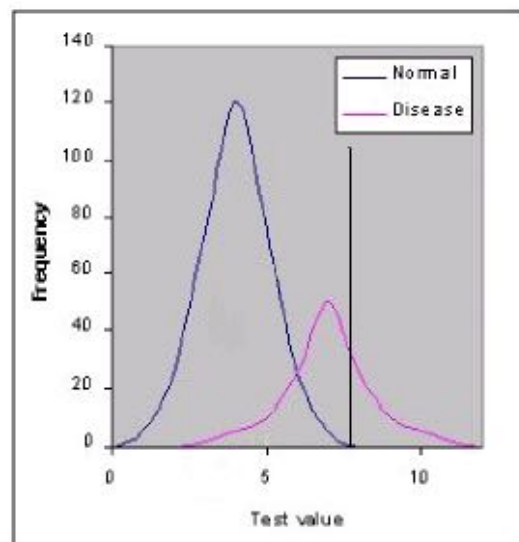
Why can't some tests be both 100% Sensitive AND Specific?:

- Certain diseases have a distribution in a population

This T4 distribution shows there is no point where the test is 100% sensitive and 100% specific



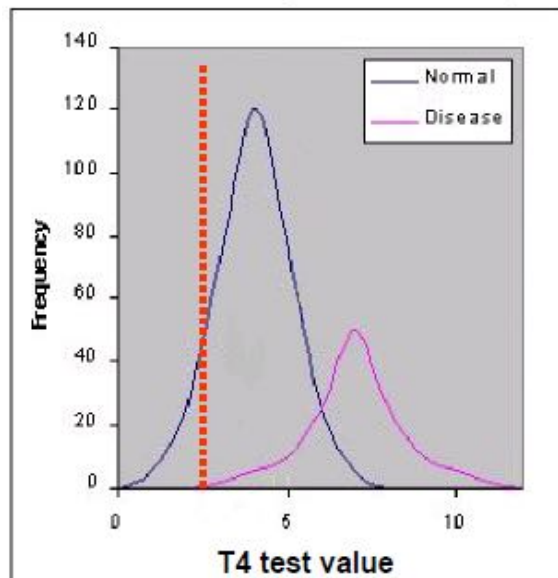
100% sensitive; lots of false positives



100% specific; lots of false negatives

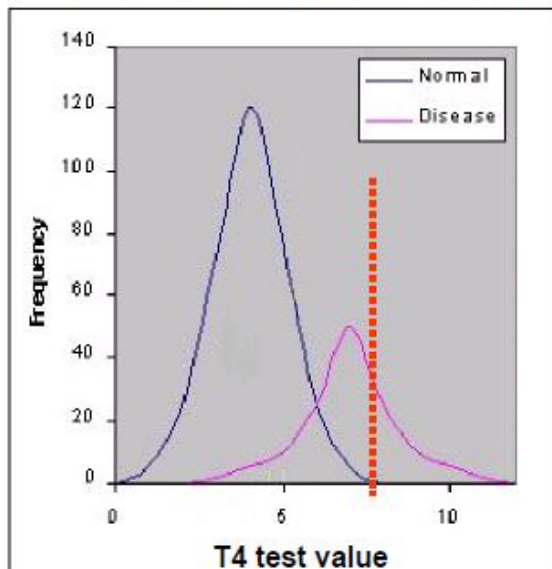
- **So where do you draw the line? Answer – Depends on the disease in question**
- **Note: When 'drawing the line', you trade Sensitivity for Specificity and vice versa:**
 - Eg: If the disease has extreme morbidity/mortality, and the treatment is cheap and harmless, then you want a highly *Sensitive* test to pick up every possible case
 - Eg: However, if the consequences of the disease are minor, but the treatment is extremely costly/invasive/risky, you want a highly *Selective* test so you only treat actual cases

The distribution of thyroid disease using T4



**Put the cut-off point here
... and everyone with the
disease will be picked up
(= 100% sensitivity)**

The distribution of thyroid disease using T4



**Put the cut-off point here
... and everyone free of
the disease will be
weeded out
= 100% specificity**

Positive Predictive Value (PPV):

- Tells us how likely a Positive Test will be a True Positive
 - o Ie: The % of Positives that were *True*
- **Calculating PPV:**

$$PPV = \frac{True\ Positives}{True\ Positives + False\ Positives}$$

Relative Risk: "The risk of getting a disease when comparing one group to another"

- Eg: Relative risk of lung-cancer in smokers is 2x that of non-smokers
- **Rate Ratio:**
 - o Derived from Cohort Studies
 - o Compares the incidence **rates** of a disease in 2 groups of people (With/Without Exposure)
 - o **Calculating Rate Ratio:**

$$Rate\ Ratio = \frac{Incidence\ Rate\ in\ Exposed}{Incidence\ Rate\ in\ Unexposed}$$

- **Odds Ratio:**
 - o Supposedly tells you what your **Odds** are of getting a disease if you are exposed to a certain risk factor
 - o **Calculating Odds Ratio:**

$$Odds\ Ratio = \frac{The\ \%\ of\ people\ with\ the\ disease\ who\ had\ Exposure}{The\ \%\ of\ people\ without\ the\ disease\ who\ had\ Exposure}$$

	With CHD	Without CHD	Total
Smokers	a. 80	b. 10	90
Non-smokers	c. 20	d. 90	110
Total	100	100	200

OR = $\frac{a/c}{b/d}$ (odds people with CHD were smokers compared to non-smokers)
(odds people without CHD were smokers compared to non-smokers)

Absolute Risk:

- The actual risk of getting the disease, over a period of time
- Eg: Assuming you live to 90, your risk of getting breast cancer is ≈12%
- This is based on the prevalence of that specific disease in that population

Numbers Needed to Treat (NNT):

- The number of patients you need to treat to prevent one additional bad outcome
- Gives insight to the effectiveness & cost of a treatment
- Ideal NNT = 1 Ie: Everyone treated improves
- Eg: A drug with an NNT of 5 → you have to treat 5 people with the drug to get 1 cure

Validity & Reliability:

- **Validity** = The ability of a test to test what it's meant to be testing
 - o (Eg: How well IQ measures intelligence)
- **Reliable** = The degree of consistency of results despite changes in external factors
 - o (Eg: Different testers, different times, different places)

HEALTH BEHAVIOUR

HEALTH BEHAVIOUR

Health Promotion:

- Promote healthy behaviours through education
- Monitor individual wellbeing and risk-taking behaviours
- **Doctor's Role:**
 - Advise the most effective way to a healthy lifestyle
 - Monitor patient's behaviour
 - Skill training
 - Reinforcement of behaviour
 - Role modelling
 - Provision of information
 - Give "expert" opinions
- **Psychologist's Role:**
 - Develop interventions at individual & community levels
- **Mass Media's Role:**
 - Educate people about health risks (AIDs, smoking, alcohol)
- **Role of Legislation:**
 - Rules enforcing healthy behaviour (seatbelts/drink-driving/smoking)

Role of Behavioural Factors in Disease & Disorder:

- **Health Behaviours:**
 - Behaviours that promote/maintain individual wellbeing (Eg: Exercise/healthy diet)
 - Either **Habitual or Intentional**
 - **Health Habits:**
 - Seatbelt/cleaning teeth etc
- **Risk Behaviours:**
 - Behaviours which are proven to increase susceptibility to a specific disease/illness

Primary Prevention:

- Instilling good health habits & changing poor ones
- **Strategies:**
 - Change current health behaviour
 - Prevent the uptake of poor health habits in the first place

Obstacles to Changing Health Behaviours:

- Pleasure (being high)
- Addiction (drugs)
- Behaviour is now habitual
- Relapse
- Factors influencing behaviour (stress → smoking)

Unrealistic Optimism & Irrational Risk Perception:

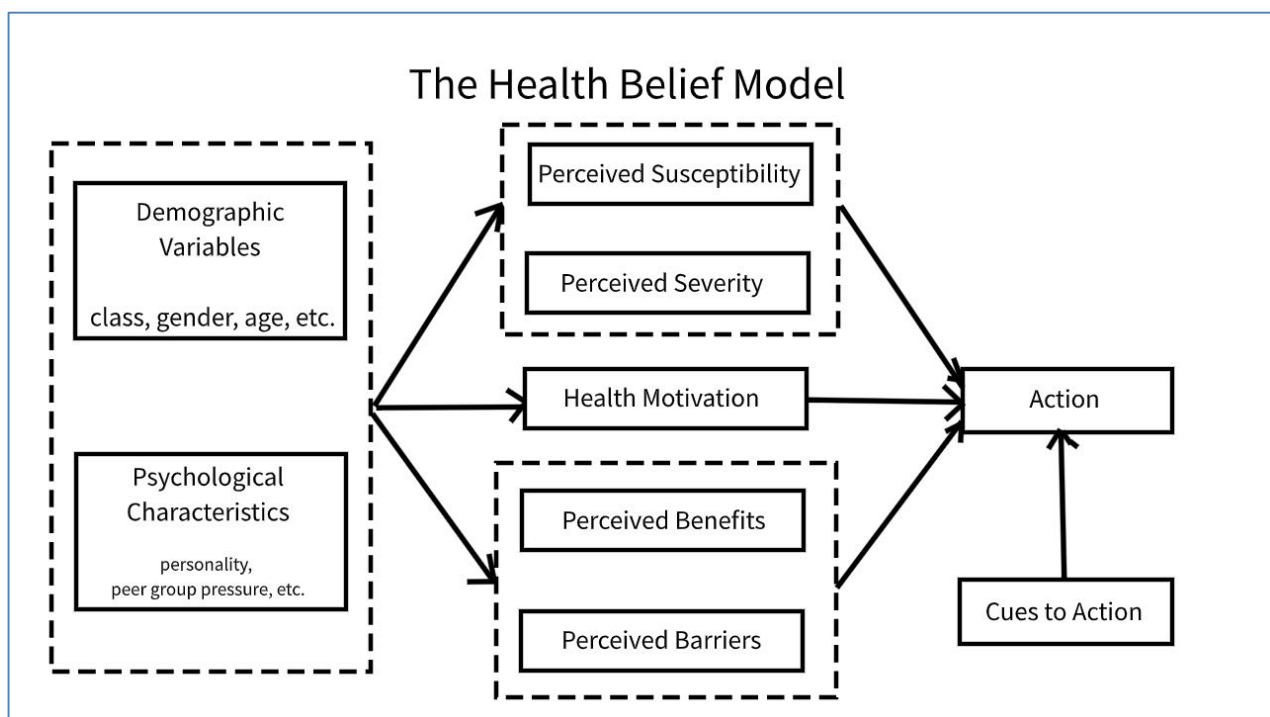
- Inaccurate perceptions of risk
- Inaccurate perceptions of susceptibility
- Lack of personal experience with problem
- There's no problem now so there won't be in the future
- Belief that problem is infrequent

Components of Motivation:

- **Patient must be...**
 - **Willing**
 - Perceived importance of change
 - **Able**
 - Self-efficacy
 - **& Ready**
 - Motivations to change outweigh motivations not to change

Health Belief Model (Factors determining health behaviour):

1. **Perceived Threat:**
 - a. **Perceived Susceptibility:** One's perceived risk of contracting a health condition
 - b. **Perceived Severity:** One's opinion of the seriousness of getting/having the condition
2. **Perceived Benefits:** The believed effectiveness of preventative measures
3. **Perceived Barriers:** Potential negative consequences of taking the preventative measures
4. **Cues To Action:** Events (symptoms/media/social) that motivate people to take action
5. **Self-Efficacy:** One's confidence in being able to undertake the preventative measure successfully

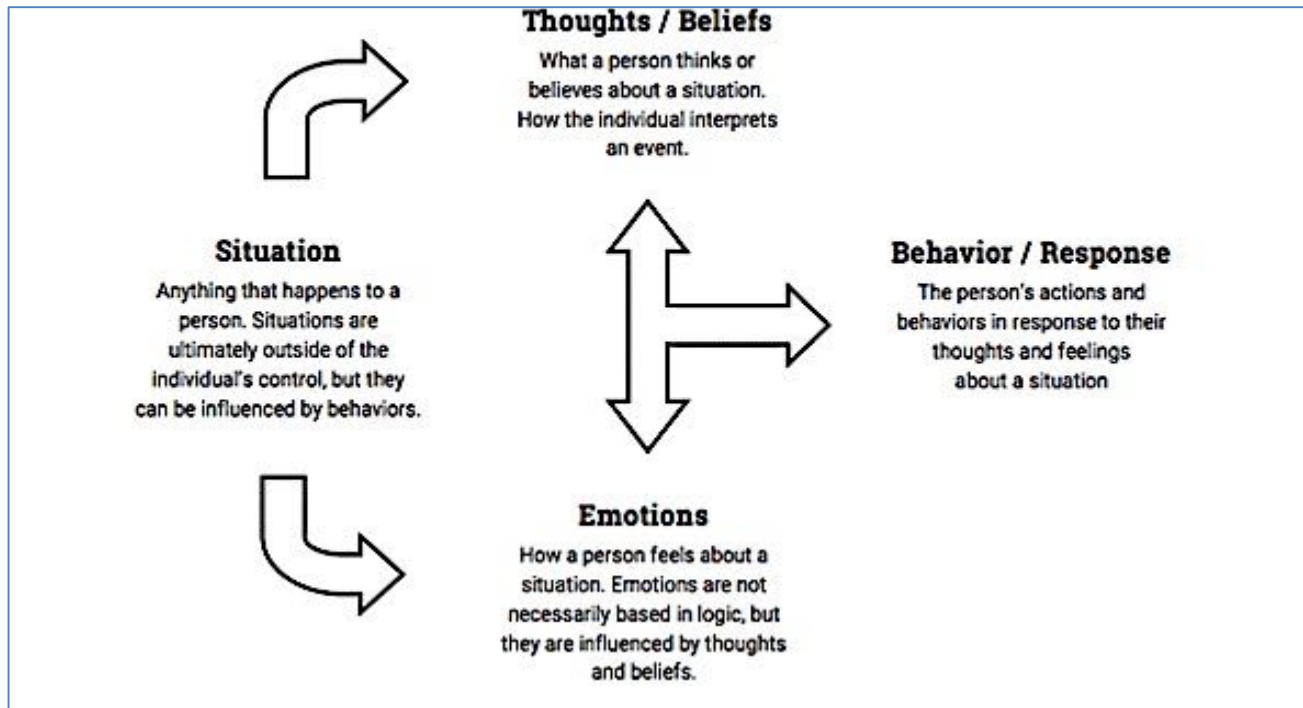


Theory of Planned Behaviour

- Assumes that behaviour is a direct result of a person's intentions
- **3 Behavioural Intentions:**
 1. **Attitude Toward Behaviour**
 - Evaluation of outcomes: Positive / Negative
 - "If I diet, I'll lose weight, improve my health & be more attractive"
 - Being healthy & looking good are desirable
 2. **Subjective Norms**
 - The individual's perception of social standard pressures
 - Pressures of significant others (family/friends/girlfriend) to change behaviour
 3. **Perceived Behavioural Control**
 - One's perceived confidence in being able to change their behaviour
 - "I think I can diet"
- **Results in an Intention:**
 - **Change** behaviour
 - Or **Continue** behaviour
 - **Results in Behaviour**

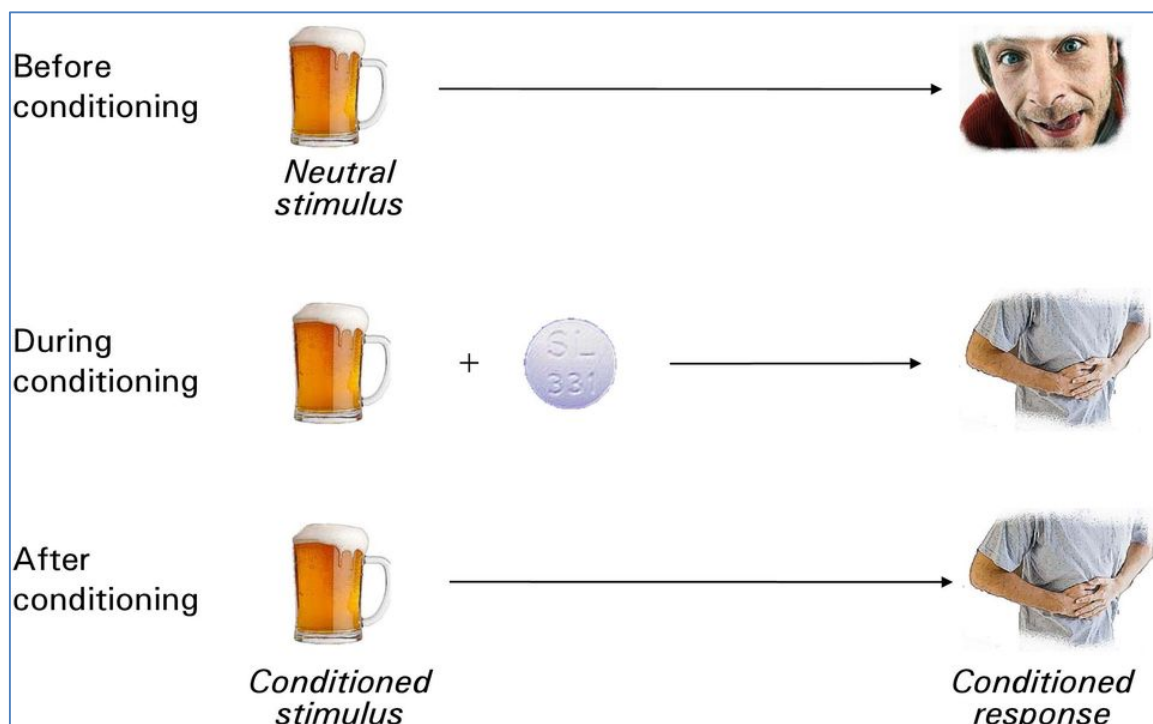
Cognitive-Behavioural Therapy

- **Behaviour** = the outcome of an interaction between the way one thinks and environmental events
- Behaviour is governed by the individual's expectations about the outcomes of engaging in it
 - Eg: Hot Stove Vs Smoking
- **Focus on:**
 - **The behaviour itself:** Ie: The conditions that elicit/maintain/& reinforce it
 - **Individual's Beliefs about their health habits:** "I will never be able to quit smoking"
 - **Self-observation & monitoring:** Record & chart behaviour



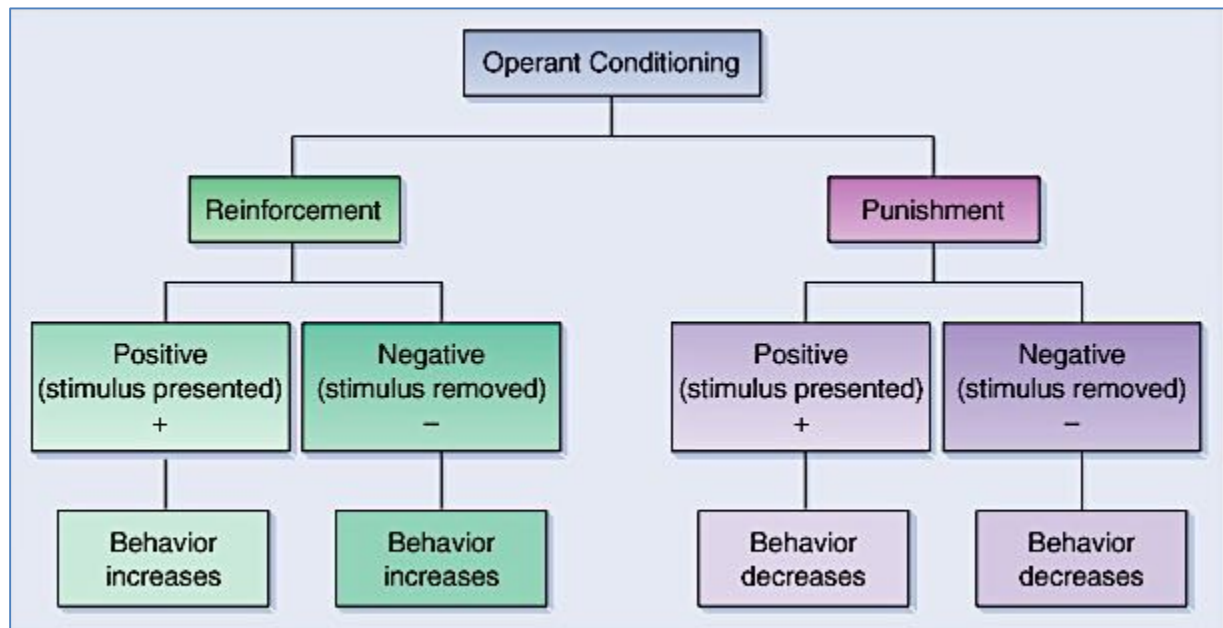
Classical Conditioning

- Where a natural stimulus acquires the ability to be evoked by another stimulus
 - Ie: Unacquiring a 'taste' for something
 - Eg: Using 'antabuse' to treat alcoholism:



Operant ('Instrumental') Conditioning

- Assumes that an individual's behaviour is a consequence of **positive** or **negative reinforcement**
- If positive, the behaviour is more likely to occur again
- If negative, it is less likely



CHANGING BEHAVIOUR

CHANGING BEHAVIOUR

How People Change:

- **Note: Patients don't change just because you say so**
 - o Ambivalence, Resistance & Defence Mechanisms are Normal
 - o **Intentional Change Occurs Gradually**
- **Requirements for Change:**
 - o Change in Thinking/Feeling about an Issue
 - o Planned Steps
 - o Time

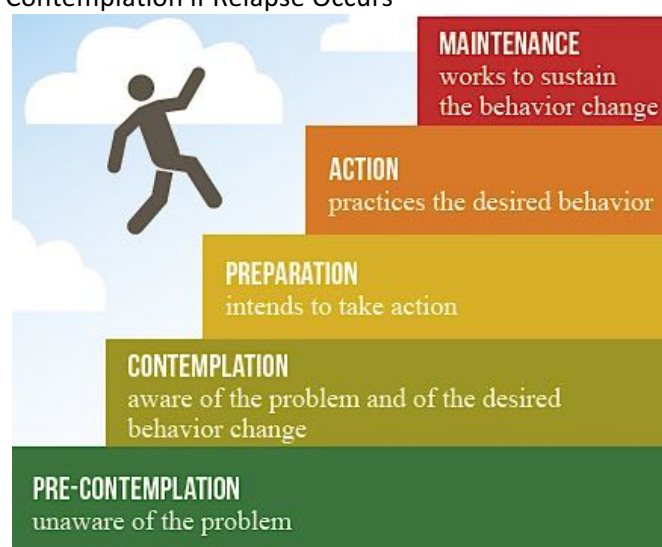
"SNAP": – Guidelines for Managing Lifestyle Risk Factors:

- **What are the Risk Factors?**
 - o **Smoking**
 - o **Nutrition**
 - o **Alcohol**
 - o **Physical Exercise**
- **5 A's Approach to SNAP:**
 - o **1: Ask:**
 - Ask which Risk Factors apply to Patient
 - Eg: Do you Smoke/Eat Healthily/Drink/Exercise?
 - o **2: Assess:**
 - Assess Level of Risk & Relevance to Patient's Health
 - Ie: Behaviour History (Smoking/Diet/Drinking/Exercise History)
 - BMI
 - ***Cardiovascular Risk Calculator** – Work out absolute risk level for CVD
 - Assess Readiness to Change
 - o **3: Advise:**
 - Advise with Written Information (Eg: Pamphlets)
 - Advise with a Lifestyle Prescription (Life Script)
 - Advise with a Brief Intervention & Motivational Interviewing
 - o **4: Assist:**
 - Assist with Pharmacotherapy
 - Assist with Self-Monitoring (Suggest Keeping a Diary)
 - o **5: Arrange:**
 - Arrange Referral to:
 - Specialist Services (Eg: Dietician/Exercise Physiologist/'ATODs')
 - o Note: ATODs = Alcohol, Tobacco & Other Drugs
 - Support Groups
 - Helplines
 - Counselling
 - Arrange Follow-Up

Risk factor	Prevalence, %
Smoking	15.9
Poor diet	85.5
Excess alcohol consumption	10.1
Insufficient physical activity	62.3

A Useful Tool: "The 5 Stages of Change Model":

- **1: Precontemplation:**
 - No intention to change behaviour
 - **Precontemplation → Contemplation:**
 - Make the patient aware of the problem (Link their Behaviour to their Health)
 - Encourage them to take ownership of the problem
 - Explain the Negative Aspects of Problem (Convince patient that the behaviour *is* a problem)
- **2: Contemplation:**
 - Person is thinking about changing behaviour
 - **Contemplation → Preparation:**
 - Get patient to Think How the Behaviour is Affecting Others
 - Change how they think & feel about the Issue
 - Note: Pushing People to Change can be Counterproductive → Resentment
 - 3 Strong Motivators:
 - Health
 - Money
 - Relationships
- **3: Preparation:**
 - Person prepares to make the change:
 - **Preparation → Action:**
 - Gathers information
 - Finds out how to achieve the change
 - Set Firm Goals & Priorities
 - Acquiring Skills Necessary for change
- **4: Action:**
 - Person makes changes (may be small steps at first)
 - **Action → Maintenance:**
 - Self-Efficacy is very important
 - Keep focussed
 - Acknowledge that Change is Difficult & Potential Relapse is Normal
- **5: Maintenance:**
 - Consistently practices new/alterd behaviour
 - Acknowledge that Change is Difficult & Potential Relapse is Normal
- **//Relapse:**
 - Person relapses back to original behaviour
 - Move back to Contemplation if Relapse Occurs



Relapse Prevention

- Common with addictive disorders (smoking, drinking, gambling, etc)
- More likely to occur in times of stress, anxiety, depression
- Once relapse has occurred, it is just as hard to 'quit' the 2nd time as it was for the 1st

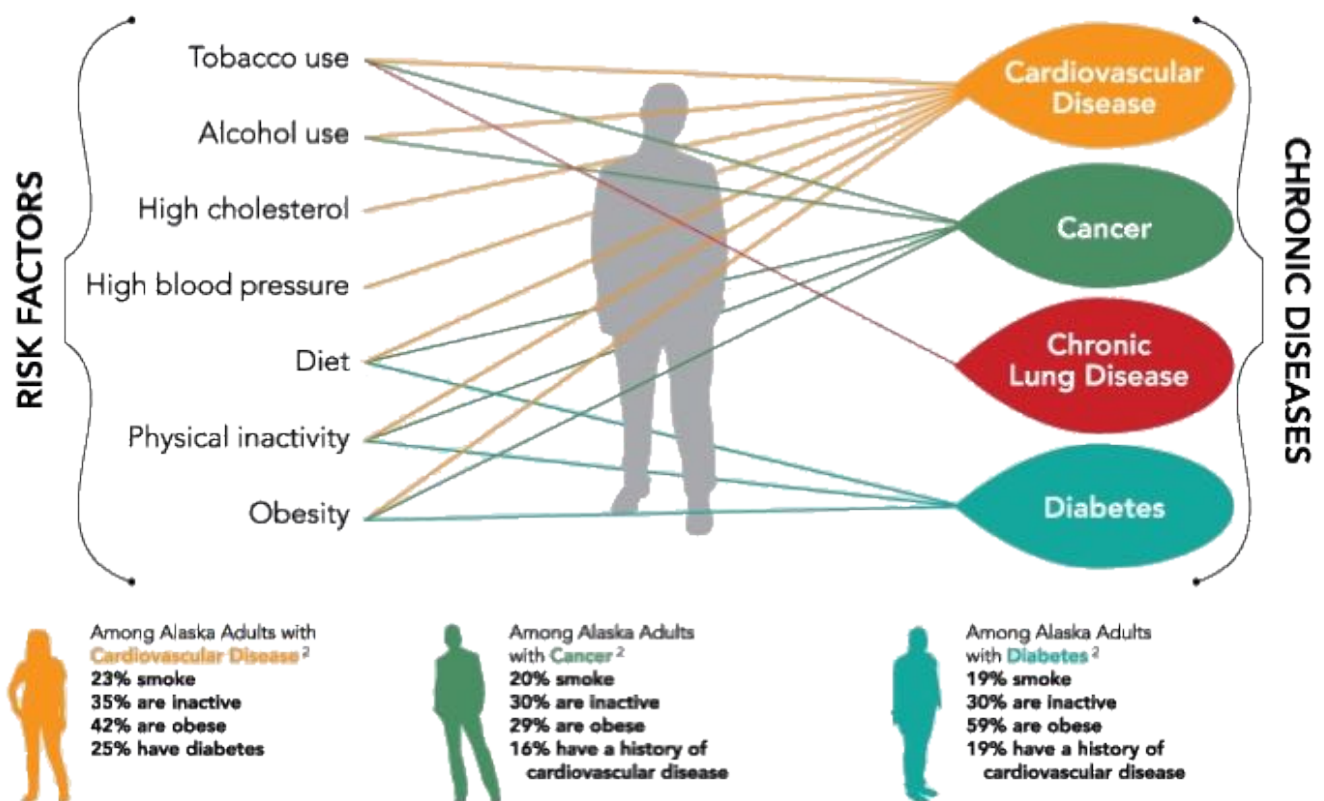
CHRONIC DISEASE & RISK FACTORS

CHRONIC DISEASE & RISK FACTORS

Chronic Disease:

- **Definition:**
 - **A Disease with One/More of the Following Characteristics:**
 - It is Permanent (I.e: Incurable) and Leaves Residual Disability (Morbidity)
 - Caused by Non-Reversible Pathological Alteration
 - Requires long-term Observation/Management /Care
- **Biggest Contributors to Burden of Chronic Disease:**
 - Cardiovascular Disease
 - Anxiety/Depression
 - Diabetes
 - Chronic Kidney Disease
- **Causes of Chronic Disease:**
 - **Patients Presenting to Doctors:**
 - ≈20% are Smoking (Decreasing)
 - ≈55% are Overweight/Obese (Increasing)
 - ≈65% Do Less than Recommended Levels of Exercise (30mins x 5days/week)
 - ≈25% Drink at Risky Levels
 - **Risk Factors:**
 - Risk Factors are often Associated with Many Diseases
 - Risk Factors shouldn't be considered in Isolation
 - Risk Factors Interact → Multiplies Risk
 - Most Risk Factors are Completely Avoidable

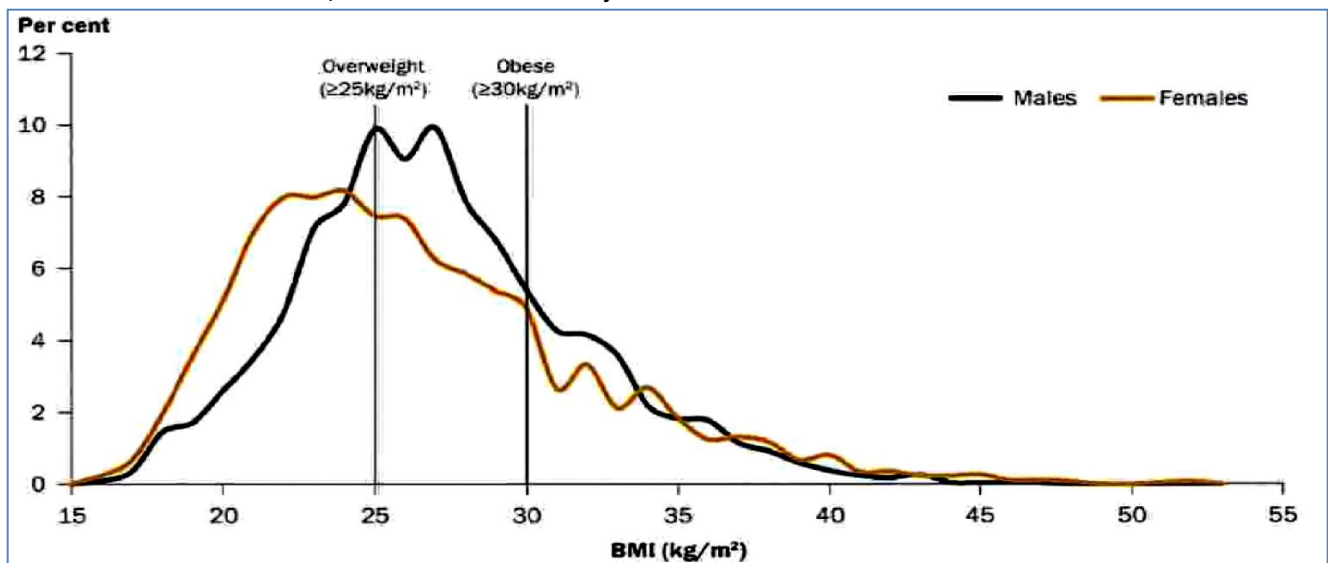
The Whole Person: The Web of Chronic Disease'



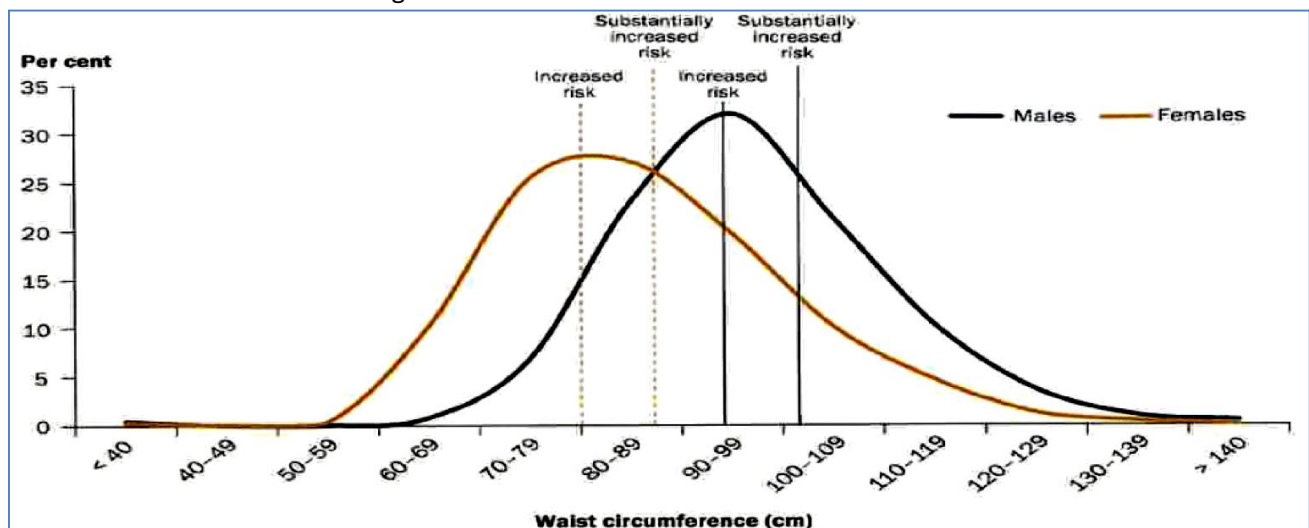
<https://dhss.alaska.gov/dph/Chronic/Pages/Publications/Default.aspx>

Overweight & Obesity:

- **Trend:**
 - ≈55% are Overweight/Obese
 - Rates are Increasing in first world countries
- **BMI:**
 - **Calculation:**
 - Kg/Height in m^2
 - **Ranges:**
 - Underweight <18.5
 - Normal 18.5 – 25
 - Overweight 25 – 30
 - Obese >30
 - **Limitations:**
 - Limited Sensitivity – Some people who are clearly overweight may be tall → False Negatives
 - Limited Specificity – Extremely muscular people will have a high BMI → False Positives
 - Hence, should be used in Conjunction with Waist Circumference



- **Waist Circumference:**
 - Males:
 - > 94cm → Increased Risk
 - > 102cm → High Risk
 - Females:
 - >80cm → Increased Risk
 - >88cm → High Risk



Physical Inactivity:

- ie: Sedentary Lifestyle
- Recommended Levels of Exercise (30mins x 5days/week)
- It's estimated ≈65% of people don't do enough exercise
- Note: Sedentary Lifestyle increases with Age
- Associated with higher cholesterol levels and risk of type 2 diabetes



No, not literally equivalent to a pack of cigarettes; but illustrates risk
<https://mobile.twitter.com/avonhmo/status/1022838331053432832>

Poor Diet:

- Inadequate Fruit & Vegetable Intake
- Most prevalent in Low Socioeconomic Status groups



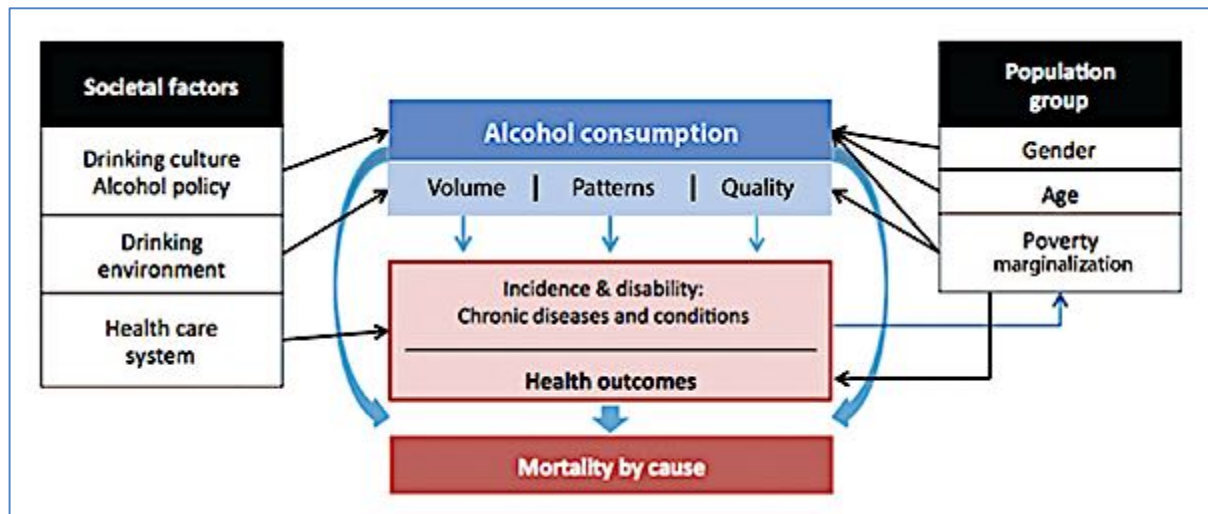
Tobacco Smoking:

- Smoking rates are ≈20% spread evenly across all age groups
- Most prevalent in Low Socioeconomic Status groups



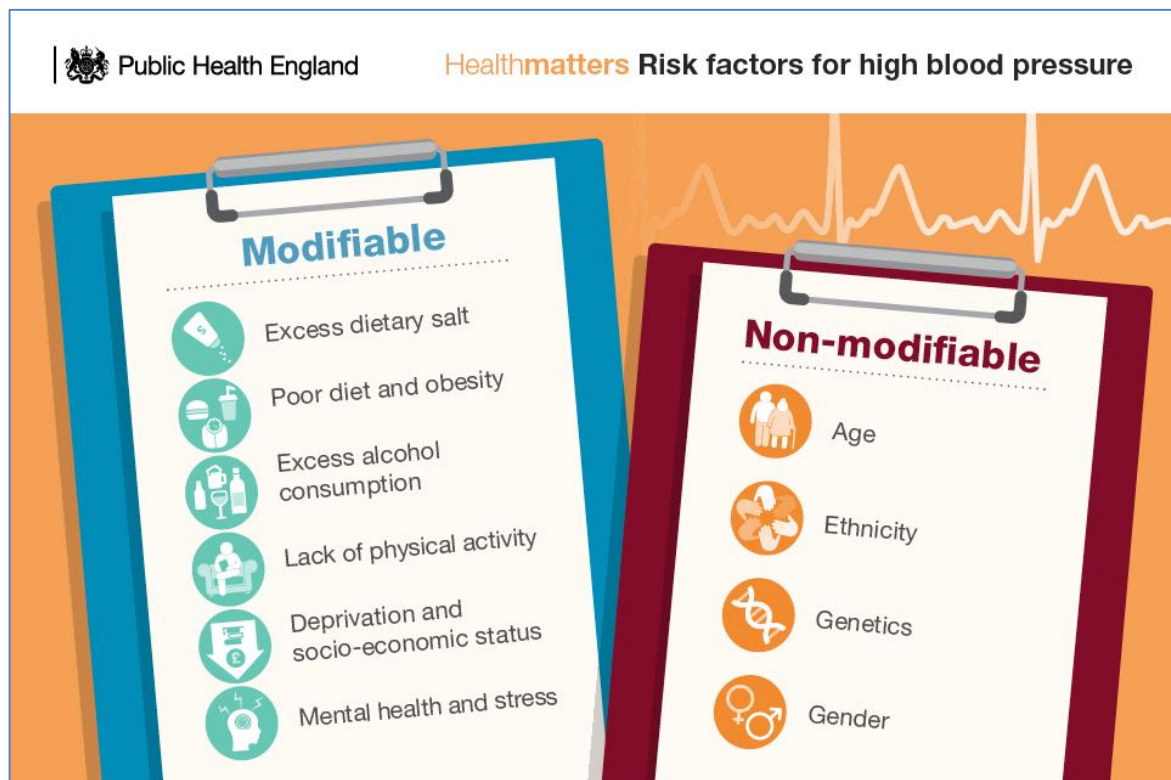
Excessive Alcohol:

- Approximately ≈25% Drink at Risky Levels
- Rates among adults are consistent with age
- Most prevalent in Rural & Remote Areas



High Blood Pressure:

- Approximately 30% of Adults over 25yrs
- Most prevalent in Males
- **What is High?**
 - o Systolic above 140mmHg
 - o Diastolic above 90mmHg

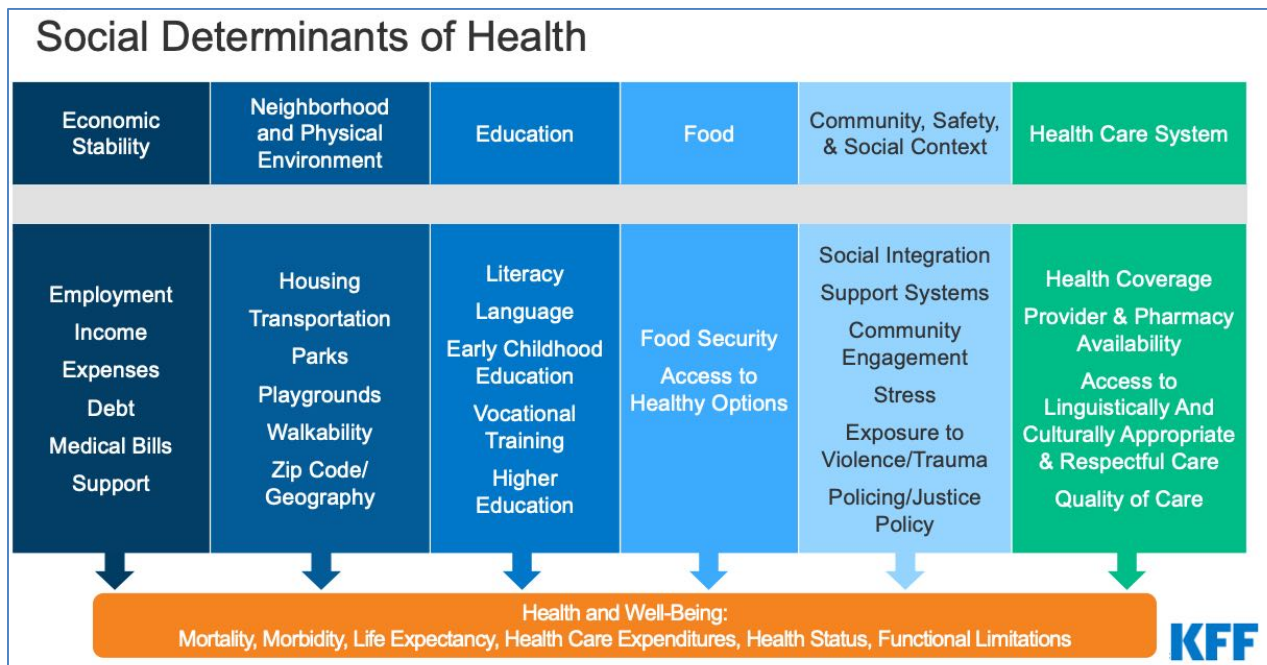


High Blood Cholesterol:

- Approximately 50% of Population have High Cholesterol (Rates are stagnant)
- ≈60% of Indigenous (Rates are stagnant)
- **What is High?**
 - o LDL:HDL Ratio

Social Determinants of Health:

- **Socioeconomic Status:**
 - o High SES people tend to live longer
 - o **Why? – They can Afford Better:**
 - Nutrition
 - Medical Care
 - Education → ↓ Risky behaviours
- **Early Life:**
 - o Eg: Low Birth Weight (Ie: From maternal smoking)
 - o Eg: Poor Nutrition
 - o Eg: Neonatal Infections
 - o Eg: Breastfed Vs Non-Breastfed
- **Stress:**
 - o Money
 - o Family
 - o Relationship
 - o Job Security
- **Employment:**
 - o Eg: Occupational Hazards
 - o Eg: Bad influences of Workmates (Eg: Drinking/Smoking)
 - o Eg: Fast foods for lunches
- **Social Networks:**
 - o Or Lack of → Depression
 - o Social Exclusion (Eg: Minorities – Racial/SES/Sexuality/Weight/etc)
- **Drug Addiction:**
 - o Direct impact on health (Eg: Hep-B/HIV/Substance-Dependence)
 - o Indirect impact through:
 - Crime
 - Compromise on nutrition etc. To save money for drugs



<https://mobile.twitter.com/avonhmo/status/1022838331053432832>

Prevention of Chronic Diseases You Will See as a Doctor:

- Hypertension:

- **Primary Prevention:**
 - ↑Exercise
 - Lose weight
 - ↓Salt intake
 - ↓Saturated Fats
 - ↓Stress
 - Coping Strategies
- **Secondary Prevention:**
 - Screening for Hypertension
 - Early Diagnosis
 - Review for other risk factors
 - Lifestyle Counselling
- **Tertiary Prevention:**
 - Antihypertensive Drug Interventions
 - Follow-up Monitoring



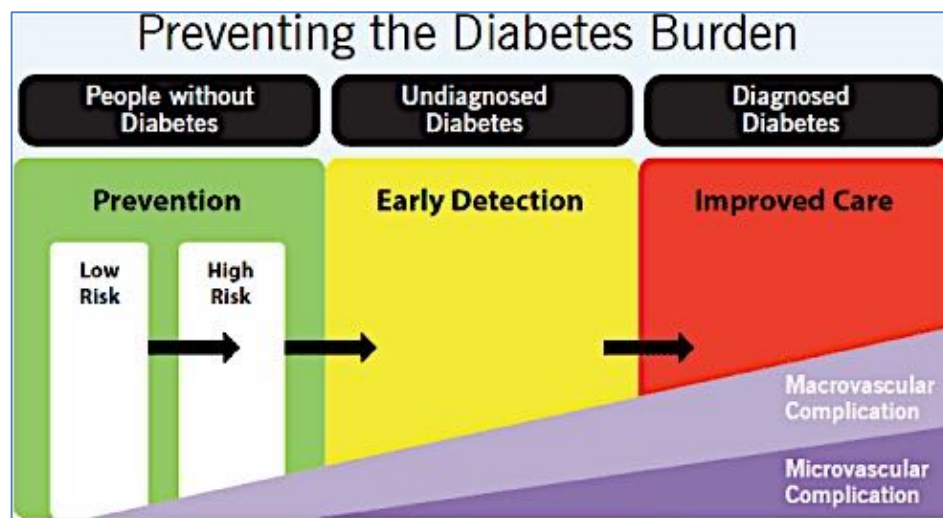
Source: WHO

- Depression:

- **Primary Prevention:**
 - Address Social Isolation/Greif/Family Problems
 - Strategies for Coping with Stress
 - Build good support networks
 - Physical Exercise → ↓Stress
- **Secondary Prevention:**
 - Screening for signs of depression
 - Early Diagnosis
 - Early Intervention
- **Tertiary Prevention:**
 - Appropriate Therapy/Counselling
 - Monitoring & Support
 - Refer to Therapist

- **Diabetes:**

- **Primary Prevention:**
 - Physical Activity
 - Weight Control & Diet
 - Find out Family History
- **Secondary Prevention:**
 - Screening blood tests in *At-Risk* patients
- **Tertiary Prevention:**
 - Referral to Diabetes Educator
 - Initiation of Treatment
 - Ongoing Monitoring



- **Lipid Disorder:**

- **Primary Prevention:**
 - Diet
 - Exercise
 - Family History
- **Secondary Prevention:**
 - Screening
 - Risk Factor Profile
 - Dietary Counselling
- **Tertiary Prevention:**
 - Start Treatment (Monitor Effects & Side-Effects)

- **Osteoarthritis:**

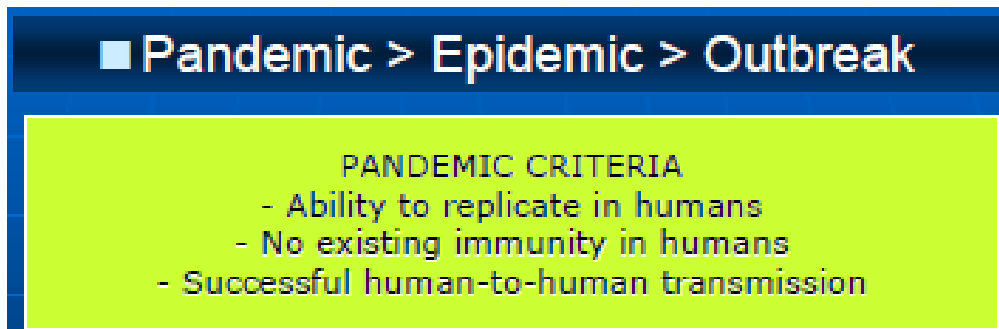
- **Primary Prevention:**
 - Avoidance of Injuries in Early Life
- **Secondary Prevention:**
 - Diagnose from Other Rheumatological Disorders
 - Provide Early Intervention
- **Tertiary Prevention:**
 - Medication
 - Physical Therapies
 - Devices & Aids
 - Surgical Referral

PANDEMICS

PANDEMICS

What is a Pandemic?

- **World Health Organization (WHO) – 3 Criteria:**
 - Disease is **New** to a population; (ie: NO Existing Human Immunity)
 - Agents infect **Humans**, causing serious illness; and
 - Agents **Spread Easily** and among humans
- **It is not a pandemic just because it is widespread or kills many people; it must also be infectious**
 - Eg: Cancer kills many people, but is not a pandemic because it is not infectious or contagious
- **Excludes “seasonal influenza” –not a new disease**



Pandemic Management:

- Potential Solutions Depend on phase of pandemic
- **One example of a pandemic management model:**
 - **ALERT:**
 - A novel zoonotic virus with **pandemic potential** causes severe disease in humans
 - There is no Human-Human Transmission
 - Novel virus has not arrived in the home country
 - **DELAY:**
 - Novel virus still not in the home country
 - OS4 Small cluster of cases in one country overseas
 - OS5 Large cluster(s) of cases in only one or two countries overseas
 - OS6 Large cluster(s) of cases in more than two countries overseas
 - **CONTAIN:**
 - Pandemic Virus Arrives in the home country
 - Small Number of Cases
 - **SUSTAIN:**
 - Pandemic virus Established in the home country and Spreading in the community
 - **CONTROL:**
 - Customised Vaccine widely available
 - Beginning to bring the Pandemic under control
 - **RECOVER:**
 - Pandemic controlled in the home country but further waves may occur if the virus drifts and/or is re-imported

Who do you treat?

- **Depends on phase-**
 - **Early phase**, treat everyone → Reduce disease transmission
 - **Later phases**, treat at risk groups
- **Who pays for it?**
 - If you treat everyone, it gets very expensive

Vaccine Prophylaxis: Theoretically Becomes Available At "Control" Phase:

- **Who Gets it?**
 - *High Risk Individuals* (Eg: old, debilitated, chronic heart/respiratory/renal disease)
 - *People in closed institutions* (Eg: Prisons/nursing homes)
 - *Groups in community service* (Eg: Doctors/Hospital Staff)
- **Types:**
 - **Inactivated Vaccines** are prepared from the appropriate strain of virus
 - **Subunit Vaccines** are prepared to reduce the content of extraneous proteins
 - **Live Attenuated Vaccines:**
 - These vaccines could be administered as a nasal spray
 - This would encourage the development of appropriate immune responses based on mucosal immunity
 - **Recombinant Vaccines:**
 - Based on Recombinant DNA Technologies

Ethical Allocation Of Scarce Resources:

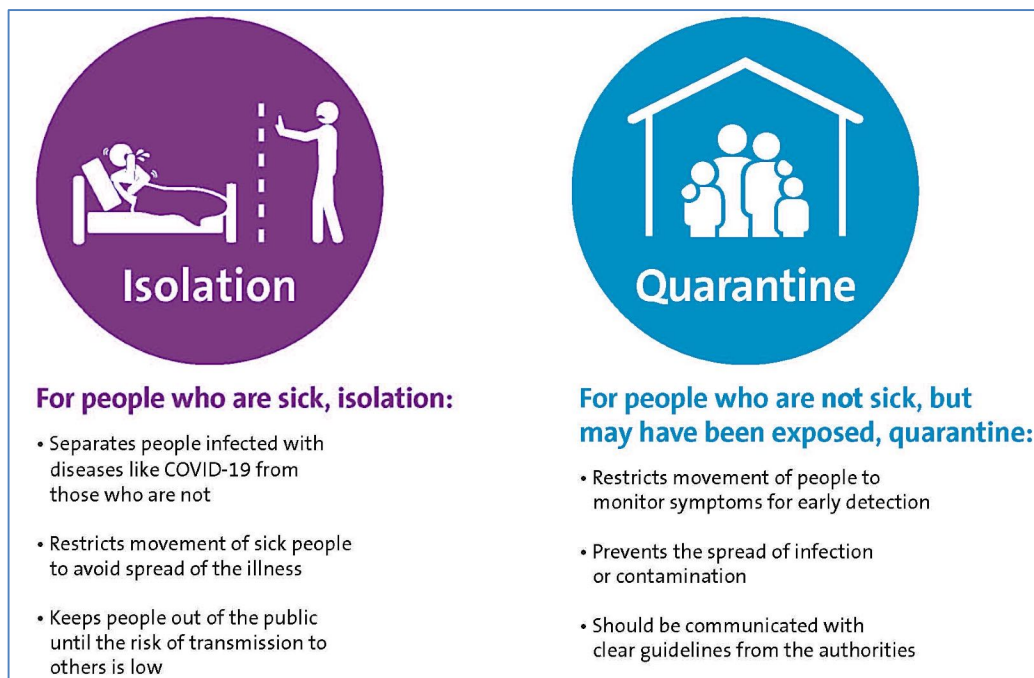
- **Single biggest question is how to ration scarce lifesaving resources:**
 - "who shall live when not all can?"
- **WHO GETS IT? (Fleming, BMJ 2005)**
 - **Blind Justice?**
 - (1st come or lottery)
 - **High risk given priority?**
 - (old, chronic disease)
 - **Healthcare workers?**
 - **Essential services?**
 - (Policemen/Firemen)
 - **Children?**
 - **Global allocation of resources**
- **Eg: In the Recent Covid-19 Outbreak:**
 - **Massive access block (Where there are insufficient beds for new patients)**
 - →ED overcrowding
 - → Not enough ICU beds
 - **→More beds were created by:**
 - →Cancellation of elective surgery
 - →Especially cancellation of elective major surgery needing ICU
 - → Erecting Field Hospitals

2 Arms of Countermeasures in a Pandemic:

- **Therapeutic countermeasures**
 - stockpiling of resources
 - vaccines
 - antiviral medications
 - access to care
 - health care workers
- **Non therapeutic countermeasures**
 - infection control
 - surveillance and contact tracing
 - social separation
 - quarantine and containment
 - international boundaries, duties and foreign nationals

Isolation And Quarantine – Essential In Early Phase:

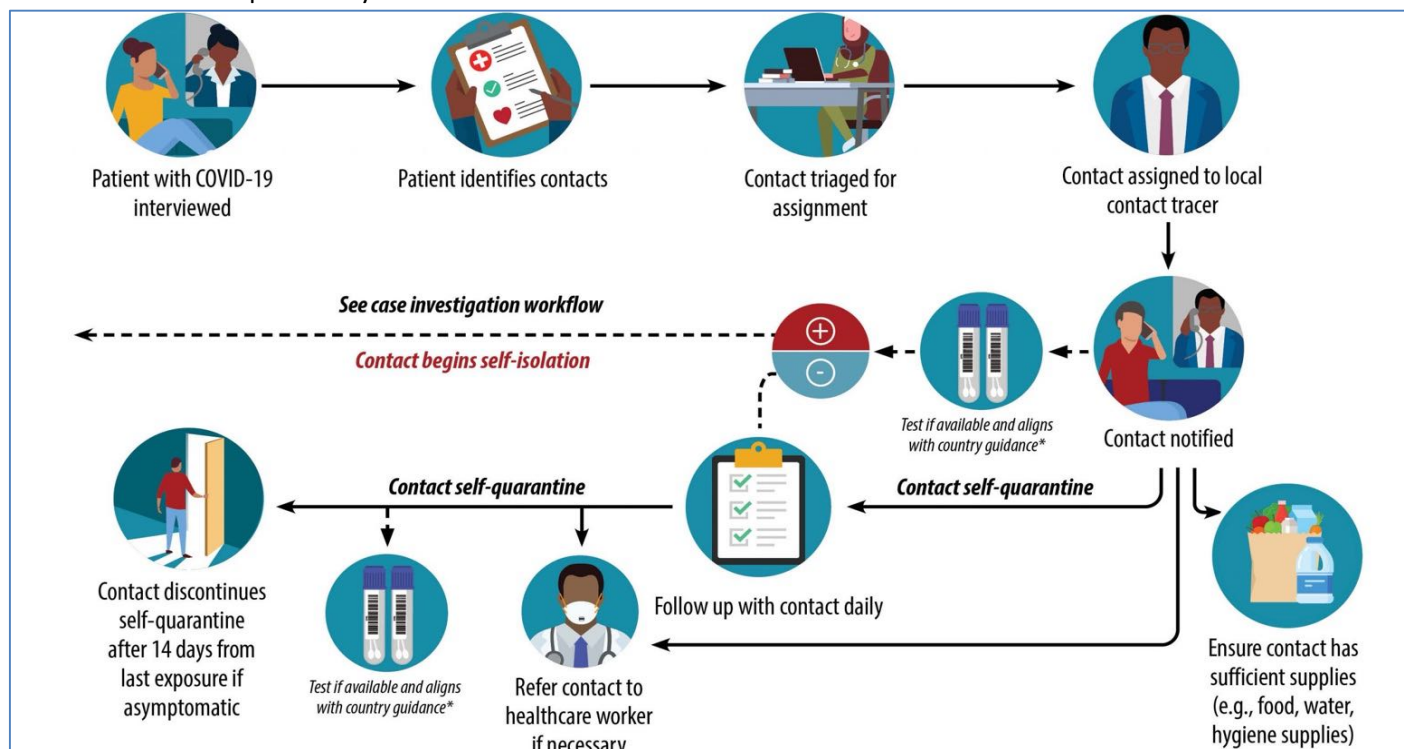
- **Isolation:**
 - Separation (for the period of communicability) of known **infected persons** to prevent or limit the spread of infection
- **Quarantine:**
 - Restriction of activities of **healthy persons who have been exposed**, to prevent disease transmission during the incubation period



<https://www.matherhospital.org/wellness-at-mather/isolation-vs-quarantine-whats-the-difference/>

Surveillance & Contact Tracing:

- **Contact Tracing** = Identification of cases by Name
- **Surveillance** is more intrusive than simply reporting names:
 - daily temperatures, health questionnaires
 - complete daily certificates



<https://www.cdc.gov/coronavirus/2019-ncov/global-covid-19/contact-tracing-workflow.html>

Conflicts of Interest:

- **Clinical Medicine:**
 - Doctors Promote best interests of a Patient
 - (Respect patients Liberty & Autonomy)
- **Public Health:**
 - Promotes best interest of the Population
 - Liberty and Autonomy of the Patient may be Overridden for the good of the public
 - Quarantine
 - Isolation
 - Closing International Borders

What will the future hold?

- Will there be another pandemic – **Yes!!**
- What serotype of virus will be responsible - ??
- How well are we prepared ?? - NOT
- How much warning will we get??
- What should I be doing to prepare ?

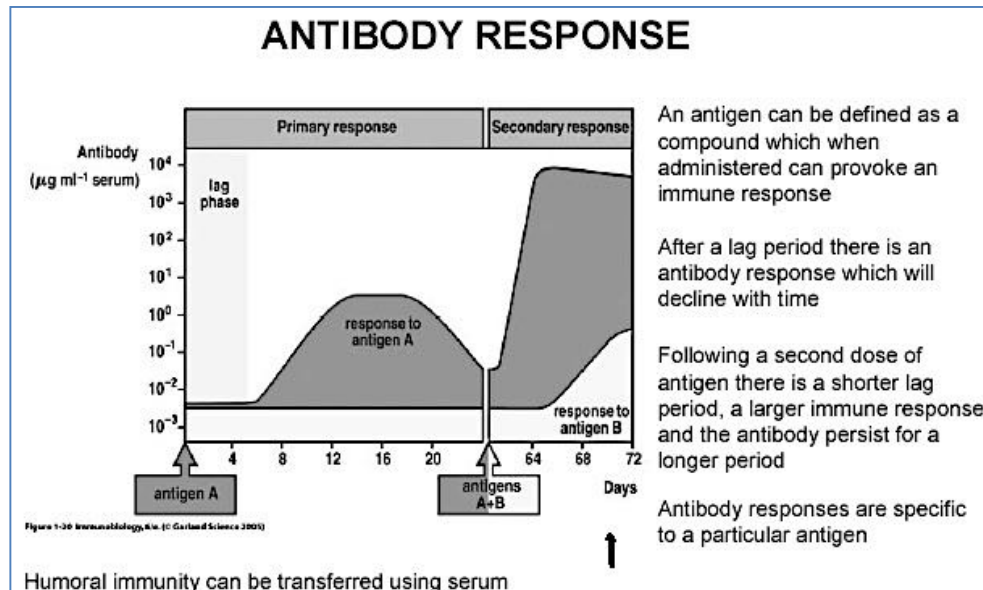
VACCINATION/IMMUNIZATION

VACCINATION/IMMUNIZATION

WHO: "The most successful & cost-effective public health intervention in history"

How Vaccines Work:

- **Individual Protection:**
 - They stimulate the immune system to create Antibodies/Memory-Cells in the absence of Disease Symptoms
 - Once immune, if the body is subjected to the actual pathogen, it mounts an even stronger attack (Secondary Immune Response) against it
- **Herd Immunity:**
 - Protects those who aren't vaccinated in the community, providing there is a high rate of vaccination within the population



Principals of Vaccine Development & Use:

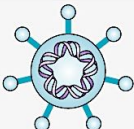
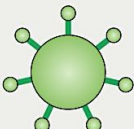
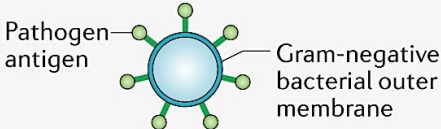
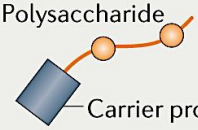
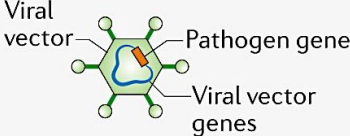
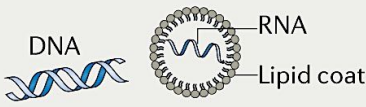
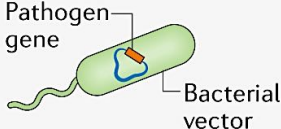
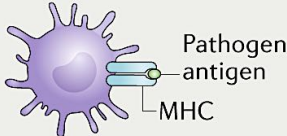
- 1: Separate the Disease-Causing effects from the Immune-Generating effects in an organism
- 2: Give it to susceptible individuals to provoke an immune response
- 3: Result is non-susceptible, immune individuals
- 4: Eventually results in herd immunity

Contraindications to Vaccination:

- **Absolute:**
 - Anaphylactic response to vaccine or component
- **Relative: (1e: Risks Vs Benefits)**
 - Immunocompromised (Live Vaccines)
 - Pregnant or Suspected Pregnancy (Live Vaccines)
 - Fever of $>38.5^{\circ}\text{C}$
 - Recent Live Vaccine (4 weeks)
 - Recently received blood/blood products
 - Guillain Barre Syndrome (GBS)
 - Influenza
- Note: All other *excuses* are not good reasons not to vaccinate!
- Note: All patients should have the freedom to choose whether or not to receive a vaccine

Types of Vaccines:

- **Live Attenuated vaccines:**
 - Live organisms that have been de-pathogenised
 - **Advantages:**
 - Robust Immune Response
 - Lifetime Immunity with 1 or 2 doses
 - **Disadvantages:**
 - Potential to cause disease (can't give to immunocompromised or pregnant women)
 - Potential for Adverse Events/Side-Effects
- **Inactivated/killed vaccines:**
 - Dead organisms containing relevant proteins but unable to replicate
 - **Advantages:**
 - No Ability to cause disease
 - Fewer Adverse Events/Side-Effects
 - **Disadvantages:**
 - Less robust immune response
 - Waning Immunity → Requires multiple doses & may require booster
- **Acellular/Toxoid/Subunit:**
 - Artificially synthesised non-toxic antigens
 - **Advantages:**
 - No Ability to cause disease
 - Fewer Adverse Events/Side-Effects
 - **Disadvantages:**
 - May require Adjuvants or Conjugation
 - Less robust immune response
 - Waning Immunity → Requires multiple doses & may require booster
- **Recombinant Protein Vaccines:**
 - Made using bacterial or yeast cells to manufacture surface proteins from the pathogen
 - **Advantages:**
 - No ability to cause disease
 - Can be highly targeted
 - **Disadvantages:**
 - Needs to be supplemented with adjuvants to stimulate generation of antibodies
- **Nucleic Acid Vaccines (Eg: RNA Vaccines):**
 - Inoculating the body with a synthetic RNA strand, which gets into body's own cells → translated into antigen proteins → stimulates immune response to antigen
 - **Advantages:**
 - Can be highly targeted
 - No risk of disease
 - **Disadvantages:**
 - Expensive
 - Labour intensive
 - Delicate
 - Highly sensitive to cold chain

Type of vaccine		Licensed vaccines using this technology	First introduced
Live attenuated (weakened or inactivated)		Measles, mumps, rubella, yellow fever, influenza, oral polio, typhoid, Japanese encephalitis, rotavirus, BCG, varicella zoster	1798 (smallpox)
Killed whole organism		Whole-cell pertussis, polio, influenza, Japanese encephalitis, hepatitis A, rabies	1896 (typhoid)
Toxoid		Diphtheria, tetanus	1923 (diphtheria)
Subunit (purified protein, recombinant protein, polysaccharide, peptide)		Pertussis, influenza, hepatitis B, meningococcal, pneumococcal, typhoid, hepatitis A	1970 (anthrax)
Virus-like particle		Human papillomavirus	1986 (hepatitis B)
Outer membrane vesicle		Group B meningococcal	1987 (group B meningococcal)
Protein-polysaccharide conjugate		<i>Haemophilus influenzae</i> type B, pneumococcal, meningococcal, typhoid	1987 (<i>H. influenzae</i> type b)
Viral vectored		Ebola	2019 (Ebola)
Nucleic acid vaccine		SARS-CoV-2	2020 (SARS-CoV-2)
Bacterial vectored		Experimental	–
Antigen-presenting cell		Experimental	–

Pollard, A.J., Bijker, E.M. A guide to vaccinology: from basic principles to new developments. *Nat Rev Immunol* **21**, 83–100 (2021). <https://doi.org/10.1038/s41577-020-00479-7>

2 Important Vaccine-Preventable Diseases:

- Measles:

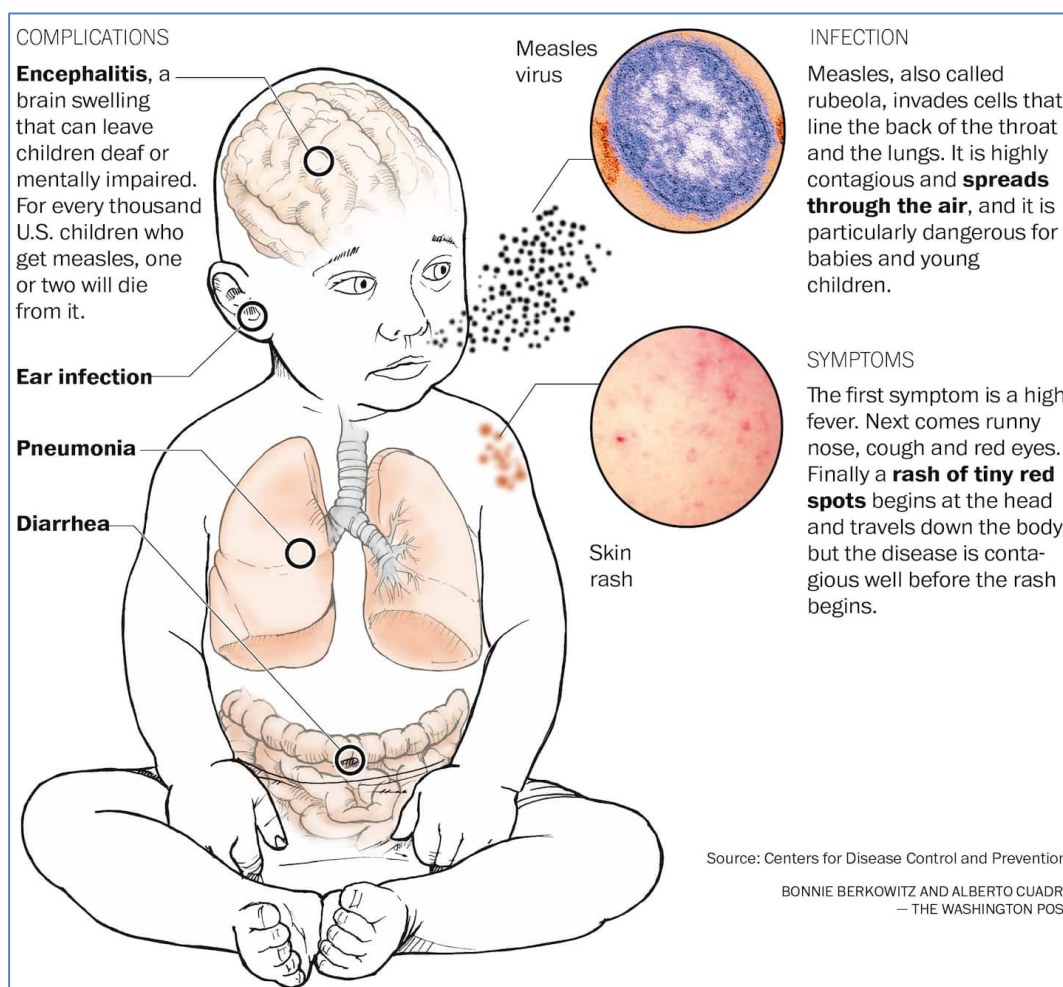
- **Extremely Virulent:** – One of the most infectious (easily Spread) Diseases known to man
- **Genus:** – Morbillivirus
- **Occurrence:**
 - Prior to Immunisation = >100 Million cases/year → 6 Million Deaths/year (Worldwide)
 - Post-Immunisation = 99% drop in cases
- **Transmission:** - Airborne Droplet Spread
- **Incubation Period:**
 - ≈10 days to onset of fever
 - ≈14 days to onset of rash
- **Period of Communicability:**
 - From 'Prodrome' (Time before symptoms) to 4 days after onset of rash
- **Susceptibility:**
 - Everyone un-sensitised people
 - Survival of Illness → Lifetime immunity
 - 1st Vaccine → 95% Immune
 - 2nd Vaccine → 99% Immune
 - Maternal Antibodies protect infant for 6-9 months
 - Malnutrition is a problem – as measles causes diarrhoea in children
- **Symptoms:**
 - Fever
 - Malaise
 - Cough
 - 'Coryza' – ("Overflowing Head")
 - Conjunctivitis
 - Rash – starting on face → Spreading to rest of body
 - Koplic Spots (Unique to Measles) – White/Blueish spots on buccal mucosa
- **Complications:**
 - Otitis Media (Middle ear infection)
 - Pneumonia
 - Diarrhoea
 - Acute Encephalitis (Rare)
- **Measles Vaccine:**
 - **Type:** - Live Attenuated Vaccine (Given in combination with Mumps, Rubella & Varicella)
 - **Note:** It interferes with other live vaccines
 - **NOT given during Pregnancy**
 - **Adverse Reactions:**
 - Fevers – common
 - Faint red rash
 - Local swelling
 - Local Knot in muscle
 - **Dosing:**
 - 1st Dose @ 12mths
 - 2nd Dose @ 18mths



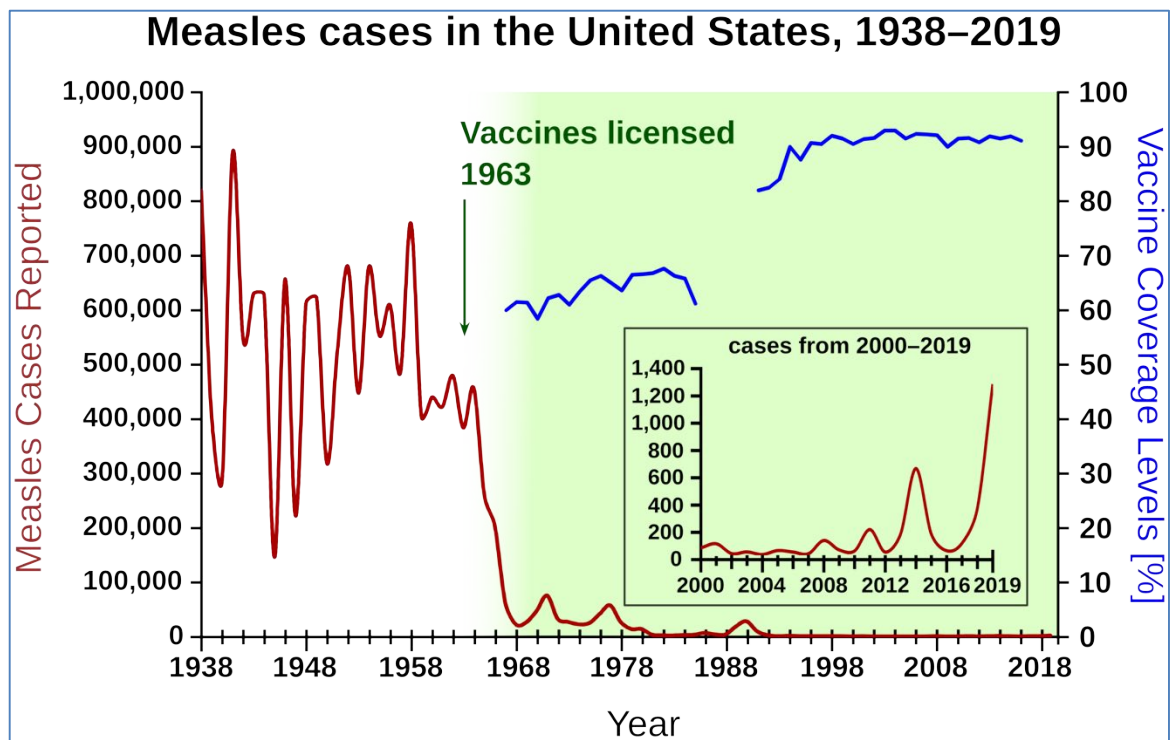
Photo Credit:Content Providers(s): CDC/Dr. Heinz F. Eichenwald, Public domain, via Wikimedia Commons



CDC, Public domain, via Wikimedia Commons



Public domain: CDC



Julius Senegal, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

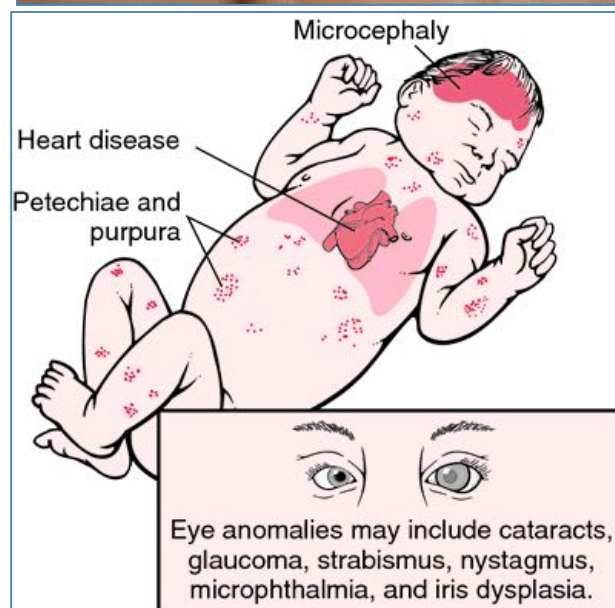
- **Rubella:**

- **Genus:** - Rubivirus
- **Occurrence:**
 - Pre-Vaccine = Worldwide Endemic
 - Epidemics every 5-9yrs
 - Post-Vaccine = Elimination of Rubella
- **Transmission:**
 - Airborne Droplet Spread
 - Contact with Mucus Membranes
- **Incubation Period:**
 - 14-21 Days
- **Period of Communicability:**
 - 1 Week Before & 4 Days after Rash
- **Susceptibility:**
 - Universal Without Vaccine/Prior Infection
 - Immunity after single dose (95-100%)
 - Maternal Antibodies protect infant for 6-9mths
- **Symptoms:**
 - 50% Asymptomatic
 - Low-Grade Fever
 - Headache
 - Malaise
 - Coryza ("Overflowing Head")
 - Conjunctivitis
 - Lymphadenopathy
 - Arthralgia
 - Rash
- **Complications:**
 - Post Viral Encephalitis (Uncommon)
 - **Congenital Rubella Syndrome (CRS)
 - Occurs in 90% of babies whose mother had Rubella during 1st Trimester
 - Multiple Defects are common (Eg: Blindness, Deafness, many more...)
 - Hence ALL WOMEN OF CHILD-BEARING AGE MUST BE IMMUNISED
- **Rubella Vaccine:**
 - **Type:** - Live Attenuated Vaccine (Given in combination with Measles, Mumps & Varicella)
 - **Note:** It interferes with other live vaccines
 - **NOT given during Pregnancy**
 - **Adverse Reactions:**
 - Fevers – common
 - Faint red rash
 - Local swelling
 - Local Knot in muscle
 - **Dosing:**
 - 1st Dose @ 12mths
 - 2nd Dose @ 18mths
 - **ALL WOMEN OF CHILD-BEARING AGE MUST BE IMMUNISED**





<https://www.cdc.gov/rubella/about/photos.html>



http://phil.cdc.gov/phil_images/20030724/28/PHIL_4284_lores.jpg

Other Notable Vaccines & the Diseases They Prevent:

- **Hep B:**
 - **Hepatitis B:**
 - An Infectious illness caused by Hep-B-Virus
 - Infects the Liver
 - Causes Vomiting, Jaundice, Liver Cirrhosis & Liver Cancer (Rarely death)
- **DTP:**
 - **Diphtheria:**
 - Upper Respiratory Tract illness
 - Characterised by sore throat, low fever & a pseudomembrane on tonsils/pharynx/nasal cavity
 - **Tetanus:**
 - Gram-Positive anaerobic bacteria infection – occurs through skin wound
 - Bacteria secrete a neurotoxin → Prolonged contraction of skeletal muscle fibres (“Tetany”)
 - **Pertussis:**
 - Whooping Cough – A highly contagious disease spread by droplet transmission
 - Bacterial Infection
 - Droplet Transmission
- **Hib:**
 - **Haemophilus Influenzae B:**
 - Actually a Bacteria (Despite the ‘viral’ name)
 - Cause opportunistic infections
 - Leads to Bacteraemia, Pneumonia, Bacterial Meningitis
- **IPV:**
 - **Inactivated Poliomyelitis (Polio):**
 - AKA: Infantile Paralysis
 - An acute viral infection
 - Faecal-Oral Transmission
 - 90% of infections are asymptomatic
 - If the virus enters the CNS, it preferentially destroys motor neurons → Muscle Weakness, Paralysis & Muscle Wasting
- **7vPCV:**
 - **Pneumococcal Conjugate:**
 - Bacterium →
 - Pneumonia, Sinusitis, Otitis Media
 - Meningitis (Most common cause of bacterial meningitis)
 - Bacteraemia → Sepsis
 - Endocarditis
 - Pericarditis
- **Rotavirus:**
 - Leading cause of Severe Diarrhoea among infants & young children
 - Also causes gastroenteritis & dehydration
 - Known as *Stomach Flu* (But no relation to influenza)
 - Faecal-Oral Route
- **MMR:**
 - **Measles:** See Above
 - **Mumps:**
 - Viral Disease
 - Droplet Transmission
 - Typically presents as painful swelling of the Salivary Glands, fever & headache
 - Can also cause painful testicular swelling & rash
 - **Rubella:** See Above
- **MenCCV:**
 - **Meningococcal C:**
 - Bacterium
 - Typically causes Meningitis & Fever, but is most dangerous when infection becomes septic

The Cold Chain:

1. What is the *cold chain*?

- Is the transport and storage of a vaccine at or below a certain temperature
- It includes the vaccine equipment, people and the procedures

2. What are the *stages* in the cold chain?

- Manufacture
- Supply
- Distribution
- Clinic
- Fridge
- Patient

3. Name 5 Vaccines that are damaged or destroyed by *freezing*?

- Tetanus
- DTP
- Hib
- Hep A & B
- Influenza
- Pneumococcal

Which vaccines are damaged by exposure to *heat or light*?

- BCG
- Oral polio
- MMR

4. What is the main requirement for *vaccine fridge thermometers*?

- Can measure max & min temperature for the previous 24 hrs

5. Which are the *best types of thermometers*?

- Mercury and digital thermometers

6. Where in the fridge would you place the temperature probe?

- In the centre of the fridge
- In the case of a multiple purpose fridge (Eg: If food is stored with it), then is best located with the immunisations in the foam box within the fridge

7. What *monitoring device* is included in vaccine transportation of vaccine supplies?

- Temperature monitors → heat & freeze cards (charts)

8. How would you *pack* vaccines for transport and what other *precautions* would you take for transportation?

- Vaccine package
 - Good icebox with a tight fitting lid
 - Store immunisations with ice block that is “sweating”
 - Shredded newspaper is recommended to allow air to circulate around vaccines
 - Layers: ice block - shredded newspaper – immunisations – shredded newspaper – piece foam – ice block
- Transport
 - Whenever being transported don’t place in direct sunlight
 - Minimise duration of journey
 - Record temperature whenever vaccines are put in/ taken out
 - Place securely in boot
 - Only deliver to surgery if someone is there
 - Check the temperature of the fridge at surgery before put new immunisations in

9. What are the *requirements of a surgery vaccine fridge?*

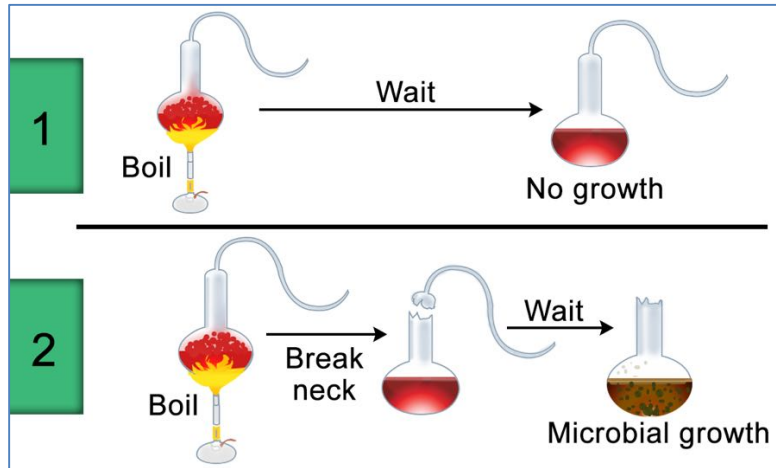
- Have a safe lockable fridge
- Try to keep immunisations in a proper immunisation fridge – if not possible separate the immunisation from the other things that are being stored in the fridge (Eg: Have separate shelf for immunisation and separate shelf for medications)
- Check the temperature of the immunisation fridge daily
- When placing new vaccines in fridge, rotate stock
- Use older vaccines first, don't use the most recent – will prevent vaccines going out of date
- Use only 50% of available space in fridge – allow air to circulation around the vaccines in the fridge
- Place immunisation in correct location in fridge – the bottom shelf and door shelves should not be used
 - recommended to place salty water in these draws to prevent shelves from being used and the water from being drank
- Don't co-store non-vaccine items in the fridge (eg food) – this will ↑ fridge door opening and may interfere with temperature
- If it is thought that cold chain may have been broken or vaccine has been tampered with contact supplier 1st before throwing out. Isolate in the fridge in with clear label stating “do not use” until ascertained whether should be thrown out

BASIC CONCEPTS OF INFECTIOUS DISEASES

BASIC CONCEPTS OF INFECTIOUS DISEASES

The Birth of 'Germ Theory':

- People recognised that meat broth became cloudy and overgrown with microbes
- **But where did these microbes come from? – 2 Schools of Thought:**
 - o 1: Spontaneous Generation
 - o 2: Formed from Seeds/Germs
- **Pasteur** Proved that Microbes exist *In the Air* through his Swan-Necked Flask experiment



<https://www.medassureservices.com/getting-to-know-germs-and-hand-sanitizers/>

- **Robert Koch** Proved that *Specific Microbes* caused *Specific Diseases* through 'Koch's Postulates':

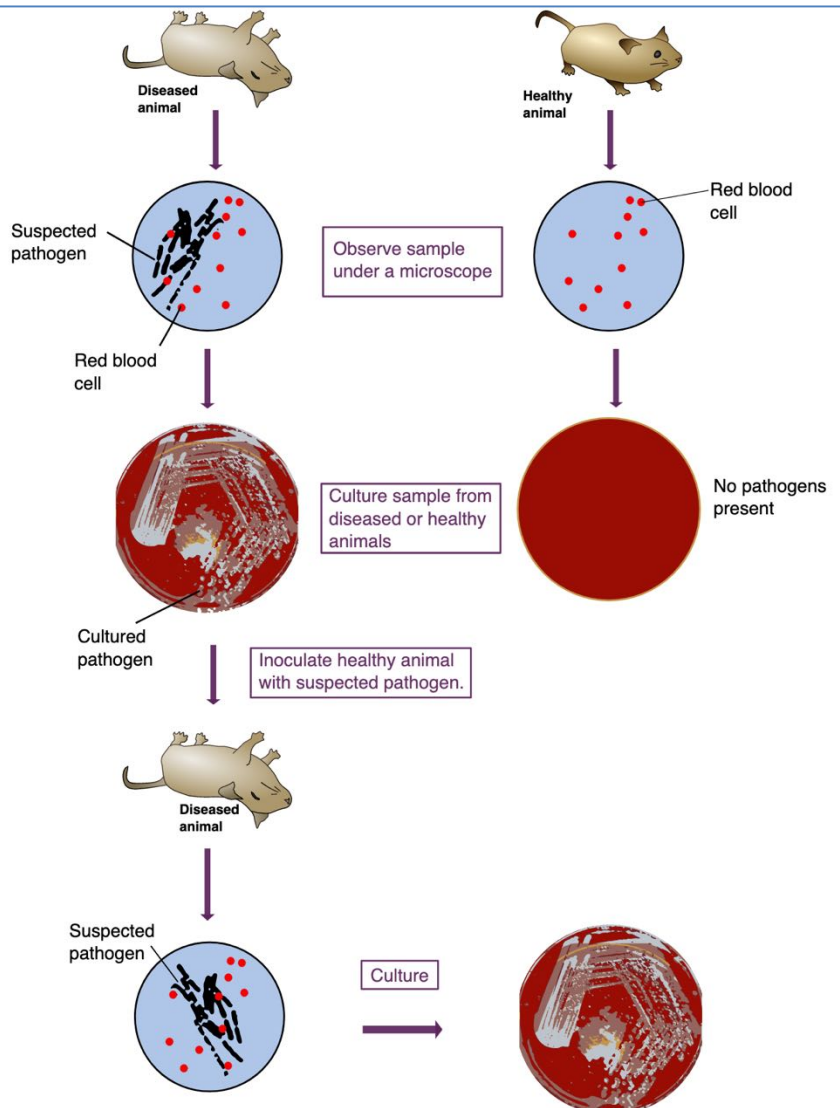
Koch's Postulates:

① The microorganism must be found in abundance in all organisms suffering from the disease, but should not be found in healthy organisms.

② The microorganism must be isolated from a diseased organism and grown in pure culture.

③ The cultured microorganism should cause disease when introduced into a healthy organism.

④ The microorganism must be reisolated from the inoculated, diseased experimental host and identified as being identical to the original specific causative agent.



Revision of Microbial Diversity:

- Some bugs are good (even essential) and some bugs are bad
- Organisms capable of causing disease are **pathogens**

Normal Flora (commensals)

- **Heavily colonise skin – armpit, perineum, interdigital areas**
 - Nose and oropharynx
 - GI Tract
 - Uro-genital tract
- **Are normal at certain places where they are not harmful**
 - However when they colonise an area where they shouldn't, they cause disease (nosocomial infection)

Pathogenesis

- The biochemical mechanisms whereby microbes (bacteria, fungi, parasites & viruses) causes disease
- **Virulence:** the propensity of a microbe to cause infection → disease

Steps to disease:

- I. **Entry**
 - Oral
 - Skin
 - Trans-placental
 - Inhalation
 - Inoculation (wound/skin penetration)
 - Sexual
- II. **Colonisation**
 - Breach of skin/epithelia/conjunctiva
 - Attachment
- III. **Persistence + avoiding host defences**
 - Beat natural barriers – flushing, mucous + cilia, stomach pH, Lysosomes in saliva, etc
- IV. **Replication**
 - Mucosal (GI tract)/systemic (blood)/nerves (viruses)/cerebrospinal fluid (meningitis)
- V. **Dissemination – (Host-Host)**
 - Faecal-oral (diarrhoea), Aerosols (sneezing), Sexual (intercourse)
 - Depends on:
 - Organism size
 - Ability to survive in external environment
- VI. **Cause Disease**
 - Can release toxins – either local effects / or systemic
 - Can cause unusual cellular activity
 - Can cause tissue damage

Host-Parasite Interactions:

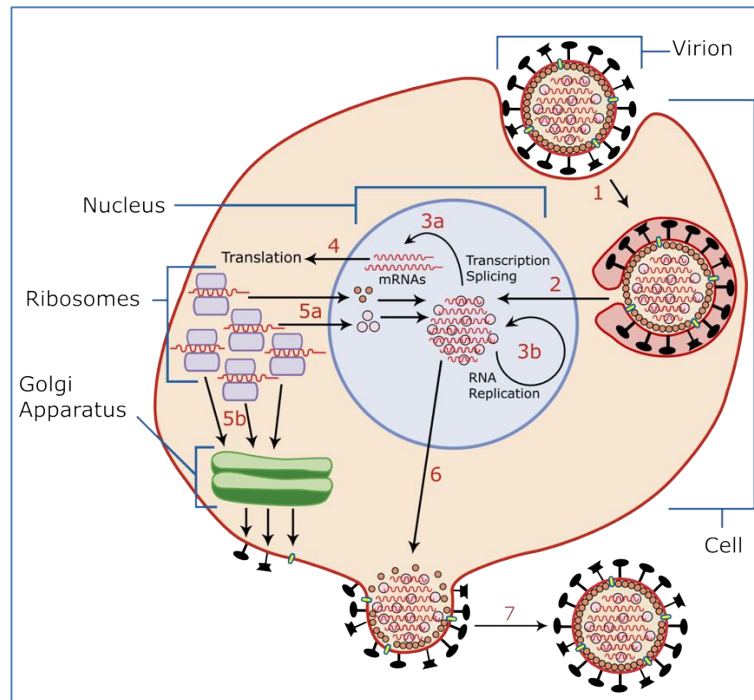
- 1) Colonised, no disease, no illness (asymptomatic)
Eg: Helicobacter – in stomach
- 2) Colonised, disease, no illness (asymptomatic)
Eg: Chlamydia & other genital tract infections
- 3) Colonised, disease, illness (symptomatic)

The Organism Classifications:

- **Prokaryotes:**

- **Viruses**

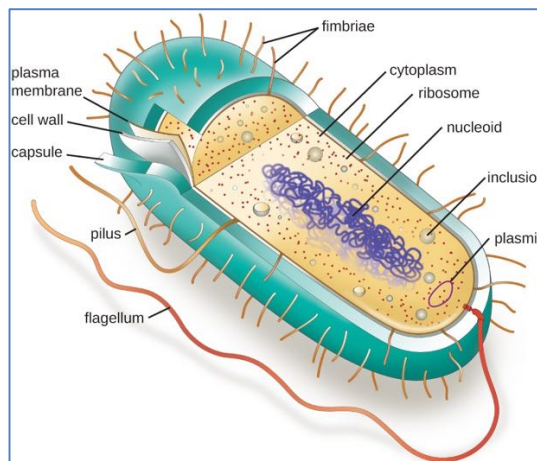
- Very small
- Nucleic acid inside protein coat (DNA or RNA)(ss or ds)
- Complete parasitic dependency
- Replicates inside cell - but metabolically inert in external environment
- Need close/direct contact
- Need a moist environment
- Lyses host cells and then infects more
- Respiratory route / oral / inoculation / sexual transmission



User:YK Times, CC BY-SA 3.0 <<http://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

- **Bacteria**

- Larger than viruses
- Visible under light microscope
- Living → replicate by binary fission
 - Can be killed
- Intracellular or extracellular
- Motile
- Can produce toxins
- Contain DNA, Ribosomes + Inclusions – no true nucleus
- Resulting disease often more severe

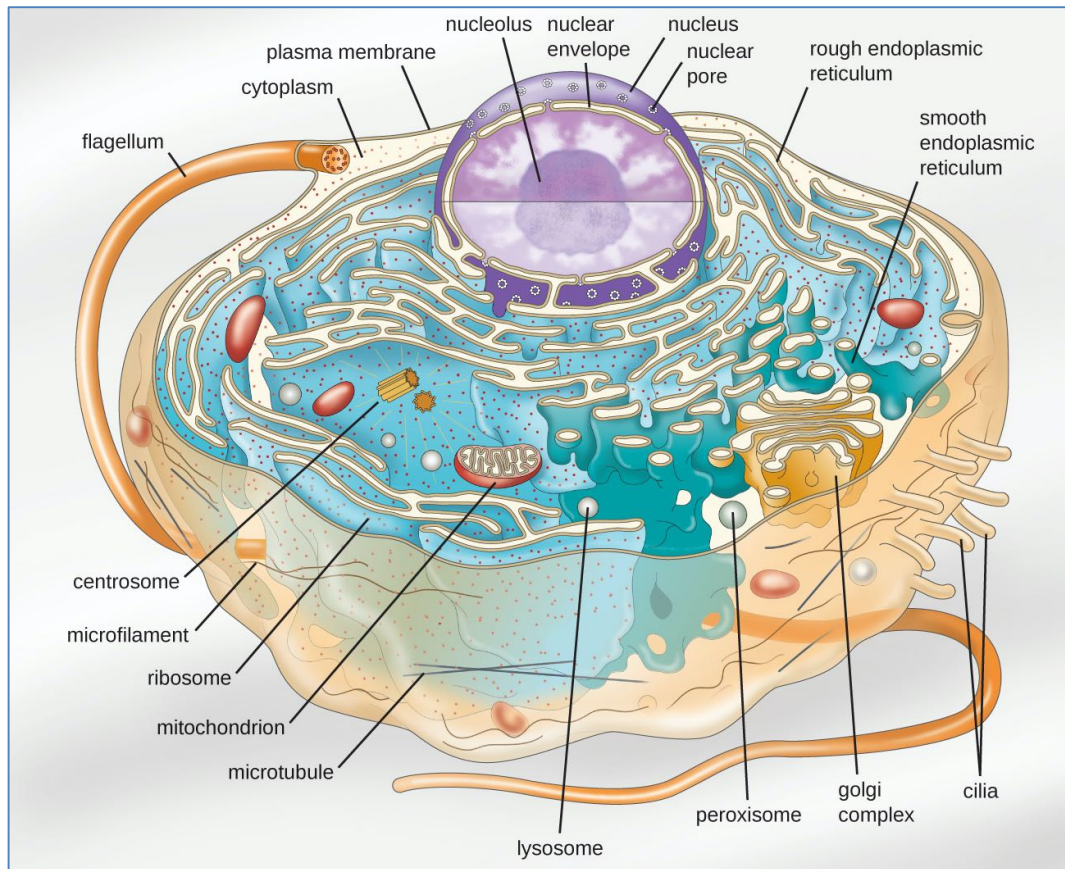


CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

- **Eukaryotes:**

- **Protozoa**

- **Single-Celled Animals**
 - Larger than bacteria – still small enough to live intracellularly
 - Can also live extracellularly
 - Vectors / faecal-oral route → most infections occur tropically



CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

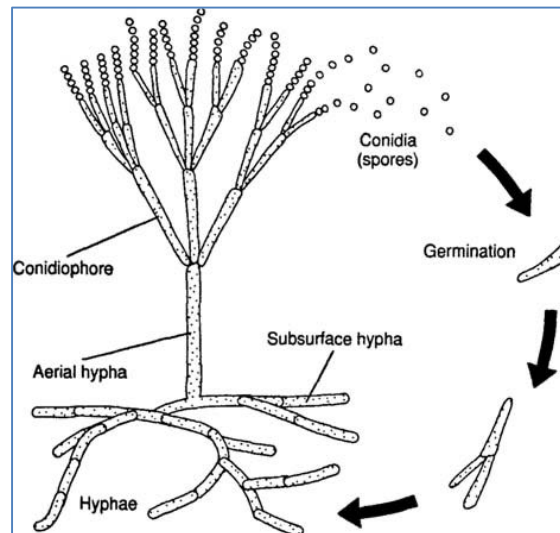
- **Helminths**

- Multi-celled, often macroscopic organisms
 - Complex body organisation and reproduction (some have sexual dimorphism)
 - Difficult for immune system to destroy – too big
 - Cause inflammation
 - Are often never eliminated



<https://upload.wikimedia.org/wikipedia/commons/b/bf/Hookworms.JPG>

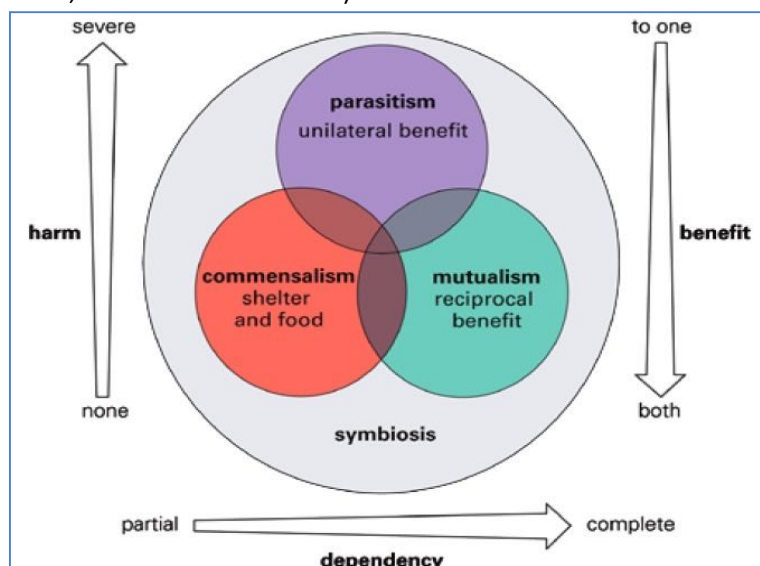
- **Fungi**
 - Thousands of species
 - Few are pathogenic to humans
 - 20ish are fatal
 - Resulting Mycoses (disease) either:
 - Superficial
 - Cutaneous
 - Subcutaneous
 - Systemic
 - Opportunistic – seen in compromised hosts
- Depending on site of infection
- Exist as branched filamentous forms, or yeasts
- Asexual spores (conidia)
- Spores commonly inhaled & cause infection



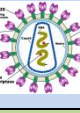
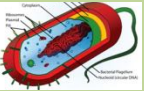
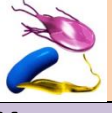
<https://aspergillusproject11.wordpress.com/2013/04/18/life-cycle/amp/>

Host-Parasite Relationships:

- **Commensalism:**
 - Colonised; But No Disease
 - (Eg: E-Coli in stomach)
- **Mutualism:**
 - Colonised; No Disease; Mutually Beneficial
 - (Eg: Digestive Bacteria in Colon; Lactobacillus in Vaginal → Acidity)
- **Parasitism:**
 - Colonised; With Disease/Damage
 - (Eg: Hookworm; Plasmodium Malariae)



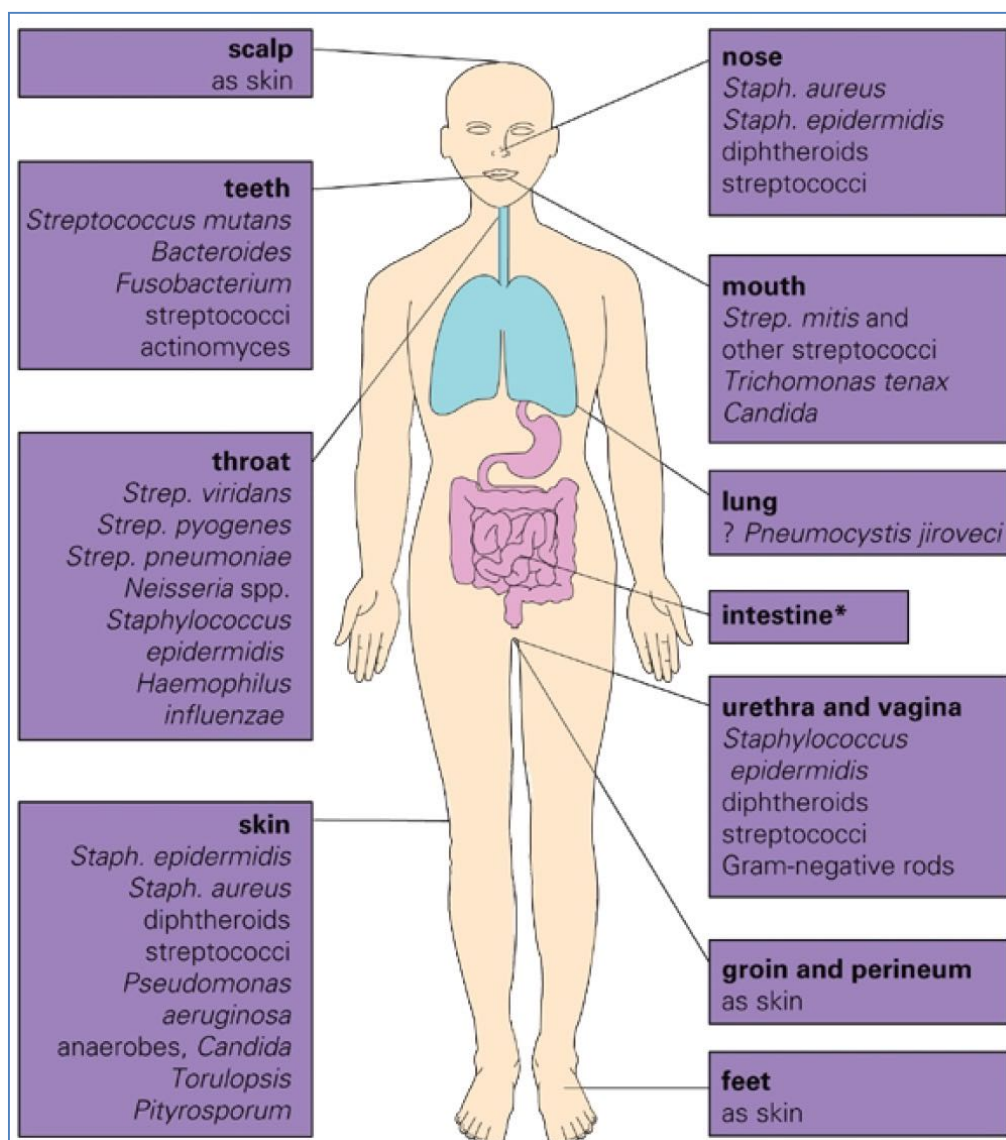
- **Differential Features Of Microbes:**

<u>Pathogen:</u>	<u>Visible Via:</u>	<u>Cellular?</u>	<u>Nuclear Material:</u>	<u>Nuclear Organisation:</u>	<u>Structural Constituents:</u>	<u>Outer Surface:</u>
Prions 	Electron Microscope	Acellular	No Nucleic Acid – Just Protein.	No Nucleus	No Membrane.	No Membrane
Viruses 			DNA or RNA		No Cellular Machinery. No Cytoplasm.	‘Enveloped’ Or ‘Non-Enveloped’
Bacteria (Prokaryotes) 	Light Microscope	Single Cell	DNA	No Distinct Nucleus. Single, Circular Chromosome. Simultaneous Transcription & Translation.	Membrane-Bound. Cellular Machinery.	Bi-lipid membrane is covered by a thick Cell Wall. Gram Pos = Peptidoglycan Gram Neg = Lipopolysaccharide
Protozoan Parasites (Eukaryotes) 				Distinct Nucleus. Several Linear Chromosomes		Simple Bi-lipid Membrane.
Metazoan (Eukaryotic) Parasites (Helminths) 	Naked Eye	Multi-Cell		Transcription in the Nucleus. Translation on Ribosomes in the Cytoplasm.		

Normal Flora Vs Pathogens:

- Normal Flora (Commensals):

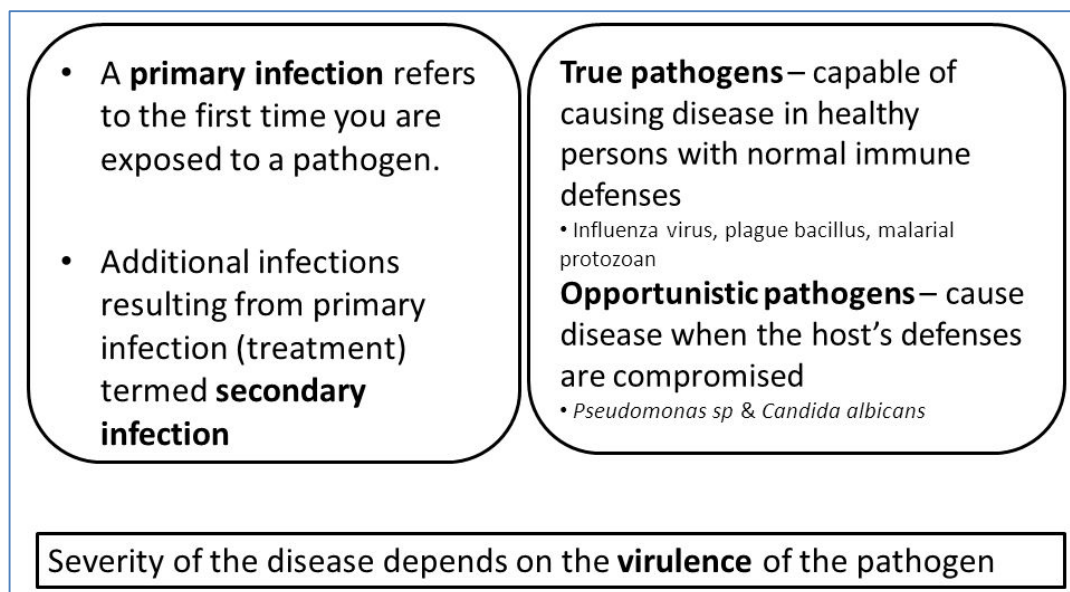
- **Can be Beneficial:**
 - Can be *Protective* by outcompeting potential pathogens for Space/Nutrients
 - If they are washed away (Eg: Vaginal Antibiotics), pathogens can colonise the area → Disease
- **Heavily Colonise Skin:**
 - Armpit, Perineum, Interdigital areas
 - Nose and oropharynx
 - GI Tract
 - Uro-genital tract
- **Heavily Colonise the GIT:**
 - *Density* of Microbes *Increases* Towards the Rectum (Stomach Acid → Low Numbers)
 - *Species* of Microbes change throughout due to different environments
- **Some Areas are Sterile:**
 - Bladder
 - Blood
 - Organs
- **Location depends on Aerobic/Anaerobic Species:**
 - Aerobic – Likes Oxygen (Eg: In Respiratory Tract)
 - Anaerobic – Cannot stand Oxygen (Eg: Found in Bowel/Necrotic Tissue/Etc)
- **Nosocomial Infection (Opportunism):**
 - If Commensals Colonise somewhere they shouldn't, they cause disease
 - Often occurs in Hospitals → Typically Highly Resistant to Antibiotics



Unattributable

- **Pathogens (4 Features of a "Pathogen"):**

- **Pathogens** = Organisms capable of causing disease
- ***They MUST do ALL 4 of the Following:**
 - 1: Gain Entry to Host
 - 2: Attach & Multiply
 - 3: Evade Host Defences
 - 4: Cause Damage to Cells/Tissues
- **Primary Vs Secondary Pathogens:**
 - **1° Pathogens:**
 - Can produce an Infection without the help of other organisms
 - → Also Encourage 2° Pathogens
 - (Eg: HIV → Immunocompromise)
 - **2° Pathogens (Aka: Opportunistic Pathogens):**
 - Only produce an Infection due to damage caused by 1° Pathogens

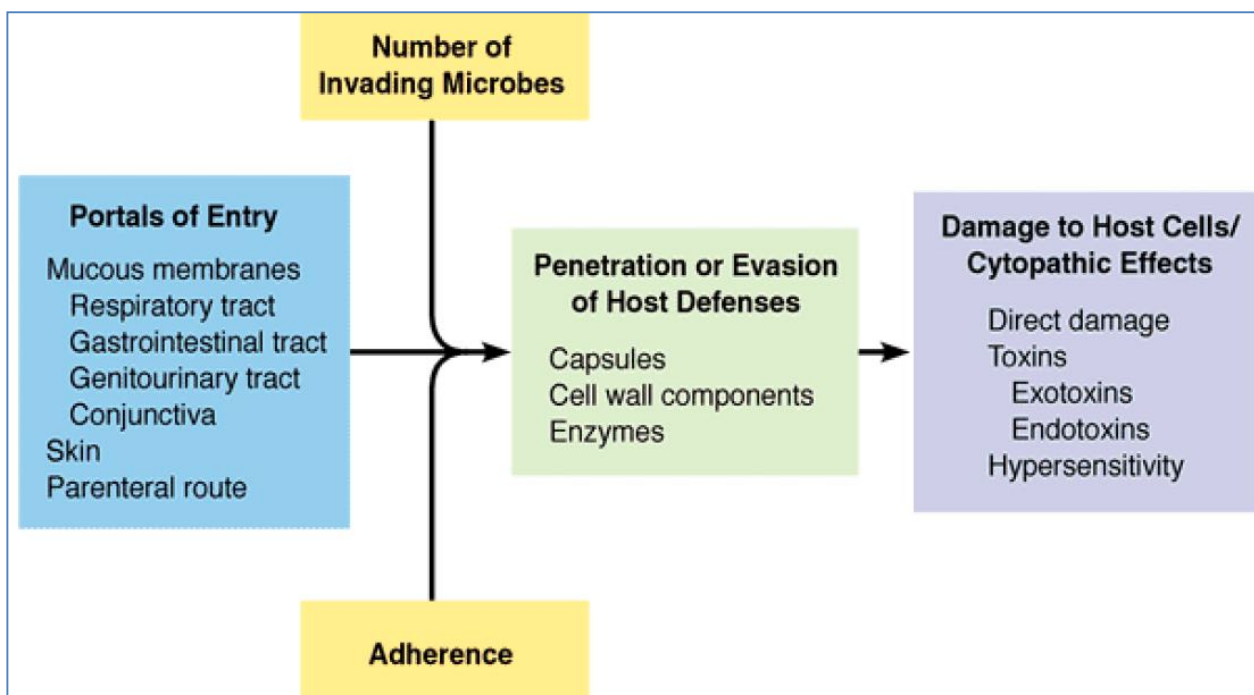


Virulence:

- **Short Definition:** The propensity of a microbe to cause infection → disease in a Definitive Host
- **Long Definition:** The degree of pathogenicity of an infectious agent, indicated by:
 - Case-fatality rates
 - Ability of the agent to invade and damage tissues of the host
 - Toxicity
 - Ability to overcome/evade body defences
- **(‘Avirulence’ = Antonym)**
- **Virulence Factors:**
 - Molecules Expressed/Secreted by Pathogens that enable them to achieve the following:
 - Colonization of a Niche in the host (this includes adhesion to cells)
 - Immuno-evasion, evasion of the host's immune response
 - Immunosuppression, inhibition of the host's immune response
 - Entry into & Exit out of cells (if the pathogen is an intracellular one)
 - Obtain nutrition from the host
 - Eg: Endotoxin (LPS) – Potent antigen
 - Eg: Exotoxins (Eg: Tetanus Toxin) → Tetanus
 - Eg: Fungal Mycotoxins (Eg: Aspergillus) → Severe Liver Damage
 - Eg: Ig-Proteases (Eg: Strep Pyogenes) → Break down Antibodies
 - Eg: Capsules (Eg: Bacterial cell walls) → Inhibits Phagocytosis

Pathogenesis (4 Stages to Infection):

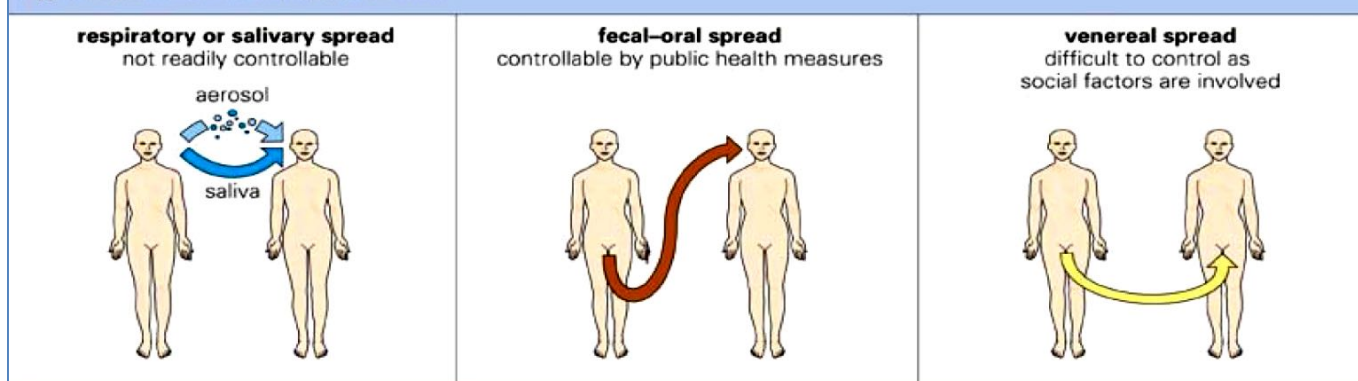
- **Pathogenesis** = The biochemical sequence of events whereby microbes (bacteria, fungi, parasites & viruses) causes disease
- **4 Stages to Infection:**
 - **1: Gain Entry to Host:**
 - Needs a **Portal of Entry & Exit**. Egs:
 - **(For Exogenous Organisms)** - Oral/Skin/Trans-placental/Inhalation/Inoculation (wound/skin penetration)/Sexual
 - **(For Endogenous Organisms)** – Organisms already present On/In Body – Requires Immunocompromise
 - **2: Attach & Multiply:**
 - **Attachment Via:**
 - Adhesion Receptors (Eg: Glycoproteins on Viruses)
 - Cellular Extensions (Eg: Fimbriae/Pili on Bacteria)
 - Physical Structures (Eg: Hooks/Suckers on Helminths)
 - **Multiplication/Spread of Infection:**
 - Local (Abscesses/Mucosal/Nerves/CSF)
 - Systemic (Blood/Sepsis)
 - **Factors Affecting Spread:**
 - **Organism Factors:**
 - Virulence Factors
 - **Host Factors:**
 - Genetic Susceptibility
 - Immune Status
 - (Age, Pregnancy, Nutrition, Etc)
 - **3: Evade Host Defences:**
 - Beat Physical Barriers (Eg: Flushing, Mucous + Cilia, Stomach pH, Lysosomes)
 - Beat Innate Cellular Defences (Eg: Inflammation, Phagocytosis, NK Cells)
 - Beat Adaptive Defences (Eg: Antibodies, Cell-Mediated Immunity)
 - **4: Cause Damage to Cells/Tissues:**
 - Physical Disruption
 - Toxic Damage
 - Aberrant Cell Activity
 - Immune-Mediated Damage



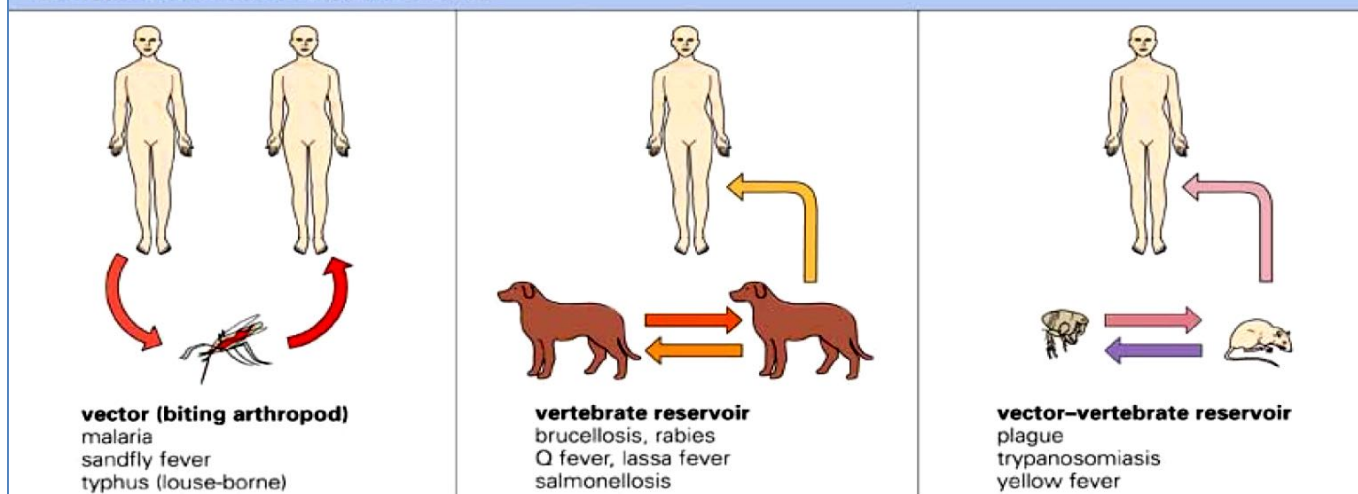
Transmission - (Exiting the infected host & Spread of Infection):

- **Successful Microbes Must Exit the Body → Transmit to a new host. They are Either:**
 - Shed in Secretions/Excretions
 - Taken up by Vectors (Eg: Mosquitoes) from the blood
- **Transmission Depends On:**
 - Number of Organisms Shed (High Numbers Needed)
 - Activities of the host (Eg: Coughing, Sneezing, Diarrhoea, Intercourse)
 - Stability in the Environment (Eg: Amoebic Cysts resist drying & heat → Survive)
 - # Required for Infective Dose (Depends on organism & route)
 - Virulence/Pathogenicity
- **Types of Transmission:** - (Requires a "Vehicle")
 - Airborne – (Must survive outside the host & in Dry Conditions)(Eg: Influenza)
 - Waterborne – (Eg: Cholera)
 - Food-borne – (Spoilage, Preformed Toxins, Faecal-Oral)

types of transmission and their control



arthropod-borne infections and zoonoses



Unattributable

- **Vertical Transmission:** Parent → Offspring
- **Horizontal Transmission:** Person → Person
- **Zoonotic Transmission:** Animal → Human (Via Contact/Inhalation/Ingestion/Bites/Scratches)

Epidemiology:

- **Epidemiology** = “The relationship between factors determining the frequency & distribution of infectious disease in a population”
- **Factors Influencing Epidemiology (Eg: Δ in # of susceptible/environment/organism/new organism):**
 - **The Organism:**
 - Δ in Properties of the Endemic Organism (Eg: Persistence; Transmissibility)
 - New Organism
 - **The Host:**
 - Δ in # of Susceptible Hosts
 - Δ in Concentration Of Susceptible Hosts
 - Δ in Behaviour
 - **The Environment:**
 - Δ in Climate (Eg: Cold $\rightarrow$ People crowd indoors $\rightarrow$ $\uparrow$ Droplet Transmission)
 - Δ in # of Vectors (Eg: Rainy season $\rightarrow$ $\uparrow$ Mosquitoes)

Immune Evasion Strategies:

- **Viruses:**
 - Persist as Latent Infections $\rightarrow$ Reactivation/Recrudescence following Immunosuppression/Stress
 - Superantigens $\rightarrow$ Inappropriate Immune Response
 - Inhibition of MHC-I Synthesis/Assembly/Ag-Loading
- **Bacteria:**
 - Depression of phagocytosis by neutrophils
 - Depress cellular immunity
 - Induction of apoptosis
 - Killing of alveolar macrophages
 - Superantigens $\rightarrow$ Inappropriate Immune Response
 - Produce superoxide dismutase, catalase or oxidase $\rightarrow$ protect it from the hydrogen peroxide of the respiratory burst of Neutrophils
 - Intracellular bacterial evasion:
 - Travel b/w cells without being exposed to extracellular fluid
 - Escape into vacuole in the cytoplasm
 - Prevent fusion of lysosomes with phagosomes
- **Parasites:**
 - **Protozoan Parasites:**
 - Antigenic Variation
 - Antigenic Drift
 - Molecular Mimicry (Expression of Host Proteins)
 - Intracellular Localisation
 - Self-Isolation in Membrane-bound Vesicle
 - Prevent fusion of lysosomes with phagosomes
 - Sequestration in privileged sites
 - Regulation of host functions
 - **Helminth Parasites:**
 - Antigen Shedding
 - Protease production $\rightarrow$ Neutralise some immune components (Eg: Antibodies)
 - Superoxide Dismutase $\rightarrow$ Neutralise Respiratory Burst by Neutrophils
 - Regulation of host functions (Immunosuppression/Maladaptive Response)
 - Skew the T-Helper Response to Favour Th1-Cells:
 - Favouring Th1 $\rightarrow$ Reduced class-switching to IgE, the AntiParasitic Antibody
 - Use Host Cytokines as Parasitic Growth Factors

MICROBIOLOGY: PRIONS

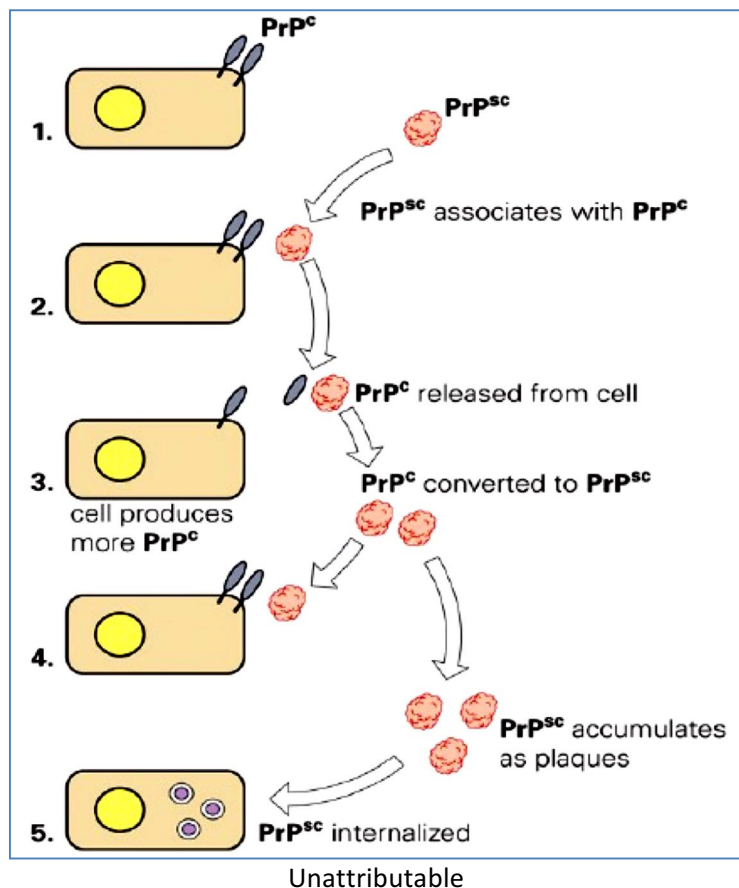
MICROBIOLOGY: PRIONS

Prions; What are they?

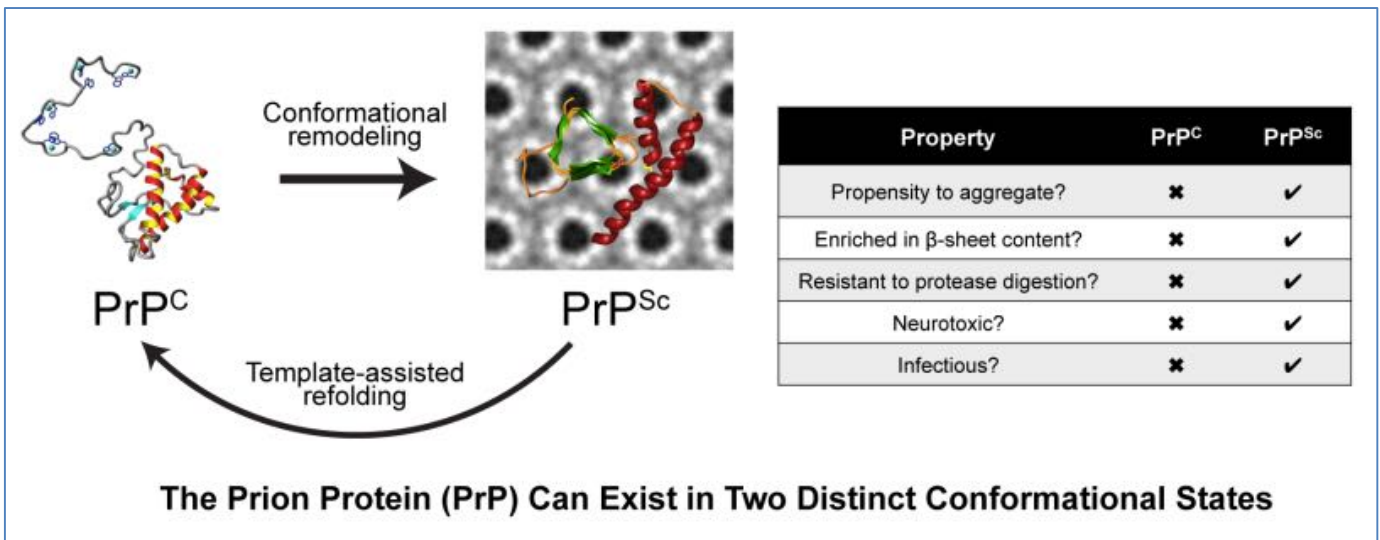
- **Abnormally folded Host-Proteins** that accumulate in the brain → Spongiform Encephalopathies
- Note: **All** known Prion Diseases **affect the Brain** and are **currently Untreatable & Universally Fatal**
- The precise structure of the Prion is Unknown
- **Derivation of the term "prion"**: Proteinaceous, Infectious + 'on'

TSE's – (Transmissible Spongiform Encephalopathies):

- **EG: CREUTZFELDT JACKOB DISEASE, GERTSMANN-STRAUSSLER SYNDROME, KURU KURU:**
 - **Aetiology:**
 - Prion Infection of the Brain
 - **"Prions" = Proteinaceous, Infectious + 'on'**
 - = **Abnormally folded Host-Proteins** that accumulate in the brain
 - **NO DNA or RNA!! (Important for Exams)**
 - **Prion Proteins (PrP):**
 - **Normal Form = PrP^c (Cellular)**
 - Normal α -Helix form (Functional & Denaturable)
 - Found throughout the body (Also in mammals)
 - **Abnormal Form = PrP^{sc} (Scrapie)**
 - Abnormal β -Sheet form (Non-Functional & Non-Denaturable)
 - Accumulates in plaques in the brain → Tissue Damage & Cell Death
 - **EXTREMELY STABLE** – Resists denaturation ∴ Difficult disposal
 - **Pathogenesis:**
 - Prions cause Neurodegenerative Disease by aggregating Extra-Cellularly in the CNS → form amyloid plaques → Plaques are Internalised → Vacuole formation in Neurons → Spongy Architecture
 - Accumulation in Neurons → Death of Neurons
 - **Propagation: Conversion of Normal Proteins (α -helix → β -sheet):**
 - Prions propagate by transmitting a **Mis-Folded Protein State**, not replicating
 - I.e: They convert *Pre-Existing, Normal* forms of the protein to the *Abnormal Form*



- **Morphology:**
 - **Macro:**
 - Empty cystic lesions in the brain → Spongiform Encephalopathy
 - **Micro:**
 - Neuronal Vacuolation & Plaque Formation
- **Clinical Features:**
 - Initially Subtle Memory & Behavioural Changes → Then Rapidly Progressive Dementia
 - Convulsions (Myoclonus)
 - Dementia
 - Ataxia, Dysarthria, Dysphagia, Nystagmus
 - Behavioural/Personality Changes
- **Prognosis:**
 - **All** known Prion Diseases **affect the Brain** and are **currently Untreatable & Universally Fatal**
 - **7mths life expectancy**



Source: <https://joelwattslab.org/prions/>

Transmission:

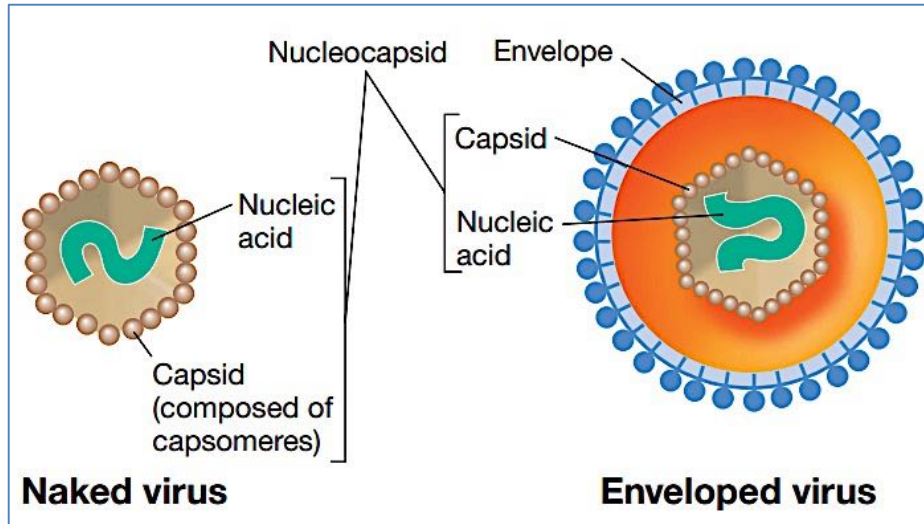
- **Acquired, Familial, or Sporadic**
- **Current Theory** – Primarily infected through ingestion. Prions may be deposited in the environment through Animals Carcasses, Urine, Saliva, other body fluid; and may linger in the soil.

MICROBIOLOGY: VIRUSES

MICROBIOLOGY: VIRUSES

Virus Nomenclature:

- **Virion** – A Complete Viral Particle
- **Capsid** – The Protein Coat made up of smaller structural Subunits (Capsomeres)
- **Capsomeres** – The Subunits of the Capsid
- **Nucleocapsid** – The Capsid + Nucleic Acid + Associated Nucleoproteins
- **Envelope** – Lipid Bilayer of *Host-Cell Origin*, imbedded with Viral Lipoproteins



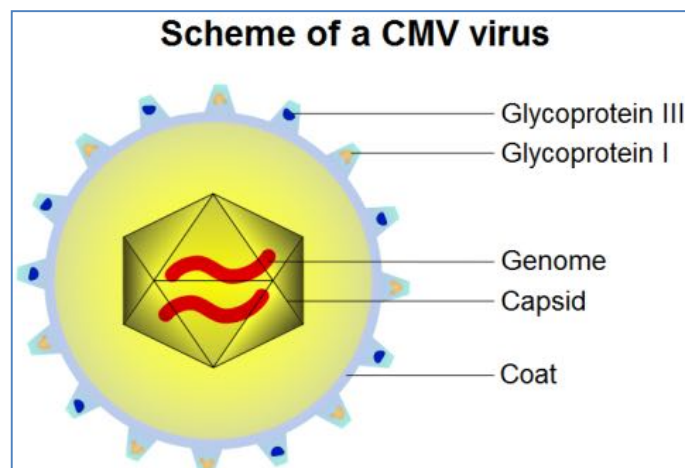
<https://microbeonline.com/virus/>

Properties Distinguishing Viruses from other Microorganisms:

- Acellular
- No Cell Membrane
- No Cytoplasm
- Can have a DNA or RNA Genome (All others only have DNA)
- No Cellular Synthetic Machinery (Metabolically Inert)
- Can Only Replicate in Living Cells

Viral Envelopes - (Construction/Origin/Proteins):

- **Origin:** Some viruses envelop themselves in a modified piece of host cell membrane (Either the Plasma Membrane, or Organelle Membranes)
- **Construction:** This membrane is studded with Viral & Host Proteins. Most enveloped viruses depend on the envelope for infection.
- **Proteins:** Viral envelopes are studded with **Glycoproteins** – Serve to identify and bind to receptor sites on the host's membrane. The viral envelope then fuses with the host's membrane, allowing the capsid and viral genome to enter and infect the host.



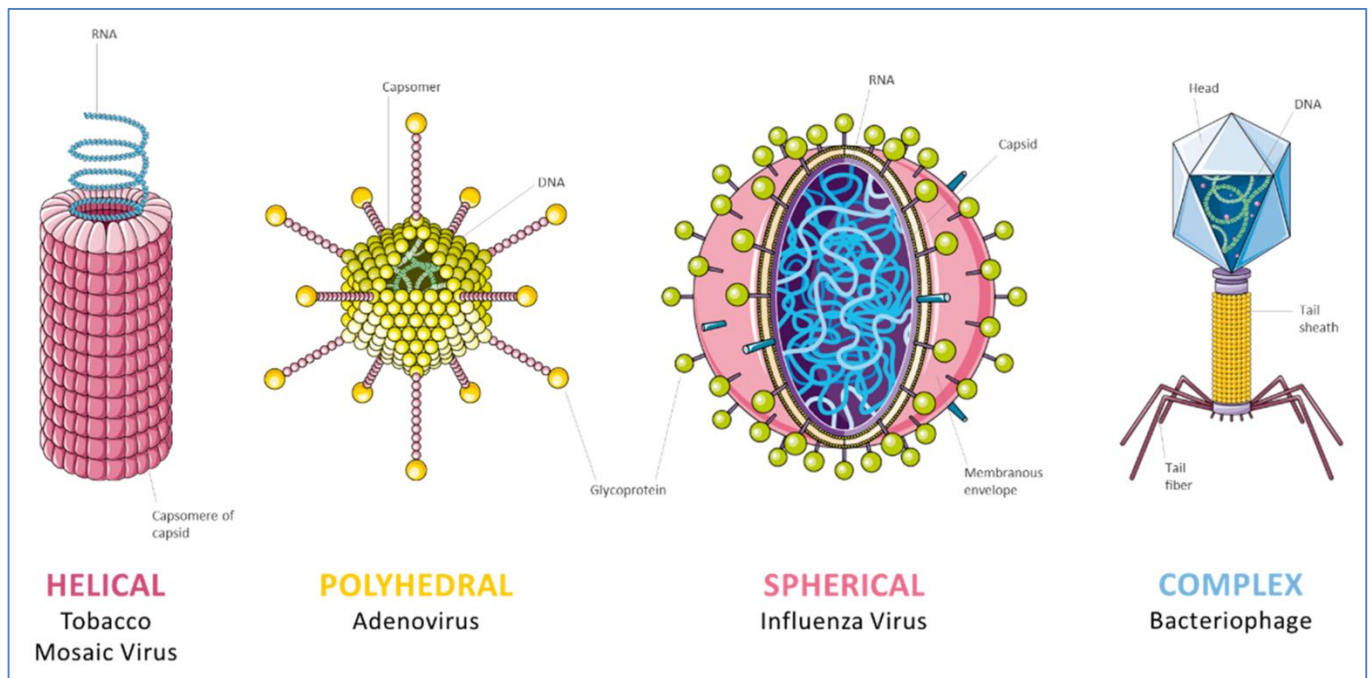
Emmanuel Boutet, CC BY-SA 2.5 <<https://creativecommons.org/licenses/by-sa/2.5/>>, via Wikimedia Commons

Structural Vs Non-Structural Proteins:

- **Structural Proteins:**
 - Proteins Encoded by a Virus that form Structural Components of the end Viral Particle
- **Non-Structural Proteins:**
 - Proteins Encoded by a Virus, but NOT part of the Viral Particle

Symmetry:

- **Helical Symmetry:**
 - Composed of a single type of capsomere stacked around a central, coiled Nucleic Acid → form a helical structure
 - Results in rod-shaped or filamentous virions: (Short and Rigid, or Long and Flexible)
- **Cubic (Icosahedral/Polyhedral) Symmetry:**
 - Icosahedron = a regular polyhedron with 20 identical equilateral triangular faces
- **Spherical Symmetry:**
 - Membranous envelope forms spherical shape. (Eg. Influenza virus)
- **Complex Symmetry:**
 - Capsid is neither purely helical, nor purely icosahedral, and that may possess extra structures such as protein tails or a complex outer wall
 - Some have a Cubic head bound to a Helical tail, with protruding protein tail fibres that attach to the host cell and then injecting the viral genome into the cell



Artasensi, A.; Mazzotta, S.; Fumagalli, L. Back to Basics: Choosing the Appropriate Surface Disinfectant. *Antibiotics* **2021**, *10*, 613. <https://doi.org/10.3390/antibiotics10060613>

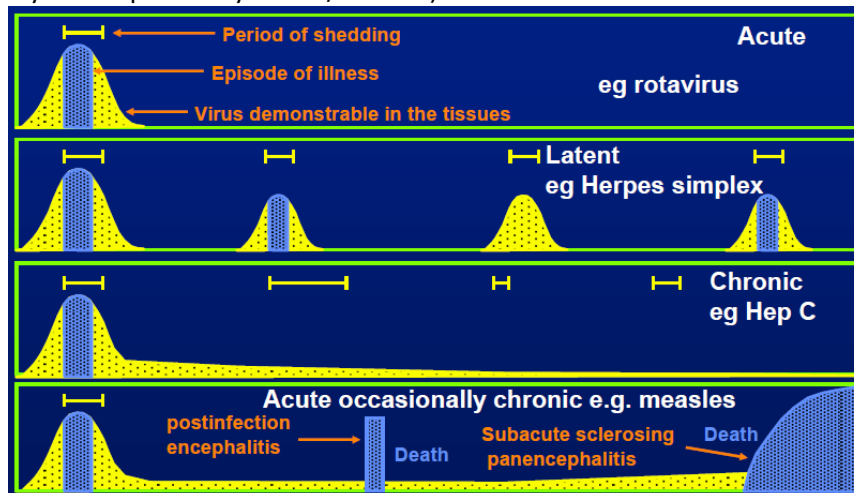
Requirements for Viability & Culturing Viruses:

- **Viability Requires:**
 - **1: Must Retain an Intact Protein Coat**
 - (Note: Enveloped Viruses are Inactivated by Detergents → Disperses Lipid Bilayer)
 - **2: Must Retain an Intact Genome**
- **Culturing Viruses Requires:**
 - Living Cells – (Because Viruses lack the cellular machinery for replication)

"Shedding" & Disease:

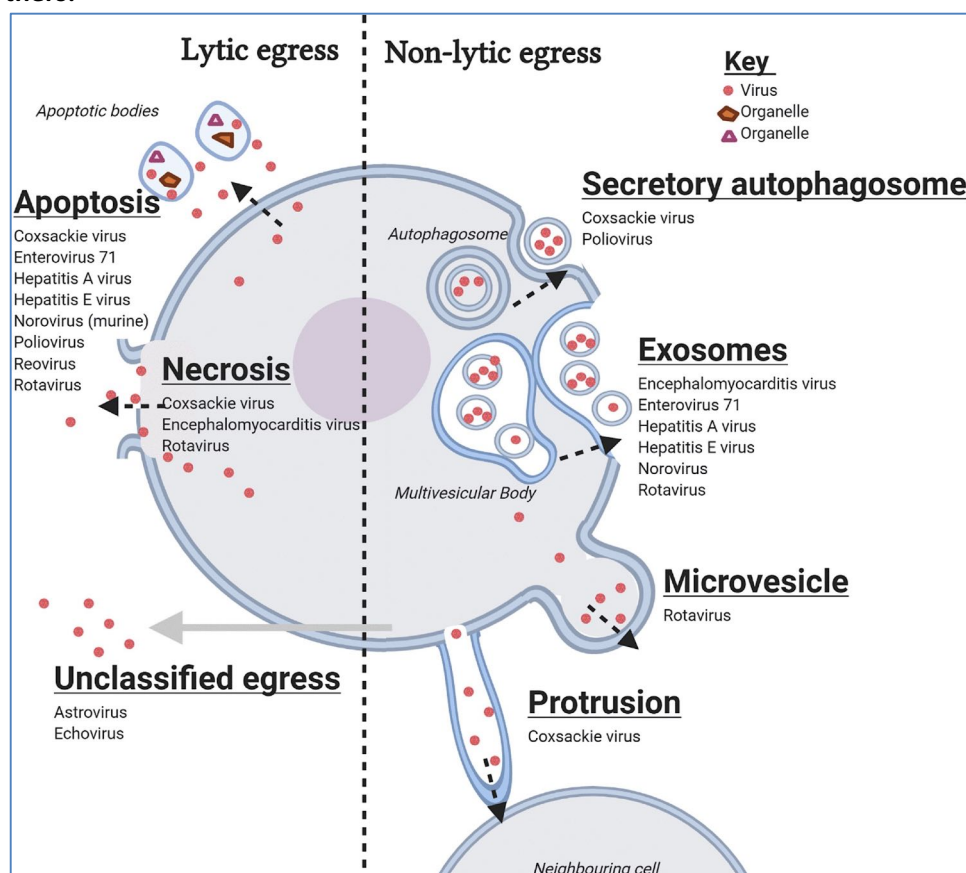
- Viral Shedding:

- The Successful Reproduction, Expulsion & Host-Cell Infection caused by Virus Progeny
- (Typically Accompanied by Illness/Disease)



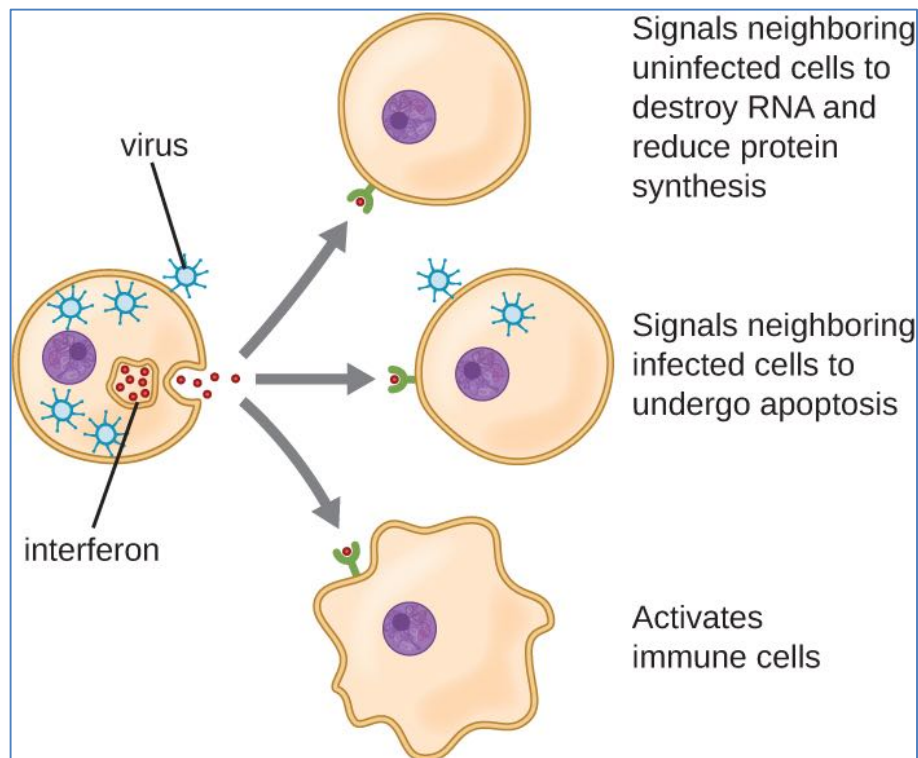
- Examples of Shedding Mechanisms:

- **Shedding Via Budding:**
 - "Budding" through the cell membrane, using it to form the viral Envelope
 - Primarily Enveloped Viruses
- **Shedding Via Apoptosis:**
 - Forcing cell into Apoptosis → Release of progeny into Extracellular Space within apoptotic bodies. Macrophages phagocytose the apoptotic bodies → Become Infected
 - Primarily Non-Enveloped Viruses
- **Shedding Via Exocytosis:**
 - Exocytotic release of Viral Progeny into the Extracellular Space
 - Primarily Non-Enveloped Viruses
- **Others:**



Innate Immunity Against Viruses:

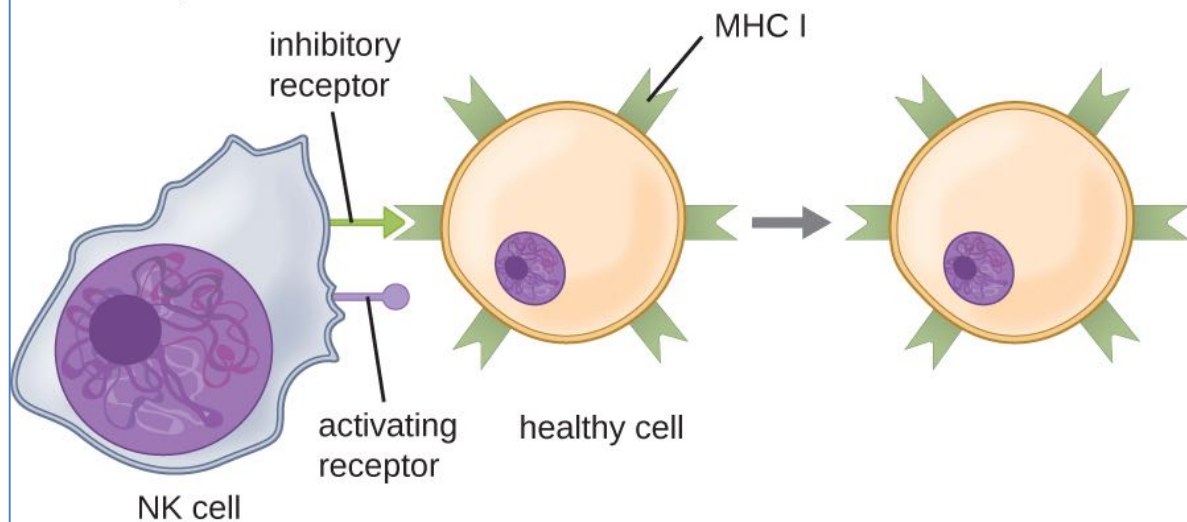
- *****Interferons (IFNs):**
 - **(Four Major Classes):**
 - IFN α – Produced by virally-infected WBCs
 - IFN β – Produced by virally-infected Fibroblasts
 - IFN γ – Produced by Ag-Stimulated Effector T-Cells (Helper & Cytotoxic)
 - IFN ω – Secreted by Embryonic Trophoblasts
 - **Early, non-specific Anti-Viral Proteins (Particularly IFN- γ)**
 - Secreted by Virally Infected Cells to protect nearby cells that haven't yet been infected
 - **Mechanism of Action → IFN results in Synthesis of Gene Products:**
 - ****Ribonuclease:**
 - Cleaves Viral mRNA → Inhibits Viral Protein Synthesis & Reproduction
 - Allows time for Adaptive Immunity to destroy infected cells
 - **Nitric Oxide Synthase:**
 - Prevents viral growth in Macrophages
 - **Protein Kinase:**
 - Prevents Elongation of Viral dsRNA
 - **Mx Protein:**
 - Can inhibit the Transcription & Translation of some viral mRNA
 - **Also Activates Natural Killer-Cells**



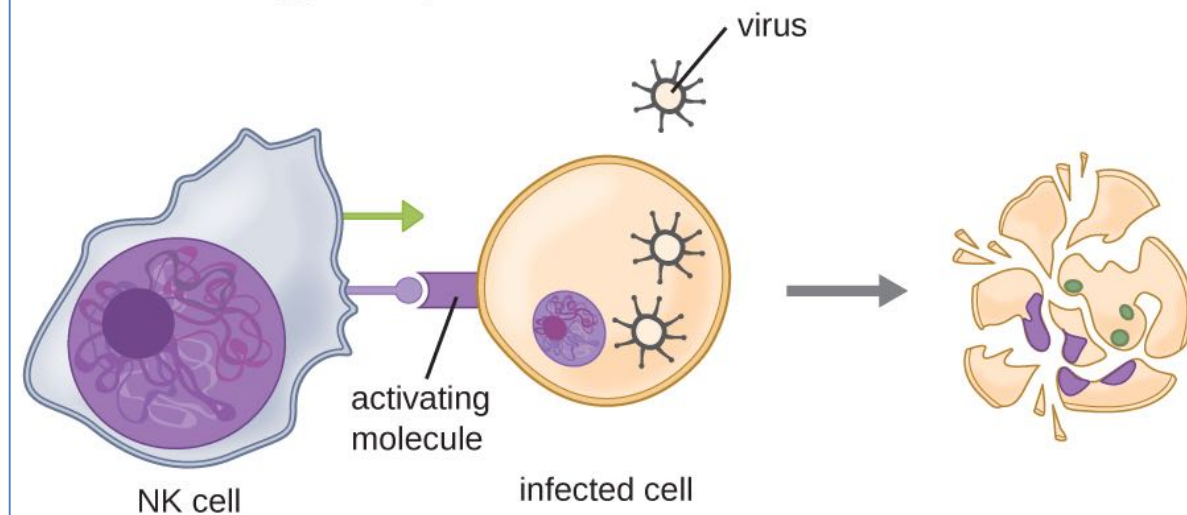
CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

- *****Natural Killer Cells:**
 - (Activated by IFN- γ)
 - Lyse some Virally-Infected Cells
 - Altered/Missing MHC-I → NK cell lyses cell

A natural killer (NK) cell recognizes MHC I on a healthy cell and does not kill it.



An infected cell does not present the MHC I, but does present ligands for the activating receptor. The NK cell will trigger a response that kills this cell.

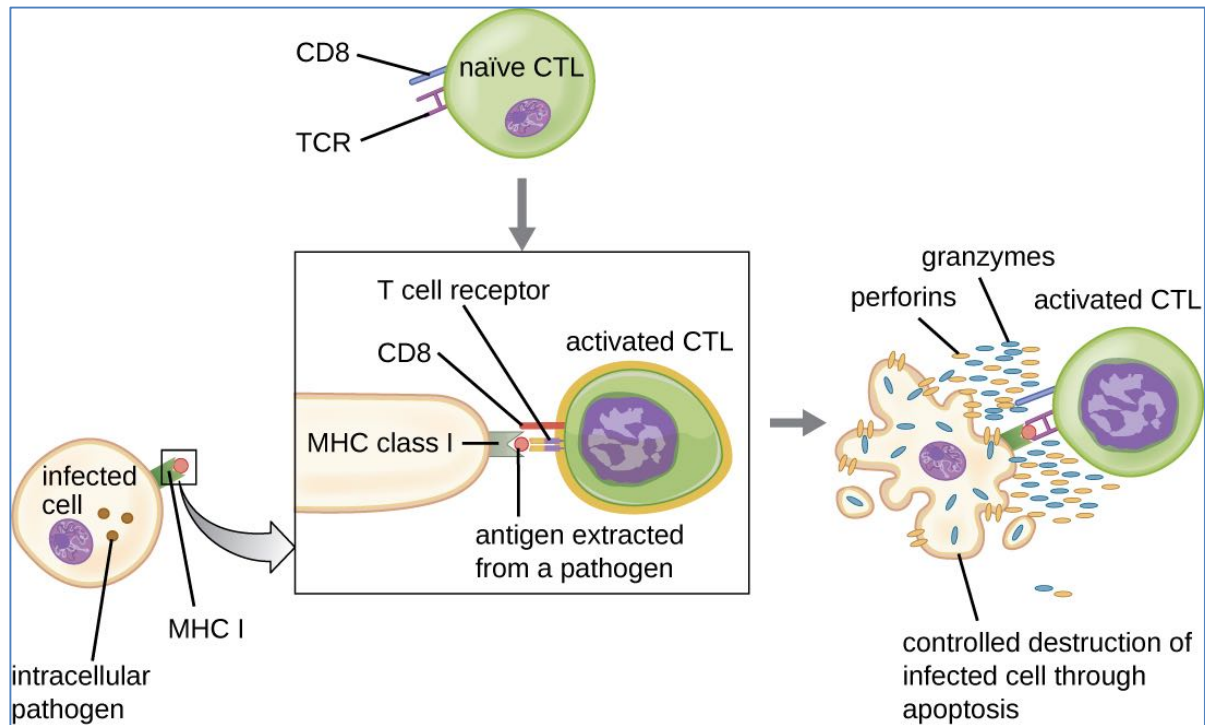


CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

- ****Compliment Activation (Alternate Pathway) & Phagocytosis of Extracellular Viruses:**
 - o C3b opsonisation → Phagocytosis
- **Lysozyme:**
 - o (in Tears/Saliva/Mucus/Neutrophils)
 - o Some viruses are susceptible
- **Stomach Acid:**
 - o Denatures some viruses
- **Intestinal Enzymes:**
 - o Degrade some viruses

Adaptive Immunity Against Viruses:

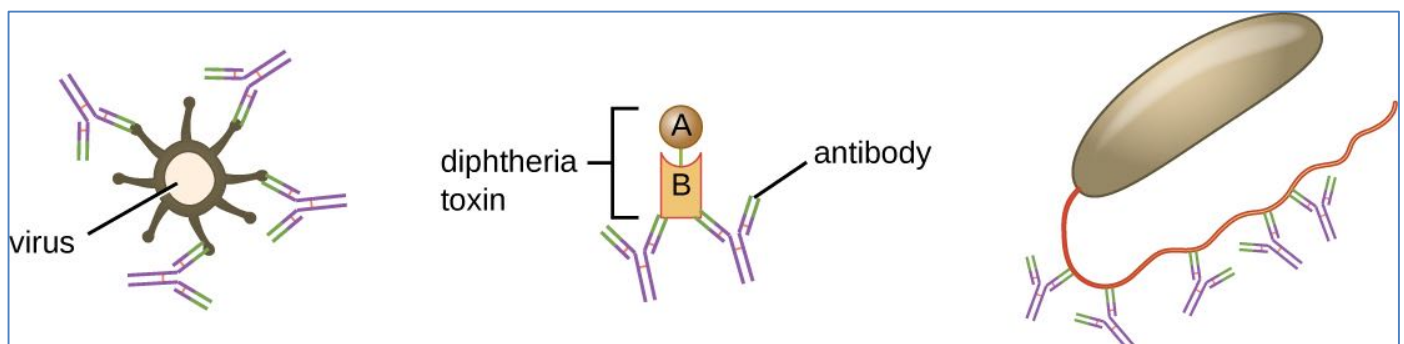
- ****Helper CD4 T-Cells:**
 - → Secretion of IFN- γ (→ Further activates NK Cells)
 - → Activates Macrophages → Kill intracellular contents
 - → Activates CD8-T-Cells → Proliferate
- ****Cytotoxic CD8 T-Cells:**
 - Recognition of **Viral Peptide:MHC-I** → Cytotoxic Granules line up @ site of cell contact
 - → Apoptosis of Virally Infected Cells
 - (also → Secretion of IFN- γ) (→ Further activates NK Cells)



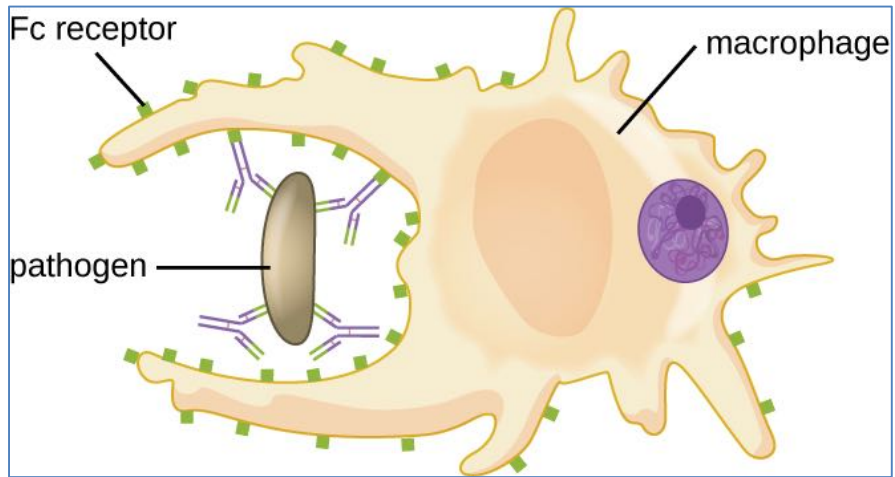
CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

- ****Antibodies:**
 - **Antibodies** → **Neutralise** Extracellular Viruses
 - (By Blocking Viral Absorption & causing Agglutination)
 - **Antibodies** → **Opsonisation** of Virus for **Phagocytosis** (Macrophages)
 - **Antibodies** → **Opsonisation** of Virus for **Antibody-Dependent Cell-Mediated Cytotoxicity**
 - (ADCC – Fc Receptors on Cytotoxic cells bind to Antibody → Lysis of Virus)
 - **Antibodies + Complement** → **Opsonisation** of Virus for Phagocytosis (Macrophages)
 - **Antibodies + Complement** → **Violysis** (NK Cells/T_c-Cells)
 - **Antibodies + Viral Ags on Cells** → Initiate Complement → CD8-mediated Lysis of infected cell
 - **Antibodies + Viral Ags on Cells** → Cell-Mediated Cytotoxicity → Lysis of infected cell

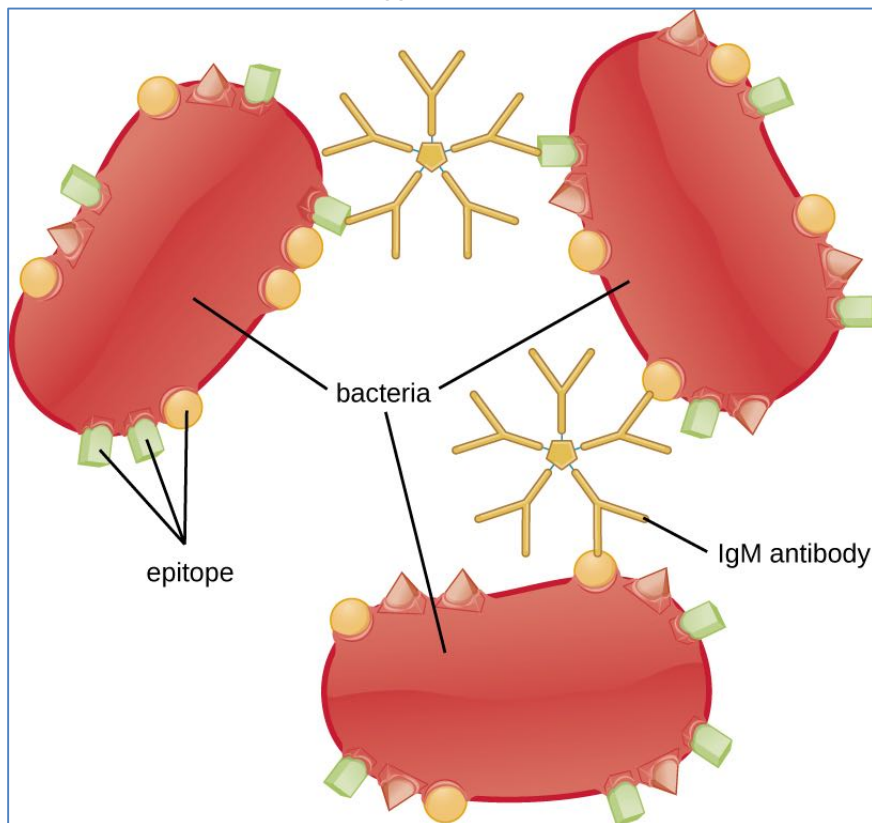
Neutralisation



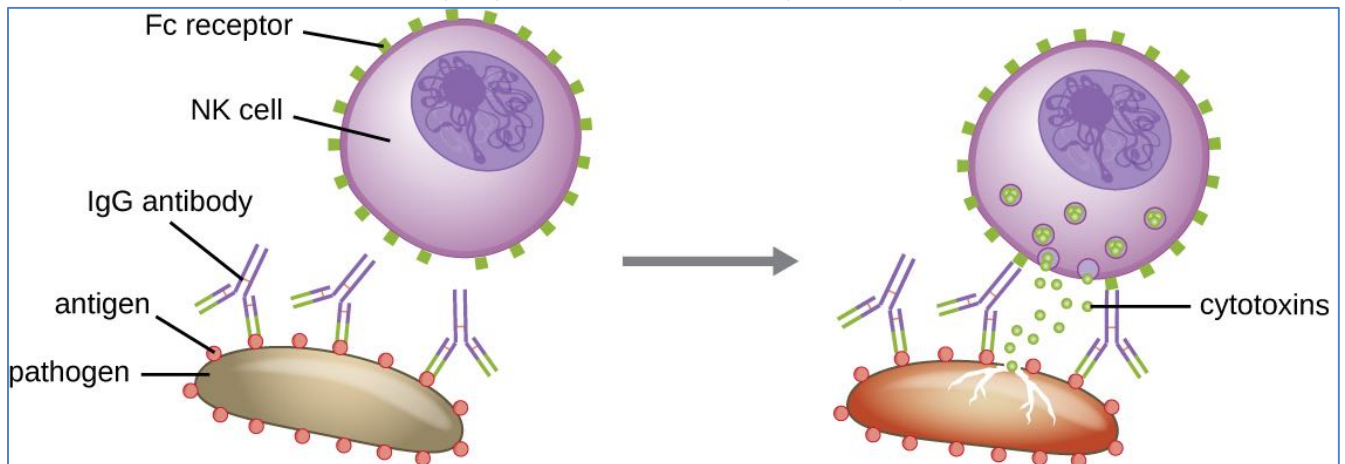
Opsonisation



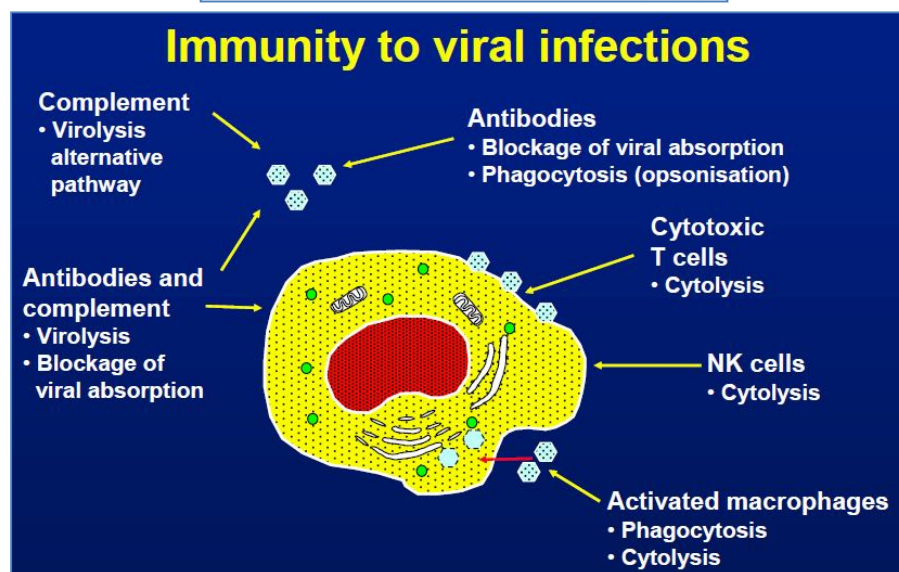
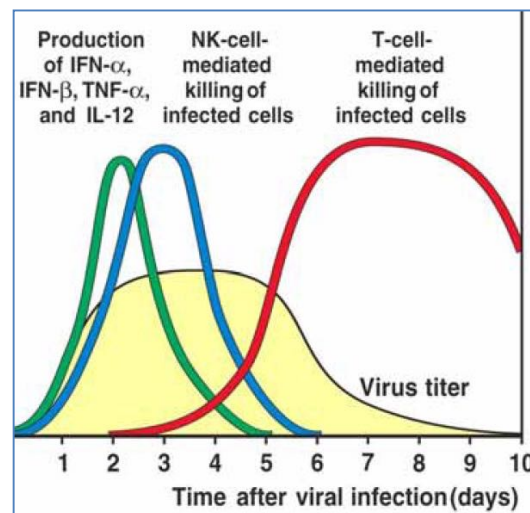
Agglutination



Antibody-dependent cell-mediated cytotoxicity (ADCC)



- **Activated Macrophages:**
 - (Via CD4 T-Helper Cells)
 - → Phagocytosis & destruction of Extracellular Viruses



Latency (Recurrence & Recrudescence):

- **Latency = The Ability of a Pathogenic Virus to lie Dormant within a cell**
- **Virus Production Ceases:**
 - NO active Viral Shedding
 - NO Pathologies/Symptoms
- **However, Latency is still an Active Process:**
 - Maintaining latency requires expression of viral genes which may function to:
 - Keep the viral genome from being digested by cellular *Ribozymes*
 - Downregulate MHC-I to hide from the immune system
 - Inhibit Apoptosis
 - Induce Cell Growth/Division
- **2 Types of Latency:**
 - **Episomal Latency:**
 - Viral genes are left floating in the Cytoplasm or Nucleus
 - (Eg: Herpes Virus)
 - **Proviral Latency:**
 - Virus genome Integrates into the Host Genome → Becomes a Provirus
 - (Eg: HIV)
- **Reactivation/Recrudescence:**
 - A Latent Virus can Reactivate
 - Triggers include Stress, Sunlight

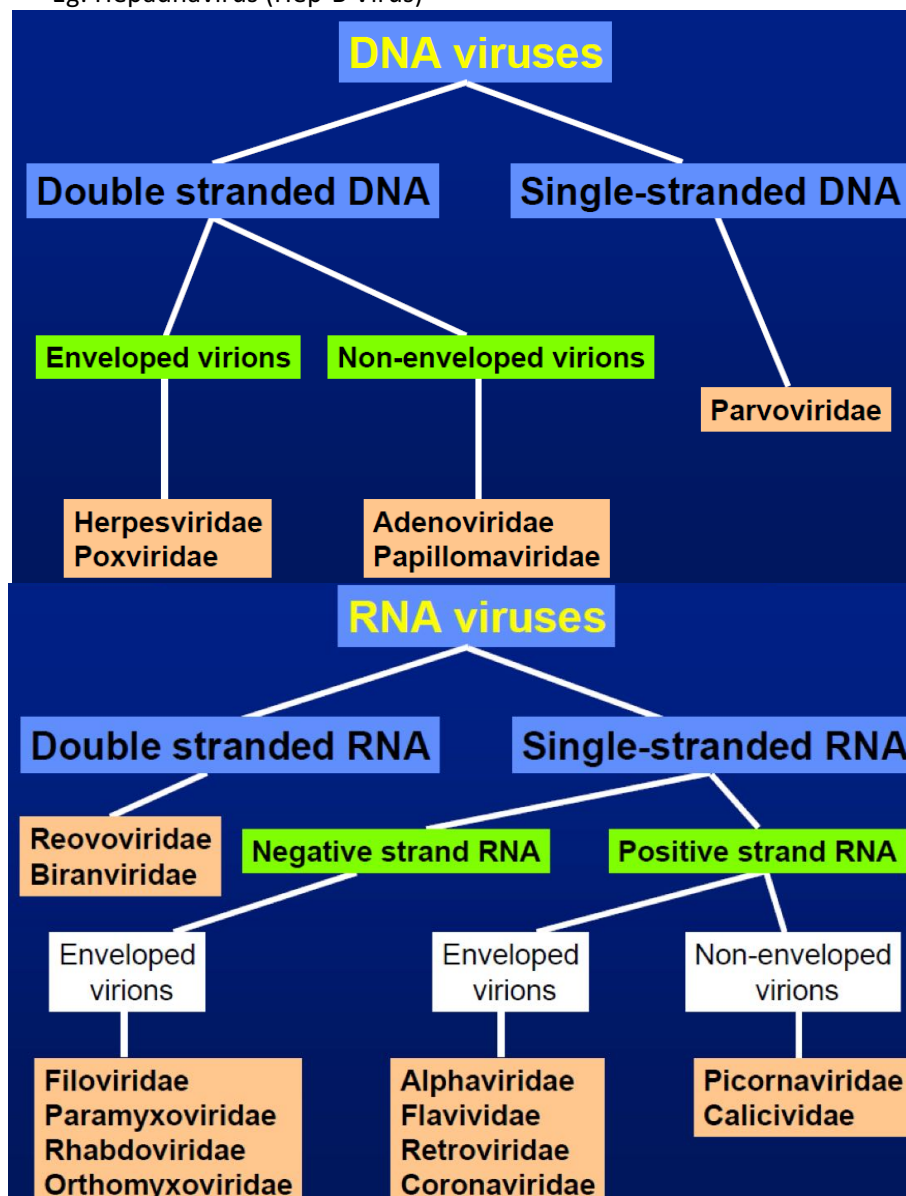
Classification of Viruses:

- The ICTV Classification:

- **Family** (“-viridae”)
 - **Genus** (“-virus”)
 - **Species**
- (I.e: Binomial Nomenclature isn't used)

- The Baltimore Classification:

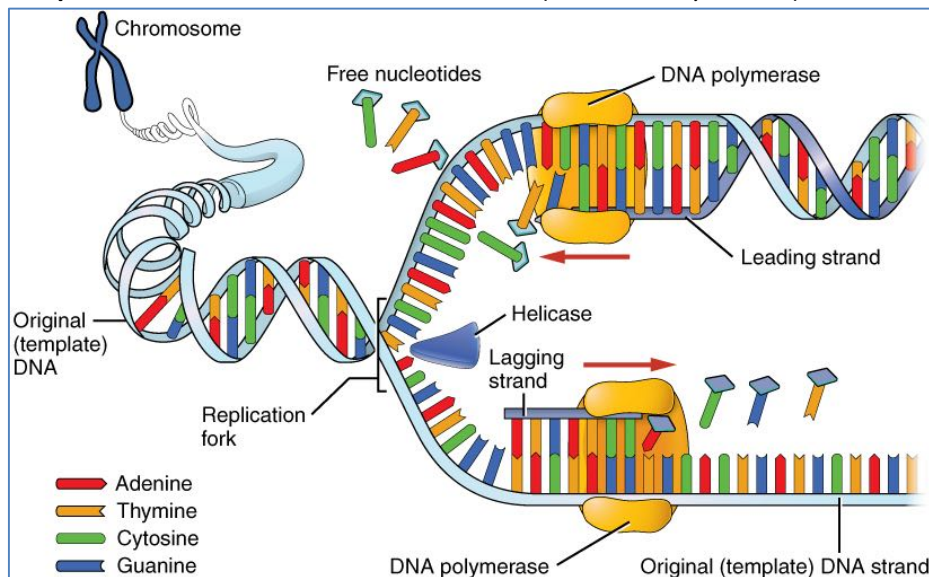
- (7 Groups of viruses based on DNA/RNA, Strandedness (Single/double), Sense/Nonsense, & Method of Replication)
- **1: dsDNA Viruses** (**double-stranded DNA Viruses**)
 - Eg: Herpesvirus, Poxvirus, Adenovirus
- **2: ssDNA Viruses** (**single-stranded DNA Viruses**)
 - Eg: Parvovirus
- **3: dsRNA Viruses** (**double-stranded RNA Viruses**)
 - Eg: Reovirus
- **4: (+)ssRNA Viruses** (**positive [sense] single-stranded RNA Viruses**)
 - Eg: Picornavirus, Togavirus
- **5: (-)ssRNA Viruses** (**negative [nonsense] single-stranded RNA Viruses**)
 - Eg: Orthomyxovirus, Rhabdovirus
- **6: ssRNA-RT Viruses** (**single-stranded RNA Reverse Transcriptase Viruses**)
 - Eg: Retroviruses (HIV)
- **7: dsDNA-RT Viruses** (**double-stranded DNA Reverse Transcriptase Viruses**)
 - Eg: Hepadnavirus (Hep-B Virus)



Revision of DNA Replication/Transcription/Translation & Enzymes (For Viral Replication Cycles):

- **DNA Replication:**

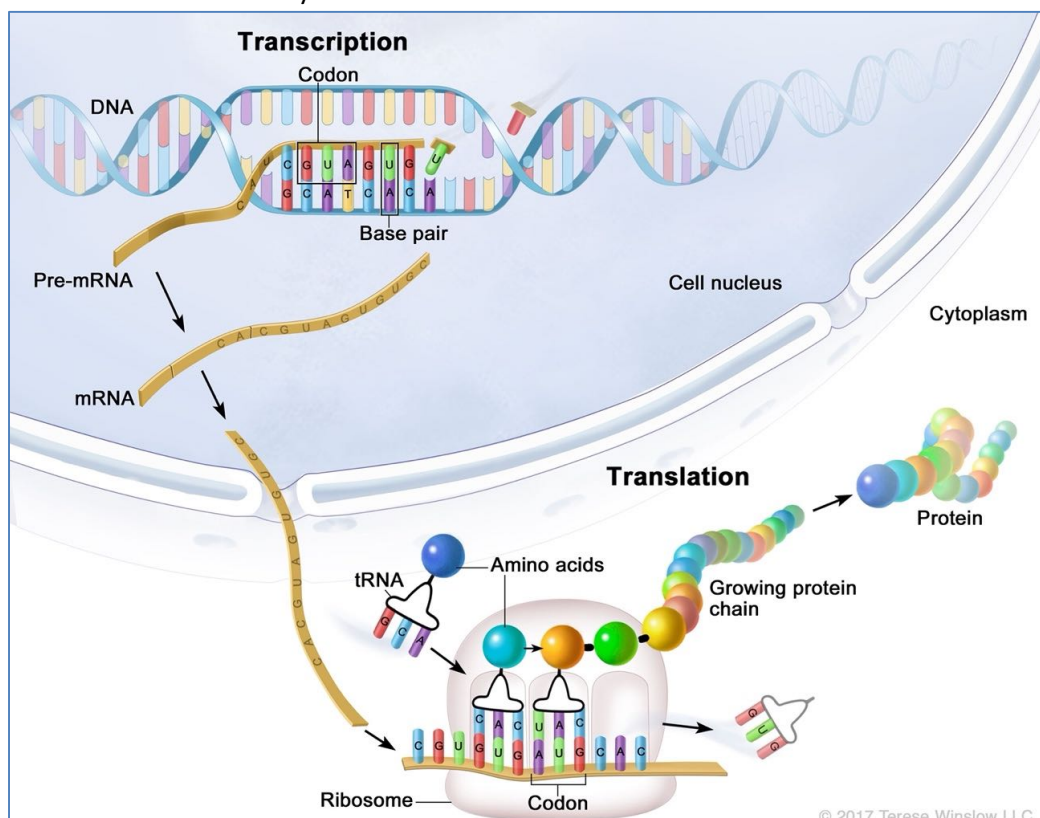
- **DNA Replication** **DNA → DNA** (Via DNA Polymerase)



CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

- **Protein Synthesis:**

- **1: Transcription:** **DNA → mRNA** (Via RNA Polymerase)
 - mRNA Exits the Nucleus → Cytosol
- **2: Translation:** **mRNA → Protein** (Via Ribosomes)
 - Occurs in the Cytosol



Public Domain: <https://www.cancer.gov/publications/dictionaries/genetics-dictionary/def/transcription>

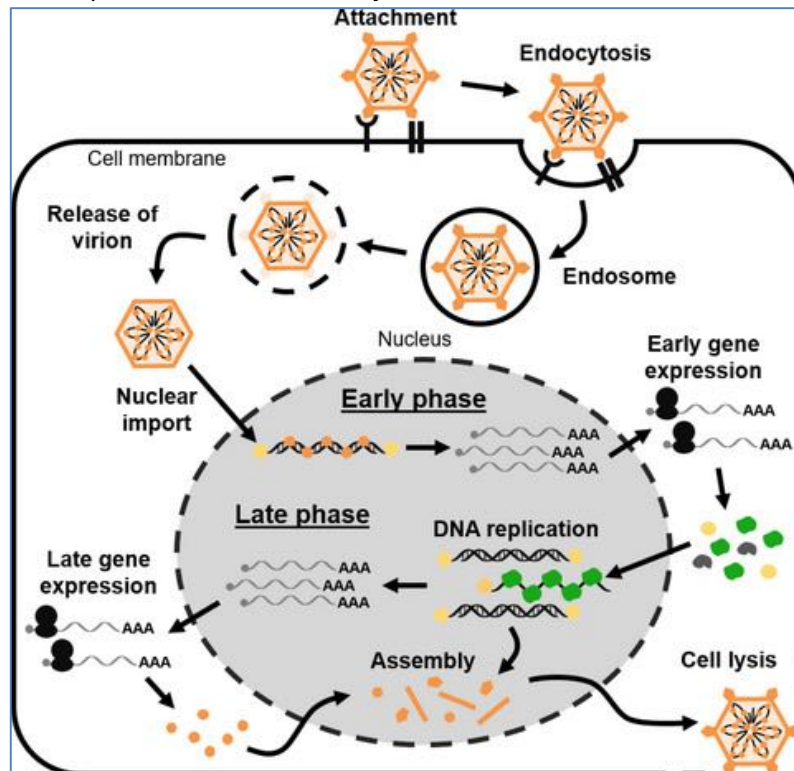
- **Enzymes: (Host Enzymes) (Viral Enzymes):**

- **DNA Polymerase** – Synthesizes new DNA from DNA
- **(DNA-Dependent) RNA Polymerase** – Synthesizes mRNA from DNA
- **RNA-Dependent RNA Polymerase (Transcriptase)** – Synthesizes new mRNA from mRNA
- **RNA-Dependent DNA Polymerase (Reverse Transcriptase)** – Converts mRNA back to dsDNA
- **Retroviral Integrase** – Allows viral DNA to be *integrated* into the DNA

Viral Replication Cycles:

- 1: dsDNA Viruses (double-stranded DNA Viruses)

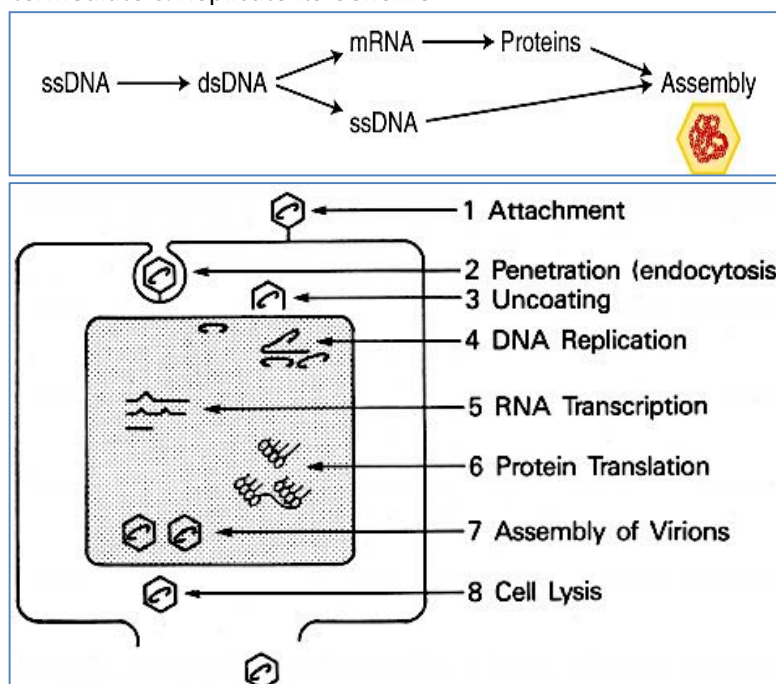
- Eg: Herpesvirus, Poxvirus, Adenovirus
- **Replication Features:** (In the Nucleus (As with all DNA Viruses))
 - Requires *Host-Cell DNA Polymerase* (& hence Host-Cell Division) to Replicate its Genome
 - Also Requires *Host-Cell RNA Polymerase* to transcribe dsDNA → mRNA for Protein Synthesis



Public Domain: https://www.ogtr.gov.au/sites/default/files/2021-06/dir180-full_risk_assessment_and_risk_management_plan.pdf

- 2: ssDNA Viruses (single-stranded DNA Viruses)

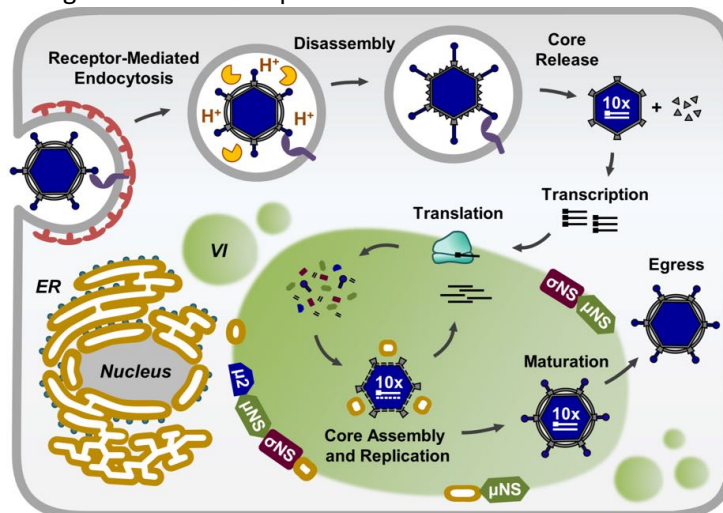
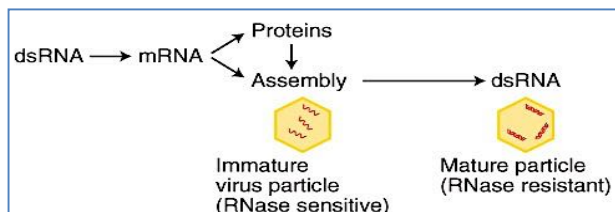
- Eg: Parvovirus
- Replication
- **Replication Features:** (In the Nucleus (As with all DNA Viruses))
 - Requires *Host-Cell DNA Polymerase* (& hence Host-Cell Division) to form a dsDNA Intermediate & Replicate its Genome



Human Parvovirus B19: https://www.researchgate.net/figure/Schematic-life-cycle-of-B19_fig3_11279270

- 3: dsRNA Viruses (double-stranded RNA Viruses)

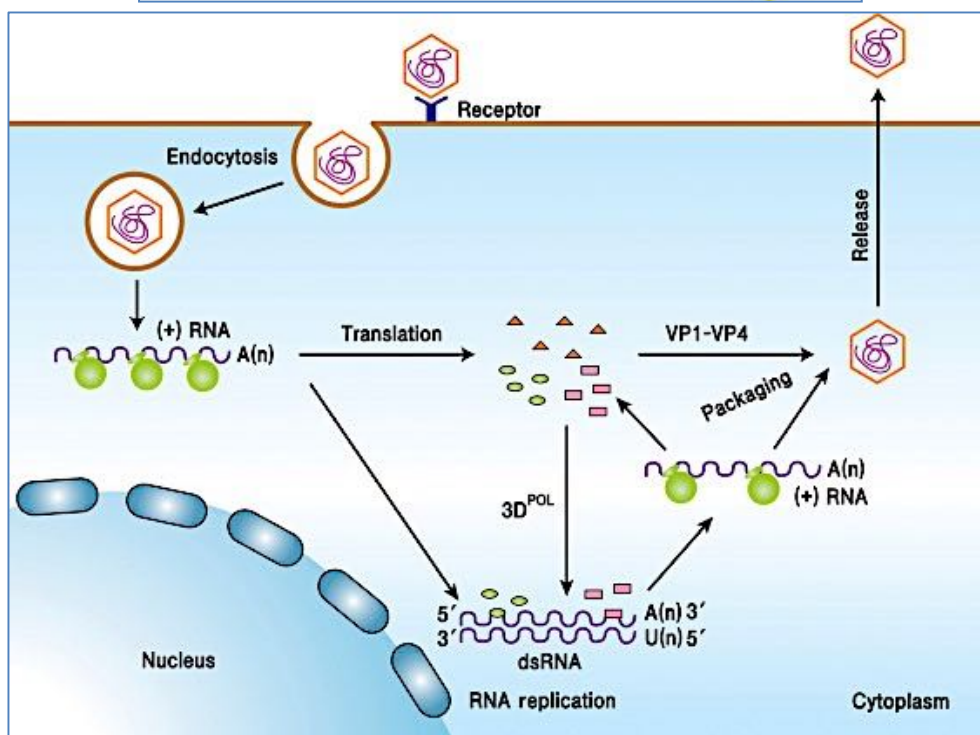
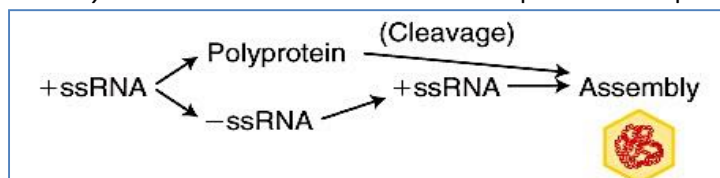
- Eg: Reovirus
- **Replication Features:** (In the Cytoplasm (As with all RNA Viruses))
 - Supplies its *Own RNA-dependent-RNA-Polymerase* for RNA Replication
 - (As opposed to RNA Polymerase which transcribes DNA → RNA)
 - Have *Segmented Genomes* – each gene codes for 1x protein



Source: https://www.researchgate.net/figure/The-reovirus-replication-cycle-VI-viral-inclusions-ER-endoplasmic-reticulum_fig1_331945223

- 4: (+)ssRNA Viruses (positive [sense] single-stranded RNA Viruses)

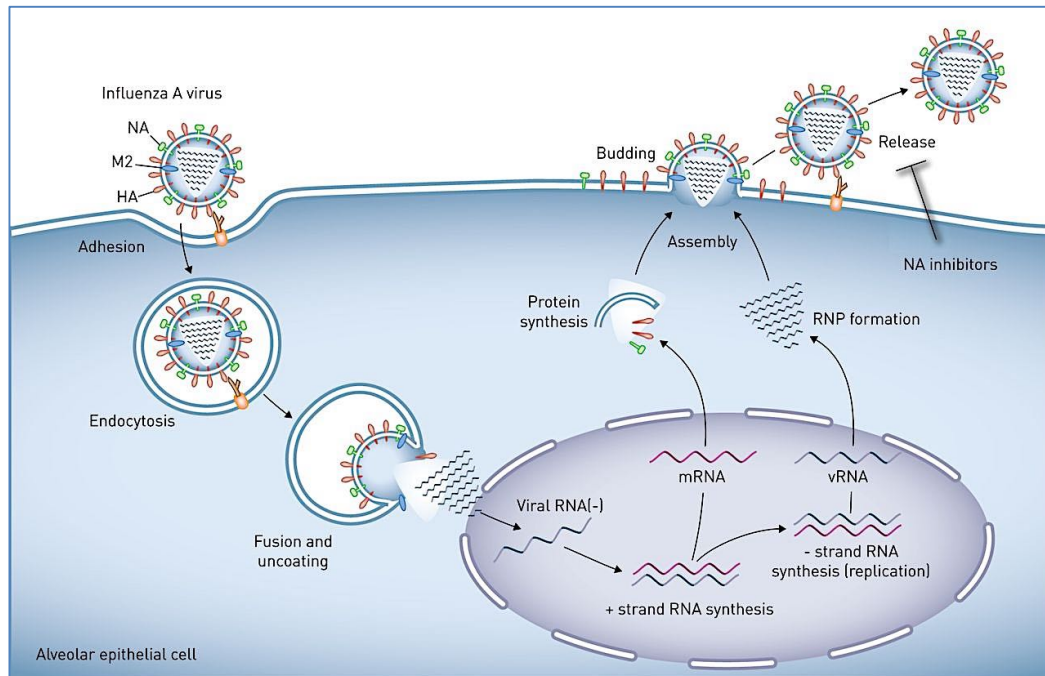
- Eg: Picornavirus, Togavirus
- **Replication Features:** (In the Cytoplasm (As with all RNA Viruses))
 - Supplies its *Own RNA-dependent-RNA-Polymerase* for RNA Replication
 - Directly access *Host Ribosomes* → Viral Poly-Protein Synthesis
 - *Poly-Proteins* are cleaved to form multiple different proteins



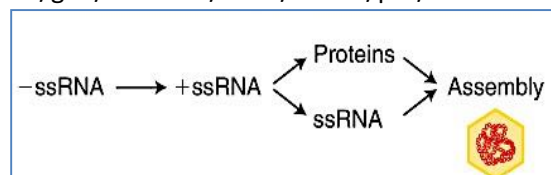
Unattributable

- **5: (-)ssRNA Viruses (negative [nonsense] single-stranded RNA Viruses)**

- Eg: Orthomyxovirus, Rhabdovirus
- **Replication Features:** (In the Cytoplasm (As with all RNA Viruses))
 - Can't directly access *Host Ribosomes* – Because it is a *Nonsense Strand*
 - Must first use its *Own RNA-dependent-RNA-Polymerase* to transcribe a *Positive (Sense)* Strand
 - Positive Strand → Accesses **Ribosomes** → Viral *Poly-Protein* Synthesis
 - *Poly-Proteins* are cleaved to form multiple different proteins

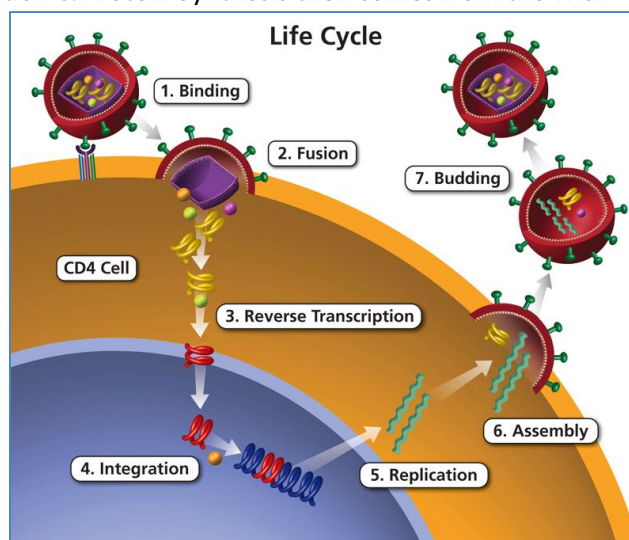


http://geb.uni-giessen.de/geb/volltexte/2017/13103/pdf/JankauskaiteLina_2017_08_17.pdf



- **6: ssRNA-RT Viruses (single-stranded RNA Reverse Transcriptase Viruses)**

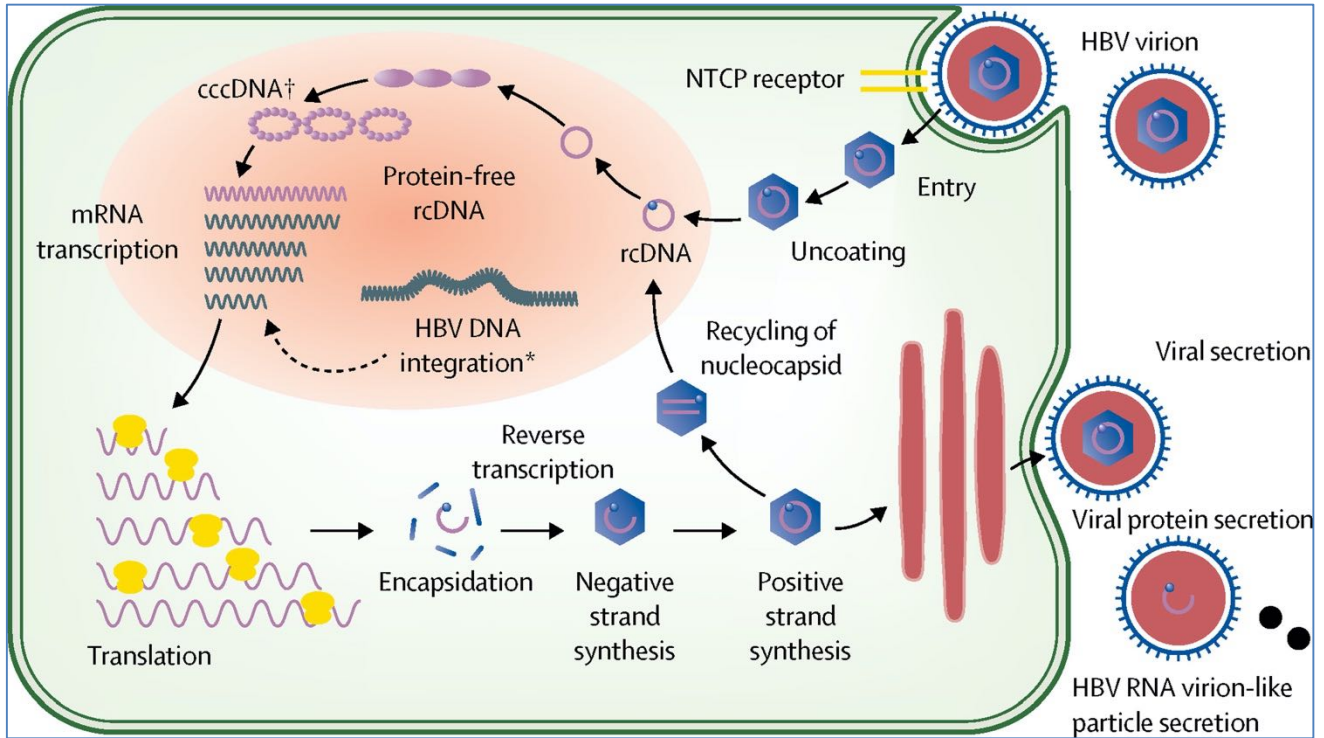
- Eg: Retroviruses (HIV)
- **Replication Features:** (In the Cytoplasm AND the Nucleus)
 - Instead of using the +ssRNA to make proteins, it converts the +ssRNA → dsDNA via **Reverse Transcriptase**
 - The resulting DNA is spliced into the *Host Genome* using **Integrase**
 - Replication & Protein Synthesis then comes from the viral DNA in the Nucleus



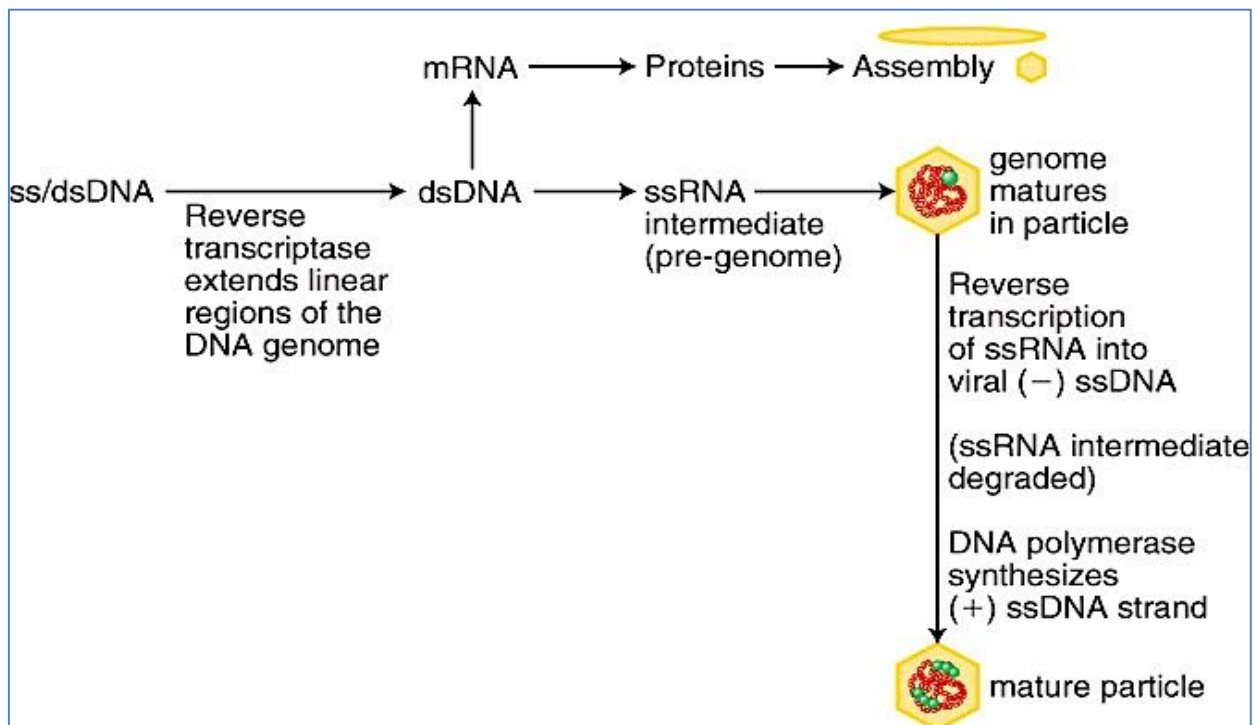
<https://clinicalinfo.hiv.gov/en/glossary/binding>

- **7: dsDNA-RT Viruses (double-stranded DNA Reverse Transcriptase Viruses)**

- Eg: Hepadnavirus (Hep-B Virus)
- **Replication Features:** (In the Cytoplasm AND the Nucleus)
 - Genome is a 'covalently closed circle' (cccDNA) → Nucleus
 - → Transcribed to mRNA (Via **RNA-Polymerase**)
 - mRNA exits Nucleus → Protein Synthesized In Cytoplasm
 - Then, **Reverse Transcriptase** converts the mRNA Back to DNA → cccDNA



https://sums.ac.ir/page-gehrchen/en/408/article/717-G118/blk-tool_article_sample_gehrchen_block44970



Unattributable

Viral Assembly:

- **Assembly** = When all of the components of the virus are assembled into a particle
 - Occurs when an appropriate concentration Of Virus Proteins & Nucleic Material is reached
 - (Note: Some particles self-assemble)

Viral Maturation:

- **Maturation** = Stage in the Virus Life-Cycle when it becomes infectious
 - Involves proteolytic cleavage of capsid or envelope poly-proteins into functional proteins

Targets for Antiviral Therapies:

- **1: Attachment**
- **2: Penetration**
- **3: Uncoating**
- **4: Replication**
- **5: Assembly**
- **6: Maturation**
- **7: Release**

Quasispecies:

- A substrain of an organism that develops in an individual by the process of evolutionary selection

MICROBIOLOGY: PARASITES

MICROBIOLOGY: PARASITES

General Features:

- Live at the expense of their host → Acquires Nutrients/Other Benefits without Reciprocal Benefits
- Complex Life-Cycle involving 2 or More Hosts (Definitive Host & Intermediate Host/s)
- Are Successful if:
 - o Produces minimal disturbance
 - o Not regarded by host as foreign
- Parasite infections tend to be Long-Term (As opposed to Bacteria/Viruses)
- Many make use of the Host's growth-factors to promote their *own* growth
- **(Including Protozoa, Metazoa [Helminths/Worms] & Arthropods):**

Protozoa

Unicellular, either intracellular (for example, malaria) or extracellular (for example, African trypanosomes).
Malaria kills over 1 million per year.



Leishmania mexicana

Helminths

Multicellular, metazoan worms; includes round-worms (nematodes), schistosomes and tape-worms.
Over 25% of global population infected.



Heligmosomoides polygyrus



Ixodes hexagonus

Ectoparasites

Lice, mites, ticks and other arthropods.

Hosts (Definitive Vs Intermediate):

- **Definitive Host:**
 - o - Harbours the Mature, Adult Form of the Parasite
- **Intermediate Host:**
 - o - Harbours the Immature, Larval Form of the Parasite

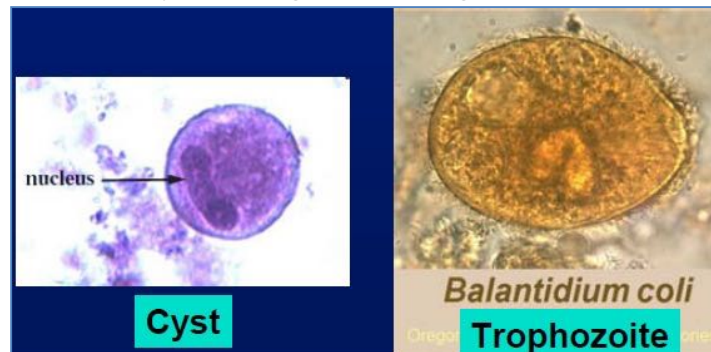
Grouping: Protozoan Vs Metazoan:

- Protozoan Parasites:

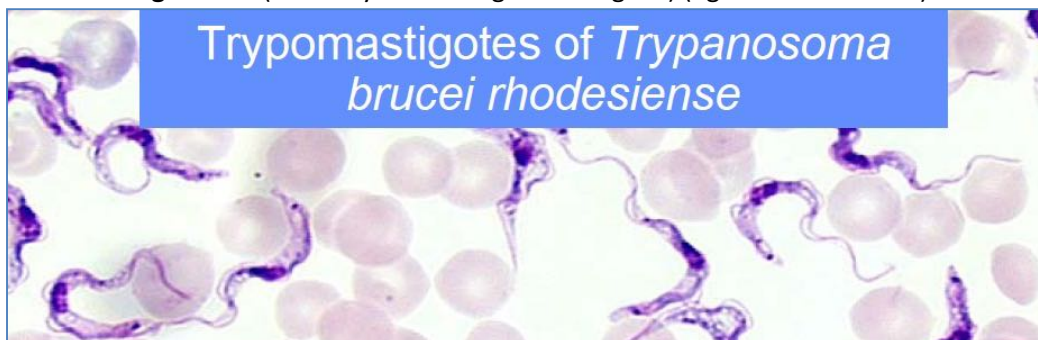
- (Single-Celled Parasites)
- **3 Categories of Locomotion:**
 - **Amoeba** – (Move by Crawling) (Eg: Entamoeba Histolytica)



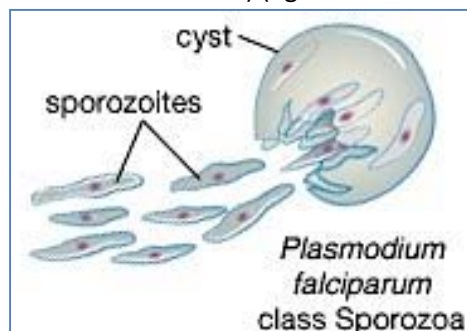
- **Ciliate** – (Move by Swimming – via cilia) (Eg: Balantidium Coli)



- **Flagellate** – (Move by Swimming – Via Flagella) (Eg: Giardia Lamblia)



- **(Sporozoa)** – (No motile structures) (Eg: Plasmodium Malariae; Toxoplasma)

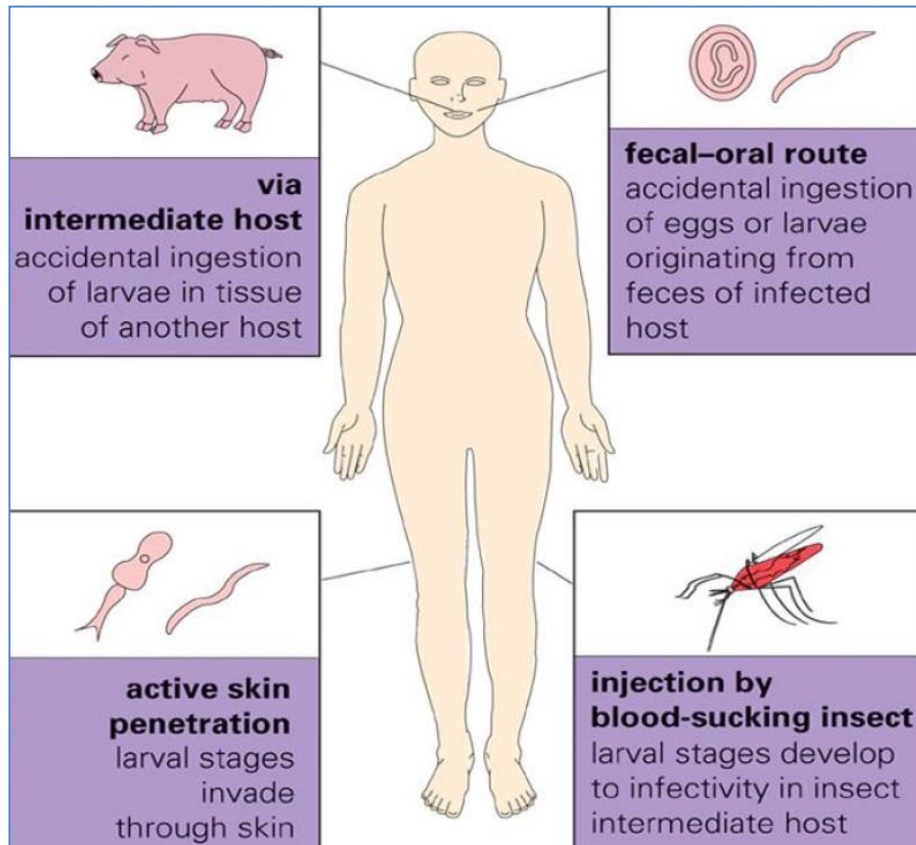


<https://www.britannica.com/science/Eimeria>

- Metazoan Parasites – (Helminths & Arthropods):

- (Multi-Celled Parasites)
- **Helminths:**
 - **3 'Phyla':**
 - **Platyhelminthes** (Flat worms)
 - ***Trematodes*** (Flukes)
 - ***Cestodes*** (Tapeworms)
 - **Nematoda** (Round Worms)
 - **Acanthocephala** (Spiny-headed worms)
- **Arthropods:** (Animals with segmented bodies, exoskeletons and jointed appendages)

Routes of Entry of Helminths:



Unattributable

Immune Evasion Strategies:

- Protozoan Parasites:

- Antigenic Variation
- Antigenic Drift
- Molecular Mimicry (Expression of Host Proteins)
- Intracellular Localisation
- Self-Isolation in Membrane-bound Vesicle
- Prevent fusion of lysosomes with phagosomes
- Sequestration in privileged sites
- Regulation of host functions

- Helminth Parasites:

- Antigen Shedding
- Protease production → Neutralise some immune components (Eg: Antibodies)
- Superoxide Dismutase → Neutralise Respiratory Burst by Neutrophils
- Regulation of host functions (Immunosuppression/Maladaptive Response)
- Skew the T-Helper Response to Favour Th1-Cells:
 - Favouring Th1 → Reduced class-switching to IgE, the AntiParasitic Ab
- Use Host Cytokines as Parasitic Growth Factors

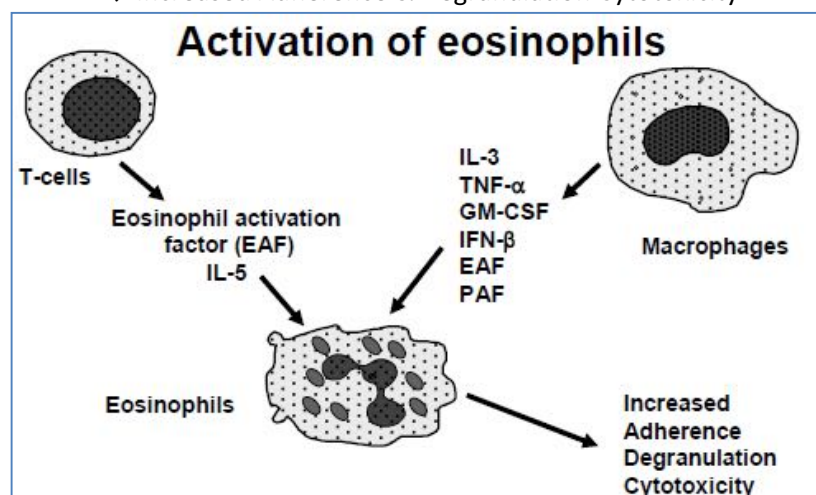
Immunity Against Parasites:

- Innate Immunity:

- **Lysozyme:**
 - (in Tears/Saliva/Mucus/Neutrophils)
 - Some parasites are susceptible
- **Eosinophils (Eosinophil Granulocytes):**
 - Combat multicellular Parasites
 - Degranulate → Release Reactive Oxygen Species → to kill parasites
- **Complement Activation:**
 - By Alternate Pathway – Complement Activation by Binding to Pathogen Surface
 - By MB-Lectin Pathway – Complement Activation by Binding to Lectin on Pathogen Surface
 - (Note: Classical Pathway is Adaptive – Complement Activation by Ab's on Pathogen Surface)
- **Phagocytes in Spleen:**
 - Infected RBCs express specific Parasite Antigens which are opsonised by antibody/complement → Recognised & Removed by Phagocytes in the Spleen

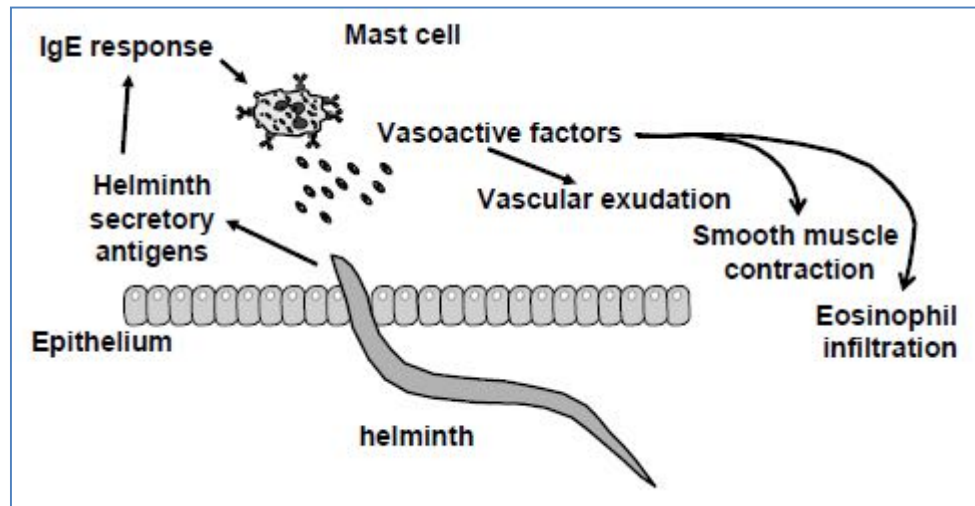
- Adaptive Immunity:

- **Antibodies (B-Cells):**
 - **Typically for Extracellular Infections (in blood/Tissues)**
 - **IgE** is the Major Isotype (Important in eliminating many helminth infections)
 - → (Hence, many infections are associated with **Type-1 Hypersensitivity** reactions)
 - → Oedema, Asthma, Urticaria
 - - Can destroy Tachyzoites (young parasites) in blood
 - - Can neutralise *Proteases* used by parasites to enter tissues
 - - Can block 'Anal Pores' of parasites
 - - Can block enzyme pathways of some helminths (Can arrest egg production)
 - (Note: However, Many parasites are unaffected by antibodies)
- **Complement Activation (By Classical Pathway):**
 - Complement Activation by Ab's on Pathogen Surface
 - - Can destroy Tachyzoites (young parasites) in blood
- **Cell-Mediated:**
 - **Typically for Intracellular Infections**
 - **Th1-Cells Activate Macrophages:**
 - Macrophages become more Phagocytic and Destroy Intracellular Parasites
 - (Note: Typically only Protozoan parasites are small enough to live intracellularly)
 - **Th2-Cells Help B-Cells produce Antibodies:**
 - (Th2 is the predominant response)
 - **Tc-Cells Destroy Infected Cells:**
 - May also directly destroy larvae
- **Eosinophils:**
 - **Note: They are the MAIN Effector Cell against Helminth Infections**
 - **Activated by:**
 - Th-Cells (IL-5) & Macrophages (TNF- α , IFN- β , IL-3)
 - → Increased Adherence & Degranulation Cytotoxicity

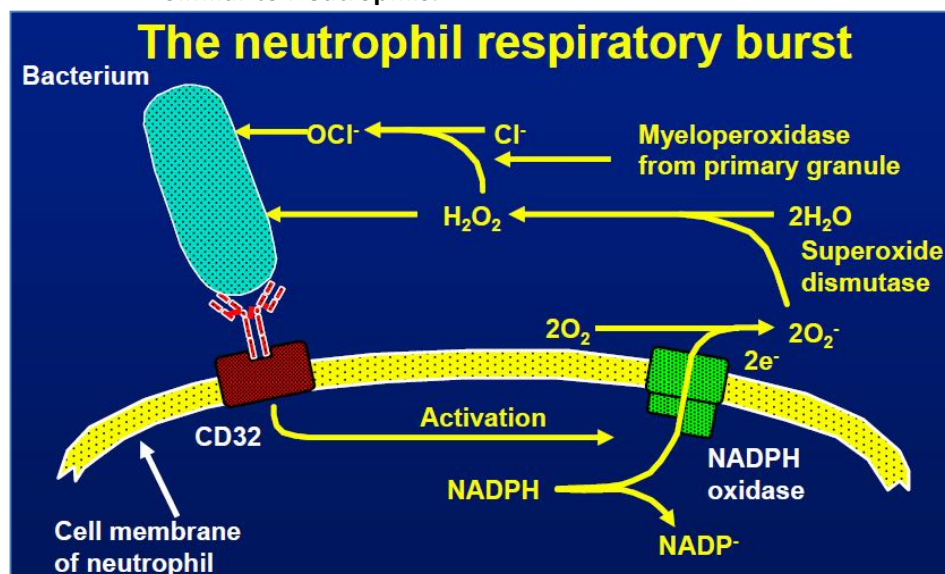


- Eosinophils have Fc receptors (Allow binding to Parasites covered with **IgE-Antibodies**)

- Binding of Antigen to Eosinophil-Bound-IgE → Degranulation
- Similar to Mast Cells:



- Release Granules onto the worm:
 - ***Major Basic Protein** (Damages Cuticle of Helminths)
 - Eosinophil-Cationic Protein (A Ribonuclease – Toxic to Helminths)
 - Lysophospholipase
 - Phospholipase D
- Respiratory Burst:
 - Superoxide
 - Chloride Ions
 - Hydrogen Peroxide
 - Similar to Neutrophils:



Immune Evasion by Parasites:

- Resistance to Immune Effector Mechanisms:

○ Eg: Molecular Mimicry:

▪ Eg: Expression of Host-Proteins:

- **Eg: Some Schistosomes** cover themselves with *Host Proteins* (Eg: Blood-Group Antigens & MHC products)
 - → **Avoids Recognition by Effector Immune Mechanisms**

○ Eg: Protease Production to Neutralise Anti-Parasite Immune Components:

- **Eg: Shistosomula (Helminth)** Produces Proteases → Cleave Antibodies
 - They also Inhibit Macrophage Function
- **Host Proteases may be Inactivated by Protease Inhibitors**

- Immunosuppression or Inappropriate Immune Responses:

○ Eg: Helminths – Secrete Soluble Immunosuppressant Factors:

- → Inhibit Lymphocyte Function
- → Inhibit Mast-Cell Degranulation

- Sheltering in Immune-Privileged Sites:

○ Eg: RBCs:

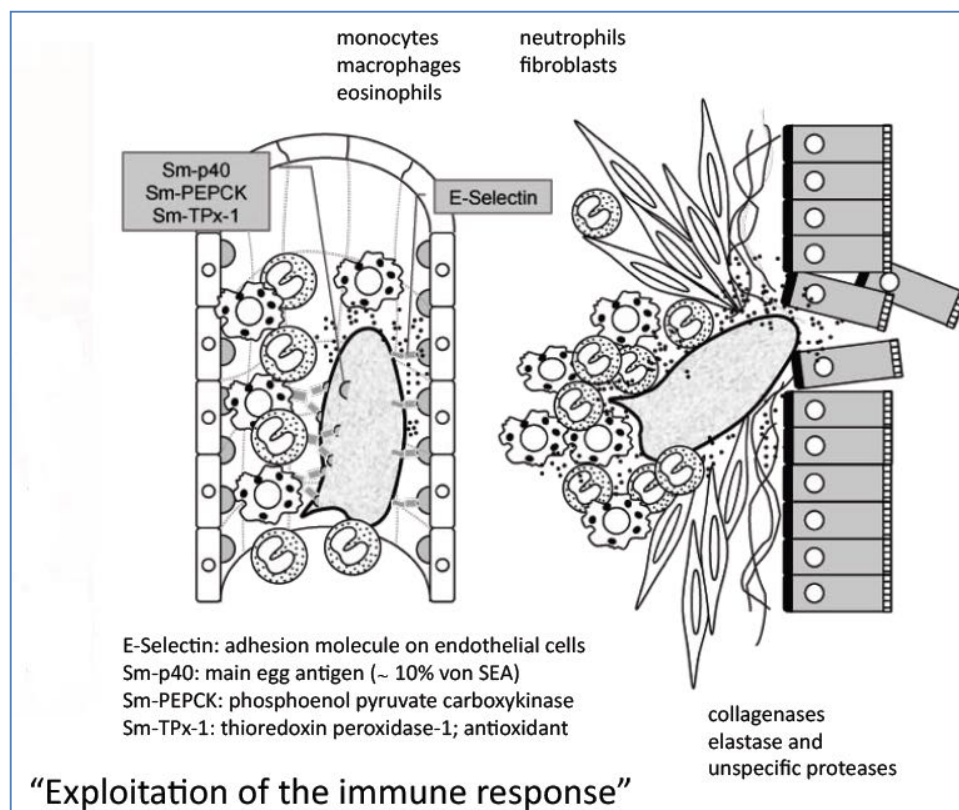
- ***Plasmodium Falciparum* (Malaria)** – lives inside RBCs which don't express MHC-I:
 - → Can't be recognised by CD8-T-Cells
 - → Are Shielded from Antibodies
- However, Infected RBCs *Can* be recognised/destroyed in the spleen:
 - To avoid this, Malaria Parasites cause the RBCs to become *Sticky* →
 - RBCs adhere to endothelium in peripheral organs
 - (Note: Can lead to peripheral vasculopathies & ischaemic organ failure)

- Exploiting The Immune System to Aid in Life-Cycle:

○ Eg: Some Helminths Exploit the Increased Expression of Cell-Adhesion-Molecules in Inflammation:

▪ Eg: Helminths which lay eggs need to get the eggs out of the Blood Vessels

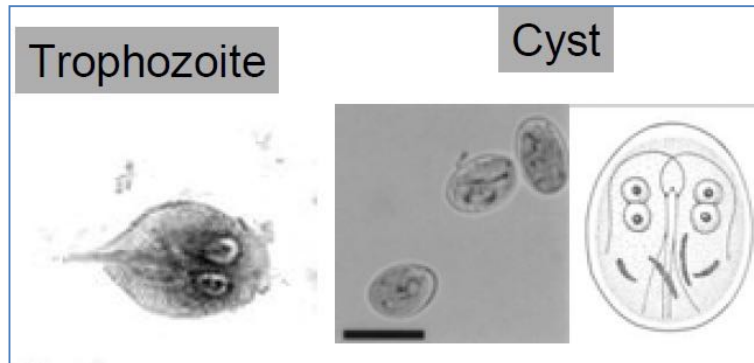
- Therefore, by causing Inflammation, Endothelial Cells Increase CAM Expression
- → Eggs use these Adhesion Molecules to adhere to the Endothelium
- → They then secrete Collagenases/Elastases/Proteases → to Exit the Blood Vessel



Parasite Replication Cycles:

- Protozoan Parasites:

- **Trophozoite Stage:** ("Tropho" = Feeding)
 - Infective, Proliferative Stage – Lives in the definitive host
 - Trophozoites *Actively Feed*
 - Protozoa can reproduce by *Fission, Sexual Reproduction*, or be *Hermaphroditic*
 - **Encystation** = Conversion of Trophozoite → Cyst
- **Cysts Stage:**
 - Hardy, thick-walled spore able to survive for lengthy periods **outside a host** (Organisms that create oocysts include [Cryptosporidium](#) and [Toxoplasma](#))
 - Resistant to heat, harmful chemicals
 - Can survive without access to nutrients, water, or oxygen
 - Often shed in the faeces (Eg: Giardia)
 - **Excystation** = Conversion of Cyst → Trophozoite

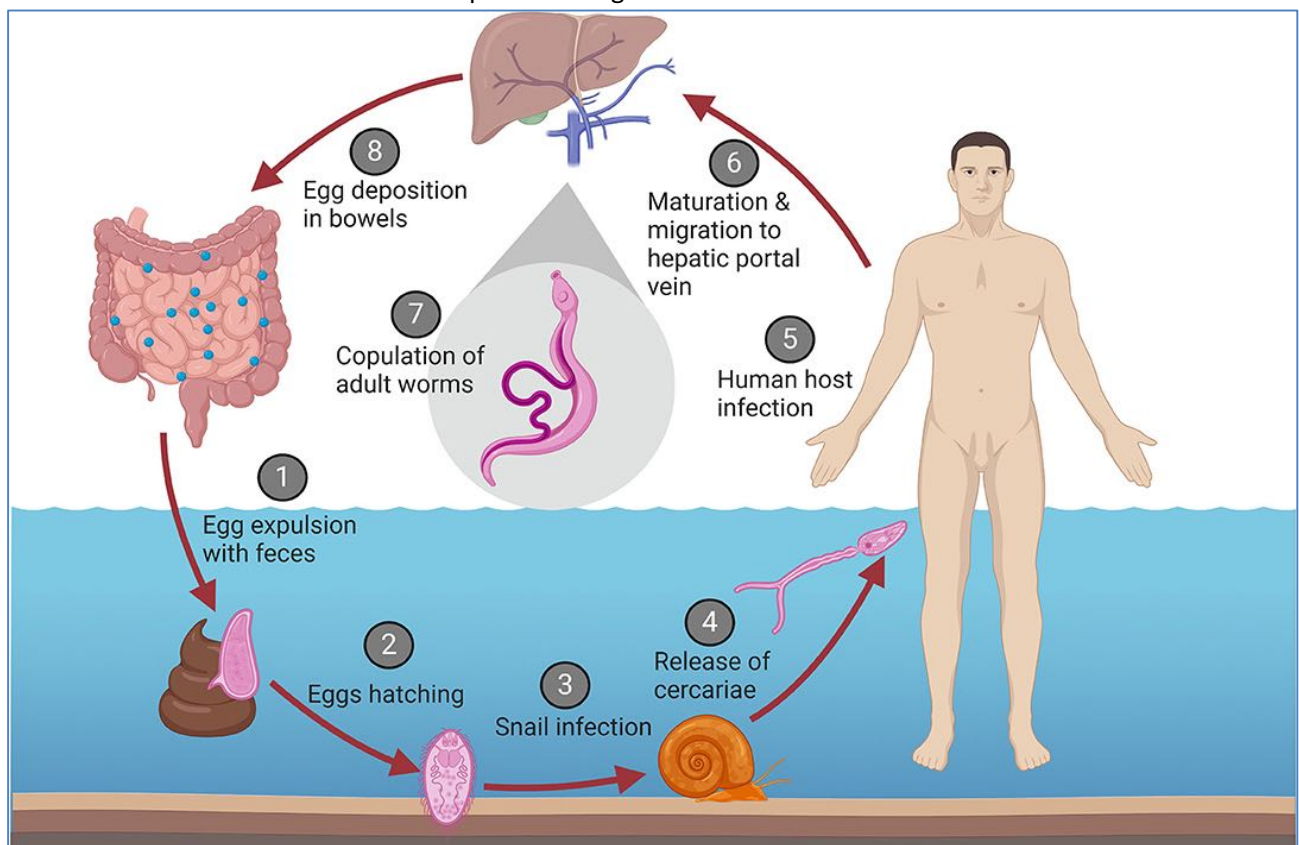


- Metazoan Parasites: (Platyhelminthes, Nematodes & Acanthocephala)

○ Platyhelminthes:

▪ Trematodes (Flukes):

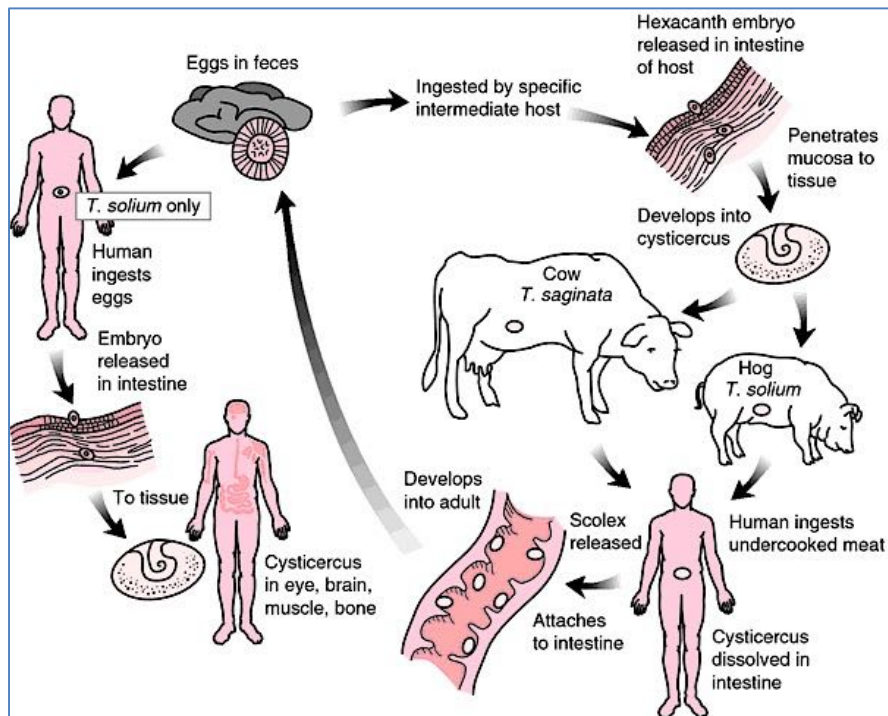
- The Eggs of Trematodes (Flukes) pass out in the Faeces, develop into larvae, which **MUST PASS THROUGH THE SNAIL (Intermediate Host)** and develop into Cercaria before the parasite is again infective to humans



Clinical use of chistosoma mansoni antigens as novel immunotherapies for autoimmune disorders; Cleenewerk, Garssen & Hogenkamp; DOI: <https://doi.org/10.3389/fimmu.2020.01821>

▪ **Cestodes (Tapeworms):**

- Cysts are shed in Human Faeces → Grass → Eaten by Cow or Pig (Intermediate Hosts). Humans are infected by eating Contaminated Beef. Adult worm attaches to SI-Mucosa.

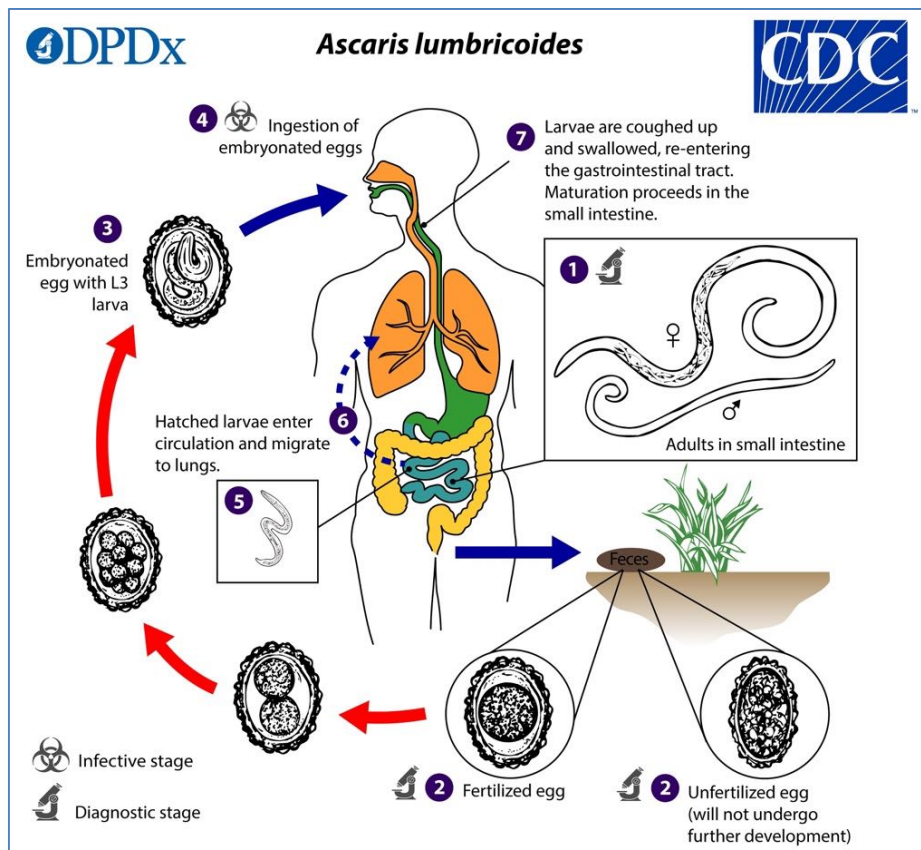


<https://medical-dictionary.thefreedictionary.com/tapeworm>

○ **Nematodes (Round worms):**

▪ **Intestinal Nematodes:**

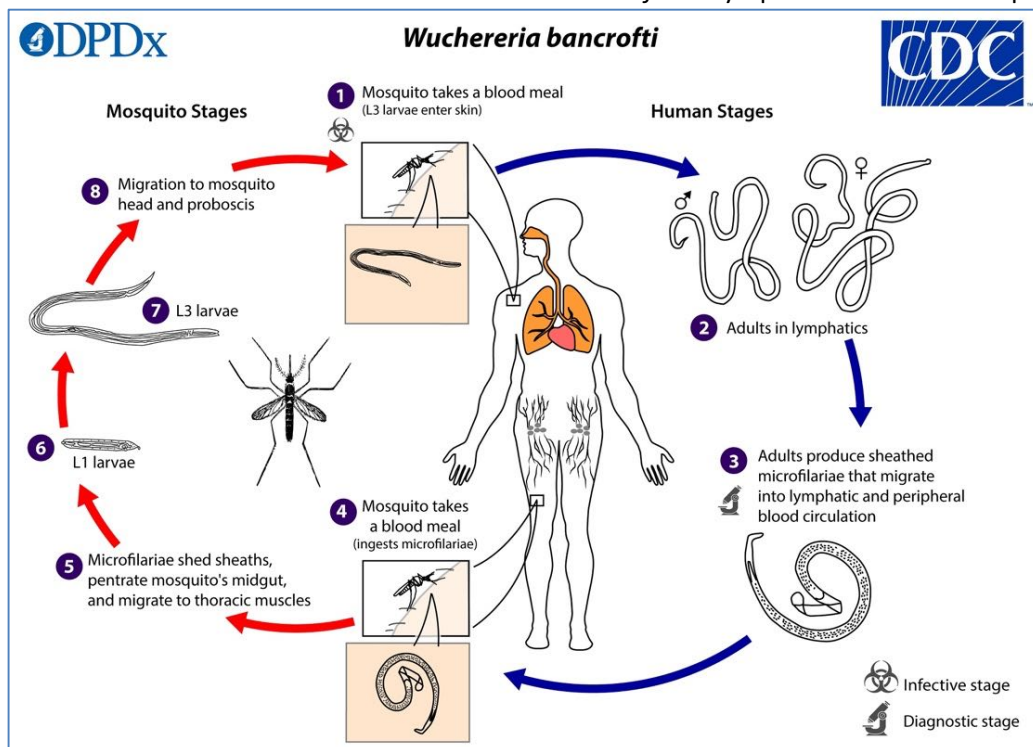
- Direct Life-cycle (Horizontal Transmission *Without* Intermediate Hosts)
- Faecal Oral – Eggs in Faeces → Ingested → Hatch in SI → Burrow into Bloodstream → Exit blood in lungs → Pass up the Trachea & are Swallowed → Adults mature in Small Intestine



Public Domain: CDC

▪ **Filarial Nematodes:**

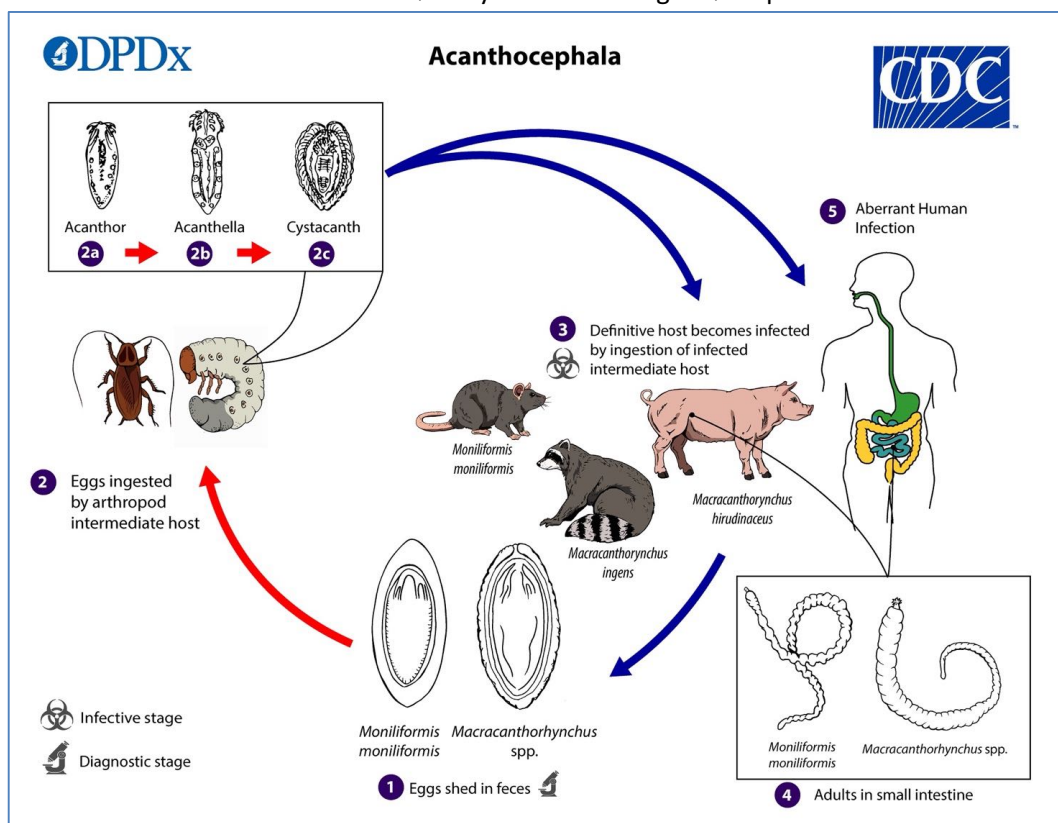
- Microfilariae in the blood are infective to Mosquitoes → Pass on the infection to other people
- Most Common = *Wuchereria Bancrofti* → Lymphatic Filariasis & Elephantiasis



Public Domain: CDC

○ **Acanthocephala:**

- (A Phylum of parasitic worms known as Thorny-headed/Spiny-Headed Worms)
- Complex Life-Cycles involving a number of hosts:
 - Embryo is released in faeces → Ingested by a Crustacean (Eg: A Mollusc) (the Intermediate Host) → Encystation occurs → Intermediate Host is Ingested by the Definitive Host → Excystation in the gut → Reproduction



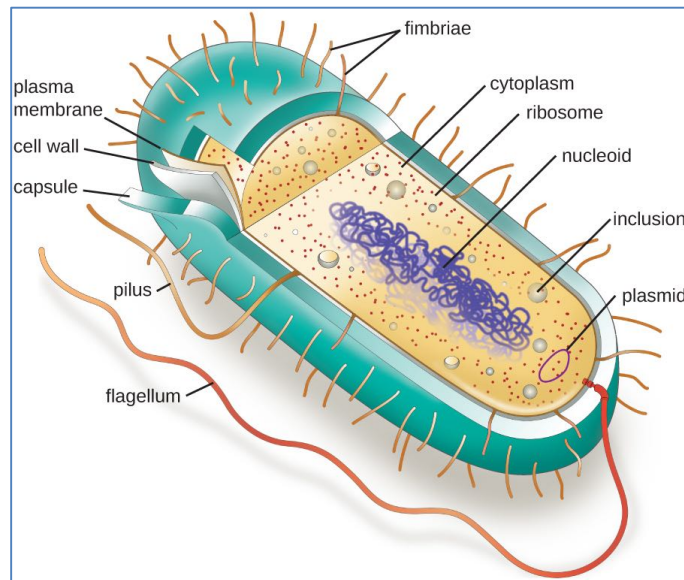
Public Domain: CDC

MICROBIOLOGY: BACTERIA

MICROBIOLOGY: BACTERIA

Structure of the Bacterial Cell:

- Prokaryotic
- Single-Celled Organisms
- DNA Based Genome, but *No Distinct Nucleus* (Circular Chromosome)
- **3 Layers:**
 - o Plasma Membrane
 - o A Thick Cell Wall covers the Plasma Membrane (Composition depends on Gram -/+)
 - **Gram Positive:**
 - Thick Peptidoglycan Layer
 - & Teichoic Acid
 - **Gram Negative:**
 - Primarily Lipid-Based (Including Lipopolysaccharide – LPS)
 - (+Thin Peptidoglycan Layer)
 - o A Polysaccharide Capsule covers the Cell Wall (Considered a Virulence Factor – Resists Phagocytosis, Detergents & Dehydration)
- **Pili/Fimbriae:**
 - o For Adherence to Cells or Other Bacteria
- **Flagellum:**
 - o For Mobility



CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

Taxonomy & Classification:

- Uses a 'Binomial Nomenclature' – (*Genus + Species*):
 - o **Genus** = Eg: Staphylococcus
 - o **Species** = Eg: Aureus
 - o (*Staphylococcus Aureus*)

staining	shape	respiration	shape/reproduction	genus	species
Gram-positive	cocci	aerobic	clusters	<i>Staphylococcus</i>	<i>S. aureus</i>
			chains/pairs	<i>Streptococcus</i>	<i>S. faecalis</i>
		anaerobic		<i>Peptococcus</i>	<i>P. magnus</i>
	bacilli	aerobic	sporing	<i>Bacillus</i>	<i>B. anthracis</i>
			non-sporing	<i>Listeria</i>	<i>L. monocytogenes</i>
		anaerobic	sporing	<i>Clostridium</i>	<i>C. tetani</i>
			non-sporing	<i>Propionibacterium</i>	<i>P. acnes</i>

Classification of Bacteria:

- Staining of Cell Wall Structure:

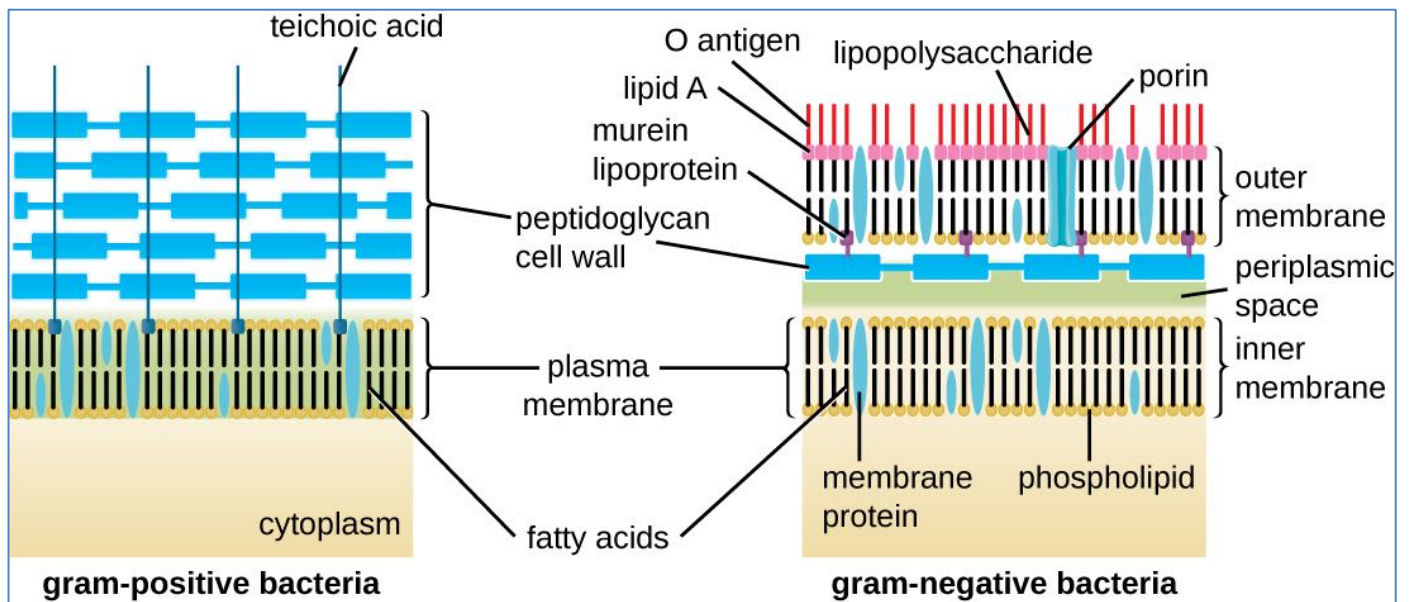
○ Gram Stain:

▪ **Gram Positive:**

- (Stain Blue/Purple)
- Thick Peptidoglycan Layer
 - (The Site of Action of β -Lactam Antibiotics)
- & Teichoic Acid

▪ **Gram Negative:**

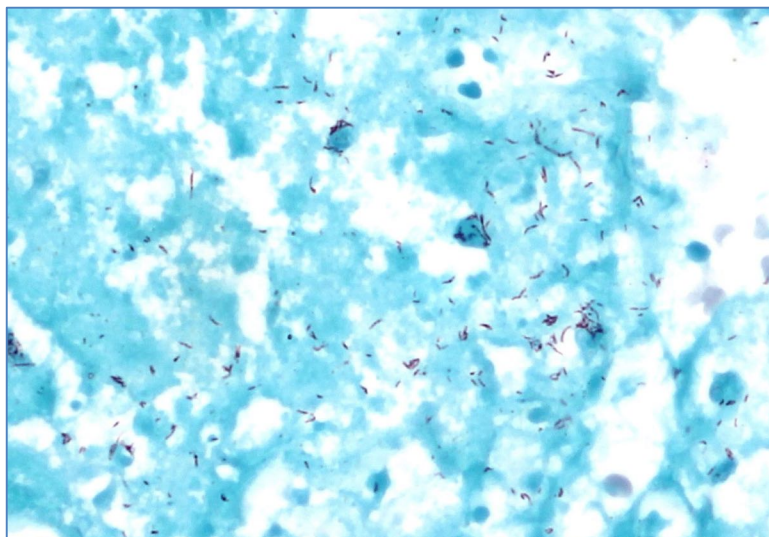
- (Stain Pink/Red)
- Primarily Lipid-Based (Including Lipopolysaccharide – LPS)
 - (Note: LPS = Endotoxin; can $\rightarrow$ *Septic Shock*)
- (+ 'Lipid A' = Endotoxin; can $\rightarrow$ *Septic Shock*)
- (+Thin Peptidoglycan Layer)



CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

○ Acid-Fast Stain:

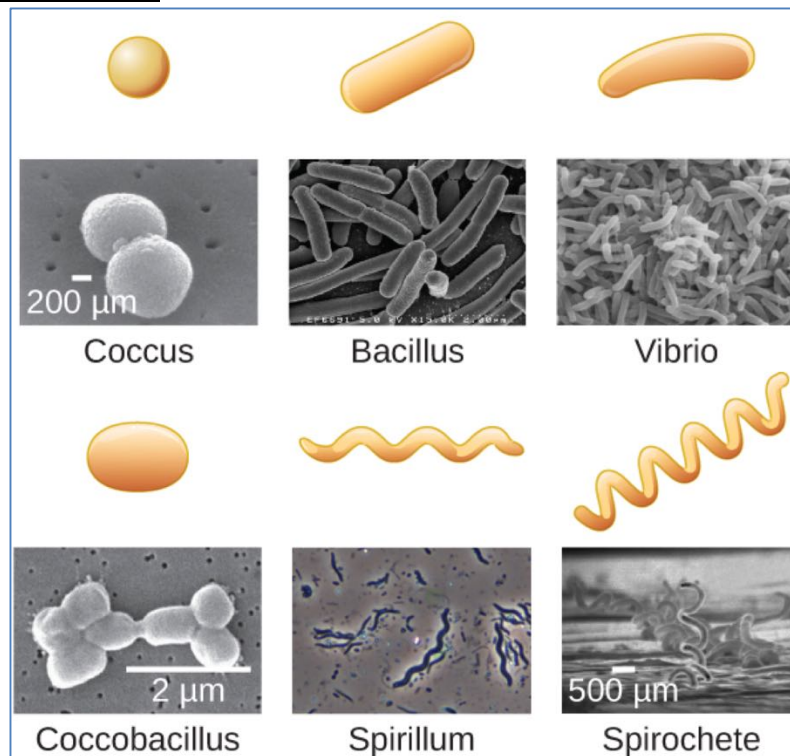
- (Eg: *Mycobacterium Tuberculosis*)
- Doesn't Stain with Gram
- Similar Cell-Wall to Gram + Bacteria, but different type of Peptidoglycan
- Stains with the "*Ziehl Neelsen Stain*" ('Acid-Fast Stain')
 - "Acid-Fast" Bacilli stain bright red in contrast to a blue background



CoRus13, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

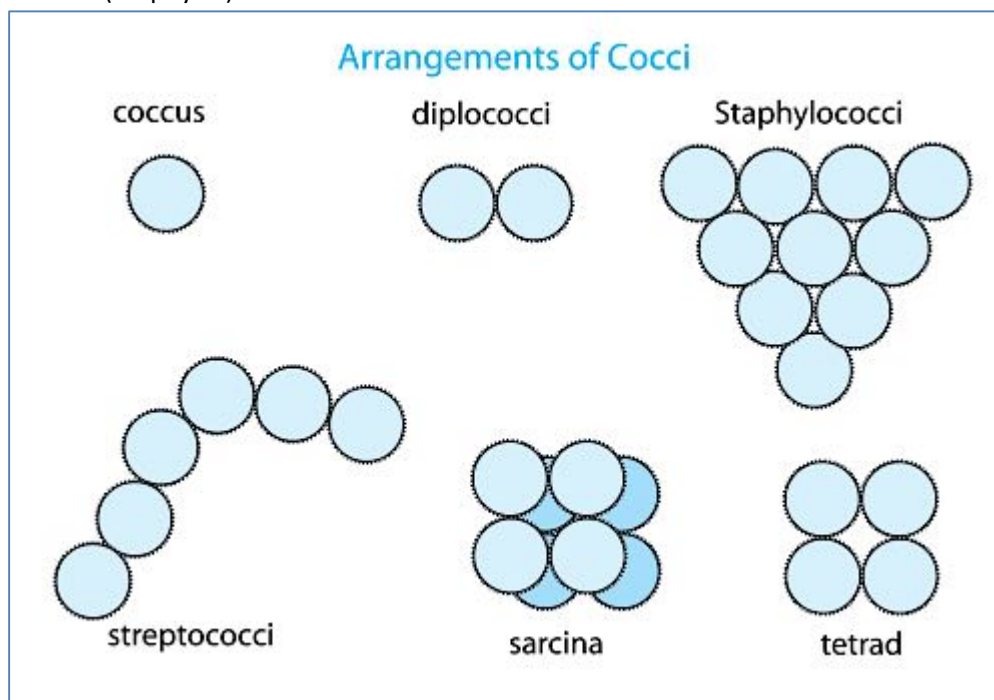
- **Respiration:**
 - o Aerobic Vs Anaerobic

- **Cellular Morphology (Shape):**



CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

- **Grouping:**
 - o Single
 - o Pairs (Diplo-)
 - o Chains (Strepto-)
 - o Clusters (Staphylo-)



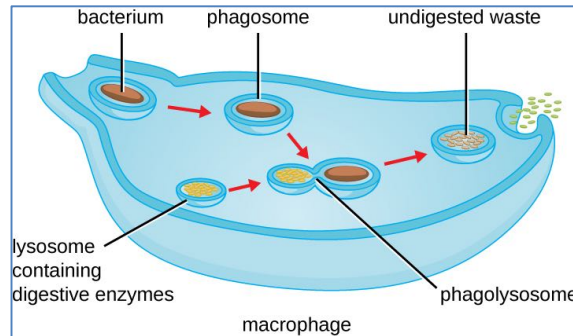
LadyofHats, Public domain, via Wikimedia Commons

Immunity Against Bacteria:

- Innate Immunity:

○ ****Phagocytosis:**

- By Macrophages/Dendritic Cells/Neutrophils
- May be Independent, Antibody or Complement Mediated



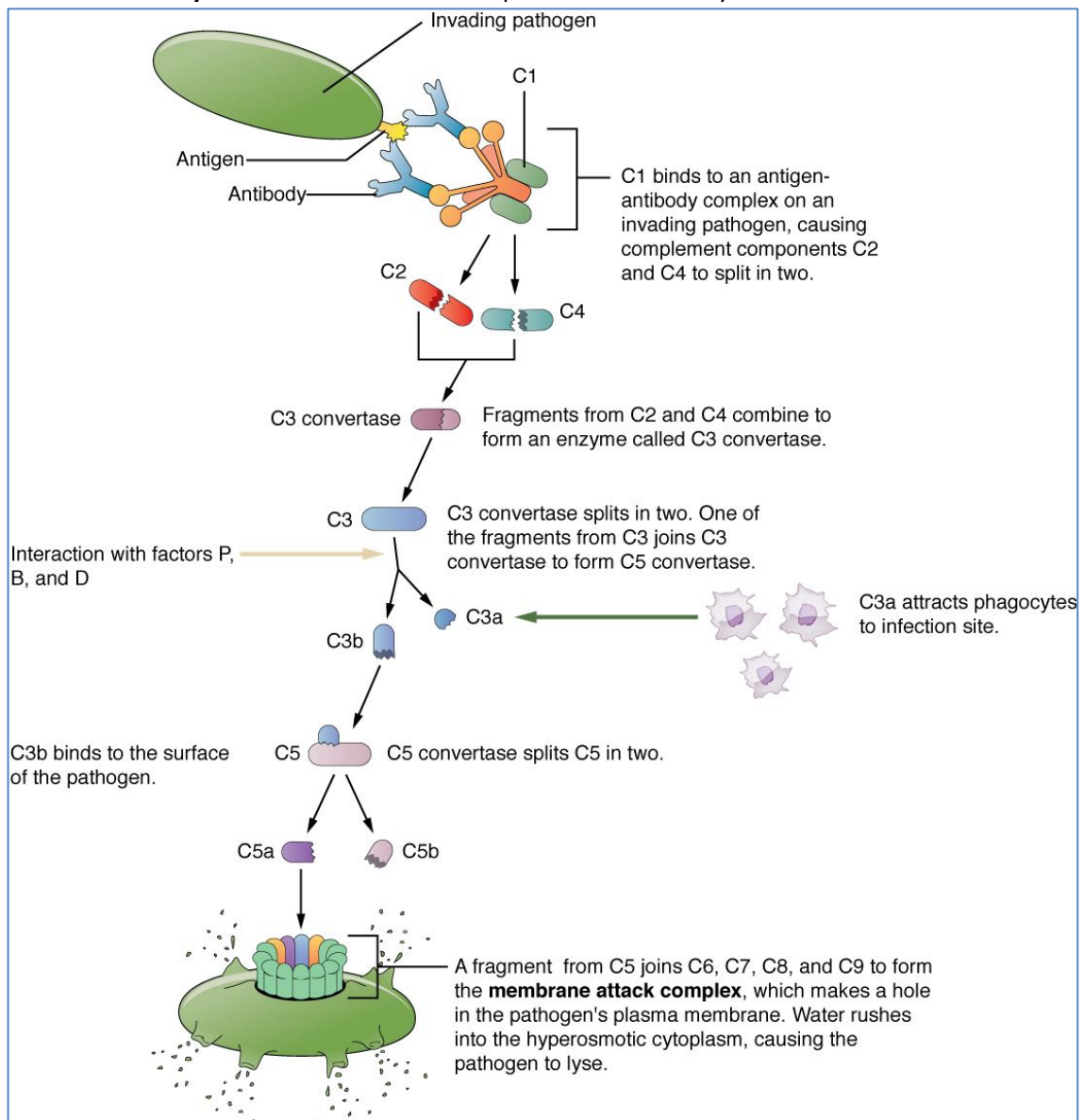
CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0>>, via Wikimedia Commons

○ ***Lysozyme:**

- An Antibacterial enzyme dissolved in bodily secretions (Tears/Saliva/Mucus/Neutrophils)
- → Splits the Cell Wall Proteoglycans of Bacteria → Lysis

○ ****Complement Activation – Via Alternative Pathway:**

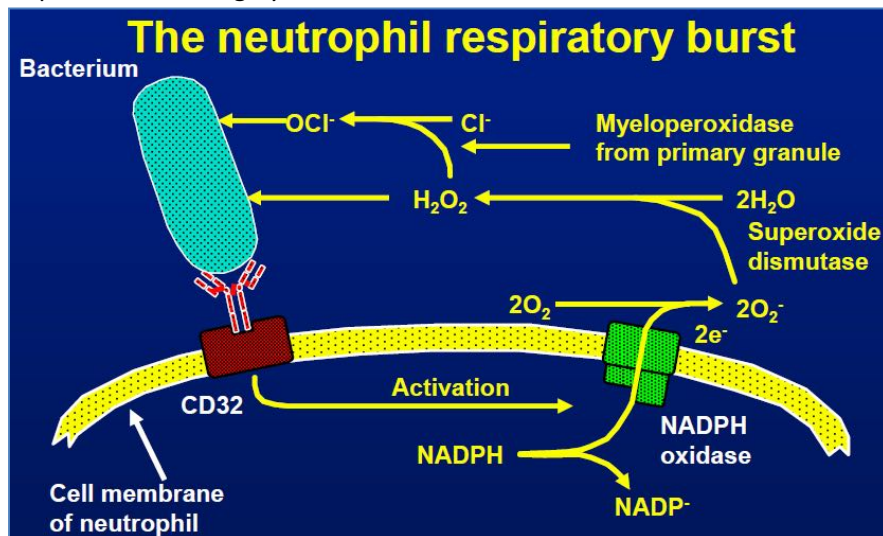
- → **Phagocytosis:** C3b opsonisation → Phagocytosis of Bacteria
- → **Lysis:** Membrane attack complex formation → Lysis of Bacteria



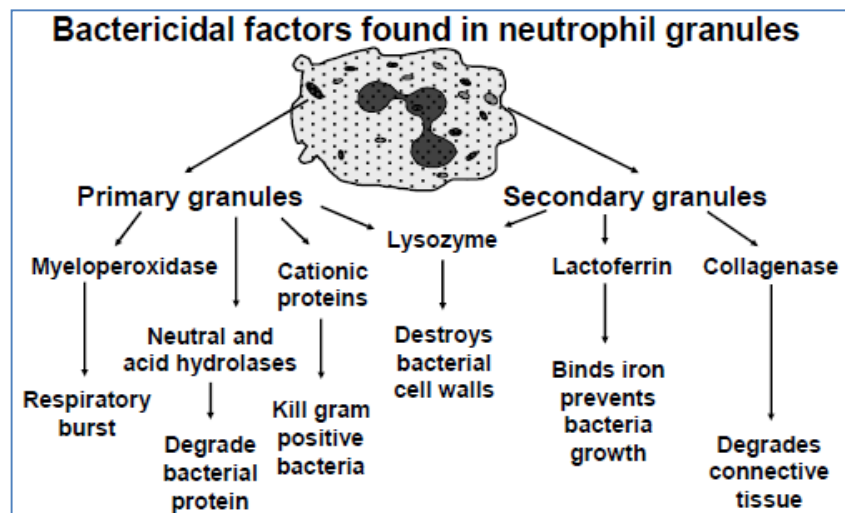
CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0>>, via Wikimedia Commons

- ****Neutrophils:**

- → **Phagocytosis:** Neutrophils ingest & kill many Microbes
- → **“Respiratory Burst”:** Binding of Fc-portion of Antibodies on opsonised Bacteria stimulate production of Highly Oxidative Molecules which kills the bacteria



- → **Bactericidal Granules:**



- **Local Inflammation – (Due to Cytokine Release after Macrophage Phagocytosis):**

- → Local Inflammation
- → Fever
- → Acute Phase Proteins

- **Acute Phase Proteins:**

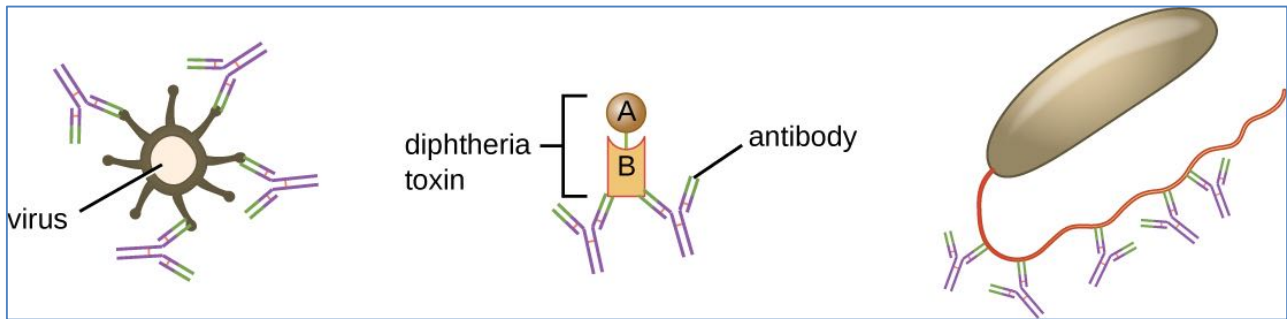
- **(C-Reactive Protein [CRP], Mannose-Binding Lectin [MBL])**
- **Both are:**
 - **Opsonising Agent for microbes** → Phagocytosis (Similar action to Antibodies – except have broad specificity for PAMPs)
 - **Complement Activators** → Activate the **Classical (CRP)/Lectin (MBL) Pathways of the Complement Cascade**

- **Adaptive Immunity:**

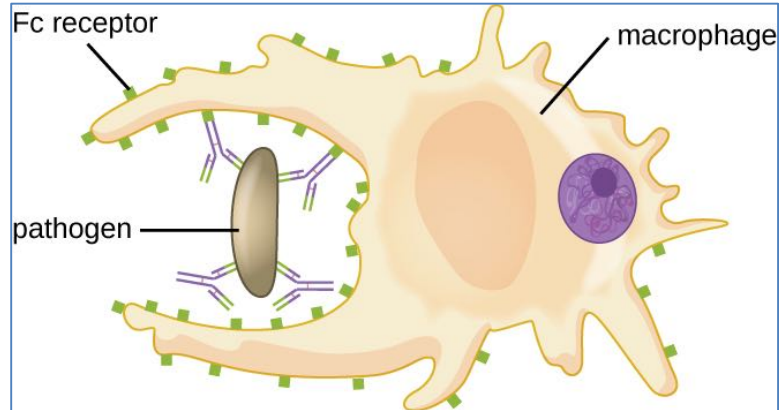
- ****Antibodies – (Produced by B-Cells):**

- → **Exotoxin Neutralisation:**
 - IgG is Entirely Responsible for this
 - IgG essentially competes with the receptor for the toxin → Prevents binding to cellular target
- → **Endotoxin Opsonisation:**
 - Fc-Receptor-Mediated Phagocytosis
- → Bacteriolysis

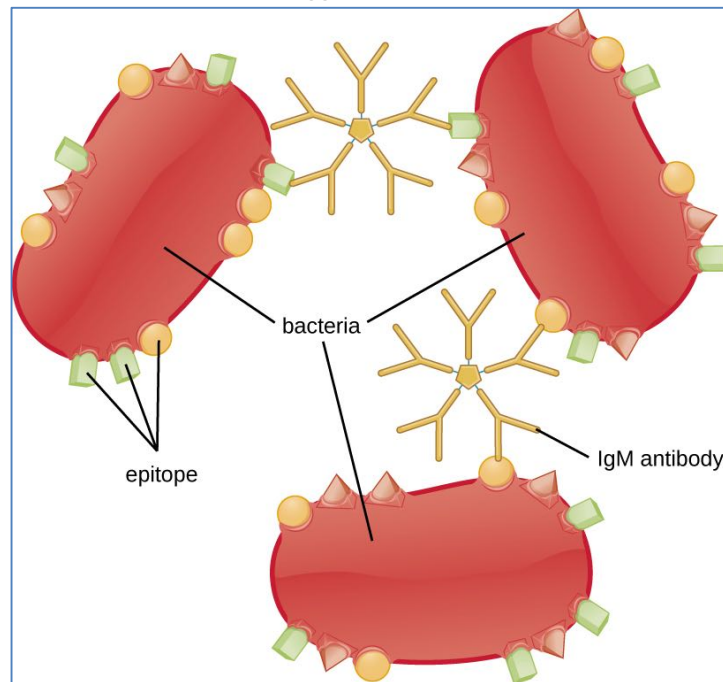
Neutralisation



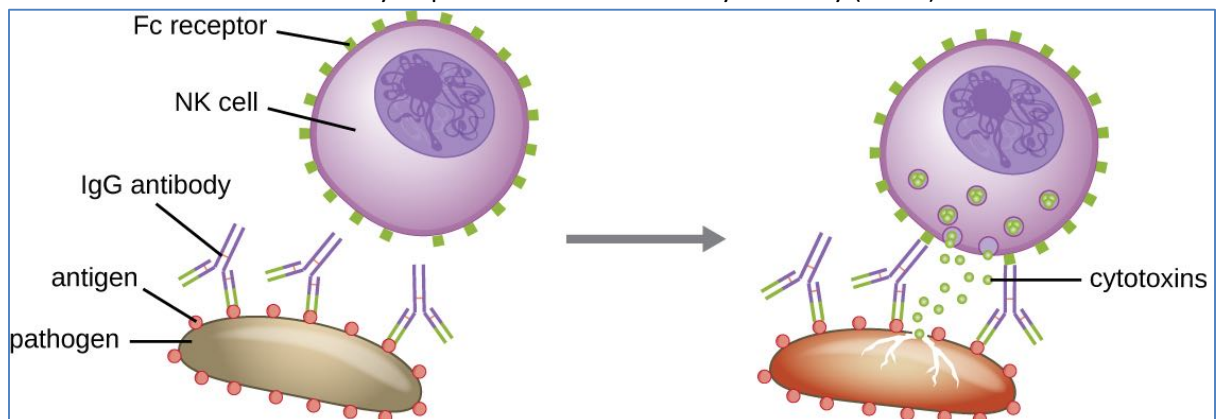
Opsonisation



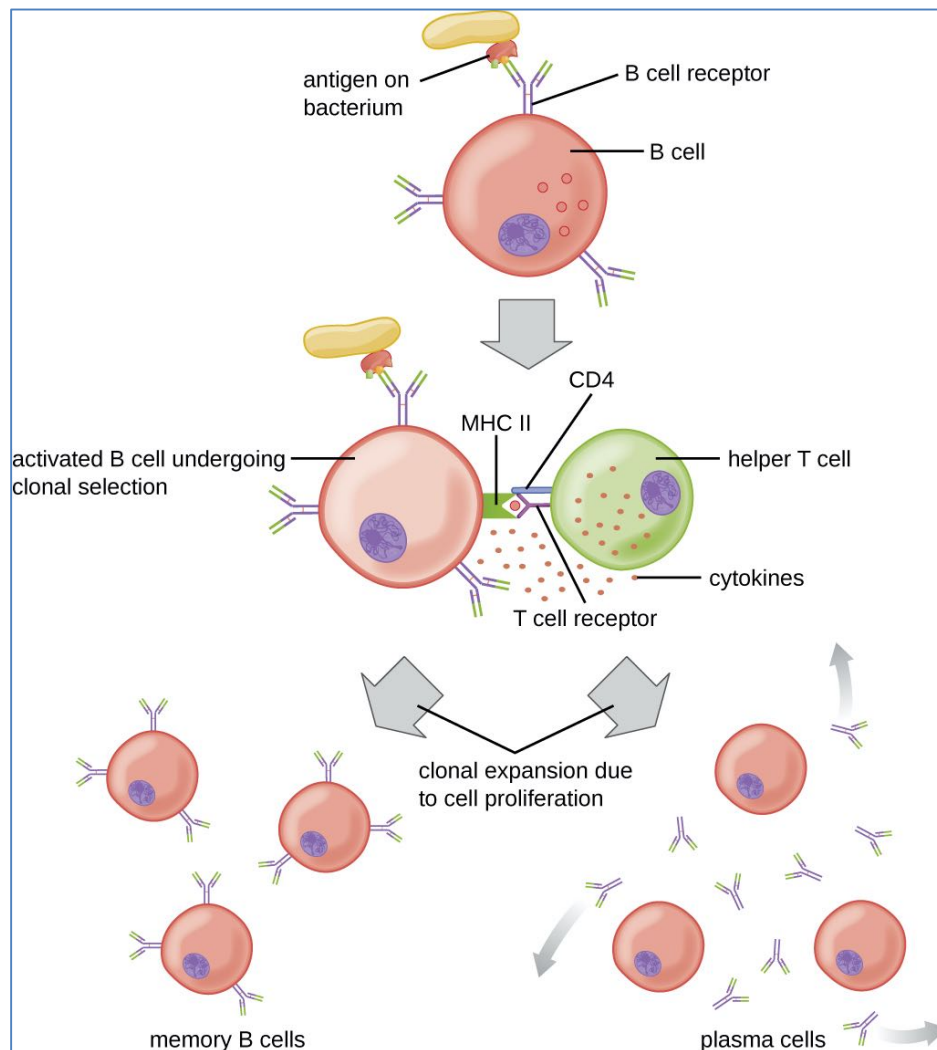
Agglutination



Antibody-dependent cell-mediated cytotoxicity (ADCC)

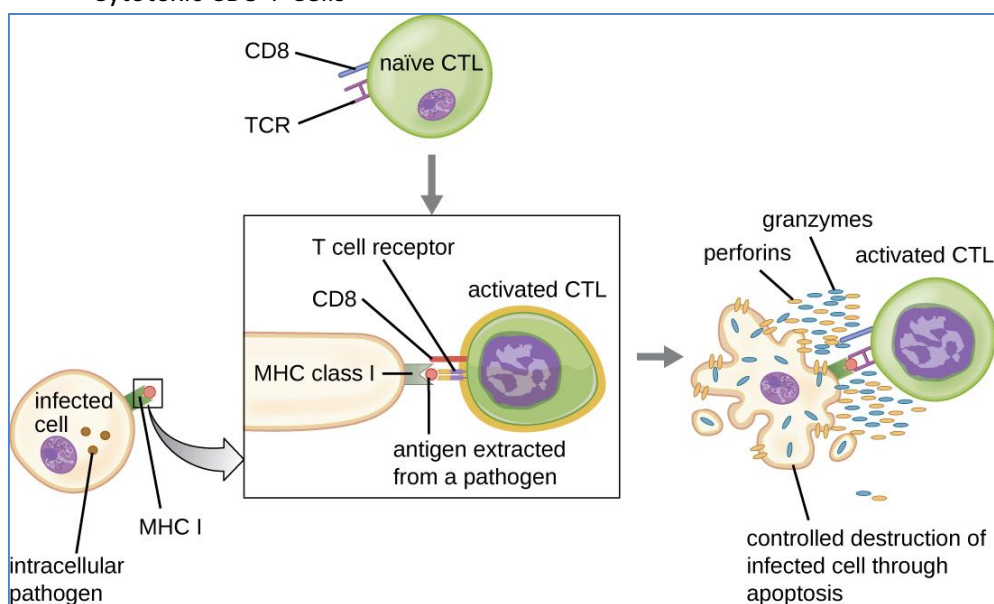


- **CD4-T-Helper Cells:**
 - → **Activate Macrophages:**
 - → Destruction of Phagocytosed Bacteria
 - → **Activate B-Cells:**
 - → Secretion of Antibodies



CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

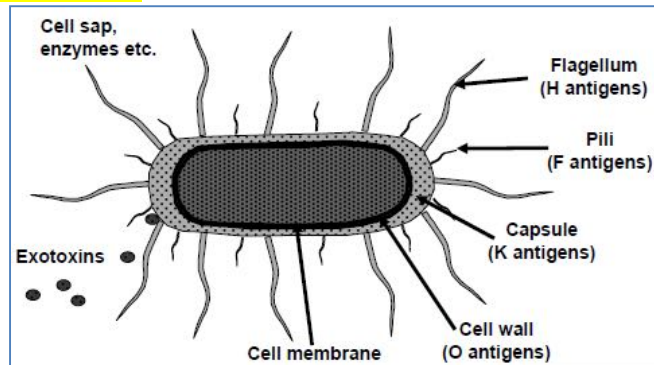
- **CD8-T-Cytotoxic Cells → Kill Infected Cells:**
 - Infected Cells displaying bacterial peptide on MHC-I are lysed by Perforins released by Cytotoxic CD8-T-Cells



CNX OpenStax, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

Bacterial Virulence Factors:

- Molecules Expressed/Secreted by Pathogens that enable them to achieve the following:
 - o Colonization of a Niche in the host (this includes adhesion to cells)
 - o Immuno-evasion, evasion of the host's immune response
 - o Immunosuppression, inhibition of the host's immune response
 - o Entry into & Exit out of cells (if the pathogen is an intracellular one)
 - o Obtain nutrition from the host
- **Eg: Endotoxins – (In the Walls of Gram Negative Bacteria) → Septic Shock:**
 - o (Note: Recognised by Toll-Like Receptors on Macrophages → Cytokines)
 - o ***Lipopolysaccharide (LPS)
 - o Surface Array Proteins (Eg: Enzymes)
 - o Flagellum
 - o Adhesion Pili
 - o Capsule Antigens
 - o Cell Wall Antigens
 - o Cell Membrane
- **Eg: Exotoxins – (Toxic Molecules Released by the Bacteria) → Toxic Shock:**
 - o Eg: Tetani Toxin
 - o Eg: Staph → Superantigen



- **Eg: Ig-Proteases (Eg: Strep Pyogenes) → Break down Antibodies**
- **Eg: Capsules (Eg: Bacterial cell walls) → Inhibits Phagocytosis**

Bacterial Immune Evasion Strategies:

- Antigenic Variation
- Inhibition of Complement Activation
- Resistance to Phagocytosis
- Produce Superoxide Dismutase → Scavenge Free Radicals from respiratory burst of Neutrophils
- Intracellular bacterial evasion:
 - o Travel b/w cells without being exposed to extracellular fluid
 - o Escape into vacuole in the cytoplasm
 - o Prevent fusion of lysosomes with phagosomes
- Depress cellular immunity
- Superantigens → Inappropriate Immune Response

(EXTRACELLULAR BACTERIA)		(INTRACELLULAR BACTERIA)	
Evasion Strategies	Examples	Evasion Strategies	Examples
Antigenic variation	<i>N. gonorrhoeae</i> , <i>E. coli</i> , <i>S. typhimurium</i>	Inhibition of phagolysosome formation	<i>M. tuberculosis</i> , <i>L. pneumophila</i>
Inhibition of complement activation	Many bacteria	Scavenging of reactive O ₂ intermediates	<i>M. leprae</i>
Resistance to phagocytosis	Pneumococcus	Disruption of phagosome membrane, escape into cytoplasm	<i>L. monocytogenes</i>
Scavenging of reactive O ₂ intermediates	Catalase-positive staphylococci		

Pathological Consequences - Damage due to:

- **Exotoxins:**
 - = **Toxins Secreted from the Bacteria into the system**
 - (∴ Organisms need not be invasive to produce illness)
 - **Typically from *Gram Positive* bacteria**
 - Eg: Botulinum Toxin
 - Eg: Tetanus Toxin
 - **Toxic Shock Syndrome:**
 - Some bacteria produce Superantigens → Widespread, Non-Specific activation of Th-Cells → Massive secretion of Pro-Inflammatory Cytokines → Massive Vasodilation/↑Capillary Permeability/Hypotension → Toxic Shock Syndrome
 - – High Fever
 - – Hypotension
 - – Potential Multi-Organ Failure → Death
- **Endotoxins:**
 - = **Structural (Cell-Wall) Components of the Bacteria that are Antigenic**
 - They are *Not Secreted* and are *Not Directly Toxic*
 - They are released into the system during *Lysis/Death* of Gram Negative Bacteria
 - **Typically from *Gram Negative* bacteria**
 - Eg: *Lipopolysaccharide* (LPS)
 - **Septic Shock:**
 - Bacteraemia & ∴ ↑LPS → LPS binds to TLR-4 on Macrophages & Dendritic Cells → Secretion of Pro-Inflammatory Cytokines & Nitric Oxide → Massive Vasodilation/↑Capillary Permeability/Hypotension → Septic Shock
 - – Fever
 - – Tachypnea
 - – Tachycardia
 - – Hypotension
 - – Potential Multi-Organ Failure → Death
- **Hypersensitivity Reactions:**
 - Due to immune response

The Difference between *Septic Shock* and *Toxic Shock*:

- **Septic Shock:**
 - From ***Gram Negative Bacteria***
 - (Mediated by ***Liberated Endotoxin*** from dead organisms – LPS (Lipid A = the toxic part of LPS) → Directly Activates CD14 & TLR-4 on Macrophages → *Cytokine Storm* (including IL-1, IL-6, IL-8, TNF-alpha and PAF) → Shock)
 - – Therefore shouldn't be treated by Bacteriocidals (As they would liberate more Endotoxin)
 - **Note: Septic Shock can be fatal even after Antibiotic Treatment. Explain why?:**
 - 1: If the Antibiotics were *Bactericidal*, they will liberate more Endotoxin from lysed bacteria and further exacerbate the septic shock → Death
 - ∴ In Septic Shock, Bacteriostatic Antibiotics are most useful, as they slow bacterial growth without lysing them
 - 2: Conversely: If the shock is in the irreversible stage, no amount of antibiotics (Even bacteriostatic) will do any good (As there is irreversible organ failure)
- **Toxic Shock:** (Eg: From Staph/Strep)
 - From ***Gram Positive Bacteria*** (Don't contain Endotoxin – Cell walls are primarily Peptidoglycan)
 - (Mediated by ***Superantigens*** secreted from live organisms → Widespread non-specific MHC-II:TCR interaction → Widespread CD4-T-Cell activation → Stimulates macrophages by γ-IFN → *Cytokine Storm* (including IL-1, IL-6, IL-8, TNF-alpha and PAF) → Shock)
 - – Therefore should be treated by bacteriocidals (Don't need to worry about Endotoxin)

Antibiotic Susceptibility:

- Selective Toxicity:

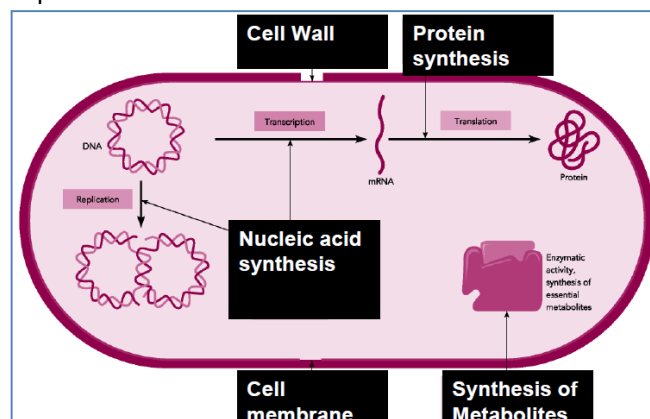
- Critical to Efficacy & Safety of Anti-Microbials
- Exploits Differences in Cell Biology between *Host* & *Pathogen* Cells
- Aim → Kill only the Pathogen Cells

- Antibiotic Action: Key Steps:

- 1: Active drug (Ie: The metabolically active form)
- 2: Present at site of infection
- 3: Bind to bacterial cell surface; **X**
- 4: Uptake into bacterial cell; **X**
- 5: Bind to target – molecular interaction with bacterial cell components; **X**
- 6: Lysis and death or growth inhibition of bacterial cell; **X**
 - (**X** = steps where bacterial activity leads to resistance)

- Classification of Antibiotics:

- **1: Bactericidal or bacteriostatic:**
 - Bactericidal → kill bacteria (Ie: Makes the organism unviable)
 - Bacteriostatic → inhibit growth → Host defences kill static population
 - Note: Some agents can be both -Eg: chloramphenicol with *E-coli* and *Haemophilus*
- **2: Target site:**
 - **Cell wall synthesis**
 - Beta lactams
 - Glycopeptides
 - **Protein synthesis**
 - Aminoglycosides
 - Tetracyclines
 - Macrolides
 - **Nucleic acid synthesis**
 - Quinolones
 - **Folic Acid Pathway:**
 - Sulphonamides
 - Trimethoprim
 - **Cell membrane function**
 - Polymixins
 - Colistin
- **3: Chemical structure**
 - Beta lactams
 - Glycopeptides
 - Aminoglycosides
 - Tetracyclines
 - Macrolides
 - Quinolones
 - Sulphonamides
 - Trimethoprim



ANTIBACTERIAL DRUG CLASSES:

- 1: Anti Cell-Wall Synthesis Antibiotics – (Bactericidal):

- Target Peptidoglycan Synthesis on Gram-Positive Bacteria

Classical Agents:	Common Uses:	Mechanism of Action:	Side Effects:
β-Lactam Antibiotics:			
Penicillins: Penicillins 'G' & 'V' Amoxicillin & Ampicillin Flucloxacillin Methicillin Ticarcillin (Suffix = "-Cillin")	Gram Positive Bacteria (Note: Bacteria Producing β -Lactamase are resistant) (Note: Flucloxacillin – for β -Lactamase Resistant) (Note: Cephalosporins – for Non- β -Lactamase Resistant)	Block "Penicillin-Binding Proteins" (Enzymes) → Inhibit Synthesis of the Peptidoglycan Layer of the Bacterial Cell Wall.	GI Upset & Diarrhoea Allergic Rash Anaphylaxis (Need Adrenaline Handy)
Cephalosporins: (Ceftriaxone)			(As above) + Mild Renal Toxicity
β-Lactamase Inhibitors: Augmentin	(In Combination with Penicillins) for Penicillin-Resistant Gram Positive Bacterial Infections	Inhibits β -Lactamase → Allows β -Lactams to work on Penicillin-Resistant Bacteria.	Nausea/Vom/Diarr Allergy
Glycopeptide Antibiotics:			
Vancomycin Teicoplanin Telavancin	Gram Positive Bacteria <i>(As a LAST RESORT for MRSA)</i> (Also if Pt. is allergic to β -Lactams)	Prevents incorporation of specific Peptide Subunits into the Peptidoglycan Layer of the Bacterial Cell Wall.	Local Pain Phlebitis (Vein Inflammation) Kidney Damage Hearing Loss

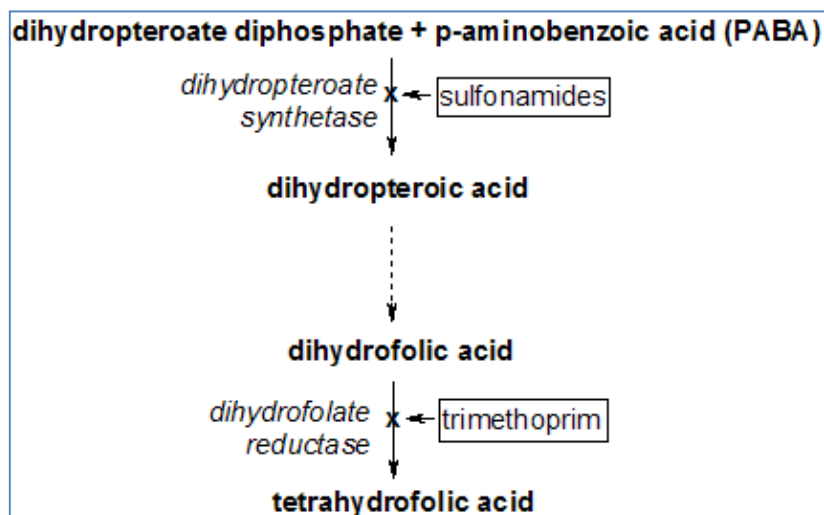
- 2: Anti Protein-Synthesis Antibiotics – (Bacteriostatic):

- Exploits differences between Eukaryotic (Human) Ribosomes & Prokaryotic Ribosomes
- Selective Toxicity** – Due to specific binding to Prokaryotic Ribosomes
- Note: Aminoglycosides are Solely eliminated by the Kidneys & Are Nephrotoxic** (Need to assess renal function first, then Dose Accordingly)

Classical Agents:	Common Uses:	Mechanism of Action:	Side Effects:
Aminoglycoside Antibiotics:			
Gentamicin Streptomycin Tobramycin	Gram Negative Bacteria (Used Synergistically with β -Lactams to ↑ drug entry into Bacteria)	Bind Specifically to <i>Prokaryotic Ribosomal</i> Subunits → Causes Misreading of mRNA → Inhibits Synthesis of Proteins vital to Bacteria.	Ototoxic (Hearing Loss & Vertigo) Nephrotoxic (Kidney Damage)
Tetracycline Antibiotics:			
Doxycycline Tetracycline (Suffix = 'Cycline')	Gram Negative Bacteria Syphilis (G ⁻), Chlamydia (G ⁻), Lyme Disease (G ⁻) (And Malaria - Protozoa)	Bind Specifically to <i>Prokaryotic Ribosomal</i> Subunits → Inhibits Binding of tRNA to mRNA → Inhibits Synthesis of Proteins vital to Bacteria.	Nausea/Vom/Diarr. Photosensitivity Staining of Teeth Renal/Liver Toxicity.
Macrolides:			
Erythromycin, Azithromycin	Gram Negative Bacteria Syphilis, Lyme Disease.	Bind Specifically to <i>Prokaryotic Ribosomal</i> Subunits → Inhibits release of tRNA → Inhibits Synthesis of Proteins vital to Bacteria.	Nausea/Vom/Diarr. Jaundice

- **3: Anti Nucleic-Acid Synthesis Antibiotics – (Bacteriostatic):**

- **Exploits differences in the Metabolic Pathways of DNA Synthesis – (Humans rely solely on *Dietary Folate*, while Bacteria have to make their own):**
 - Eg: Competitive Inhibition of *Dihydropteroate-Synthase*, a key Enzyme involved in Folate Synthesis in Bacteria
 - Eg: Competitive Inhibition of *Dihydrofolate-Reductase*, a key Enzyme involved in Folate Synthesis in Bacteria (Note: Humans share this pathway, but bacteria require it 100x more than humans)
 - Eg: Inhibition of Bacterial DNA Gyrase/Topoisomerase → Stops DNA Replication/Transcription



Classical Agents:	Common Uses:	Mechanism of Action:	Side Effects:
Sulphonamides:			
Sulfasalazine (Prefix = "Sulfa")	Urinary Tract Infections	Competitive inhibition of <i>Dihydropteroate-Synthase</i> , a key Enzyme involved in Folate Synthesis. (Folate is necessary for Nucleic Acid Synthesis → & Hence DNA Synthesis.	Nausea/Vom/Diarr Allergy Precipitation in Urine –Kidney Failure Leukopenia Photosensitivity
Trimethoprim:			
Trimethoprim	Urinary Tract Infections	Competitive inhibition of <i>Dihydrofolate-Reductase</i> , a key Enzyme involved in Folate Synthesis. (Folate is necessary for Nucleic Acid Synthesis → & Hence DNA Synthesis.	Nausea/Vom/Diarr Allergy Precipitation in Urine –Kidney Failure Leukopenia Photosensitivity (BIRTH DEFECTS)
Quinolones:			
Ciprofloxacin Norfloxacin (Suffix = "Floxacin")	Urinary Tract Infections Community Acquired Pneumonia Bacterial Diarrhoea Gonorrhoea	Inhibits bacterial DNA Gyrase or Topoisomerase → Inhibits DNA Replication & Transcription.	Nausea/Vom/Diarr Allergy

- **4: Antimycobacterial Drugs:**

- **2 Main Types of Mycobacterial Diseases:**
 - Tuberculosis
 - Leprosy
- **Why are they a Problem?**
 - Because Mycobacteria can live inside Macrophages following Phagocytosis
 - Also, Multi-Drug-Resistant strains are on the rise
- **Compound Drug Therapy:**
 - A frequent strategy to decrease the probability of the emergence of resistant organisms
 - Also requires Long-Term Therapy

Classical Agents:	Common Uses:	Mechanism of Action:	Side Effects:
Isoniazid:			
Isoniazid	Combination Treatment of M. Tuberculosis	MOA unknown. (Bacteriostatic & Bactericidal)	Allergic Skin Eruptions Fever Hepatotoxicity Haemolysis (in G6PD Deficiency)
Rifampicin:			
Rifampicin	Combination Treatment of M. Tuberculosis	Binds to & Inhibits DNA-Dependent <i>Prokaryotic RNA-Polymerase</i> → Inhibits DNA Transcription & therefore Inhibits Protein Synthesis. (Bacteriostatic & Bactericidal)	Allergic Skin Eruptions Fever Hepatotoxicity
Ethambutol:			
Ethambutol	Combination Treatment of M. Tuberculosis	MOA Unknown. (Bacteriostatic)	Optic Neuritis Visual Disturbances Colour Blindness.
Pyrazinamide:			
Pyrazinamide	Combination Treatment of M. Tuberculosis	Active in Low pH–(In Phagolysosomes) (Bacteriostatic)	Gout GI Upset Hepatotoxicity

- **Antibiotic Resistance:**

- = “Bacteria *Isn’t* Inhibited/Killed by an Antibacterial @ Normal Dosage Concentrations”
- **Note: Bacterial “Resistance Genes” exist**, and *Mutation Potential* is HIGH!
 - (Due to huge numbers of rapidly proliferating bacteria)
- **Antibiotic Usage *Preferentially Selects* these resistant strains** → Transmission of “Resistance Genes” to offspring
- **THEREFORE** – “*Restraint of antimicrobial use is the best way to ensure their efficacy*”

- **Categories of Organism Susceptibility:**

- **Susceptible:**
 - Patient is likely to respond to treatment with that Antibiotic @ Normal Doses
 - (I.e: Organism is Killed/Inhibited by the drug @ Normal Doses)
- **Resistant:**
 - Patient *NOT* likely to respond to treatment with that Antibiotic @ Normal Doses
 - (I.e: Organism *NOT* Killed/Inhibited by the drug @ Normal Doses)
- **Intermediate:**
 - *Higher Doses are Needed* to ensure treatment success

- **Intrinsic Vs Acquired Resistance:**

○ **1: Intrinsic resistance**

- Normal genetic, structural or physiologic state -lack target site or impermeable to agent
- Predictable within genus or species
- Eg: All Gram Neg bacilli are intrinsically resistant to Vancomycin
- Eg: Bacteria secreting β -Lactamase are resistant to β -Lactams

○ **2: Acquired resistance**

- Changes in usual genetic state of bacteria > altered cellular physiology or structural changes
- Unpredictable > why laboratory methods are necessary to detect resistance
- Resistance arises from:
 - 1) chromosomal mutation
 - 2) transmissible plasmids
 - 3) Combination of mutation and gene transfer events
- Multiple mechanisms may exist in 1 organism

Common Acquired Resistance

- Penicillin – *Staph aureus*: 90% R.
- Ampicillin – *Ecoli*: 45% R
- Tetracycline- Group B Strep: 91% R
- Methicillin -*Staph aureus* : 29%
- Timentin- *Pseudomonas* 19% R
- Nitrofurantoin- *Kleb pneumoniae*: 54% R

This is why you need to test for resistance to ensure efficacy of drugs

- **Common Resistant Organisms:**

○ **MRSA:**

- (Methicillin Resistant Staphylococcus Aureus)
- Due to an alteration in PBP (Penicillin binding protein) in cell wall
- Also produce β -Lactamase (therefore resistant to all β -lactams, including β -lactamase Inhibitors)
- Nosocomial (hospital) strains – Typically Multi-Resistant
- Community Acquired – Typically not Multi-Resistant

○ **VRE:**

- (Vancomycin resistant Enterococci)
- Due to altered Target Site
- 2 Types:
 - Van A – Resistant to both Vancomycin And Teicoplanin
 - Van B – Just resistant to Vancomycin

○ **VISA:**

- (Vancomycin Intermediate/Resistant Staph Aureus)
- Have Thick Cell walls → Trap Vancomycin
- Very difficult to detect

○ **ESBL:**

- (Extended Spectrum Beta Lactamase)
- Resistance due to β -Lactamase enzymes
 - → Hydrolyse β -lactam ring → Inactivate β -Lactam Antibiotics
- Now many ESBLs exist → influence affinity for β -Lactams

○ **MDR-GNote:**

- (Multi Drug Resistant Gram Negative bacilli)
- Resistant to all commonly used antibiotics → limited treatment options

Bacteraemia & Intravascular Infection

A few definitions...

- **Bacteraemia:** The Presence of viable Bacteria in the Bloodstream
- **Septicaemia:** (old term) The *Spread* of Microbes from Wound →Lymphatics→Bloodstream
- **Sepsis:** Physiological term; A condition where Bacteraemia is Associated with an Inflammatory Response from the body (→systemic inflammatory response syndrome), characterised by Fever or Hypothermia, Tachypnoea, Tachycardia and Hypotension

The Human Eco-system:

- **Commensal Flora Exist on:**
 - o Skin (Eg: Staphylococcus)
 - o Pharynx & bronchial tree (Eg: Streptococcus)
 - o Gut (Eg: E-Coli)
 - o Vagina (Eg: Lactobacilli)
- **Sterile Sites:**
 - o Lungs
 - o Uterus & fallopian tubes
 - o Urethra, bladder, Ureters & Kidneys
 - o Peritoneal Cavity
 - o Solid organs and tissues
 - o Blood
 - o CSF

The Bloodstream Can be a Home for Microbes:

- **Favourable Conditions:**
 - o Contains Oxygen, Water & Nutrients
 - (all things required for life – [except For anaerobes])
 - o Has a neutral pH
 - o Appropriate temperature for Microbial Growth
 - (Ie: Most cultures are incubated at 37 degrees)
- **Unfavourable Conditions: (To balance out those favourable conditions)**
 - o Blood is Constantly Moving
 - →Inhibits Adherence
 - o Antimicrobial Defence Mechanisms
 - Phagocytes
 - Complement
 - Antibodies
 - Interferon
 - o Blood recirculates through spleen & liver
 - →Foreign things Get filtered out

Origins of Organisms in Blood Infections:

- Commensal Flora (Ie: Opportunistic Endogenous Organisms)
 - o Skin
 - o Nose and pharynx
 - o Gut
- Sites of infection/Introduction of Pathogens (Ie: Exogenous Organisms)

Things that Can Cause Bacteraemia:

- Chewing food/Brushing Teeth/Dental work (Eg: fillings, extractions):
 - o Can Introduce mouth flora into blood
- Minor injuries:
 - o Can Introduce Skin Flora into blood
- G-I Endoscopy, Polypectomy:
 - o Can introduce Intestinal Flora into blood
- Urinary Catheterisation:
 - o Can introduce perineal flora into blood
- Abscess Rupture:
 - o Skin and soft tissues
 - o Bone
 - o Visceral abscesses
- Significant infection anywhere:
 - o Pneumonia
 - o UTI
 - o Wound Infection
- Contaminated IV lines or catheters

Conditions Required for Infection:

1. Large Numbers of Organisms
2. Anatomical Defect Facilitating Colonization:
 - a. Eg: Faulty Heart Valves – Slows down local blood flow, or increases turbulence, giving the organisms more chance to hang on
 - b. Eg: Break in Epithelium – No barrier to infection
3. Organisms have protective mechanism/s:
 - a. Ie: Virulence Factors:
 - i. Eg: A Capsule [polysaccharide layer outside normal cell wall] → Not Immunogenic & Resists Phagocytosis
 - ii. Eg: Secrete Proteinases → Aid in penetrating into tissues
4. Impaired host defence (Ie: Immunocompromise)

At risk patients (May require some form of prophylaxis against infection)

1. Disruption or penetration of anatomical barriers (Ie: Bypassed physical barriers):
 - a. Wounds
 - b. IV catheters
 - c. Contaminated IV drugs
2. Devitalised tissue:
 - a. Eg: Necrotic tissue has no blood supply (Ie: No way the immune system can get to that area)
3. Defective granulocyte function:
 - a. Eg: In Chemotherapy/Diabetes
 - b. (Note: Chemo patients Don't have many polymorphonucleocytes)
4. Complement defects/deficiency (Immunodeficiencies)
5. Splenic Malfunction/Absence:
 - a. Ie: Poor filtering of bacteria out of the blood (Especially Encapsulated Bacteria)

Safe Havens For Bacteria:

- Damaged Heart Valves & Endocardium
- Organisms can form Bio-films on foreign material because there is no immune system to prevent their growth. Eg:
 - o Catheters
 - o Prosthetic valves, joints

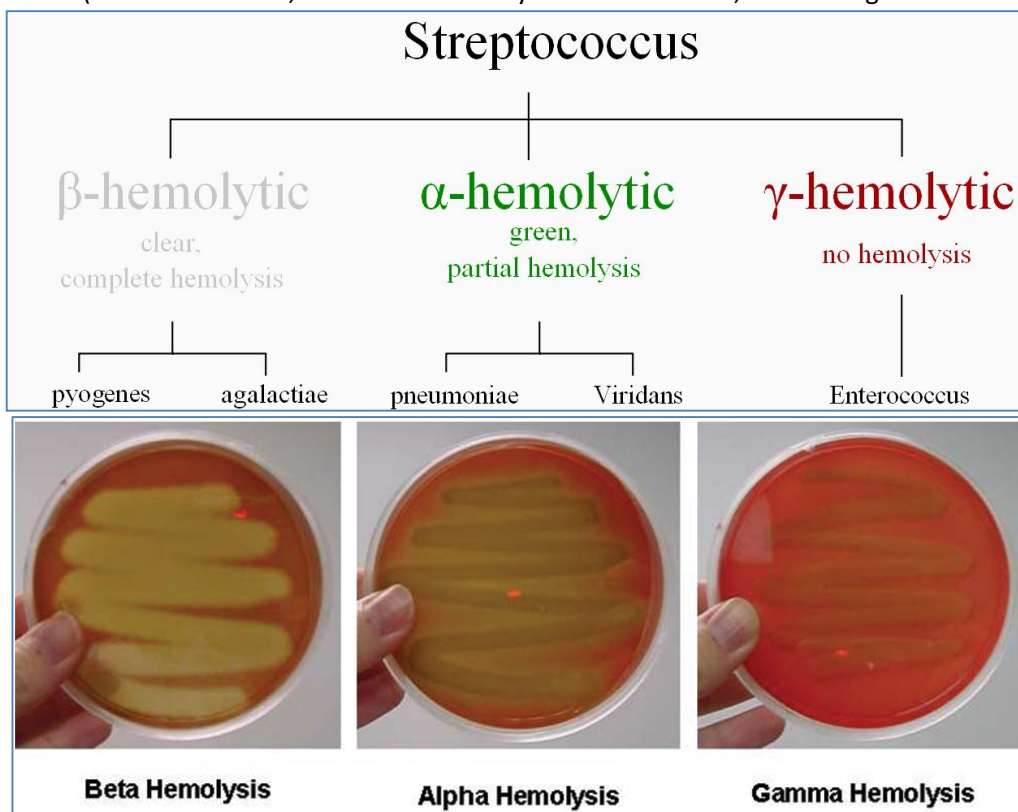
Diagnosis Of Bacteraemia:

- Blood culture (off antibiotics) (The best)
- Imaging to identify primary & secondary foci of infection
- Histology / Culture of any pathological foci

Common Bacteria to Know:

Gram +ve Bacteraemia:

- **Staphylococcus**
 - **-Aureus** (Common flora of Skin & Nasopharynx)(Note: Has many Virulence Factors – Eg: a capsule, toxins, antioxidants)
 - **-Epidermidis** (Common skin flora)
 - (Coagulase Positive Vs Coagulase Negative):
 - **Coagulase Positive Staphylococcus: (Eg: Aureus)**
 - Ie: Produce Coagulase → Converts Fibrinogen to Fibrin → Forms a Fibrin Coat around Bacteria → Resists Phagocytosis → More Virulent
 - **Coagulase Negative Staphylococcus: (Eg: Epidermidis)**
 - Ie: Don't Produce Coagulase
- **Streptococcus**
 - (**α-Haemolytic**)(**α-Haemolysis = Oxidation of Haemoglobin → Greenish colour on Blood-Agar**)
 - **-Pneumoniae** (a Leading cause of Bacterial Pneumonia)(Occasionally causes meningitis)
 - **-Viridans** (Common flora of Mouth)(Can cause Endocarditis in Bacteraemia)
 - (**β-Haemolytic**)(**β-Haemolysis = Complete rupture of RBCs → Wide, clear areas around bacterial colonies on Blood-Agar**)
 - Further Grouped by Serotyping (Based on Cell wall Antigens) – Types: A/B/C/D
 - **Group A Streptococcus** – (Implicated in Rheumatic Fever and Post-Strep Glomerulonephritis)
 - (**Non-Haemolytic/γ-Haemolytic**)
 - Enterococcus
- **Enterococcus** (Normal in bowel; doesn't have many virulence factors, but has high antibiotic resistance)



<http://iws2.collin.edu/dcairn/CCCCD%20Micro/hemolysis.htm>

Gram –ve Bacteraemia:

- **Neisseria:**
 - **Meningitides** (a common URT/Epithelial flora; cocci)(The only cause of Bacterial Meningitis → Headache & Neck Stiffness)
 - **Gonorrhoeae** (Responsible for the STI: Gonorrhoea)
- **Escherichia Coli** (a common Intestinal Flora)(Usually harmless, but can cause Food Poisoning)
- **Klebsiella Pneumonia** (Gram Negative Rods)(Normal Flora of Skin, Mouth & Intestines)(Can cause Pneumonia)

NOTABLE INFECTIVE DISEASES

INFECTIVE ENDOCARDITIS

= Infection of the Endothelial Lining of the Heart (including the heart valves)

- Risk Factors:

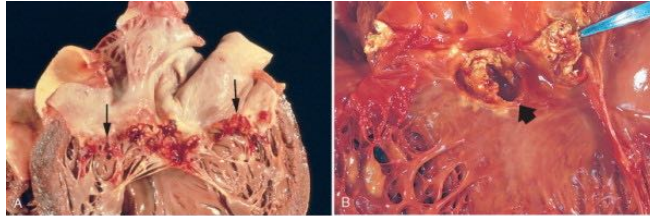
- Valve Abnormality – (Valve Murmurs, Calcification, Congenital, Artificial)
- Open-Heart Surgery
- Poor Dental Hygiene (Source of Bacteria)
- Bacteraemia
- IV Access – (Haemodialysis, IVDU, Surgery)
- Immunosuppression

- Aetiologies:

- **Subacute Bacterial Endocarditis (Most Common - 50-60% of Cases):**
 - *(Oral) Strep Viridans/(Surgical) Strep Epidermidis* (Low Virulence)
 - **Epi:** Recent Oral Surgery, or Post-Prosthetic Valve Insertion
- **Acute Bacterial Endocarditis (Rare – 10-20% of Cases):**
 - *Staph Aureus* (High Virulence - 50% Mortality)
 - **Epi:** IV Drug Users

- Pathogenesis:

- **Bacterial Infection of Valves/Endocardium** → Vegetations on Valve Cusps
 - Typically *Strep Viridans* (**Subacute-BE**) or *Staph Aureus*/MRSA (**Acute-BE**)
 - **Affects Aortic & Mitral Valves;** (RH-Valves may be affected in IV Drug Users)



- Clinical Signs:

- **Symptoms:**
 - ****Fever + New Murmur** = Endocarditis until proven otherwise**
 - +Fatigue, Malaise, Weight Loss
- **Physical Signs:**
 - **Septic Emboli → Infarcts:**
 - **Splinter Haemorrhages** (In the nail bed)
 - **Osler's Nodes** (painful erythematous nodules in pulp of digits)
 - **Janeway Lesions** (Red, nontender lesions on palms/soles)
 - **Roth Spots** (Retinal Haemorrhages - red ring lesions with a yellow centre)
 - Splenomegaly
 - Arrhythmia
- **Complications – (Begin ≈2wks after onset):**
 - **Renal Failure** (Renal Emboli/Immune Complex Deposit → Glomerulonephritis, Haematuria)
 - **TIA** (Cerebral Septic Embolism → Ischaemia → TIA/Stroke)
 - **Septicaemia**
 - **CCF**

- Investigations:

- **Clinical** – (Fever + New Systolic Murmur +/- Septic Emboli)
- **3x Blood Cultures** – (@ Different Times & From Different Sites – Eliminate Contamination)
- **ECG** – (Rule out Ischaemia/MI/Arrhythmias)
- **Echo** – (Valvular Vegetations & Mitral Regurgitation)

- Management:

- **2-6wks of High Dose IV *Vancomycin*** - (Initially Empirical; Then Culture-Directed Therapy)
- **Refer to Cardiac Surgeon – (For ?Valve Replacement Surgery?):**
 - If IV-ABs are Unsuccessful
 - Or If Valve is Destroyed (Ie: In Acute-BE) → Heart Failure

- Prognosis:

- 30% Mortality with Rx

LYMPHANGITIS:

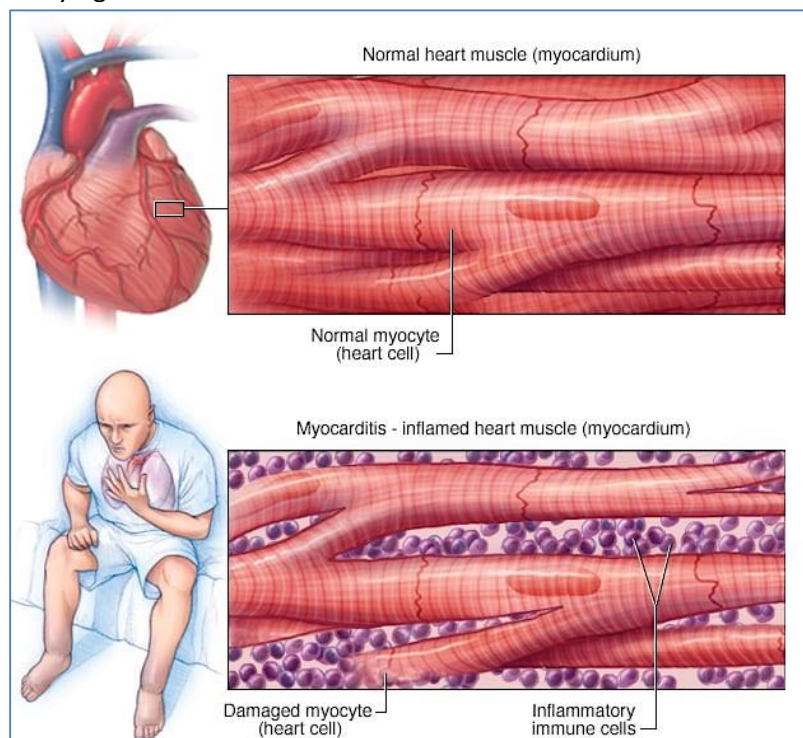
- **Aetiology:**
 - Commonly ***β -Haemolytic-Strep*** or ***Staphylococcus Aureus***
- **Pathogenesis:**
 - Bacterial Infection Spread to Lymphatics → Acute Inflammation
 - **If Severe** → Cellulitis/Abscesses
 - **If Very Severe** → Bacteraemia/Sepsis
- **Clinical Features:**
 - Fever/Chills/Malaise
 - Painful Red Subcutaneous Streaks
 - Painful Lymphadenitis (Swollen draining lymph nodes)
- **Complications:**
 - Abscesses
 - Cellulitis
 - **Sepsis**
- **Investigations:**
 - Blood Culture + Swab any open wounds
 - FBC +/- CRP
- **Management:**
 - Immobilisation of Limb
 - Antibiotics
 - Analgesia



James Heilman, MD, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

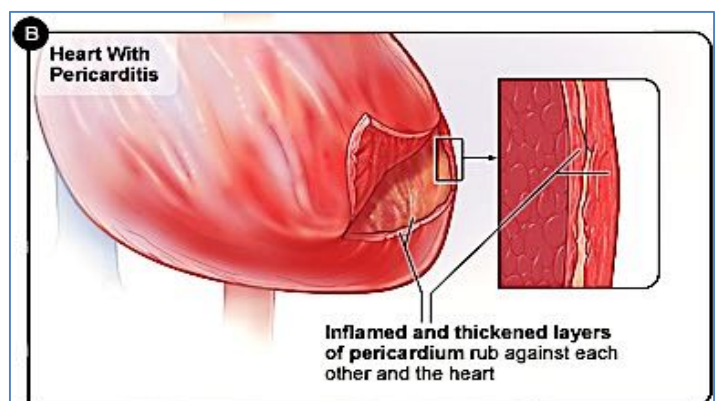
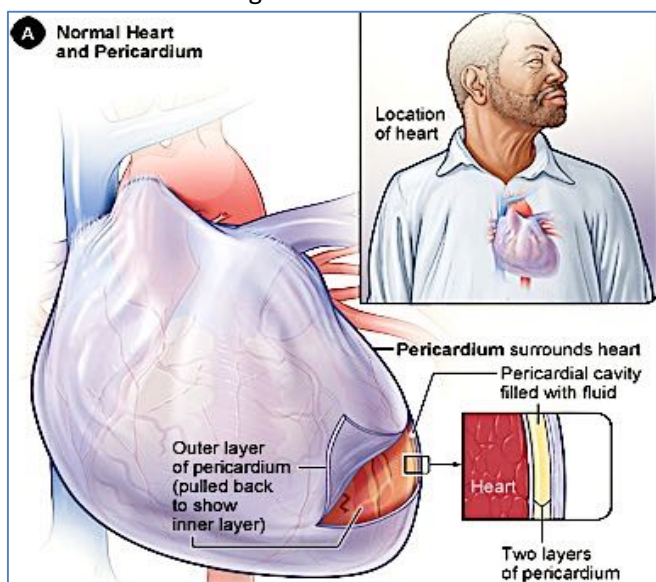
MYOCARDITIS – VIRAL & TOXIC:

- **What is it?**
 - “Inflammation of the Heart Muscle”
 - + Characterized by Myocyte Necrosis – (Positive Troponin I results seen in 35% of Myocarditis)
- **2 Main Aetiologies:**
 - **VIRAL MYOCARDITIS** (Eg: Coxsackievirus, Rhinovirus, Influenza, Parvovirus B19, etc)
 - Either Direct Myocardial Injury OR 2° Autoimmune Response
 - → Heart Thickens & Weakens → Systolic Heart Failure
 - **TOXIC MYOCARDITIS** (Eg: Chemo Drugs, Cocaine, Alcohol, Diuretics, Antibiotics, Venom, CO, etc)
 - Myocardial Damage & Inflammation due to Either:
 - Hypersensitivity to Drugs
 - Direct Toxic Damage
- **Clinical Features:**
 - (May be Asymptomatic)
 - **Symptoms:**
 - **Flu-Like Syx** - (Fever, Fatigue, Malaise)
 - **LV-Failure** – (Dyspnoea/Orthopnoea/PND/Cough)
 - **Chest Pain** – (Due to Myocarditis +/- Pericarditis)
 - **Palpitations** (Arrhythmias)
- **Complications:**
 - **Cardiomyopathy** → Heart Failure
 - **Arrhythmias** → **Sudden Death**
 - **Pericarditis** → Pericardial Effusion
- **Investigations:**
 - **ECG & Continuous Telemetry**
 - **Serial Troponins I/T** - (Immediately, then @6hrs, then @24hrs)
 - **FBC** (↑WCC), **CRP** (↑), **ESR** (↑)
 - **CXR** (Cardiomegaly)
 - **Echo** (Dilated, Poor Vent-Function)
- **Management:**
 - ****Bed Rest**
 - ****CCF Triple Therapy** (ACEi/ARB + B-Blocker + Diuretics)
 - ****Warfarin** (Prevent Thromboembolism)
 - Supportive Rx (Fluids, Analgesia)
 - Treat Underlying Cause if Possible



PERICARDITIS:

- **Aetiology:**
 - Usually **Secondary** to:
 - **Infection** (**Viruses**, Bacteria, Fungi, Parasites)
 - **Immuno** (Rheumatic Fever, SLE, Post-MI, Drug Hypersensitivity)
 - **Other** (MI, Post-Cardiac Surgery, Neoplasia, Trauma, Radiation)
- **Classification:**
 - According to **Composition of Pericardial Exudate:**
 - **Serous** (Non-Infectious Inflammatory Diseases – SLE, Uraemia, Tumours)
 - **Purulent** (Infective - by Microbes)
 - **Fibrinous/Serofibrinous** (Due to Acute MI, Chest Radiation, SLE, Trauma)
 - **Caseous** (Tuberculosis)
 - **Haemorrhagic** (Due to Metastasis, Cardiac Surgery)
- **Pathogenesis:**
 - Various Aetiologies → Inflammation of the Pericardium
 - → Thickening of Pericardium → Pericardial Exudate (Serous Fluid + Pus/Fibrin/Blood)
 - → Rubbing of Parietal & Visceral Pericardium → Further Inflammation & Exudate
- **Clinical Features & Complications:**
 - **Symptoms:**
 - Pleuritic Chest Pain (Better on Sitting Forward; Worse on Inspiration & Lying Down)
 - Fever, Fatigue
 - Dry Cough
 - Sx of CCF (Dyspnoea, Fatigue)
 - **Signs:**
 - Fever, Tachycardia
 - Muffled Heart Sounds
 - Friction Rub
 - ↑JVP
 - Heart Failure Signs if Tamponade
 - **Complications:**
 - Cardiac Tamponade/Pericardial Effusion
 - If >3mths → Chronic → Constrictive Pericarditis (Requires Surgery)
- **Diagnosis:**
 - **ECG** – (Classical PR-Depression + ST-Elevation + Tachycardia)
 - **CXR** – (Pulmonary Congestion)
 - **ECHO** – (?Pericardial Effusion)
- **Management:**
 - Rx Underlying Cause
 - Anti-Inflammatories (NSAIDs / Steroids)
 - Analgesia



IMPETIGO (SCHOOL SORES)

- **What is it?**
 - Superficial Bacterial Skin Infection
 - Most Common in school kids
 - Very Contagious – (Spread by Close Contact & Poor Hygiene)
 - Usually resolves slowly
- **Organism:**
 - **Mostly *Staphylococcus Aureus***
 - **Sometimes *Streptococcus Pyogenes***
 - Can lead to Glomerulonephritis or Rheumatic Fever if it's Strep
 - Staph Aureus (Bullous) - (Pic 1)
 - Streptococcus (Non-bullous) – (Pic 2)
- **Presentations:**
 - Occur most commonly on face
 - Fragile vesicles rupture & crust
 - Can be confused with HSV
 - **1: Nonbullous/Crusted Impetigo:**
 - (Most common)
 - Yellow crusts and erosions
 - Itchy/Irritating (but not painful)
 - **2: Bullous impetigo:**
 - Always due to *Staph Aureus*
 - → Mildly irritating blisters that erode rapidly leaving a brown crust
 - **3: Ulcerative lesions:**
 - Always due to *S-pyogenes*
 - Most common in Aboriginal Communities
- **Very Infectious**
 - Epidemic in young children
 - Transmitted through skin contact
 - Outbreaks associated with poor hygiene / crowded living conditions
- **Treatment:**
 - **Cover Affected Areas**
 - **Abstain from School**
 - Systemic or Topical Antibiotics



1. James Heilman, MD, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

2. Evanherk, CC BY-SA 3.0 <<http://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

ERYSIPELAS

- **Aetiology:**
 - Group A Strep (GAS) / Staphylococcus Aureus
 - Breaks in skin (Eg: Insect bites/ulcers/cracking skin conditions/eczema)
- **Epidemiology:**
 - Mostly infants & older people
- **Pathophysiology:**
 - Potentially serious bacterial infection of the skin
 - → Infection of the upper dermis → extends to superficial cutaneous lymphatics
- **Clinical Features:**
 - A superficial form of Cellulitis
 - 'St Anthony's fire' = Intense rash associated with erysipelas
 - Rapid onset fevers, chills, rash
 - Affected skin has sharp, raised border
 - Bright red, firm & swollen
 - May blister or become necrotic
 - may spread to deeper lymphatics (lymphangitis)
- **Management:**
 - Wound care
 - Oral/IV penicillin antibiotic
 - Erythromycin/Roxithromycin if Penicillin allergic
 - Vancomycin if MRSA



CDC/Dr. Thomas F. Sellers/Emory University, Public domain, via Wikimedia Commons

CELLULITIS

- **What is it?**
 - o Bacterial infection of the Dermis and Sub-Cutaneous Tissues
- **Organism:**
 - o **Adults:** 90% due to *Staph Aureus*/GAS
 - o **Children:** *H-Influenzae b*
 - o **Associated with cat/dog bite:** *Pasturella multocida*
- **Presentation:**
 - o Painful, raised and Oedematous Erythema (Most commonly on Lower Leg)
 - o Possible Blistering
 - o Lymphadenopathy - & Malaise & Fever
- **Distribution:**
 - o **Children** – Periorbital Area
 - o **Adults** – Lower Legs
- **There's typically an underlying cause:**
 - o Lymphedema
 - o Tinea, Herpes simplex infection, Chronic sinus infection
 - o Chronic dermatitis
 - o Poor lower leg circulation
 - o Wounds
- **Treatment:**
 - o Oral/IV penicillin antibiotic
 - o Erythromycin/Roxithromycin if Penicillin allergic
 - o Vancomycin if MRSA



Source: <https://www.nhs.uk/conditions/cellulitis/>

SCABIES

- **Organism:**
 - o *Sarcoptes scabii* (Scabies Mite)
- **Epidemiology:**
 - o Human infestations originating from pigs, horses and dogs are mild and self-limiting
 - o Scabies infestations from other humans never cure without intervention
- **Ecology:**
 - o Mites live in stratum corneum (Don't get any deeper)
 - o Eat stratum corneal Keratinocytes
 - o Make "tunnels" by eating
 - o Mating occurs on the hosts skin
 - o Fertilized Female Mites Burrow into the Stratum Corneum (1 mm deep)
 - o Salivary Secretions contain Proteolytic Enzymes → Digest Keratinocytes
- **Transmission:**
 - o High prevalence in children (50%) and adults (25%) in tropical remote communities
 - o Spread by close physical contact
- **Presentation:**
 - o Itch (Exacerbated at night and after hot showers)
 - o Itchy, Excoriated Rash on Trunk, associated with Scaly Burrows on the fingers and wrists
 - o Often vesicles and pustules on the palms and soles and sometimes on the scalp
- **Diagnosis:**
 - o **Clinical Diagnosis:**
 - Chronic itch with Symmetrical Rash
 - Burrows
 - o **Skin Scraping - Look for Scabies Mites:**
 - Intact larvae, nymphs or adults
 - Unhatched or hatched eggs
 - Moulded skins of mites
 - Fragments of moulted skins
 - Mite faeces
- **Treatment:**
 - o **Topical** Permethrin
 - o **Or Oral** Ivermectin (But not on PBS – Very Expensive)
 - o **Environmental Measures:**
 - Mites can contaminate bedding, chairs, floors, and even walls
 - (Usually only a problem with crusted scabies)
 - Wash, sun, vacuum, surface insecticide
 - o **Community Prevention:**
 - Treat all close contacts – Especially in Indigenous Communities
 - Simultaneous Effective Treatment
 - o **TREAT AGAIN IN 7 DAYS**



Gzzz, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons
Penarc, CC BY 3.0 <<https://creativecommons.org/licenses/by/3.0/>>, via Wikimedia Commons
Joel Mills, CC BY-SA 3.0 <<http://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

LICE (PEDICULOSIS)

- **3 Types:**
 - **1: Head Lice: *Pediculus Humanus Capitis***
 - **Epidemiology:**
 - Common in Primary School Children in the Tropics
 - Higher prevalence in Aboriginal Children
 - **Diagnosis:**
 - Conditioner + Fine-Tooth Comb
 - Wipe combings on white tissue paper
 - **2: Body Lice: *Pediculus Humanus Corporis***
 - Live on clothes, and come to the body to feed
 - **3: Pubic Lice: *Phthirus Pubis***
 - Largely sexually transmitted
 - Blood Feeder
 - Can infect any Body Hair (Pubic/Trunk/Legs/Axilla/Beard) but rarely head
- **Lifecycle:**
 - Eggs laid in hair (knits)
 - Larvae grow into adults
 - Adults – **blood sucking** (live in hair)
- **Transmission:**
 - head-head contact
- **Presentation:**
 - Scalp and Neck can be Itchy
 - Nits are noticeable on the hairs
- **Diagnosis:**
 - **Best Method = 'Conditioner & Comb Technique':**
 - Very Practical for parents
 - Cost Effective
 - High Sensitivity
 - Conditioner 'Stuns' the lice by suffocating them → Prevents them from running away
- **Management/Treatment:**
 - Conditioner & Nit Comb
 - Physical Removal
 - Cut Hair
 - Topical Insecticidal Cream
 - Good idea to wash pillows and hats though – Hot Wash
 - (Treat all body hair – for Pubic lice)
 - **Reasons for Treatment Failure:**
 - Inadequate application of the product
 - Lice are resistant to insecticide
 - Failure to retreat to kill nymphs emerged from eggs
 - Reinfection



1.Public domain: https://commons.wikimedia.org/wiki/File:Pediculus_humanus_var_capitis.jpg

2.<https://www.pcds.org.uk/clinical-guidance/pediculosis>

HERPES SIMPLEX

- **What is it?**
 - Common Mucosal Viral Infection that presents with localised blistering
 - Can reside in a latent state
- **2 Types:**
 - **Type 1: Typically facial/oral infections** (Cold sores/fever blisters)
 - Occur mainly in infants & young kids
 - **Type 2: Mainly Genital**
 - Occur after puberty (often transmitted sexually)
- **Presentation:**
 - **Stages of Infection:**
 - 1: Prodromal Stage Vesicle or "blister" stage
 - 2: Ulcer stage
 - 3: Crust stage
 - The virus grows down the nerves and out into the skin → Localised Blistering
 - Neuralgia
 - Lymphadenopathy
 - High Fever
- **Recurrences can be triggered by:**
 - Minor trauma/Other infections/UV radiation/Hormonal factors/Emotional stress/Operations/procedures on face
- **Treatment:**
 - Mild cases require no treatment
 - Sun protection to prevent
 - Oral Antiviral Drugs (Stop the virus multiplying)
- **Complications:**
 - Encephalopathy
 - Trigeminal Neuralgia (Neurogenic Pain)



Public Domain: CDC

CHICKEN POX (VARICELLA ZOSTER):

- **What is it?**
 - Highly contagious disease
 - Typically childhood disease (before 10yrs)
 - One infection thought to confer lifelong immunity
- **Organism:**
 - Varicella zoster virus (HHV3) (AKA: Chicken Pox Virus, Varicella, Zoster)
- **Transmission:**
 - Highly Infectious
 - From person to person
 - Aerosol Droplets
 - Direct contact with fluid from open sore
- **Pathophysiology:**
 - Incubation Period ≈ 2wks
 - **(Chicken Pox)** Initial Mucosal Infection → Viremia → Epidermal Lesions
 - May lead to → Latent infection of Dorsal Ganglion Cells of Sensory Nerves
 - **(Shingles)** Reactivation of latent Varicella Zoster Virus in Peripheral Nerves
- **Signs/symptoms:**
 - Itchy rash or red papules
 - Begins on the Trunk → Face and Extremities
 - May cover entire body
 - High fever/headache/cold-like symptoms/vomiting/diarrhoea
- **Diagnosis:**
 - Clinical Diagnosis
 - Immunofluorescence
 - Test for Elevated VZV-Specific Antibodies
 - (IgM – Primary Infection; IgG – Second Infection)
- **Treatment:**
 - Symptomatic
 - Resolves on its own
- **Complications:**
 - **Varicella During Pregnancy can → Congenital Varicella Syndrome:**
 - **Spontaneous Abortion** (3-8% in 1st Trimester) or IUGR
 - **Skin:** Cutaneous Defects, Hypopigmentation
 - **Neuro:** Intrauterine Encephalitis, Brain Damage, Seizures, Developmental Delay
 - **Eye:** Chorioretinitis, Cataracts, Anisocoria
 - **MSK:** Limb Hypoplasia
 - **Systemic:** cerebral cortical atrophy
 - **Renal:** Hydronephrosis, Hydroureter
 - **GI:** GORD
 - **CVS:** Congenital Heart Defects
 - **Perinatal Varicella Infection:**
 - severe → mortality rate of 30%



1. Camiloaranzales, Public domain, via Wikimedia Commons

2. F malan, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

HERPES ZOSTER (SHINGLES)

- **What is it?**
 - Reactivation of Latent Herpes Varicella Zoster Virus
- **Pathophysiology:**
 - Incubation Period $\approx$ 2wks
 - **(Shingles)** Reactivation of latent Varicella Zoster Virus in Peripheral Nerves
- **Presentation:**
 - Painful blistering rash along 1/more Dermatomes
 - Virus is seeded to nerve cells in spinal cord
 - Fever, malaise and headache
 - Lymph nodes draining affected area are often enlarged/tender
 - Can also result in nerve palsy
- **Diagnosis:**
 - Clinical Diagnosis
 - Test for Elevated VZV-Specific Antibodies
 - PCR
- **Transmission:**
 - Shingles are infectious
 - From person to person
 - Direct contact with fluid from open sore
- **Treatment:**
 - Antiviral treatment
 - Rest & analgesia
 - Oral Antiviral



Fisle, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

MEASLES VIRUS:

- **Organism:**
 - Morbillivirus
- **Transmission:**
 - Respiratory Route (Aerosol)
 - Contact with fluids from infected person's nose/mouth
- **Pathogenesis:**
 - Typically a Respiratory Infection
 - → Produces a Viremia → Rash
- **Presentation:**
 - Fever
 - URTI - Cough, Rhinorrhoea, Red Eyes
 - Maculopapular Erythematous (Morbilliform) Rash
 - "Koplik's Spots" – Seen on the Inside of the Mouth
- **Complications Include:**
 - Croup
 - Otitis Media
 - Enteritis with diarrhoea
 - Febrile convulsions
 - Encephalitis (Serious)
 - Subacute Sclerosing Panencephalitis (very rare)
 - (Chronic, progressive Encephalitis caused by persistent infection with immune-resistant Measles Virus)
 - No Cure
 - Fatal
- **Diagnosis:**
 - Clinical Diagnosis (Generalised Maculopapular Rash + Fever)
 - Presence of Measles IgM Antibodies
 - PCR of Respiratory Specimens
- **Treatment:**
 - No Specific Treatment
 - Prevented by MMR Vaccine
- **Prevention:**
 - Attenuated MMR Vaccine (Admin at 12mths & 4yrs)
 - Developing Countries: Low Herd Immunity → Higher Prevalence
 - Relatively High Death-Rates in Non-Immune



Photo Credit: Content Providers(s): CDC/Dr. Heinz F. Eichenwald, Public domain, via Wikimedia Commons

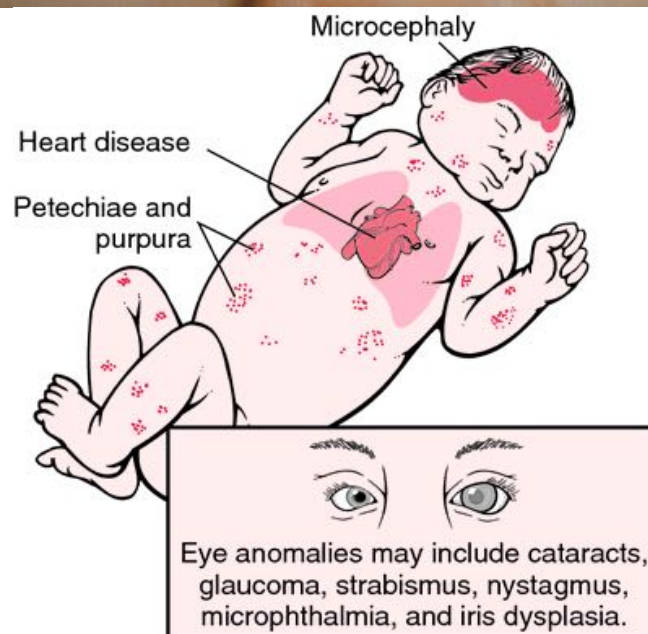
RUBELLA VIRUS ("GERMAN MEASLES):

- **Organism:**
 - Rubella Virus
- **Transmission:**
 - Respiratory Route
 - (Human Reservoir Only)
- **Presentation:**
 - Initial Flu-Like Symptoms
 - * Rash on Face → Spreads to Trunk & Limbs
 - Pink-Red, Itchy
 - Low-grade Fever, Lymphadenopathy, Joint Pains, Headache, Conjunctivitis
- **Prognosis:**
 - Typically Benign
 - Typically Lasts 1-3 Days (Children Recover Quicker)
 - Complications may include arthritis, thrombocytopenia purpura, and encephalitis
 - ***HOWEVER, Maternal Infection During PREGNANCY can be SERIOUS!!**
 - If Infected in the 1st 20wks of Pregnancy → ***Congenital Rubella Syndrome***
 - →Abortion
 - →Cardiac/Cerebral/Ophthalmic/Auditory Defects
 - Specific Foetal Damage Depends on Organ Development @ the Time:
 - The 1st Trimester is Worst, as *Organ Development* occurs during this time
 - After 1st Trimester, *Organ Growth* is the main process
- **Diagnosis:**
 - Clinical Diagnosis
 - Presence of Virus-Specific IgM Antibodies
- **Treatment:**
 - No Specific Treatment
 - Controlled by vaccination (MMR Vaccine)
 - Test pregnant women for immunity early
- **Prevention:**
 - (Note: Rubella *Itself* is relatively Benign, so why bother Vaccinating?)
 - **MMR Vaccine:**
 - **(Live Attenuated)**
 - **#1 Aim:** Prevent Rubella in Pregnant Women → ↓Congenital Rubella Syndrome
 - Aimed at *BOTH* Males & Females to ↓Male Transmission to Pregnant Females





<https://www.cdc.gov/rubella/about/photos.html>



http://phil.cdc.gov/phil_images/20030724/28/PHIL_4284_lores.jpg

HUMAN PARVOVIRUS B19 ("5TH DISEASE")

- **Organism:**
 - Parvovirus B19
- **Transmission:**
 - Respiratory Droplet
 - Blood-Borne
- **Pathophysiology:**
 - **Virus Replicates in Rapidly-Dividing Cells (Eg: Bone Marrow RBC Precursors)**
 - → RBC Haemolysis
 - → Severe Anaemia
 - → Can Result in ***Haemolytic Crisis***
 - The receptor for the virus is a globoside, which is abundant on tissues of mesodermal origin
 - **Can cross the placenta into the foetus**
 - → Foetal Anaemia
- **Presentation:**
 - Fever/Malaise
 - Characteristic Rash
 - Teenagers: 'Papular Purpuric Gloves & Socks Syndrome'
 - Children: 'Slapped Cheek Syndrome'
- **Note: Foetal Infection → Foetal Damage or Abortion**



1. Andrew Kerr, Public domain, via Wikimedia Commons

2. Sandyjameslord, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

SCARLET FEVER:

- **Organism:**
 - Certain strains of ***Strep pyogenes*** (Which carry a Bacteriophage – A virus infecting the bacteria → Produce an Eruthrogenic toxin)
- **Epidemiology:**
 - Mostly occurs in kids aged 4-8yrs
- **Pathogenesis:**
 - GAS infection of Tonsils/Pharynx/Skin
 - **Exotoxin** Released by Strep Pyogenes → Local effect on Tonsils/Pharynx/Skin
 - → Abnormalities of tongue
 - Initially covered with white exudate
 - Exudate is shed
 - inflammation of underlying tissue
 - → Diffuse, Erythematous Exanthem
- **Treatment:**
 - **Antibiotics** (Usually penicillin for 10days; or single IM dose; or erythromycin if penicillin allergic)
 - Antipyretics
 - Fluids
 - Oral antihistamines to relieve the rash
- **Longer-Term Complications:**
 - Rheumatic fever
 - Glomerulonephritis



1 - Estreya at English Wikipedia., CC BY 2.5 <<https://creativecommons.org/licenses/by/2.5>>, via Wikimedia Commons

2 - www.badobadop.co.uk, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0>>, via Wikimedia Commons



Afag Azizova, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0>>, via Wikimedia Commons

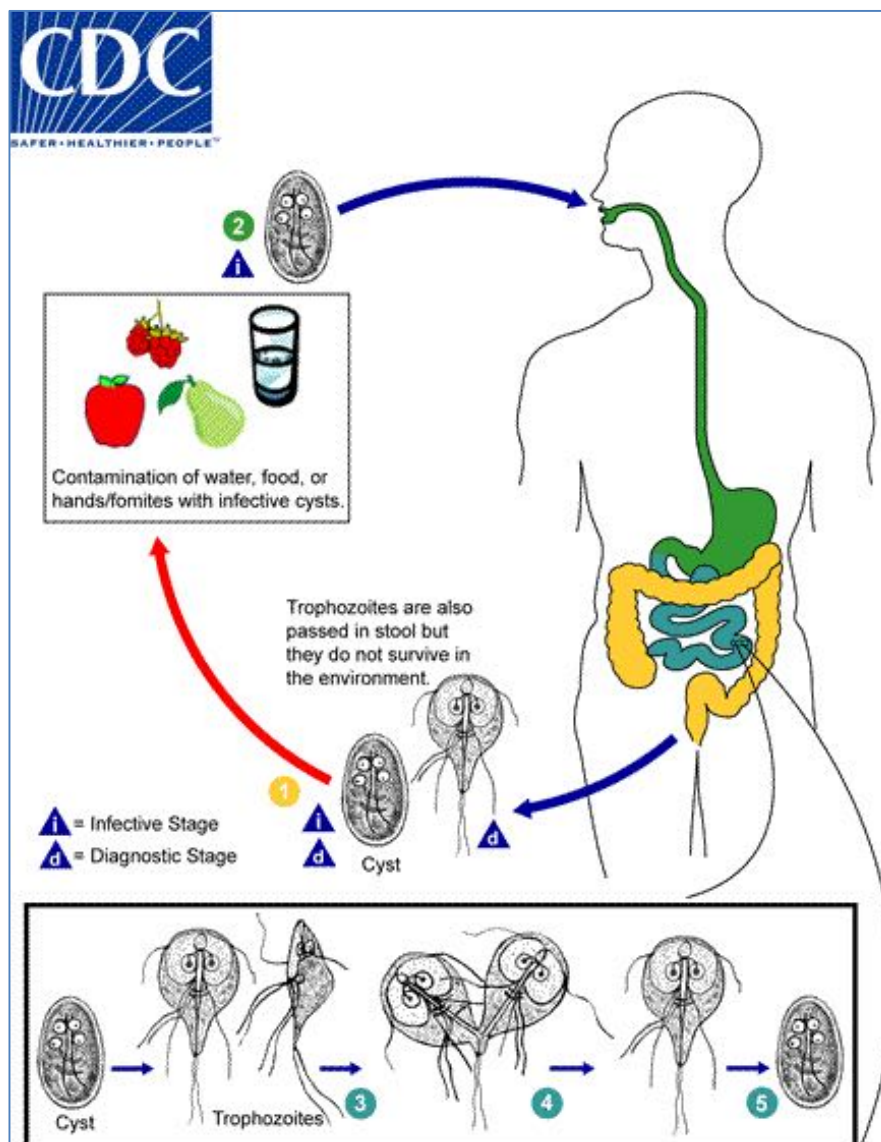
DERMATOPHYTOSIS - "RINGWORM"/"TINEA":

- **Organism:**
 - **3 Genera Are Important:**
 - *Trichophyton*
 - *Microsporum*
 - Note: *Microsporum Canis* – (From Dogs/other animals)
 - Note: Fluorescent under Wood's Lamp
 - *Epidermophyton*
- **Pathogenesis:**
 - **Fungi ONLY Metabolizes Keratin:**
 - ∴ Only infect the Stratum Corneum
 - Note: Can Also Invade Hair Shafts
- **Epidemiology:**
 - Common In Rural Indigenous Populations
- **Conditions Named Based On Location of Infection:**
 - Tinea Corporis (On Body)
 - Tinea Capitis (On Head)
 - Tinea Crura (Pubic Area)
- **Symptoms:**
 - Well Circumscribed lesions with central clearing
 - Focal hair loss due to infection of Hair Follicle
 - Focal pityriasis (Skin Flaking)
 - Usually not pruritic
 - Can infect any keratinised structure
 - Nail infections can be severe
 - "Tinea Versicolor" (Depigmentation of the Skin)
 - "Tinea Imbricata/Concentricum" (As the ringworm grows, it produces concentric silvery rings)
 - Caused by *Trichophyton Concentricum*
- **Diagnosis Of Dermatophytosis:**
 - Clinical Diagnosis
 - Woods lamp – only *Microsporum canis* fluoresces
 - Microscopy of hairs/nail shavings/skin shavings
 - Culture for dermatophyte on Sabouraud's agar
- **Treatment:**
 - **Topical Antifungals:**
 - Clotrimazole, Miconazole, Econazole, Tolnaftate, Terbinafine
 - **Oral Antifungals:**
 - Griseofulvin for 4 weeks
 - Or Itraconazole / Fluconazole / Terbinafine



PARASITIC GUT INFECTIONS (Protozoa & Helminths):

- **Transmission:**
 - Faecal-Oral – (Ingestion of Dormant Cysts in Contaminated Food/Water)
- **Diagnosis:**
 - Stool Samples (Looking for cysts) under Direct Microscopy
 - Antigen Testing
- **Prevention:**
 - Boiling Water to Eliminate Cysts
 - Good Hygiene
 - Avoiding Faecal Contact
- **Examples:**
 - **GIARDIA:**
 - **Pathogenesis:**
 - Not Toxicogenic; Rather, it covers the brush border → Malabsorption
 - **Diagnosis:**
 - Cysts in Stools
 - **Complications:**
 - Chronic Infection
 - Malabsorption
 - → Malnutrition
 - → Fatty Stools
 - **Treatment:**
 - **Metronidazole**



<https://www.cdc.gov/dpdx/giardiasis/index.html>

○ **CRYPTOSPORIDIUM:**

▪ **Transmission:**

- Ingestion of oocysts (Contaminated Drinking Water/Public Pools)
- Can survive Chlorination

▪ **Pathogenesis is mostly unknown**

- Possibly induces inflammatory response → Disrupts absorptive surface
- Damages Villi → Crypt Cells Replicate faster to replace them → Immature cells in the villus → Poor absorption

▪ **Treatment:**

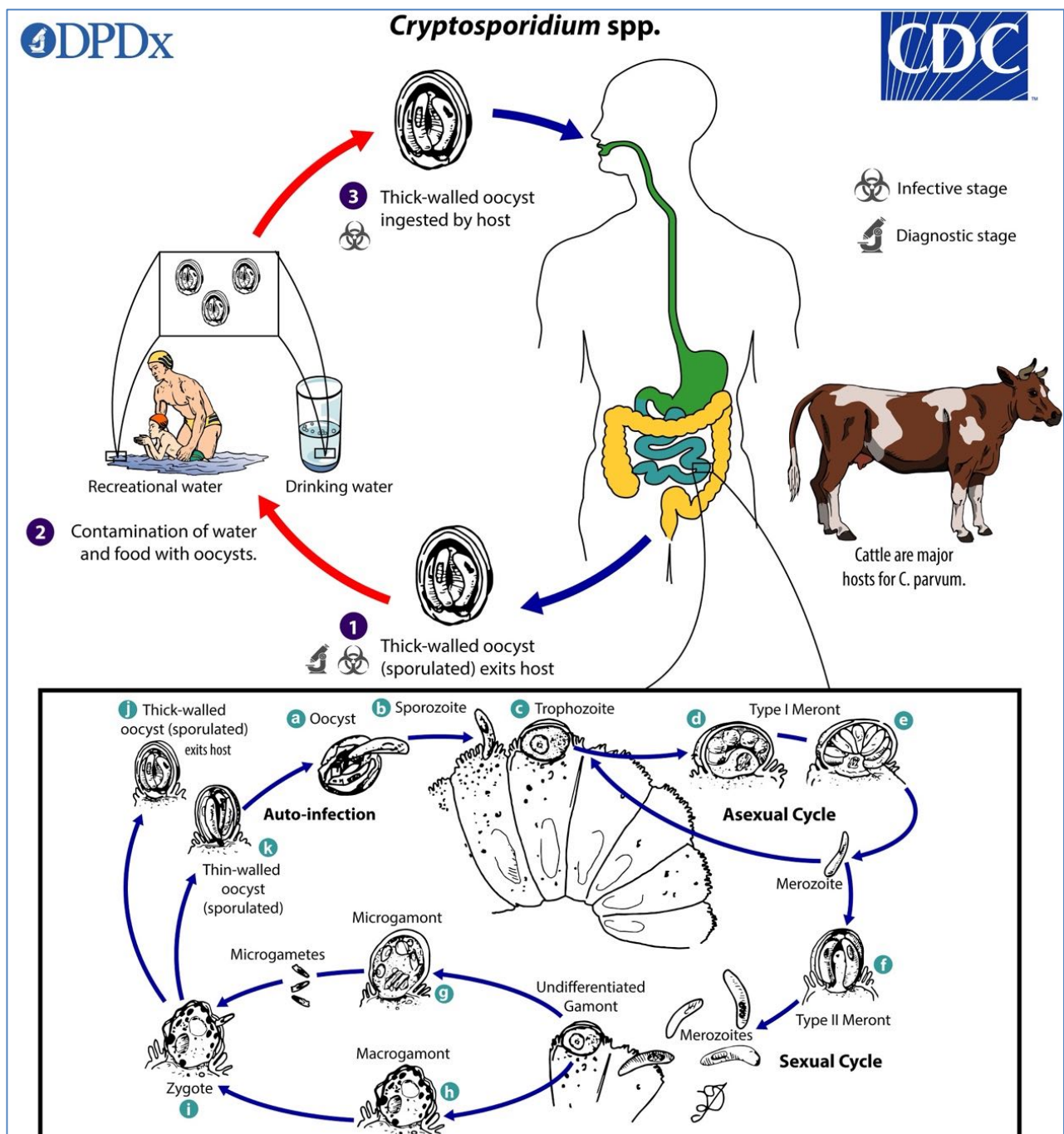
- **Nitazoxanide** (Normally Self-Limiting if Immunocompetent)

▪ **Long term Effects:**

- AIDs patients don't recover → Chronic Infection

▪ **Diagnosis:**

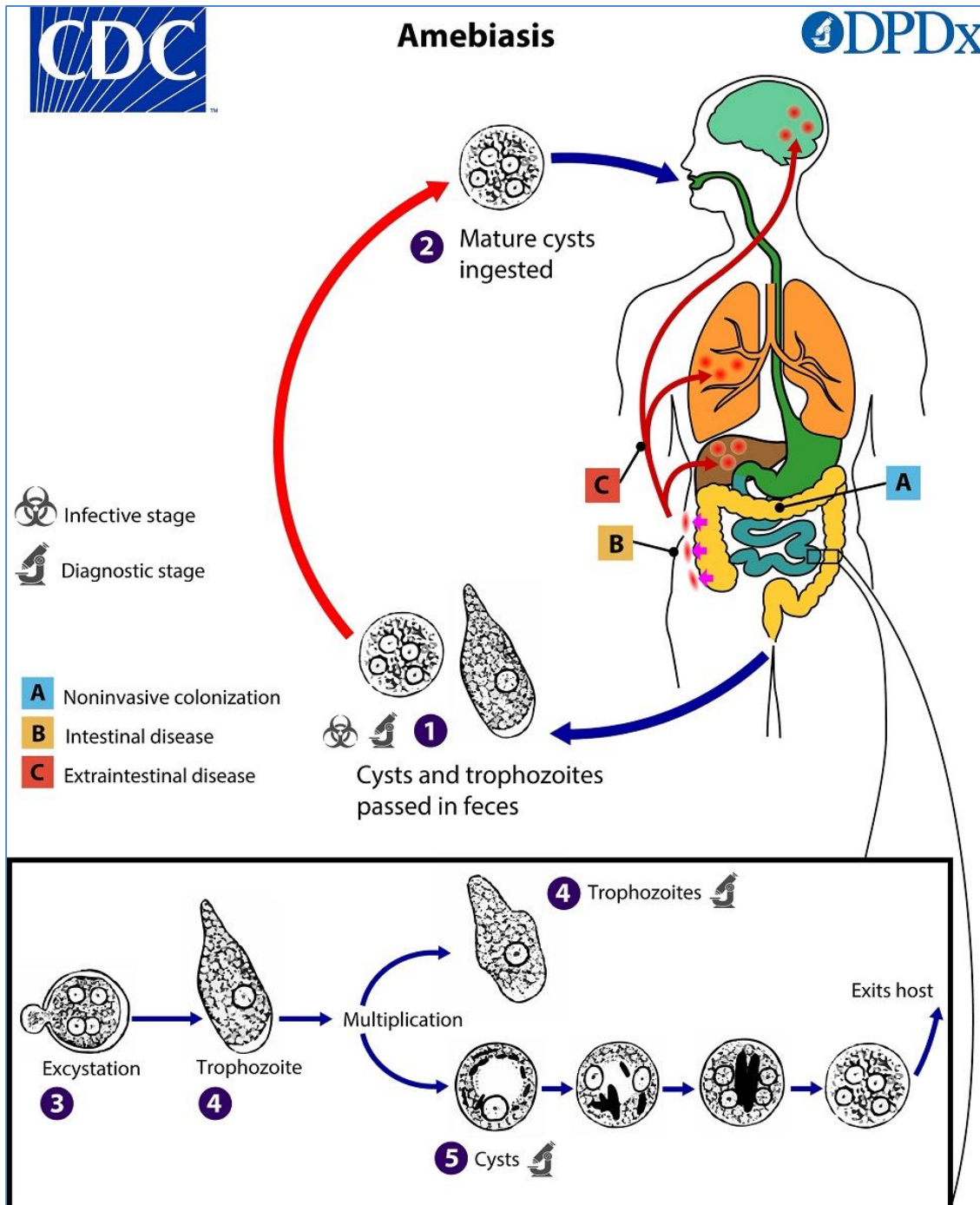
- Cysts in Stools



<https://www.cdc.gov/dpdx/cryptosporidiosis/index.html>

○ **ENTAMOEBIA HISTOLYTICA - (The Amoebic Dysentery):**

- **Transmission:**
 - Ingestion of oocysts (Faecal Oral)
- **Pathogenesis:**
 - Intestinal Invasions → Ulcerations → Dysentery (Bloody Diarrhoea)
- **Diagnosis:**
 - Cysts in Stools
- **Management:**
 - **Metronidazole**



<https://www.cdc.gov/dpdx/amebiasis/index.html>

POLIOMYELITIS:

- **Aetiology:**
 - Poliovirus Infection
- **Epidemiology:**
 - Eradicated in many countries (A single case would be an epidemic)
- **Prevention:**
 - **Vaccination Available**
 - **Live Attenuated (Oral Polio Vaccine):**
 - Advantages:
 - Easy Administration - Given Orally
 - Disadvantage:
 - Rarely causes paralysis (1 in 2.5million)
 - **Inactivated Polio Vaccine (IPV):**
 - Advantages:
 - Carries NO risk of Vaccine-Associated Polio Paralysis
 - Disadvantage:
 - Difficult Administration - Has to be injected
- **Pathogenesis:**
 - **Transmission:**
 - Faecal-Oral
 - or Respiratory
 - Initially Enteric Infection → Spreads to Bloodstream → Spinal Cord → Preferentially Infect & Destroy **Motor Neurons**
- **Clinical Features:**
 - **90% Asymptomatic**
 - **<10% Minor Viral Illness:**
 - Headache
 - Neck/Back pain
 - Abdominal Pain
 - Fever, Lethargy, Vomiting
 - **1% CNS Infection → Paralysis**
 - Acute Asymmetrical Flaccid Paralysis + Areflexia
 - If 'Spinal Polio' → Paralysis of Legs(unilateral)
 - If 'Bulbar Polio' → Cranial Nerve Paralysis (Eg: Dysphagia, Dysphasia, Dyspnoea)
 - Or Combination of Both
- **Treatment:**
 - Self-Limiting, but Lasting Disability – Only Supportive Rx (Eg: Ventilation, Physiotherapy)
 - But Vaccine Preventable



MENINGITIS:

- **Aetiology:**
 - Bacterial/Septic Meningitis – **Neisseria meningitides**, **Haemophilus influenza**, **Group B Streptococci**
 - Adults = **Neisseria meningitides** (Note: Vaccine preventable – Meningococcal A & C)
 - Children = **Haemophilus influenza** (Vaccine Preventable – Hib Vaccine)
 - Neonates = **Group B Streptococci (or E-coli)**
 - Viral/Aseptic Meningitis – **HSV**, **Enteroviruses (Echo/Coxsackie)**, **Influenza**, **Mumps**, **HIV**
 - Chronic Meningitis – **Miliary Tuberculosis**
 - Fungal Meningitis - Typically in immunosuppressed patients
- **Pathogenesis:**
 - Meningeal Infection → Inflammation & Oedema → ↑ICP → Vomiting, Drowsiness
 - Note: Meningococcal Sepsis can → **Thrombocytopenia** → Maculopapular Rash → DIC
- **Morphology:**
 - Bacterial → Exudate within Meninges (Pus beneath the meninges)
 - Viral → No pus
 - Engorged Meningeal Vessels
- **Clinical Features:**
 - *****Meningism:**
 - ***1: Neck Stiffness** (Due to Inflammation of the Meninges)
 - ∴ Brudzinski's Sign Positive (Flex the Neck → Pt bends knee)
 - ∴ Kernig's Sign Positive (Flex the hip and attempt knee extension → Pain)
 - ***2: Photophobia**
 - ***3: Headache**
 - <1% Papilloedema = Swelling of the Optic Disc secondary to the ↑Intracranial Pressure
 - + Constitutional Syx:
 - Fever/Malaise
 - Nausea/Vomiting
 - May eventually have loss of consciousness (Rare)
 - Irritability
 - Poor Feeding
 - Features Suggestive of Aetiology
 - Non-Blanching Maculopapular Rash → **Suggests Meningococcus**
 - CSF Rhinorrhoea/Otorrhoea - basal skull fracture → **Suggests Pneumococcus, Hib, Strep**
- **Diagnosis:**
 - ****Clinical Suspicion:** (Meningism +/- Rash +/- Fever/Malaise/Vomiting +/- Headache/ALOC)
 - +/- (Brudzinski's Sign +, Kernig's Sign +)
 - **Blood Cultures BEFORE IV Antibiotics!!**
 - **L3-L5 Lumbar Puncture → CSF Examination:**
 - LP can → Coning if ↑ICP ∴ DO NOT do LP if:
 - 1: Papilloedema
 - 2: Cushing's Response (Triad – ↑BP, ↓HR, Irregular Breathing)
 - 3: Unresponsive Pupils
 - Can → "Cerebral Herniation" (Aka: Cistern Obliteration) → Often Fatal
 - **CSF Samples (Take 3):**
 - Sample 1 → Serology (or PCR)
 - Sample 2 → Biochemistry (Glucose, Protein)
 - Sample 3 → Bacteriology – Most Precious (Gram Stain + Culture)
 - CSF Interpretation:

	<u>Normal</u>	<u>Bacterial Meningitis</u>	<u>Viral/Aseptic Meningitis</u> <u>(Usually Herpes Virus)</u>
CSF Pressure	Normal	Normal-Raised	Normal-Raised
White Cell Count	Normal	Raised (Polymorphs)	Raised (Lymphocytes)
Glucose	Same as Serum	Lower than Serum (Hungry Bacteria)	Normal
Protein	Normal	Raised (produced by the organisms)	Raised (produced by the organisms)
Gram Stain	None	Presence of Bacteria	Nothing ("Aseptic Meningitis")

- **Treatment:**

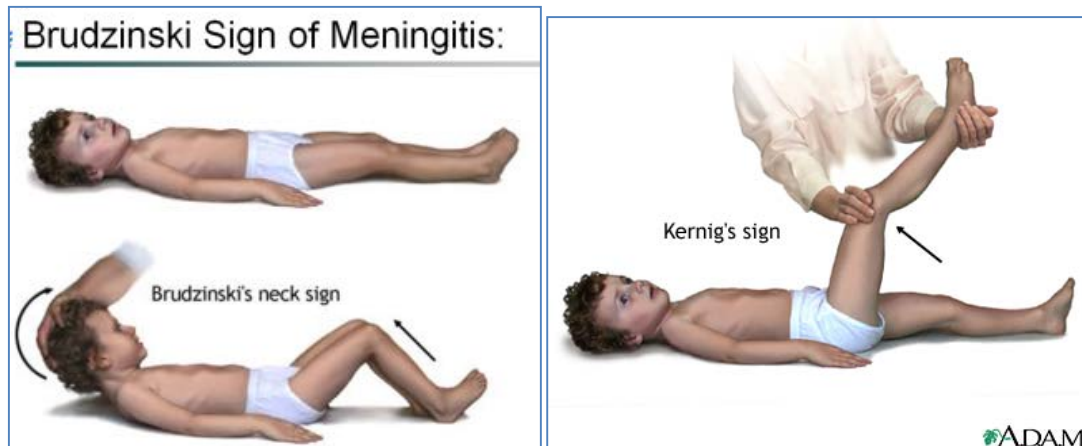
- (Bacterial Meningitis = Emergency – Can be Fatal)
- (Viral Meningitis = Usually Self-Limiting & Less Fulminant Clinically)
- *****Treat on Suspicion!!** – (Don't wait for lab results!)
- **1: Blood Cultures BEFORE IV Antibiotics!!**
- **2: Early Antibiotic Therapy is Essential for Good Outcome!!!**
 - **IV Benzylpenicillin G, or IV Ceftriaxone** (why? – Because they can enter the BBB)
- **3: Corticosteroids (Dexamethasone) WITH the Antibiotics → ↓CNS Inflammation:**
 - →Improves Neurological Outcome of bacterial meningitis
- **4: Fundoscopy, Then Lumbar Puncture** – (Check for Papilloedema before doing LP)
 - CSF – MCS
- (+ Prophylactic Measures for Close Contacts):
 - Meningitis Prophylaxis: **Rifampicin, Ceftriaxone or Ciprofloxacin:**
 - Offered to Household, child care and **CLOSE CONTACTS**

- **Prognosis:**

- **Good prognosis** with Aggressive Treatment
 - ∴ **Treatment on Suspicion:** Empirical Antibiotics (or Antivirals)

- **Complications:**

- **Acute:**
 - Encephalitis
 - Cerebral infarction
 - Oedema
 - Herniation
 - **Waterhouse-Frederichson Syndrome (Acute Adrenal Infarction)**
 - (→Petechial Haemorrhages, DIC, Septic Shock)
- **Late:**
 - Abscess
 - Subdural Empyema
 - Epilepsy
 - Leptomeningeal Fibrosis & Consequent Hydrocephalus



<https://medlineplus.gov/ency/imagepages/19069.htm>



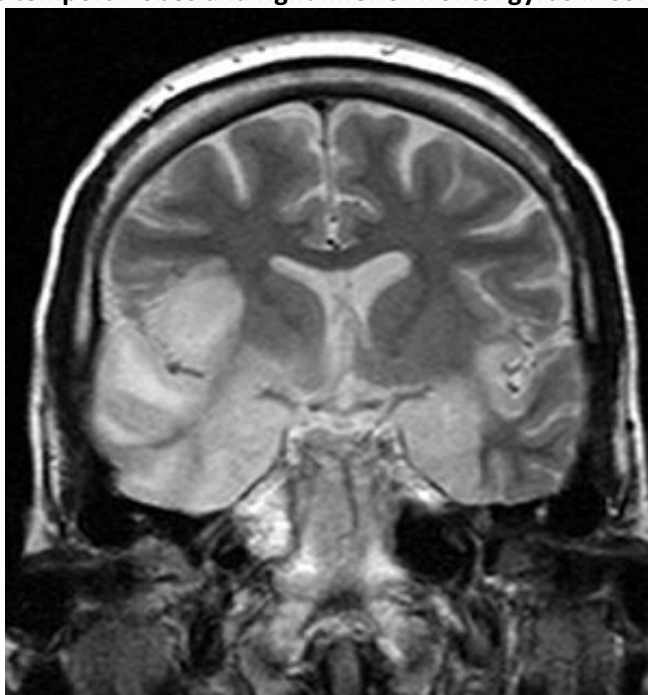
<https://www.abc.net.au/news/2021-07-21/sa-meningococcal-case/100311578>

ENCEPHALITIS:

- **Aetiology:**
 - o Almost Always Viral – (****Herpes Simplex Virus**, VZV, CMV, Poliovirus, Rabies [Rhabdovirus], JEV)
 - o Parasites such as Toxoplasma gondii and Plasmodium falciparum
 - o Fungi such as Cryptococcus neoformans
 - o Bacteria such as Treponema pallidum
- **Pathogenesis:**
 - o Viremia → Crosses BBB → CNS Infection →→ Cerebral Oedema → ↑ICP → Neurological Signs
- **Clinical Features:**
 - o **Infective Syx** – Fever, Nausea, Vomiting
 - o **+ Cerebral Syx** – **Encephalopathy** – (Altered Mental State/Abnormal Behaviour/ALOC/Drowsiness)
 - +/- Seizures
- **Treatment:**
 - o Treat on Suspicion – (**Acyclovir** + **Dexamethasone**)
- **Prognosis:**
 - o Poor - Once symptomatic, rapid inflammation & necrosis → Brain-Death or Neurological Deficit
 - o 70% Mortality Untreated
- **Investigations:**
 - o FBC – (Lymphocytosis)
 - o LP – (↑Lymphocytes, Normal Glucose, ↑Protein, Negative Cultures)

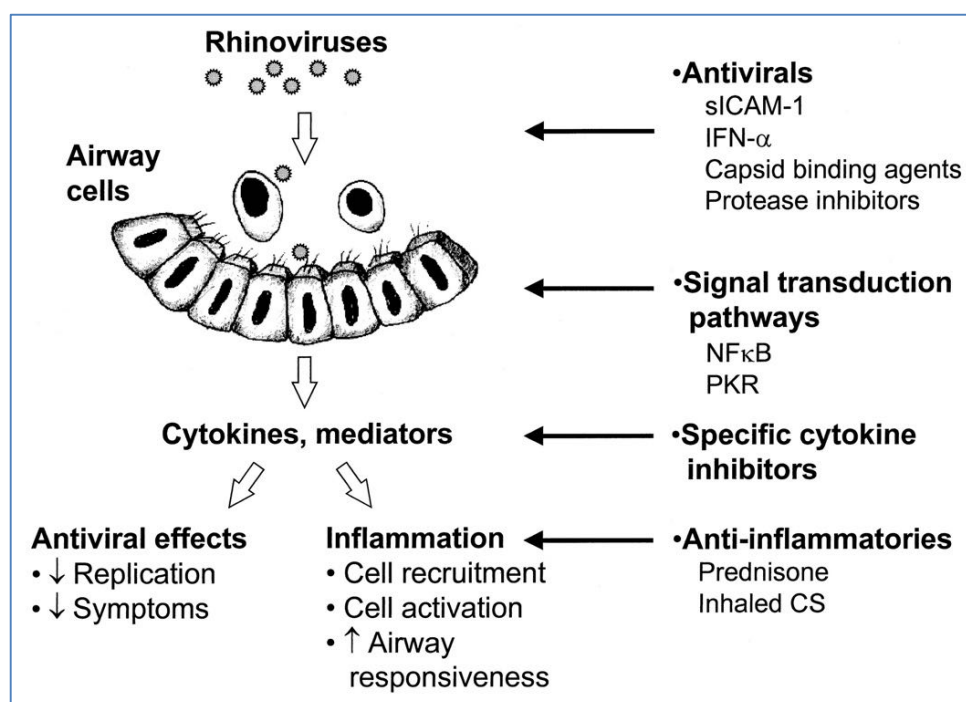
	<u>Normal</u>	<u>Bacterial Meningitis</u>	<u>Viral Meningitis (Usually Herpes Virus)</u>	<u>Encephalitis (typically viral)</u>
CSF Pressure	Normal	Normal-Raised	Normal-Raised	Markedly Raised
White Cell Count	Normal	Raised (Polymorphs)	Raised (Lymphocytes)	Raised (Lymphocytes)
Glucose	Same as Serum	Lower than Serum (Hungry Bacteria)	Normal	Normal
Protein	Normal	Raised (produced by the organisms)	Raised (produced by the organisms)	Raised
Gram Stain	None	Presence of Bacteria	Nothing ("Aseptic Meningitis")	Nothing

MRI shows high signal in the temporal lobes and right inferior frontal gyrus in someone with HSV encephalitis.



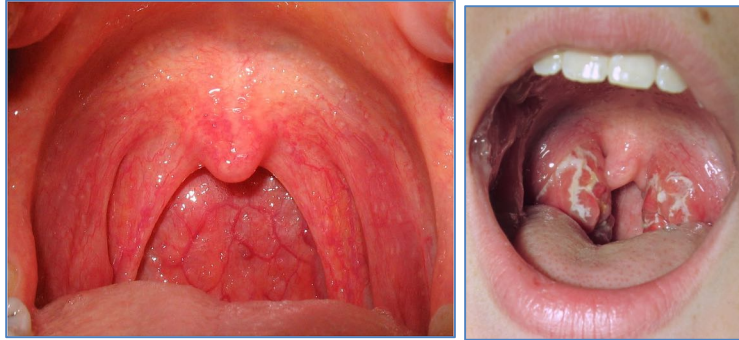
COMMON COLD (ACUTE RHINITIS)

- **Aetiology:**
 - **Rhinoviruses**, Adenoviruses, Paramyxoviruses, Influenza viruses, Myxoviruses,
- **Pathogenesis:**
 - **Transmission** – (Droplet Transmission/Contact Secretions)
 - **Viral Infection of URT Mucosa** → URT Inflammation → Mucous Hypersecretion
 - (Note: No Cross-Protection between Serotypes → Possibility of Repeated Infections)
- **Clinical Features**
 - **Short Incubation Period (2-3 days)**
 - **1wk Of Symptoms:**
 - **Local** - Nasal Congestion, Sneezing, Sore Throat, Hoarseness, Cough, Conjunctivitis
 - **General** - Malaise, Headache, Myalgias, Mild Fever
 - **Signs**
 - Rhinorrhoea
 - Inflamed Nasal/Oropharyngeal Mucosa
 - Lymphadenopathy
 - **Note: Normal Chest Exam**
 - **Complications**
 - **Secondary Bacterial Infection:** (Otitis Media, Sinusitis, Tonsillitis, Bronchitis, Pneumonia)
 - **Asthma/COPD Exacerbation**
 - **Benign Inflammatory Nasal Polyps**
- **Diagnosis:**
 - **Differentials:**
 - Allergic Rhinitis, Pharyngitis, Influenza, Laryngitis, Croup, Sinusitis, Bacterial Infections
 - **Clinical Diagnosis** – (Symptoms + Nasal Exam + Inflamed Mucosa + Watery Discharge)
 - **Laboratory Diagnosis** – ONLY if *Other Conditions are Suspected*
- **Management:**
 - **Patient Education**
 - **No Antibiotics Indicated Because Of Viral Etiology**
 - Consider 2° Bacterial Infection if NO Resolution after 3-10 Days
 - ***Symptomatic Relief:**
 - **Paracetamol**
 - **Decongestants** (*Phenylephrine/Pseudoephedrine*), Antihistamines
 - + Rest, Hydration, Gargling Warm Salt Water, Steam
 - +(↑ Dependence On Bronchodilators/Inhaled Steroids For Asthmatics & COPD)



PHARYNGITIS (SORE THROAT)

- **Definition**
 - = Inflammation of the Oropharynx (*Without* inflammation of the tonsils)
- **Aetiologies:**
 - **Viral (40-60%) – Most Common:**
 - **Adenovirus**, Coxsackie, HSV, **EBV**, **Influenza** Virus (Orthomyxovirus),
 - **Bacterial**
 - **“Strep Pyogenes” (GABH-Streptococcus)** – (*Rh-Heart Disease, PSGN & Scarlet Fever)
 - *Neisseria gonorrhoea*, *Chlamydia pneumoniae*, *Mycoplasma pneumoniae*, ***Corynebacterium diphtheria***
- **Morphology:**
 - Red, Inflamed Oropharynx
 - May have white lesions
 - May have pus



Viral: <https://commons.wikimedia.org/wiki/File:Pharyngitis.jpg>

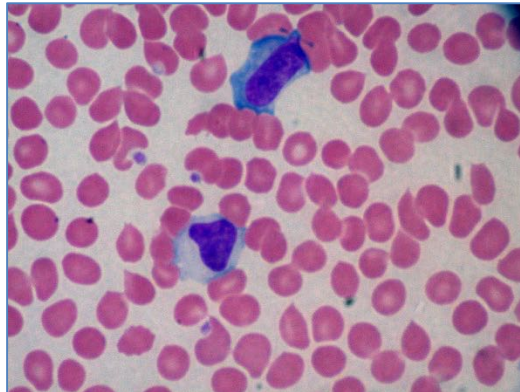
Strep: https://commons.wikimedia.org/wiki/File:Pos_strep.JPG

- **Clinical Features**
 - Typically a self-limited infection with no significant sequelae
 - **Bacterial - Group A Beta-Hemolytic Streptococcus**
 - **Absence Of Cough**, Pharyngitis, + Flu-Like Illness
 - **Signs: Fever + Tonsil Exudate + Lymphadenopathy + <15yo + NO Cough**
 - **Complications!!** - *Rheumatic Fever, Glomerulonephritis, Meningitis*
 - **Viral - Adenovirus**
 - **Cough** – (Due to Rhinorrhoea), Pharyngitis, + Flu-Like Illness
 - **Viral - Ebv (Infectious Mononucleosis)**
 - Pharyngitis, Fever, **Lymphadenopathy, Fatigue, Rash**
- **Investigations**
 - **Suspected GABH-Strep:**
 - **Throat Culture = Definitive** (But TOO SLOW in the real world!!)
 - **RDT For Streptococcal Antigen**
 - **ASOT (Anti-Streptolysin-O-Titres)** – (But only shows recent infection)
 - **Suspected EBV (Infectious Mononucleosis):**
 - **Peripheral Blood Smear** – (Reactive Lymphocytes)
 - **“Monospot” Test** (IE: The Latex Agglutination Assay, Or "Monospot")
 - **EBV Serology**
- **Management**
 - **If ?GABH-Strep:**
 - ****Throat Swab if: Fever + Tonsil Exudate + Lymphadenopathy + <15yo**
 - **Antibiotics!!:** **Penicillin-V/G or Erythromycin if Penicillin Allergic**
 - **If ?Viral Pharyngitis:**
 - **Antibiotics NOT indicated**
 - **Paracetamol/NSAIDs**
 - **Decongestants (Phenylephrine)**
 - **If ? Infectious Mononucleosis (EBV):**
 - **Antibiotics NOT indicated; Note: Penicillin will → Rash (Pathognomonic)**
 - **Self-Limiting Course;** Rest During Acute Phase Is Beneficial
 - **Supportive Treatment:** NSAIDs for fever, sore throat, malaise

Other Notable Pharyngitis's:

- (EPSTEIN BARR VIRUS) – INFECTIOUS MONONUCLEOSIS (GLANDULAR FEVER):

- Aetiology:
 - Epstein Barr Virus
- Pathogenesis:
 - Transmitted through Saliva (Ie: Kissing Disease)
 - Incubation period <8wks
 - **Preferentially Infects B-Cells → Reactive B-Lymphocytes → “Mononucleosis”**
- Morphology:
 - Tender Cervical Lymphadenopathy
 - Blood Smear – Lymphocytosis with *Atypical Lymphocytes*



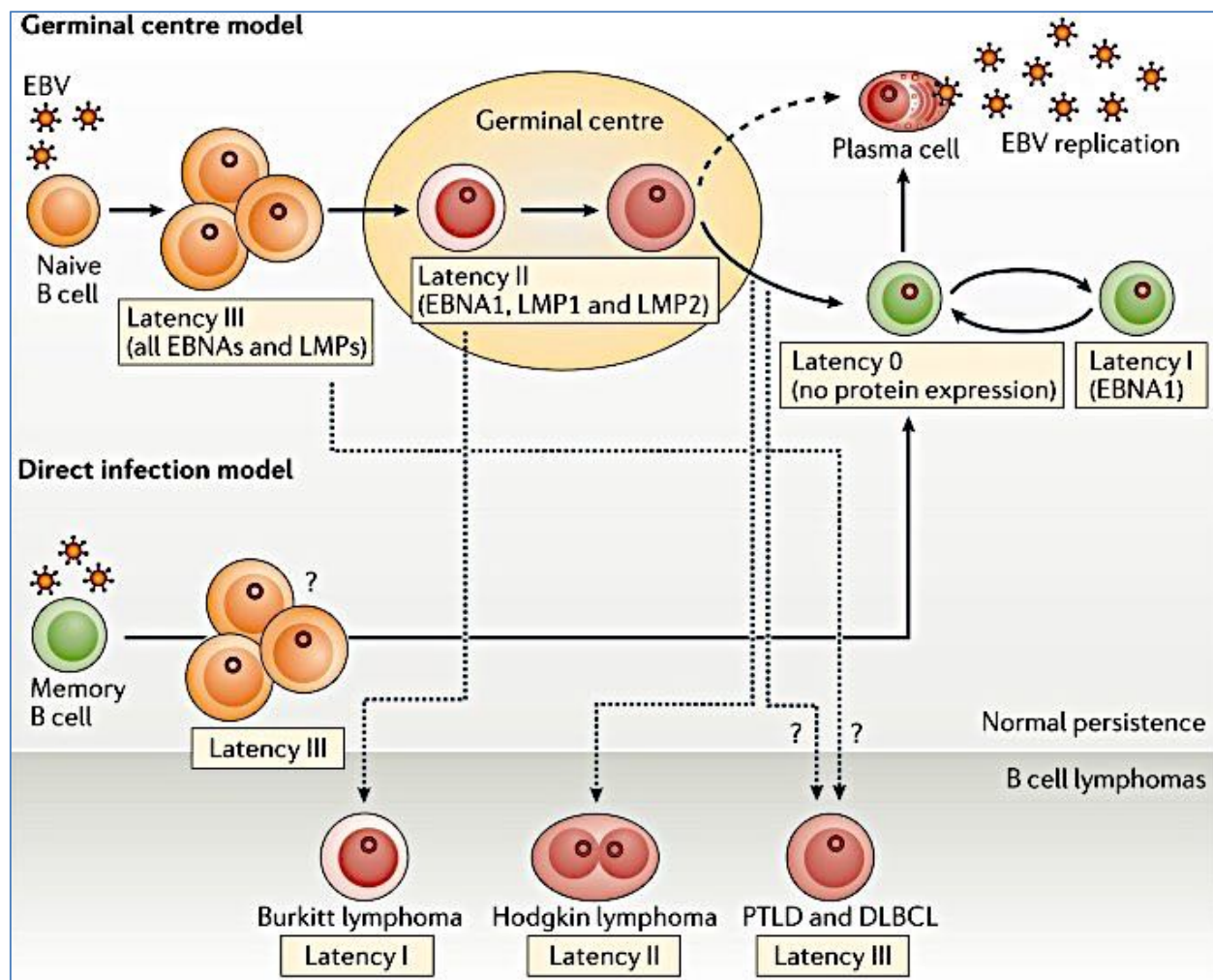
Creative commons: <https://www.flickr.com/photos/euthman/145052721>

- Clinical Features:
 - **Signs/Symptoms:**
 - **Fever +**
 - **Glandular Fever Triad:**
 - **Fatigue/Malaise** (Anorexia/Lethargy)
 - **Pharyngitis** (Sore Throat)
 - **Lymphadenopathy** (Especially Cervical)
 - **Others –** (Splenomegaly, Hepatitis, Haemolysis, Jaundice)



James Heilman, MD, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

- **Diagnosis:**
 - Typically Clinical
 - Peripheral Blood Smear – (Reactive Lymphocytes)
 - “Monospot” Test (IE: The Latex Agglutination Assay, Or "Monospot")
 - EBV Serology
 - + LFTs
- **Treatment:**
 - Antibiotics NOT indicated; Note: Penicillin will → Rash (*Pathognomonic*)
 - Self-Limiting Course; Rest During Acute Phase Is Beneficial
 - Supportive Treatment: NSAIDS for fever, sore throat, malaise
- **Complications:**
 - EBV is an Oncogenic Herpesvirus → Tumours:
 - → **Burkett's Lymphoma**
 - → **Hodgkin's Lymphoma**
 - → **Nasopharyngeal Carcinoma**



Creative Commons: https://media.springernature.com/lw685/springer-static/image/art%3A10.1038%2Fncr.2016.92/MediaObjects/41568_2016_Article_BFncr201692_Fig3_HTML.jpg

- **DIPHTHERIA:**

- **Aetiology:**
 - Gram Positive Bacterium – *Corynebacterium Diphtheriae*
- **Pathogenesis:**
 - **Transmission** – Aerosol, Physical Contact
- **Morphology:**
 - Adherent Whitish Pseudomembrane Over Pharynx & Tonsils (May → Obstruction)
- **Clinical Features:**
 - High Fever, Sore Throat, Fatigue, Nausea & Vomiting
 - **Pseudomembrane on Tonsils & Pharynx** - May have Airway Obstruction & Dysphagia
- **Complications:**
 - **Systemic Exotoxin →**
 - **Myocarditis** (Potentially fatal toxigenic Cardiomyopathy → Heart Failure)
 - Peripheral Neuritis
 - Chronic Non-Healing Ulcers
- **Diagnosis:**
 - **Swab M/C/S**
 - **+ Toxin Detection**
- **Treatment:**
 - **Penicillin or Erythromycin (if Penicillin Allergic)**



User:Dileepunnikri, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

- **SCARLET FEVER ("STRAWBERRY TONGUE"):**

- **Aetiology:**
 - Certain strains of ***GABH-Strep "Pyogenes"*** (Which are infected with a "Bacteriophage" [Virus] → Produce an Eruthrogenic toxin)
- **Pathogenesis:**
 - **GABH-Strep Infection → Exotoxin → Local effect on Tonsils/Pharynx/Skin**
 - →Tongue
 - Initially covered with white exudate
 - Exudate is shed
 - inflammation of underlying tissue
 - → Skin
 - Diffuse, Erythematous Rash
- **Complications:**
 - Rheumatic Heart Disease
 - PSGN
- **Diagnosis:**
 - ****Throat Swab if: Fever + Tonsil Exudate + Lymphadenopathy + <15yo**
- **Treatment:**
 - **Antibiotics!!:** **Penicillin-V/G or Erythromycin if Penicillin Allergic**



1 - Estreya at English Wikipedia., CC BY 2.5 <<https://creativecommons.org/licenses/by/2.5>>, via Wikimedia Commons

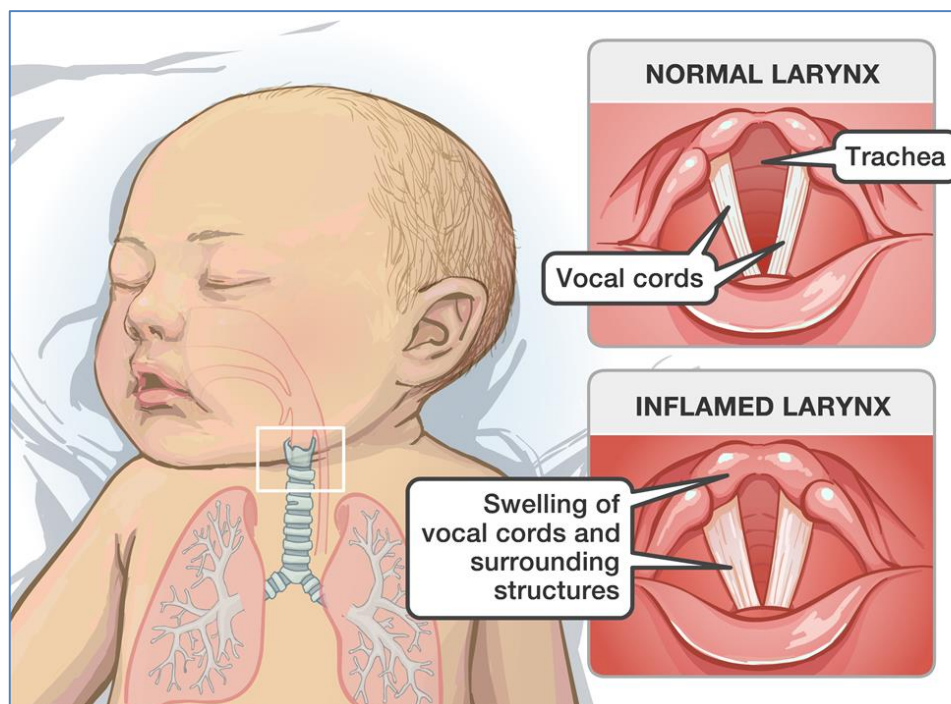
2 - www.badobadop.co.uk, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0>>, via Wikimedia Commons



Afag Azizova, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0>>, via Wikimedia Commons

ACUTE LARYNGOTRACHEOBRONCHITIS (CROUP)

- **What is it?**
 - Inflammation Of Tissues In Subglottic Space ± Tracheobronchial Tree
 - + Thick, Viscous, Mucopurulent Exudates Which Compromises Upper Airway → Barking Cough
- **Etiology – Viral:**
 - ***RSV** or **Parainfluenzae** (Most Common), II, III, Influenza A And B
- **Pathogenesis:**
 - URTI
 - → Inflammation Of Tissues In Subglottic Space
 - → Thick, Viscous, Mucopurulent Exudates Which Compromises Upper Airway → **Barking Cough**
- **Morphology:**
 - Inflamed Upper Airways + Larynx
- **Clinical Features**
 - Typically Children <5yrs
 - **Signs of Croup - the 3 S's**
 - **1: Stridor**
 - **2: Subglottic swelling**
 - **3: Seal bark cough**
 - +/- Cyanosis & Respiratory Distress
- **Treatment**
 - (Note: Viral ∴ NO Antibiotics)
 - **Oral/IM Corticosteroids** (**Dexamethasone** / **Prednisone**)
 - **Nebulised Epinephrine**
 - **Humidified O₂**
 - +/- Intubation If Severe



<https://www.aboutkidshealth.ca/croup>

ACUTE EPIGLOTTITIS

- **Etiology**
 - **HiB – (*Haemophilus Influenzae* type B)** (Uncommon due to HiB vaccine)
 - (Gram neg coccobacillus)
- **Clinical Features**
 - Typically Children 1-4yo
 - High Fever & Unwell
 - Sore Throat, Dysphagia, Anorexia
 - **Obstructive Symptoms – MEDICAL EMERGENCY → INTUBATE:**
 - **Difficulty Swallowing, DROOLING, cyanotic/pale, inspiratory stridor, slow breathing,**
- **Investigations:**
 - **Preparations For Intubation Or Tracheotomy** Must Be Made Prior To Any Manipulation
 - **Lateral Neck XR** - Cherry-Shaped Epiglottic Swelling ("Thumb Sign") - Only If Stable
 - **WBC (Elevated)**
 - **Blood And Pharyngeal Cultures After Intubation**
- **Treatment**
 - ***Admit to ICU**
 - **Urgent Intubation → Secure Airway**
 - + Humidified O2
 - **Antibiotics – (*Ceftriaxone* + *Clindamycin*)**
 - **Extubate When Afebrile**
 - **Watch For Meningitis**



https://commons.wikimedia.org/wiki/File:Epiglottitis_endoscopy.jpg

PERTUSSIS - WHOOPING COUGH:

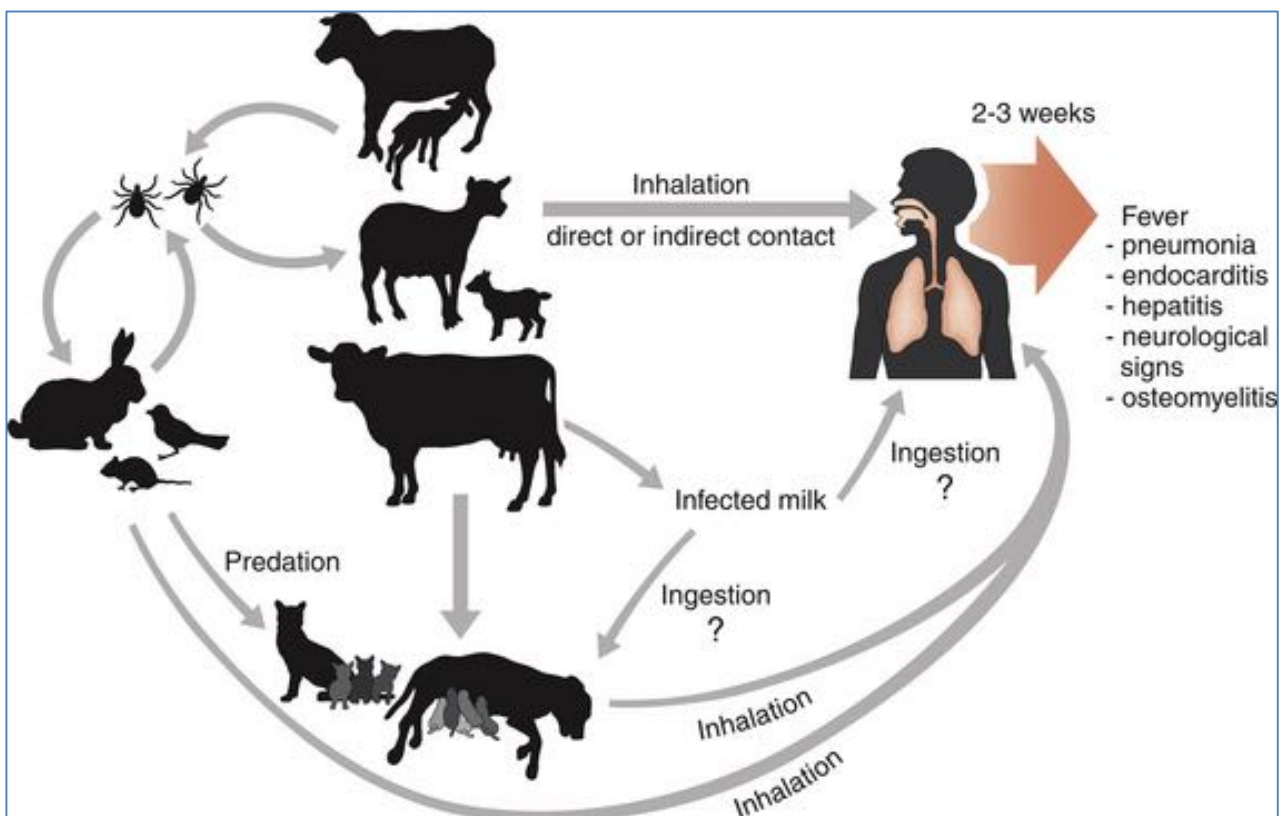
- **Aetiology:**
 - *Bordetella pertussis* – (Only a human pathogen)
- **Pathogenesis:**
 - Infection of Trachea & Bronchi → **Toxins** → Widespread Trachea/Bronchi Inflammation
 - Pertussis toxin → increased secretions → Cough
 - Dermonecrotic toxin → vasoconstriction, Ischaemia
 - Tracheal cytotoxin → inhibition of cilia movement
- **Clinical Features:**
 - Severe childhood disease
 - → Dyspnoea
 - → Chronic, Severe Coughing Fits
 - Highly Contagious (infants <12mths)
- **Investigations:**
 - Diagnosed on Clinical Suspicion
 - (Culture takes <2wks – TOO Long!)
- **Management:**
 - Empirical Antibiotics – (**Azithromycin** / **Clarithromycin** / **Erythromycin**)
 - + Booster Vaccination (Unvaccinated / Adolescents / Adults)
 - + Vaccinate close contacts (DTP Vaccine)
 - +/- Post-Exposure Prophylaxis in Close Contacts (**Azithromycin**)

Pertussis vaccine immunity wanes over time	Young infants are most at risk	How to prevent infection
Pertussis, or whooping cough, is a respiratory tract infection that can be prevented with vaccination. PROTECTION  However, protection takes several shots to establish and then protection wanes over time.	Infants less than three months of age have only received one dose of the vaccine.  Infants get the most sick A recent study found most infants less than three months with pertussis had severe disease and of those¹: • 92% required hospitalization • 28% intensive care admission	Make sure you and your family are up to date with your vaccines²: 1. ROUTINE CHILDHOOD IMMUNIZATION SCHEDULE - 2, 4, 6, 12-23 months and 4-6 years 2. ADOLESCENT BOOSTER - at 14-16 years 3. ADULT BOOSTER 4. WITH EVERY PREGNANCY - Protective antibodies are transferred to the baby - Best between 27-32 weeks 

<https://www.canada.ca/en/public-health/services/reports-publications/canada-communicable-disease-report-ccdr/monthly-issue/2018-44/issue-9-september-6-2018/article-1a-pertussis-whooping-cough-still-danger-infants-infographic.html>

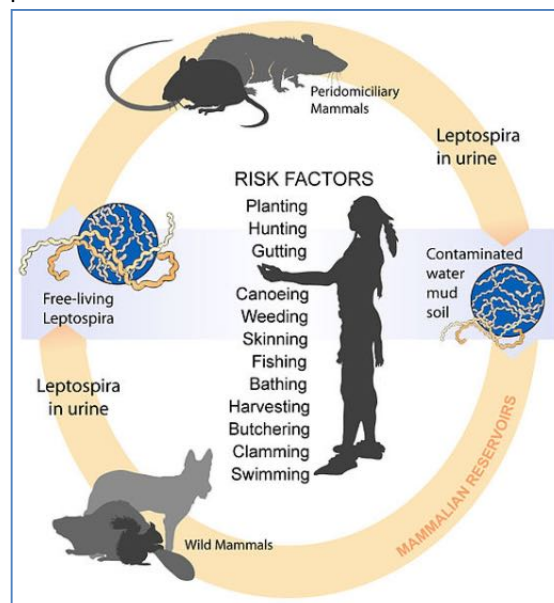
Q-FEVER

- **Aetiology:**
 - o *Coxiella Burnetii* - (found in cattle, sheep, goats and other domestic mammals, (cats and dogs))
- **Transmission:**
 - o Inhalation of Endospores / Contact with Unpasteurised Milk, Urine, Faeces of infected animals
- **Pathogenesis:**
 - o **2-3wk Incubation**
 - o **Two-Stage Disease:**
 - **Acute Stage** (Headaches, chills, and respiratory symptoms)
 - **Chronic Stage** (Asymptomatic, Insidious)
- **Clinical Features:**
 - o **Acute Symptoms:**
 - **Flu-Like Symptoms:** Abrupt Onset Fever, Chills, Sweats & Malaise
 - **Respiratory** - Dry Cough, Pleuritic Pain
 - **GI Symptoms** - Nausea, Vomiting And Diarrhoea
 - **Neuro:** +/- Severe Headache & Confusion
 - **MSK:** +/- Myalgia & Arthralgia
- **Diagnosis:**
 - o Serology
 - o PCR
 - o TOEcho – (If Suspected Endocarditis)
 - o LFT – (↑ALT & AST)
- **Treatment:**
 - o Antibiotics – **Doxycycline**
- **Complications:**
 - o Progression to **Atypical Pneumonia** → life threatening **ARDS**
 - o Rarely **Granulomatous Hepatitis** which can → hepatomegaly and RUQ pain
 - o Chronic form of Q fever → **Endocarditis**
- **Prevention:**
 - o **Q-Vax** (Whole-cell, killed vaccine via intradermal injection)
 - o (Note: Skin and blood tests should be done first to identify pre-existing immunity; vaccinating subjects who already have an immunity can result in a severe local reaction)



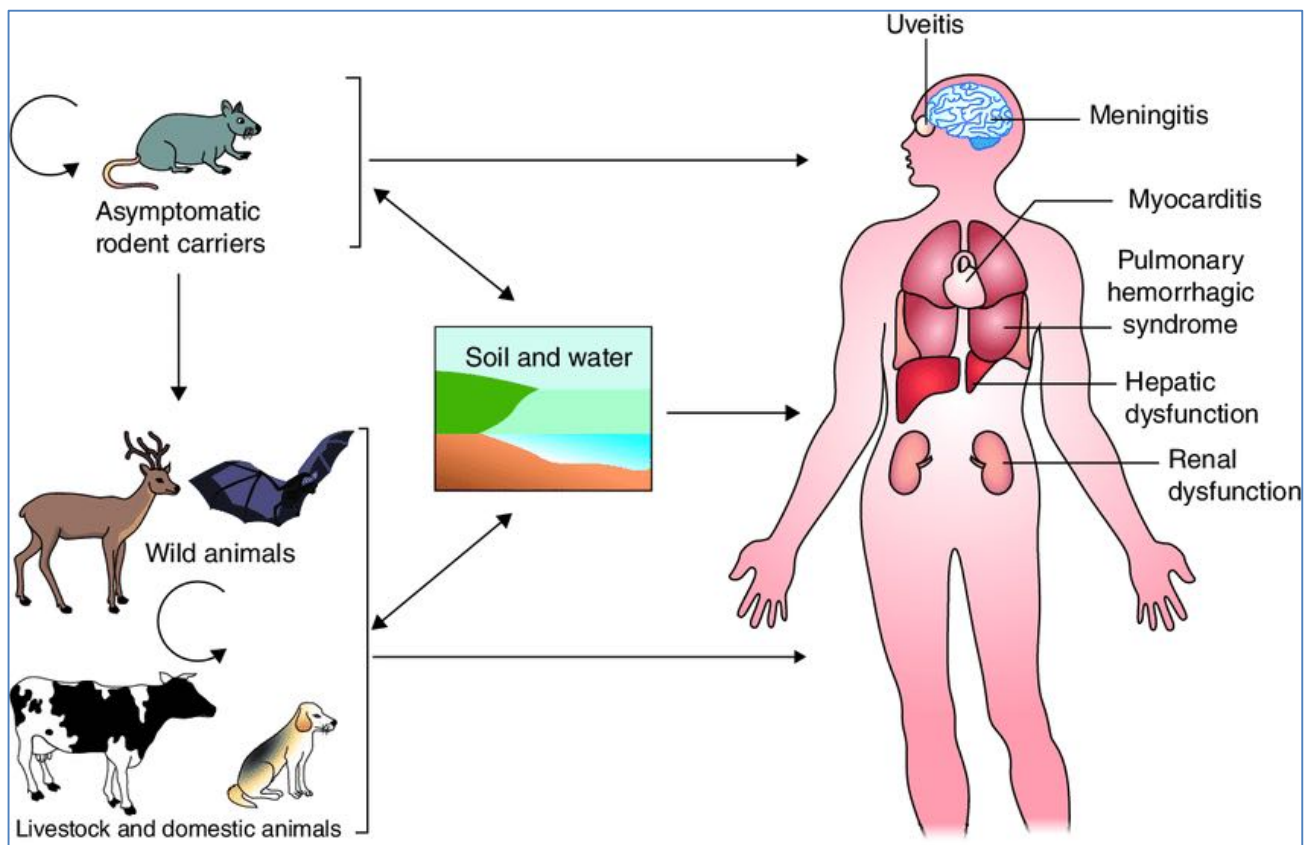
LEPTOSPIROSIS (“Weil's syndrome”, “Cane field Fever”, “7-Day Fever”, “Rat Catcher’s Fever”):

- **Aetiology:**
 - Spirochete Bacteria – *Leptospira* spp
- **Pathogenesis:**
 - **Zoonotic Disease** – (Mammals, Rats, Birds, Reptiles) – Transmission through water, food, soil containing urine of infected animals
 - NO person to person transmission
- **Morphology:**
 - **Micro:**
 - Spirochete Bacteria
- **Clinical Features:**
 - **Symptoms – Note: **Biphasic Presentation**:**
 - **1st Phase (First 7-10 Days):**
 - Flu like symptoms – Fevers, Chills, Myalgias, Headache & Leptospiral rash. Note: Resolves after 1 wk
 - ***Brief Asymptomatic Period...then**
 - **2nd Phase (After 10 Days):**
 - Meningitis (*Photophobia), Liver Damage (*jaundice), Renal Failure, Red Eyes (Uveitis)
 - **Signs:**
 - Fever (PUO)
 - Palmar Erythema
 - Leptospiral rash
 - Jaundice
 - Hepatomegaly/Splenomegaly
 - Costovertebral Angle Tenderness (Nephritis)
 - **Complications:**
 - Myocarditis, Pericarditis
 - Meningitis
 - Liver Failure
 - Renal Failure
 - Respiratory Distress
- **Diagnosis:**
 - Blood Cultures – if in 1st phase
 - **Urine Cultures or Serology – if in 2nd phase
 - (DDX's: Dengue, Hepatitis, Meningitis, Malaria, Typhoid)
- **Treatment:**
 - Penicillin/Cephalosporins



New Hypothesis for Cause of Epidemic among Native Americans, New England, 1616–1619 - Available from:

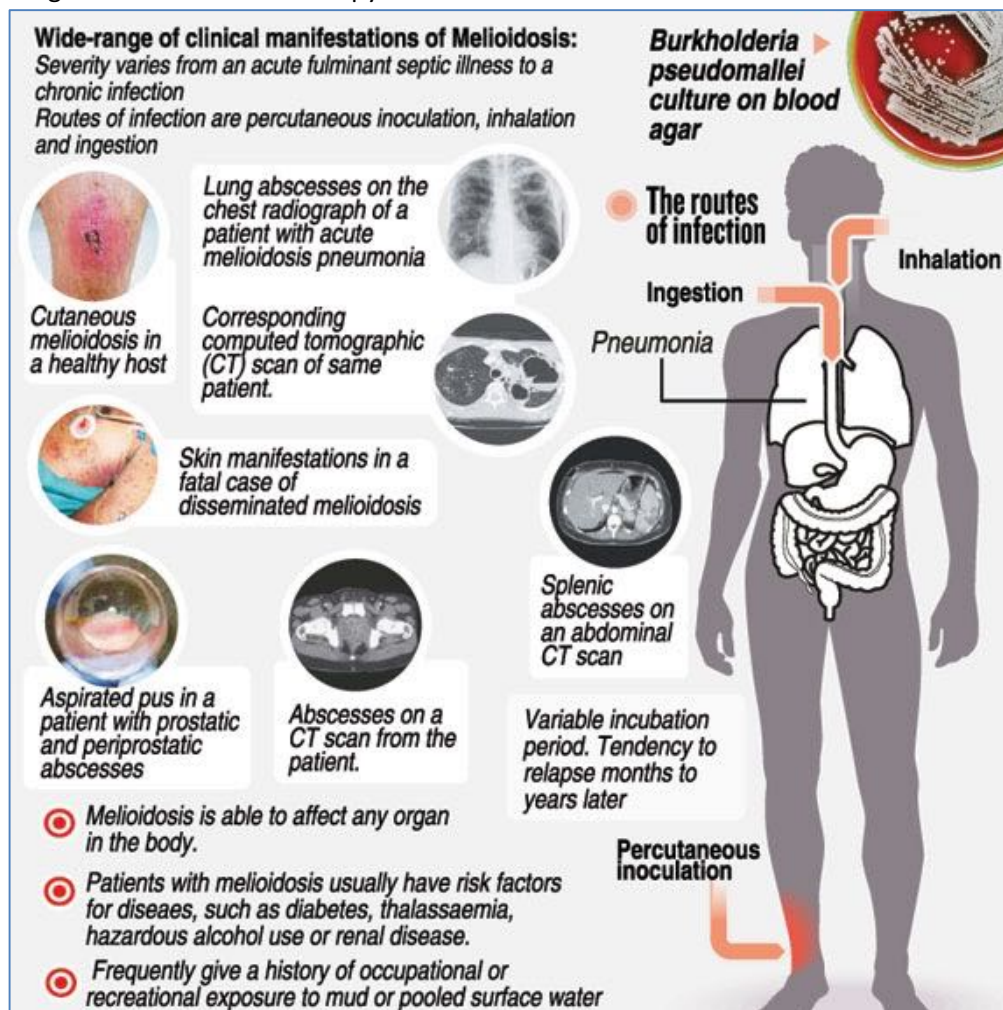
https://www.researchgate.net/figure/Leptospiral-life-cycle_fig3_41188404



Epidemiology: From Recognition to Results - Available from: https://www.researchgate.net/figure/Leptospirosis-reservoirs-and-transmission-to-humans-Source-Ko-et-al-2009-Reproduced_fig3_323678367

MELIOIDOSIS

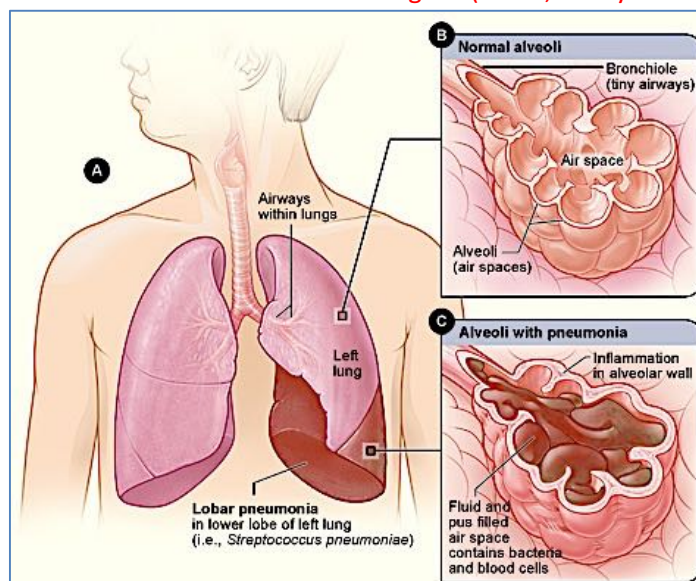
- **Aetiology:**
 - Burkholderia Pseudomallei (Intracellular Gram Negative Bacteria)
 - Lives in soil & fresh surface water (Seasonal in wet seasons)
- **Pathogenesis:**
 - **Transmission** – Percutaneous Inoculation from Wet Soils/Surface Water. Or inhalation
 - Risk factors – Immunosuppression, chronic lung disease
 - **Immune Mechanisms** - Pseudomallei lives *Intracellularly*:
 - Cell-Mediated Immunity = Most Important
 - Humoral Immunity is *Ineffective*
- **Morphology:**
 - **Macro:**
 - Cavitary Lesions in Upper Lung Lobes
 - Skin Abscesses
 - **Micro:**
 - Fluorescence stain – Rod-shaped, gram negative, bacilli
- **Clinical Features:**
 - **Typical Presentation - Pneumonia:**
 - Pneumonia + (Cavitary Lesions in the upper lung lobes (SIMILAR TO TB))
 - ∴ Cough, Sputum, Respiratory Distress
 - + PUO (Fever), Chills, Rigors
 - + Skin Ulcers/Abscesses
 - (May → Sepsis → Death)
- **Diagnosis:**
 - Cultures
- **Treatment:**
 - Note: Organism is resistant to *Broad Spectrum Antibiotics*
 - Long-Course Antibiotic Therapy



Source: New England Journal of Medicine

PNEUMONIAS ("Infections of the Lung"):

- **Aetiology:**
 - **Community Acquired:**
 - Usually Gram-Positive – (**Strep pneumonia** [90%])
 - Occasionally Gram-Negative – (**H-Influenzae**)
 - **Hospital Acquired (Nosocomial - >48hrs POST Admission):**
 - Usually Gram-Negative – (**Pseudomonas aeruginosa**, **E-coli**, Klebsiella)
 - **Atypical/Interstitial Pneumonia ("Walking Pneumonia"):**
 - Intracellular Bacteria – (**Mycoplasma**, Chlamydia, Legionella, Coxiella Burnetii)
 - **In Immunocompromised:**
 - **Cytomegalovirus**
 - **Pneumocystis jirovecii**
 - Fungal (Candida/Aspergillus)
- **Clinical Features:**
 - **General Pneumonia Triad (WHO):**
 - **Fever**
 - **Tachycardia**
 - **Tachypnoea (+/- Breathlessness)**
- **Types of Pneumonias - Based on Morphology:**
 - **LOBAR-PNEUMONIA (Well Defined; One Lobe):**
 - **Aetiology:**
 - Typically **Strep Pneumoniae** (Gram Positive Diplococci)
 - (Or *Klebsiella* in Aged)
 - **Pathogenesis:**
 - Whole Lobe Involvement
 - Exudate *Within Alveolar Spaces* → Alveolar Consolidation
 - **Morphology:**
 - Follows Anatomical Boundaries (Physically & on CXR)
 - Entire Lobe Consolidation/Opacity on CXR
 - **Clinical Features:**
 - **Symptoms:**
 - Abrupt onset High Fever + Chills
 - Productive Cough (Occasionally Rusty Sputum &/or Haemoptysis)
 - Pleuritic Chest pain + Pleural Rub
 - **Signs:**
 - Usually Unilateral
 - Exudation – Entire Lobe Consolidation
 - **Cardinal Pneumonia Signs –(Fever, Tachycardia, Tachypnoea)**



○ **BRONCHO-PNEUMONIA (Patchy; Multiple Lobes):**

▪ **Aetiology:**

- Secondary to Debilitating Diseases, Extremes of Age, or Post-Surgery:

- Gram Pos - **Strep Pneumoniae, Staph Aureus**
- Or Gram Neg – **H-Influenzae**

▪ **Pathogenesis:**

- Patchy Areas of Acute Suppurative Inflammation → Patchy Consolidation
- Basal Lower Lobes Common (Due to gravity – bacteria settle in the lower lungs)

▪ **Morphology:**

- Doesn't follow anatomical boundaries – Often *Multi-Lobar & Bilateral*
- Usually Bilateral Patchy Consolidation → Scattered Opacities on CXR

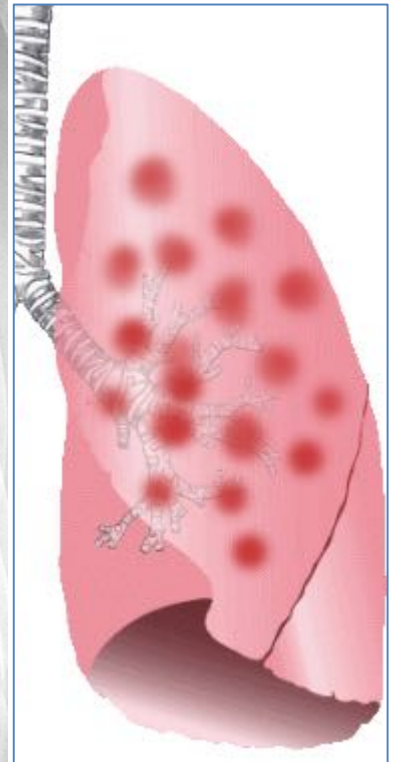
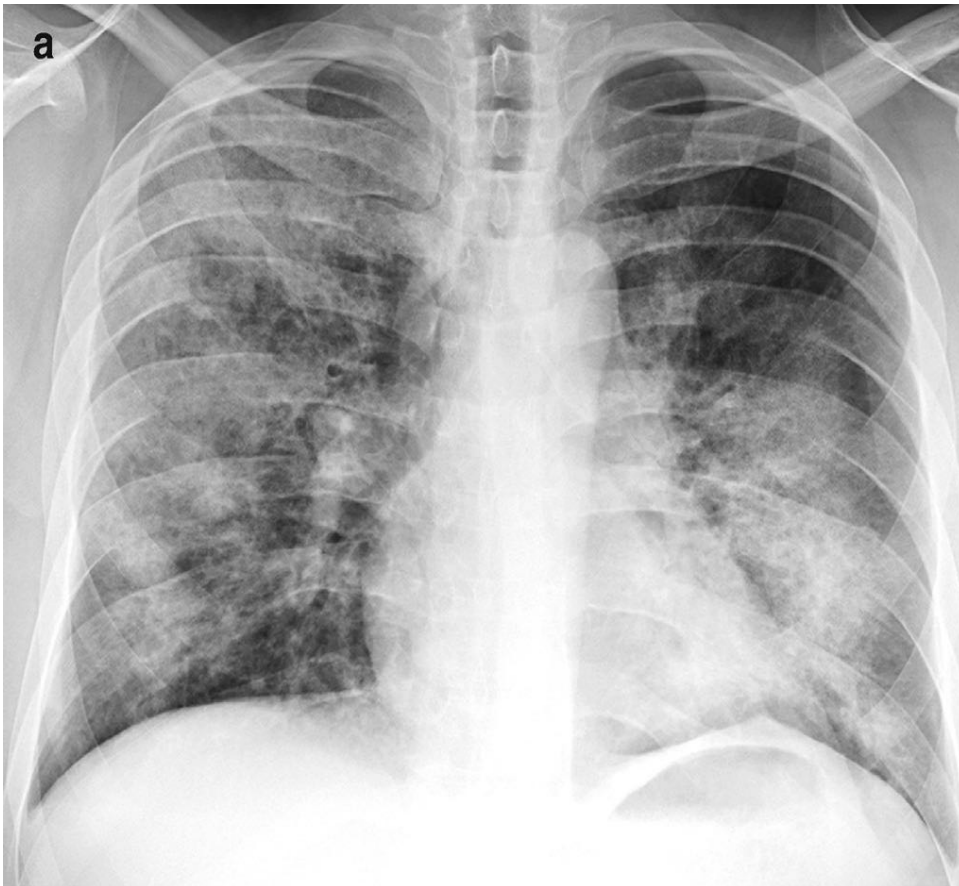
▪ **Clinical Features:**

• **Symptoms:**

- Abrupt onset High Fever + Chills
- Productive Cough (Occasionally Rusty Sputum &/or Haemoptysis)
- Pleuritic Chest pain + Pleural Rub

• **Signs:**

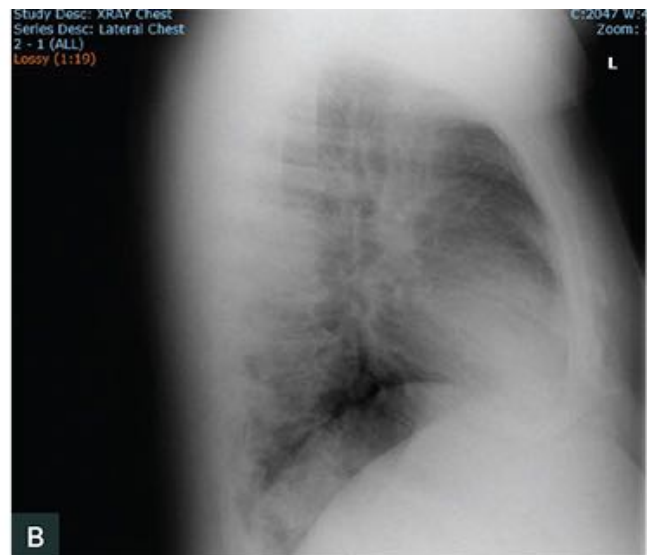
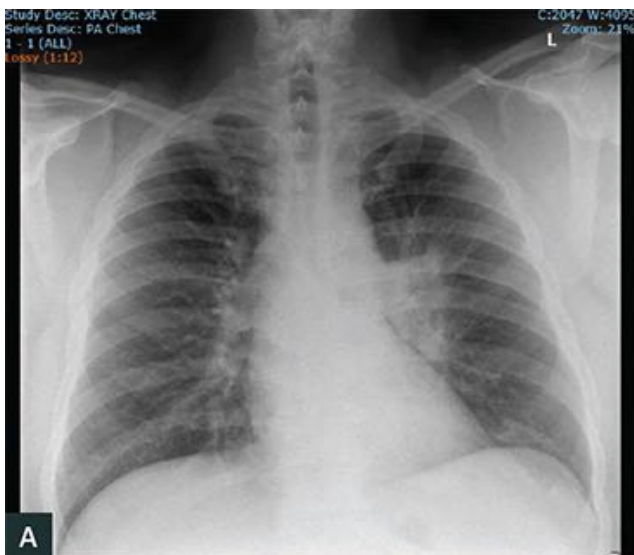
- Usually Bilateral
- Patchy Consolidation – Usually Bilateral
- **Cardinal Pneumonia Signs – (Fever, Tachycardia, Tachypnoea)**



1. Franquet T., Chung J.H., CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons
2. Suraj at Malayalam Wikipedia, Public domain, via Wikimedia Commons

○ **ATYPICAL, INTERSTITIAL PNEUMONIA ("Walking Pneumonia"):**

- **Aetiology:**
 - **Typically Intracellular Bacteria:**
 - **Mycoplasma**, Chlamydia pneumonia, Legionella, Q-Fever (Coxiella burnetii)
 - **Or Viral:**
 - **Influenza A/B, RSV – Respiratory Syncytial Virus**, Corona Virus (SARS)
- **Pathogenesis:**
 - Interstitial Inflammation (NOT within the Alveolar Spaces)
 - Note: 2° Bacterial Pneumonia (Typically Strep/Staph) may follow
- **Morphology:**
 - Inflammation localised to Alveolar Wall/Septa (Interstitium); NO Alveolar Exudate
 - Typically Bilateral
- **Clinical Features:**
 - **Symptoms:**
 - Initial URTI → SLOW Onset (Days-Weeks)
 - Symptoms more *General* & 'Flu-like'
 - Few Localizing Symptoms:
 - Often NO Cough
 - Wheezing (Not seen in other pneumonias)
 - **Signs:**
 - No *Physical Signs* of Consolidation
 - *Unresponsive* to Common Antibiotics



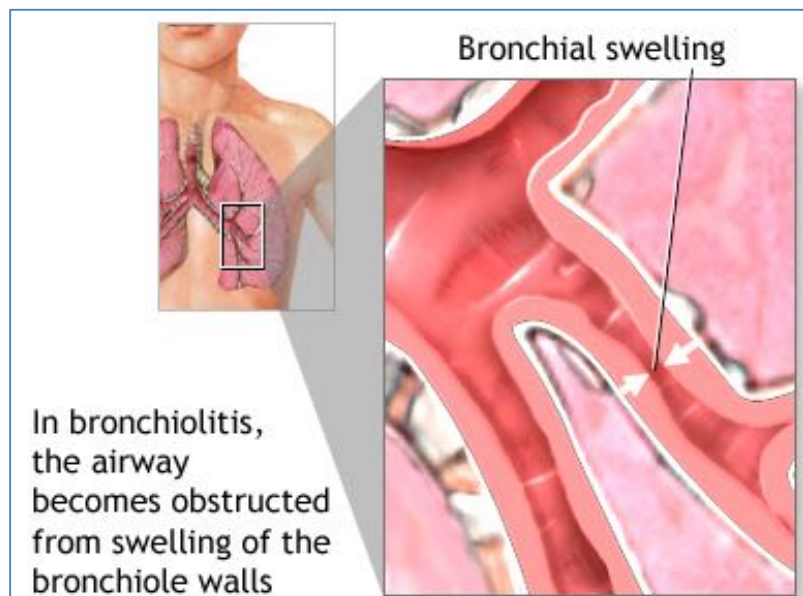
<https://www1.racgp.org.au/ajgp/2018/march/an-atypical-case-of-typical-pneumonia>



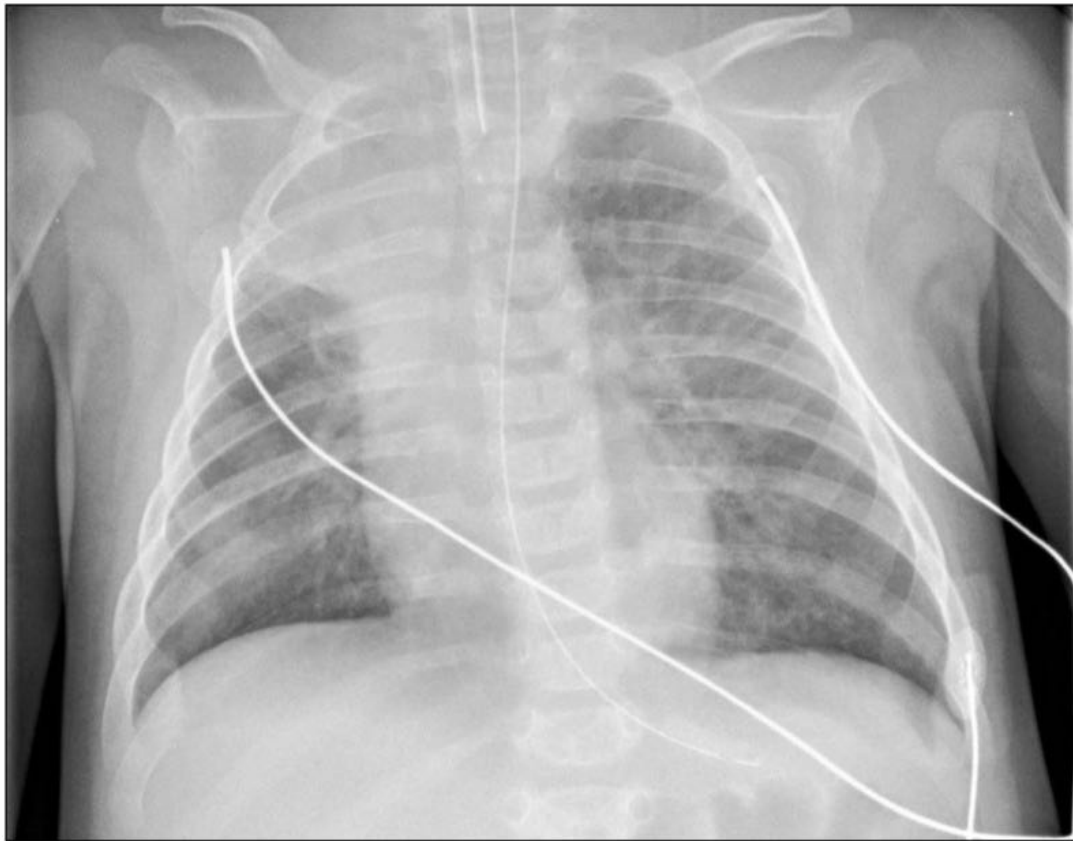
- **Investigations For Pneumonia:**
 - CXR – (Consolidation Lobar/Broncho/Interstitial)
 - Sputum MCS – (Sputum / NPA – Nasopharyngeal Aspirate / BAL – Bronchio-Alveolar Lavage)
 - Blood Culture if ?Septic
 - Serological Testing – (If ?Atypical Pneumonias)
- **Management:**
 - ?Admit to ICU? – **CURB-65** – (Score >3 → ICU):
 - Confusion
 - Uraemia
 - Resp Rate >30
 - BP <90/60
 - >65yo
 - Antibiotics:
 - Empirical:
 - ?G-Pos: Amoxicillin / Benz-Penicillin-V / Doxycycline / Clarithromycin
 - ?G-Neg: Gentamicin / Ceftriaxone
 - Severe: + Meropenem / Imipenem
 - But Ultimately Dictated by MCS
 - Fluids
 - O2 if Sats <92%
 - +/- Ventilation
- **Possible Complications of Pneumonia:**
 - ARDS – Acute Respiratory Distress Syndrome:
 - Severely Impaired Gas Exchange → Hypoxia & Confusion
 - Rx: Mechanical Ventilation and ICU
 - Lung Abscesses
 - Pleuritis/Pleural Effusion/Empyema
 - Inflammation of the pleura (Strep Pneumoniae)
 - Blood Rich Exudate/Pus in Pleural Space
 - Rx: Drainage + MCS → IV Antibiotics
 - Septicaemia, Meningitis
 - Fibrosis, Scarring, Adhesions
 - Rarely Adenocarcinoma

BRONCHIOLITIS:

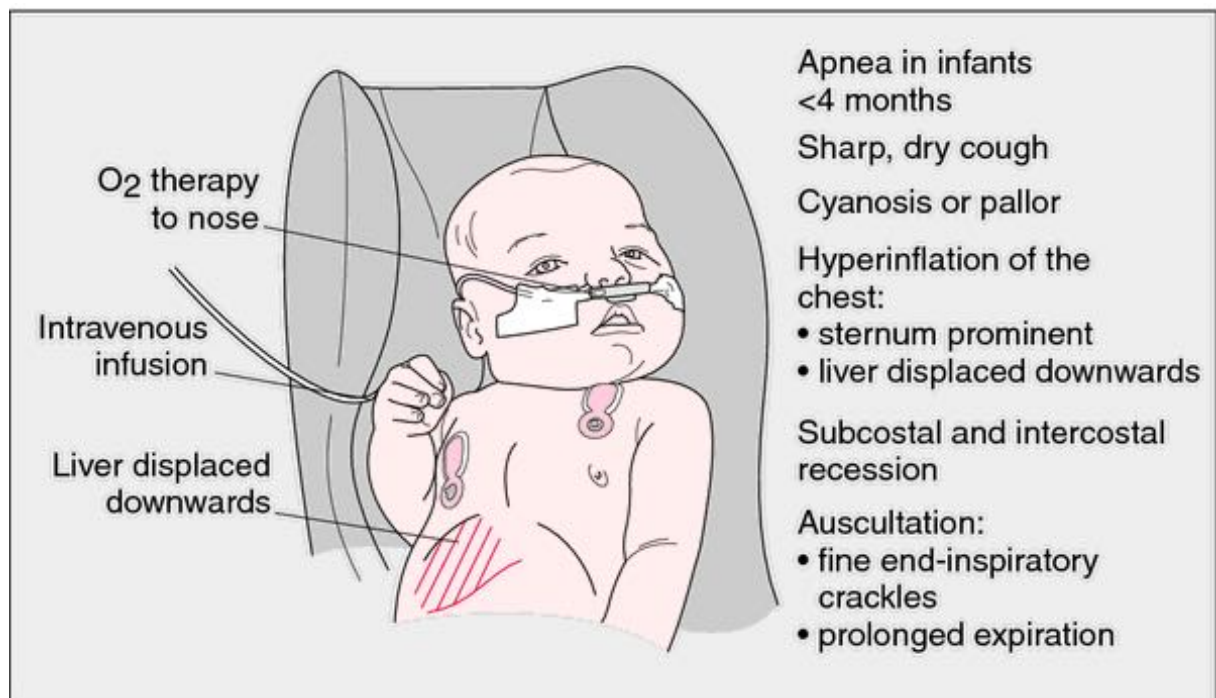
- **Aetiology:**
 - **Respiratory Syncytial Virus (RSV)** (>50%)
 - parainfluenza, influenza, rhinovirus, adenovirus, rarely *M-pneumoniae*
- **Clinical Presentation**
 - Common, **affects 50% of children in first 2 years of life**
 - **Initial URTI with cough and fever → Respiratory Distress**
 - **Wheezing, Tachypnea, Tachycardia**
 - **Intercostal Recessions, Tracheal Tug, Supraclavicular Recessions, Rib Flaring**
 - **+ Feeding difficulties, irritability**
- **Investigations**
 - **CXR** (Air trapping, peribronchial thickening, atelectasis, increased linear markings)
 - **NPA for PCR**
 - **FBC** (Lymphocytosis)
- **Treatment**
 - **Fluid** Rehydration
 - **Paracetamol** (fever)
 - **Humidified O2**
 - Bronchodilator (Ventolin [**Salbutamol**])
 - **If Severe → Intubation and Ventilation**
 - **Indications For Hospitalization**
 - **Hypoxia:** SpO2 <92%
 - **Resting Tachypnea** >160/minute
 - **Respiratory Distress even after Salbutamol**
 - **<6 months old**
 - **Feeding Problems**



<https://www.healthing.ca/other/respiratory-syncytial-virus-rsv/>



<https://www.ncbi.nlm.nih.gov/books/NBK442240/figure/ch5.f3/>



<https://m.theindependentbd.com//magazine/details/159068/Bronchiolitis:-Not-to-be-treated-lightly>

SEASONAL FLU (INFLUENZA A & B):

- **Aetiology:**
 - Influenza Virus A & B
- **Pathogenesis:**
 - **Transmission:** airborne spread; droplet
 - **Incubation Period:** 1-4 days
 - **Contagious for:** 1 day Before Sym Onset, and the next 7 days
 - Viral-Induced Epithelial Dysfunction & Destruction
- **Clinical Features:**
 - **Symptoms:** Chills, Fatigue, Cough, Myalgias, Arthralgias, Headache
 - **Signs:** High Fever (<42C); But Chest Clear (Unless 2° Bacterial Pneumonia)
 - **Complications:** 2° Bacterial Pneumonia, Otitis Media, Sinusitis
- **Diagnosis:**
 - **Clinical Diagnosis (Signs & Symptoms)**
 - +/- Nasopharyngeal Swabs
 - +/- Serology
 - **Note:** CXR is usually *Normal*
- **Treatment and Prevention**
 - **Primarily Supportive Treatment:**
 - Bed Rest, Fluid, Paracetamol/Analgesics, Antitussives, Decongestants
 - **+/- Antivirals (Effective within 48 hours of onset):**
 - *Oseltamivir* (Tamiflu TM) / *Zanamivir* (Relenza TM) → **Reduce <24hrs of Symptoms**
 - **Vaccine:**
 - **FluVax** is recommended **Annually** for **Everyone**
 - (Note: Vaccine is reformulated each year to include current serotypes)

What to know about seasonal influenza

Influenza or "the flu", is an illness caused by seasonal influenza viruses. These viruses are spread from one person to another.

How to recognize the flu?



Sudden high fever



Headache



Cough or sore throat



Muscle pain

What to do when you have the flu?



Cover your cough or sneeze with your arm/elbow or tissue



Wash your hands frequently



Get plenty of rest



Drink plenty of water and eat nutritious food



Seek medical advice if you are not getting better or if you are in a high risk group



World Health Organization

HEALTH
EMERGENCIES
programme

Source: WHO

BIRD FLU (H5N1):

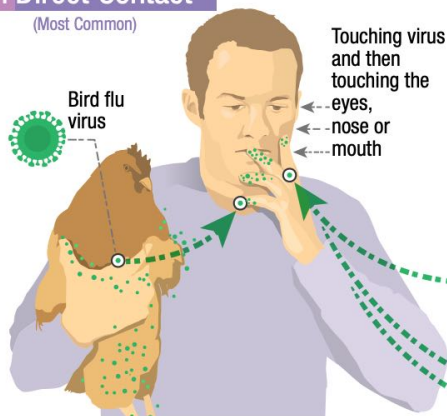
- **Aetiology:**
 - Influenza H5N1
- **Pathogenesis:**
 - **Transmission** – Aerosol/Direct Contact
 - **Incubation Period Generally 2-8 Days**
 - **Infection with Influenza H5N1** → Viral Replication → Virus-induced Epithelial Dysfunction
 - **Mortality Rate** ≈63%
- **Clinical Features:**
 - **Symptoms:** High Fever (>38°C), Headache, Myalgias, Cough (± Sputum), Dyspnoea + Others
 - **Pneumonia:** Consolidation, Tachypnoea, Tachycardia
 - **Often Progresses To ARDS** → Multi-Organ Failure → Death
- **Investigations:**
 - **NPA** → PCR
 - **CXR** - (Infiltrates +/- Pleural Effusions)
- **Treatment**
 - **ICU** - (Ventilation, Fluids)
 - **Antivirals** – (*Oseltamivir* (Tamiflu TM) / *Zanamivir* (Relenza TM))
- **Prevention**
 - No Vaccine
 - Hygiene Precautions
 - Post-?Exposure-Prophylaxis – (*Oseltamivir* / *Zanamivir*)
 - **Notify Public Health**
 - **Contact Tracing and Quarantine**

How Infected Backyard Poultry Could Spread Bird Flu to People

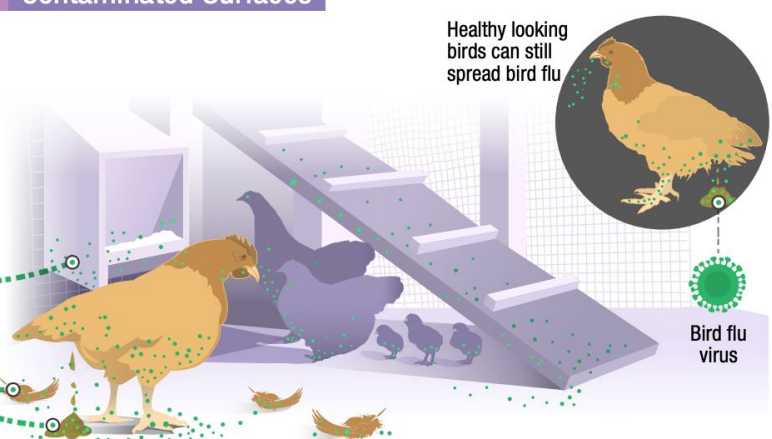
Human Infections with Bird Flu Viruses Rare But Possible

1 Direct Contact

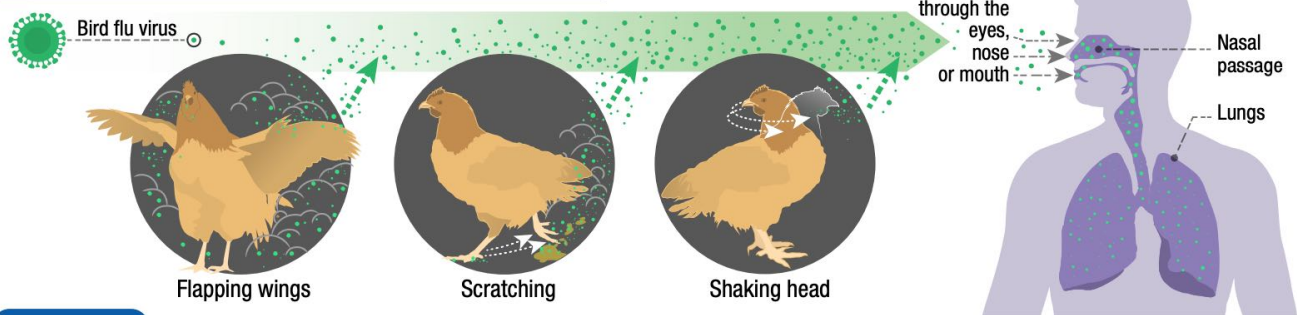
(Most Common)



2 Contaminated Surfaces



3 Bird Flu Virus in the Air (in Droplets or Dust)



U.S. Department of
Health and Human Services
Centers for Disease
Control and Prevention

www.cdc.gov/flu/avianflu/avian-in-humans.htm

CS261152

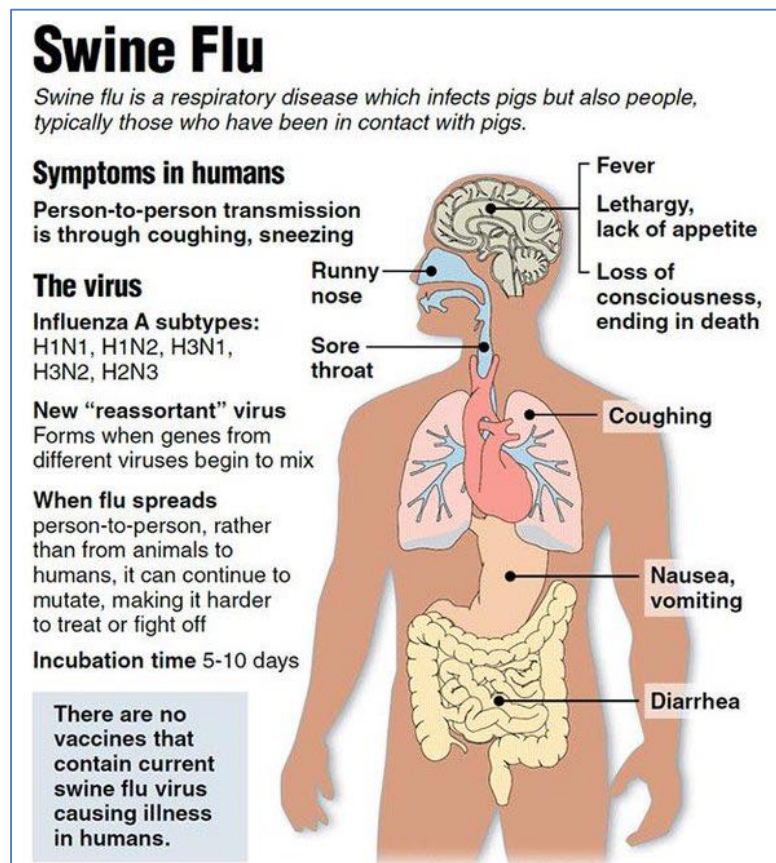
Source: CDC

SWINE FLU (H1N1):

- **Epidemiology**
 - HUMAN to HUMAN - NOT by pigs; documented mass pig slaughtering was unnecessary
 - Incubation Period 24--48 Hours
- **Aetiology:**
 - H1N1 – (A Novel strain - genes from 5 different flu viruses)
- **Pathogenesis:**
 - Droplet Transmission – Human to Human
 - Respiratory Tract Infection
- **Clinical Features:**
 - **Low Mortality Rate** - 2 deaths in first 600 cases in the US
 - **Infects The Young (<5yrs) And Old (>65yrs)**
 - **Transmission:** Aerosol/Contact (Human:Human)
 - **Symptoms:** Fever, Cough, Sore throat, N/V/Dia (25%), Myalgia/Arthralgia, Headache

Emergency warning signs	
In children	In adults
• Laboured breathing • Cyanosis • Dehydration • Irritability • Fever with rash • Quiet, not interacting	• Shortness of breath • Pain in chest or abdomen • Confusion • Persistent or severe vomiting

- **Diagnosis:**
 - Clinical Suspicion
 - PCR – (Nasal/Nasopharyngeal/Oropharyngeal)
 - Notify Public Health
 - Contact Tracing and Quarantine
- **Treatment**
 - Antivirals – (*Oseltamivir* (Tamiflu TM) / *Zanamivir* (Relenza TM))
 - +Supportive



SARS & COVID – SEVERE ACUTE RESPIRATORY SYNDROME:

- **Definition**
 - Rapidly progressing viral pneumonia caused by the SARS-associated coronavirus (SARS-CoV)
- **Aetiology:**
 - **SARS-Associated Coronavirus**
 - Incubation: 2-7 days
- **Pathophysiology**
 - Droplet Transmission – Human to Human
 - Respiratory Tract Infection with SARS-Associated Coronavirus
 - → Atypical Pneumonia +/- Respiratory Distress Syndrome
- **Clinical Features**
 - **Difficult To Differentiate SARS from other Community-Acquired Pneumonias Because:**
 - **Initial Symptoms Are Not Specific:**
 - Fever, Chills, Malaise,
 - Headache, Myalgia,
 - Cough, Sore Throat, Productive Cough
 - **However, 2/3 Of Patients Deteriorate with:**
 - Persistent Fever,
 - ↑SOB & Desaturation
 - **20% Require ICU Admission and Mechanical Ventilation**
- **Complications**
 - Respiratory failure
 - Liver failure
 - Heart failure
- **Diagnosis:**
 - Clinical Suspicion – Symptoms, Hx of Travel, Hx of Contact
- **Investigations:**
 - **CXR** – Features of Atypical Pneumonia
 - **Lab** – Neutrophilia, Lymphopenia, ↑CRP, & ↑LDH
 - **RT-PCR** – from Blood/Sputum/NPA/Swabs
 - **Serology** – (antibody detection via ELISA)
- **Treatment**
 - **Notify public health**
 - **Quarantine** (negative-pressure room, N95 Mask, gown, gloves, eye protection)
 - **Antivirals** – (*Ribavirin*)
 - **Steroids** - (To prevent immune mediated lung damage)



<https://www.cdc.gov/ncidod/eid/vol9no6/03-0264.htm>

GENITAL HERPES SIMPLEX:

- **Aetiology:**
 - **HSV2** in Genital Herpes (**12.5% Prevalence!!**)
 - (HSV1 in Cold sores; but can still cause genital infections) (**70% Prevalence!!**)
- **Pathogenesis:**
 - **Contact Transmission**
 - **1:Lives in Neurons → Latent...2:Reactivation → Travels down Axon into Skin → Lesions**
- **Morphology:**
 - Papular/Vesicular lesions on external Genitalia
- **DDXs of Genital Ulcers:**
 - **Infection:** Herpes/Syphilitic Chancre/Donovanosis/Lymphogranuloma Venereum
 - **Trauma:** Mechanical/Chemical
 - **Allergic:** Contact Wet Dermatitis
- **Clinical Features:**
 - 2F:1M
 - **Symptoms:**
 - **Course:**
 - **<3wks Incubation**
 - **Prodrome** – Paraesthesia, Itching, Redness
 - **Symptoms last for <2wks if untreated**
 - Clusters of PAINFUL, ITCHY, Papules/Vesicles on External Genitalia
 - Vesicles may Rupture → Painful Ulcerations
 - **Recrudescences:**
 - Typically milder than 1st presentation
 - 1-2 day prodrome (Paraesthesia)
 - **+/- Proctitis/Cervicitis**
 - (Note: *ANY genital ulcer, scabbed, red-edged, multiple, and painful = Think Herpes!*)
- **Diagnosis:**
 - **Clinical Diagnosis**
 - **Swab Vesicle → HSV 1&2 PCR**
 - **Tzanck Smear** (Typical intranuclear inclusion bodies & multi-nucleated giant cells)
 - HSV Serology (limited use)
- **Treatment (NO CURE; Symptomatic & Suppressive Therapy ONLY):**
 - **Valaciclovir/Famciclovir/Aciclovir – (Nucleoside Analogue Anti-Virals) (BD 10 days)**
 - Note: “Suppressive Therapy” → 50% Reduction in Transmission
 - **Analgesia – Lignocaine Gel**
 - **Counselling & Sex-Education**
 - 90% of HSV2 will have recurrences >5x/year
 - (Note: HSV1 have annual recurrences)
 - **Advise Abstinence in the Prodrome or when Lesions are Present**
 - BUT Note: *Asymptomatic Viral Shedding Still Occurs!!!*



Creative Commons: <https://en.wikipedia.org/wiki/File:SOA-Herpes-genitalis-female.jpg>

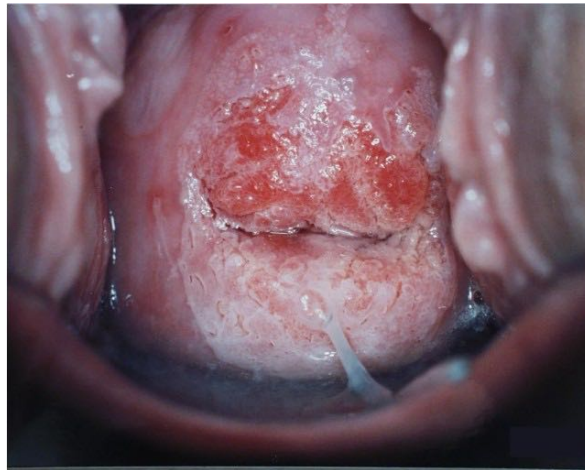
HUMAN PAPILLOMA VIRUS:

- **Aetiology:**
 - ***HPV Types 6 & 11** → Genital Warts (Preventable by **Gardasil**)
 - **HPV Types 16, 18 & 45** → Cervical Cancer (Somewhat preventable by **Gardasil**)
- **Transmission:**
 - (Direct Contact/Sexual Transmission – Highly Contagious)
- **Pathogenesis:**
 - Contact & Fomite Transmission
 - **3mth Incubation Period**
 - HPV Infection → Cell-Cycle Dysregulation → Benign Overgrowth
- **Morphology:**
 - **Macro:**
 - **Genital/Cervical Warts (6/11)** - Warty Papillomas – External Genitalia/Oral/Anal



<https://www.ncbi.nlm.nih.gov/books/NBK441884/figure/article-22202.image.f2/>

- **Cervical Ca (16/18/45)** – Abnormal looking cervix (Loss of normal smoothness, obvious dysplasia)



<https://oacapps.med.jhmi.edu/OBGYN-101/Text/Pap/Moderate%20Dysplasia.htm>

- **Micro:**
 - **Genital/Cervical Warts (6/11)** – **"Koilocytosis"** = Cells with "halo" cytoplasm
 - **Cervical Ca (16/18/45)** – *Squamous Cell Carcinomas*, or *Adenocarcinomas*
- **Clinical Features:**
 - **Symptoms:**
 - Infection is long-term, latent, and usually asymptomatic
 - **Genital Warts (6/11)** → Painless, papillary outgrowth on external genitalia
 - **Cervical Ca (16/18/45)** → Abnormal Vaginal Bleeding, Dyspareunia, Weight-Loss, Fatigue, Pelvic Pain (May be Asymptomatic)
- **Diagnosis:**
 - **Pap smear &/or Cervical Biopsy**
 - DNA detection
 - Tam Pap (Self-sampling HPV DNA test)

- **Complications:**
 - *Cervical Cancer* - Metastasis
- **Treatment:**
 - **Genital Warts (6/11)** – **Podophyllin** Cream, Aldara (**Imiquimod**) Cream, **Excision** or **Cryotherapy** – BUT Will Recur
 - + Counselling
 - +/- Refer to Gynae if – Extensive, Chronic/Recurrent, Cervical or Rectal
 - **Cervical Ca (16/18/45)** – Surgical Excision +/- Chemotherapy +/- Radiotherapy
- **Prognosis:**
 - **Genital Warts (6/11) – Benign**
 - 70% clear by 12mths (Note: Warts may disappear, but virus may persist)
 - **Cervical Ca (16/18/45) - Malignant**

SYPHILIS:

- **Aetiology:**
 - Treponema Pallidum (Spirochete)
- **Transmission:**
 - Contact, Sexual, & Blood (IVDU) Transmission
 - **!!Vertical – 100% Transmission if mother is untreated!!**
- **Pathogenesis:**
 - **Four Stages** – *Primary, Secondary, Latent, Tertiary (CVS/Neurosyphilis)*
- **Clinical Features:**
 - **Primary Syphilis:**
 - **10d-10wks Post-Infection** → Painless Chancre (ulcer) + Lymphadenopathy



https://jetem.org/syphilis_chancere/

- **Secondary Syphilis** – (Note: Most contagious during secondary syphilis):
 - **4-8wks Post-Chancere** → Characteristic Rash (Palms, Feet), Lymphadenopathy, Hepatosplenomegaly, Flu-like Illness & “Condylomata Lata” (Wart-like Growths)



<https://www.nejm.org/doi/full/10.1056/NEJMmic1502476>

- **Latent Syphilis:**
 - **Mths-Lifetime Post-Secondary-Stage** → Asymptomatic but positive serology
 - ¼ of cases → Tertiary Syphilis (Most remain latent for life)
- **Tertiary Syphilis:**
 - **>1yr Post-Infection** → Formation of 'Gummas' (Highly-Destructive → bones, skin, nervous tissue, heart & arteries) → Serious complications are Cardiovascular (Aneurysms) & Neurosyphilis (Dementia/Psychosis/Paresis/etc)



<https://pharmaceutical-journal.com/article/ld/syphilis-diagnosis-and-management-options>

- **Syphilis in Pregnancy:**

- **Note: Transmission to the Foetus Typically occurs in the 3rd Trimester of Pregnancy**
 - Trans-Placental Transmission
 - Can → Miscarriage/Premature labour
- → **Early Congenital Syphilis:**
 - Snuffles – Profuse Runny Nose
 - Cutaneous Lesions (Often on Palms and Soles)



CDC/ Dr. Norman Cole, Public domain, via Wikimedia Commons

- → **Late Congenital Syphilis:**
 - Frontal bossing
 - Short maxilla
 - High palatal arch
 - Deafness

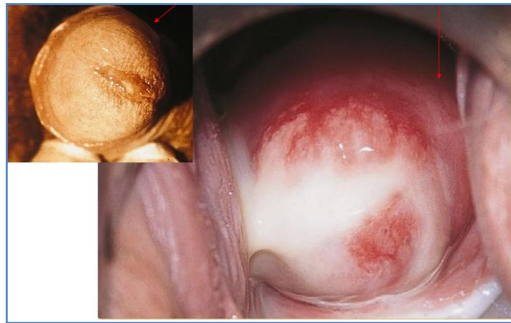


CDC/Susan Lindsley, Public domain, via Wikimedia Commons

- **Diagnosis:**
 - **Organism can't be cultured**
 - **Dark-Field Microscopy**
 - (Too small for Gram stain)
 - 1: Dark field Microscopy
 - 2: Fluorescence (Ag labelling)
 - **Serology (May remain +ve for years after recovery)**
 - **1: TPHA:** *T-pallidum* haemagglutination assay
 - **2: FTA-AB:** Fluorescent Treponemal Antibody Absorption
 - **3: VDRL:** Venereal Disease Research lab tests
 - **4: RPR – Diagnostic Standard: Rapid Plasma Reagen**
 - Tests for Non-Specific Antibodies in the blood
 - Good Sensitivity, Poor Specificity
 - **Interpretation:**
 - A 2 Titre rise Indicates infection
 - A 2 Titre fall indicates effective treatment
- **Complications:**
 - *Neurosyphilis* → Meningitis, paresis, personality change, ataxia, dementia
 - *Cardiovascular Syphilis* → Typically Syphilitic Aortitis → Aneurysm
 - ***Congenital Syphilis – 25% Miscarriage; 25% Neonatal Death; The rest are DEFORMED!!***
 - → **Early Congenital Syphilis:**
 - Snuffles – Profuse Runny Nose
 - Cutaneous Lesions (Often on Palms and Soles)
 - → **Late Congenital Syphilis:**
 - Frontal bossing
 - Short maxilla
 - High palatal arch
 - Deafness
- **Treatment:**
 - **Azithromycin/Doxycycline**
 - Or Single Dose **IM Penicillin-G**
 - **Treatment of Early Syphilis:**
 - Benzathine Penicillin
 - If Truly Allergic to Penicillin –(Azithromycin)
 - **Treatment of Late/Latent/Unknown Duration of Syphilis:**
 - Benzathine Penicillin (Intramuscular Injection)
 - (Painful)
 - **(Treatment Failure):**
 - Treatment Failure = Failure to achieve a 4x Fold drop by 6 months
 - Failure is more common in late syphilis & most common with neurosyphilis
 - **(Why treat syphilis?):**
 - To prevent transmission to others
 - Sexual
 - Neonatal
 - To Prevent long term complications
 - 1e: Tertiary syphilis
 - (30% chance of tertiary syphilis if untreated)
 - To reduce chance of transmission of HIV
 - HIV transmission increases greatly with concomitant transmission

CHLAMYDIA:

- **Aetiology:**
 - Chlamydia Trachomatis
- **Pathogenesis:**
 - Vaginal, Anal, Oral & Vertical Transmission
 - **Obligate Intracellular Replication** – (Ie: Replicate like Viruses → Shed by Infected cell lysis)
- **Morphology:**
 - **Micro:** Obligate Intracellular Bacteria → Chlamydial Intracellular Reticulate Bodies
- **Clinical Features:**
 - **Symptoms:**
 - **Males** – The COMMONEST cause of Urethritis
 - (May also → Epididymitis, Orchitis, Prostatitis & Proctitis)
 - (Note: A *Non-Gonococcal Urethritis*: Ie: Clear, Watery Discharge)
 - **Females** – Asymptomatic, or Urethritis
 - (May → Cervicitis, Salpingitis/**PID**)
 - **Neonates:**
 - Neonatal conjunctivitis (similar to Gonorrhoea)
 - Chlamydial pneumonia



Unattributable

- **Diagnosis:**
 - **Sample for PCR:**
 - **1st Catch Urine (Unisex)...or**
 - **Women** – Endocervical/High-Vaginal Swab
 - **Men** – Swab of Urethral Discharge
 - **+/- Throat Swabs:**
 - → **Antigen Detection Tests – PCR**
 - → **Gram stain & Immunofluorescence** - Intracytoplasmic inclusion bodies – Replicate intracellularly
 - **(Note: All Females <25 are screened for Chlamydia) – (Via Non-Invasive PCR)**
- **Complications:**
 - **Trachoma** – (Chlamydial Conjunctivitis)
 - **Lymphogranuloma Venereum** - (Lymphatic Chlamydial infection) → Groin Abscesses/Buboes → May become ulcerative
 - **PID** – can → Infertility, ↑ Risk of Ectopic Pregnancy, Chronic Pelvic Pain
 - **Reiter's Syndrome Triad** - Reactive Poly-Arthritis + Conjunctivitis + Urethritis

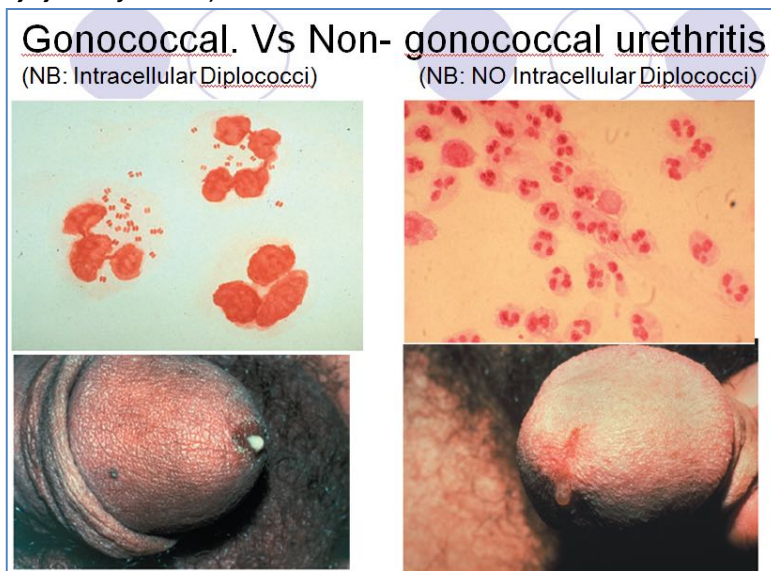


<https://www.cehjournal.org/article/who-simplified-trachoma-grading-system/>

- **Treatment:**
 - **1 Dose Azithromycin 1g**
 - or **Doxycycline** 10days 100mg BD
 - Note: Resistant strains may exist in certain communities and susceptibility-directed therapy is recommended.

GONORRHOEA:

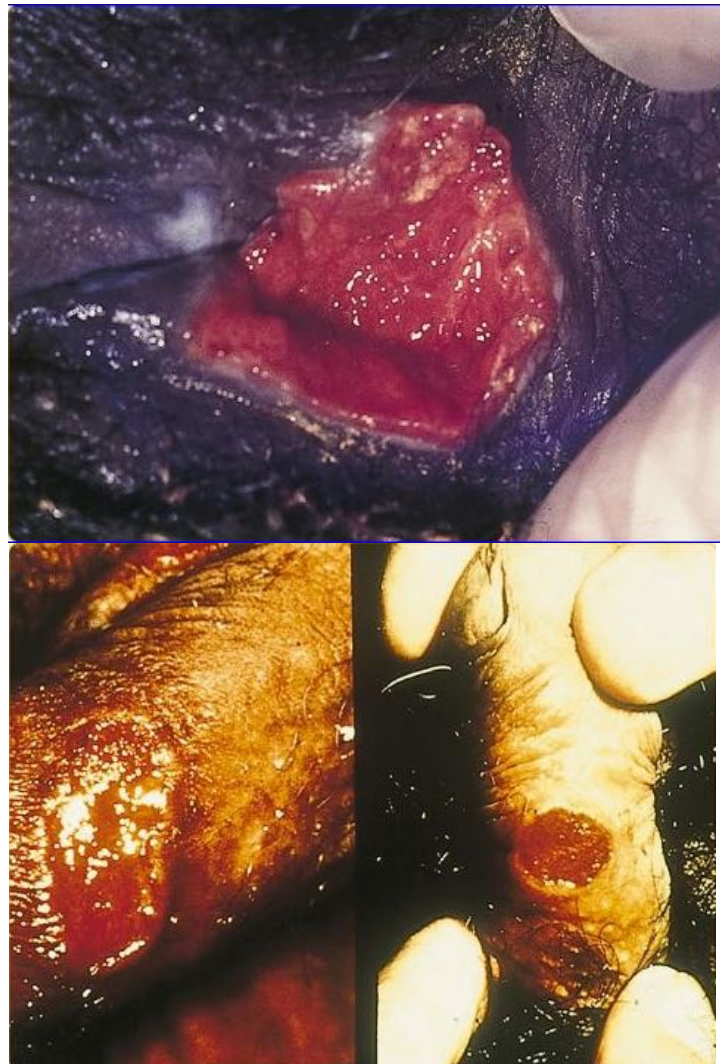
- **Aetiology:**
 - Neisseria Gonorrhoeae (Gram Negative)
- **Transmission:**
 - **Horizontal via Direct Sexual Contact:**
 - **Vertical – (During childbirth; not trans-placental [like syphilis & hep B])**
- **Pathogenesis:**
 - **Virulent**, Fastidious (Delicate), aerobic, gram negative diplococcus
 - **Pili** – anchors to urethral epithelium → Resists Flushing → Infiltrates Epithelium
 - **Gonococcal Toxin** – Endotoxin
 - **Protease** – Destroys secretory IgA
- **Morphology:**
 - **Macro** - Inflamed Urethra + Thick, Milky-white Discharge
 - **Micro** - Intracellular Diplococci on Gram Stain (Typically inside neutrophils)
- **Clinical Features:**
 - **Symptom Onset within <1wk of Infection**
 - **Men** → Acute **Gonococcal Urethritis** + Dysuria + Discharge (Thick & milky)
 - **Women** → Acute **Gonococcal Cervicitis** + Vaginal Discharge (May also be Asymptomatic in Women)
+ (Note: Can → PID in females)
- **Diagnosis:**
 - **Clinical:**
 - **Note: Differentiating Gonococcal Urethritis Vs Non-Gonococcal Urethritis:**
 - **Gono** – Thick, milky, Penile discharge. Gram Negative Diplococci on gram stain of discharge
 - **Non** – Thin, watery discharge. No organisms on Gram Stain (Typically Chlamydia)
 - **Sample for PCR:**
 - **1st Catch Urine (Unisex)...or**
 - **Women – Endocervical Swab**
 - **Men – Swab of Urethral Discharge**
 - **Men + Women – Throat Swabs**
- **Complications:**
 - **PID** (Females)– can → Infertility
 - **Urethral Stricture** → Urinary Obstruction → Hydronephrosis
 - **Epididymitis, Prostatitis**
 - **Endocarditis**
 - **Gonococcal Arthritis**
 - **Ocular Infections, Neonatal Conjunctivitis**
- **Treatment:**
 - **Stat Dose IM Ceftriaxone + Stat Dose PO Azithromycin**
 - **(Or BD Doxycycline for 1wk)**



Unattributable

DONOVANOSIS:

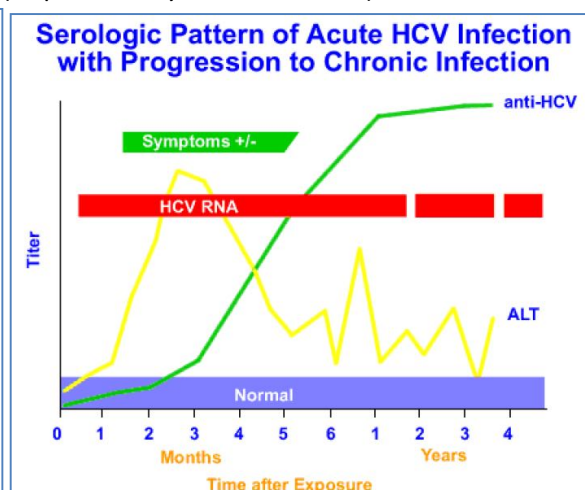
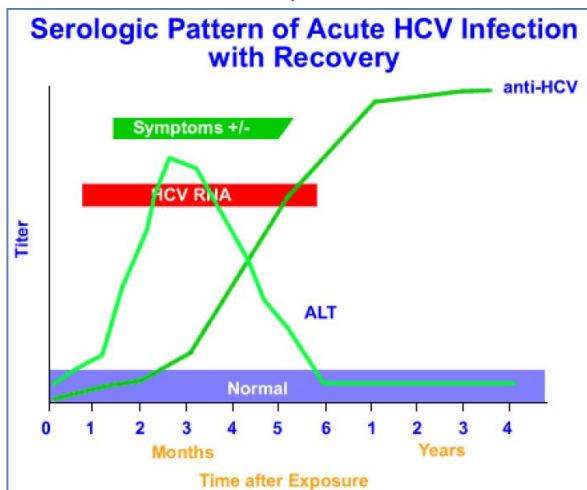
- **Aetiology:**
 - Klebsiella Granulomatis (Gram Neg)
 - (Formerly: *Calymmatobacterium granulomatis*)
- **Pathogenesis:**
 - Direct Contact Transmission with OPEN sores
- **Morphology:**
 - **Macro:**
 - Painless, Oozing, Red Ulcers with Characteristic *Rolled Edges* of Granulation Tissue
 - **Micro:**
 - Donovan Bodies = Intracellular Rod-Shaped, Oval Organisms seen inside Phagocytes
- **Clinical Features:**
 - **Symptoms:**
 - → Chronic, painless, **offensive, oozing** genital ulcers (As opposed to Syphilis = dry) + genital disfigurement (Lesions occur on Penis, Labia, or Perineum)
 - Note: **NO Lymphadenopathy** (As opposed to Syphilis = Lymphadenopathy Present)
- **Diagnosis:**
 - Thorough history and examination
 - Scrape → Microscopy (Donovan Bodies)
 - Swab → PCR
 - + Rule out Syphilis (RPR, VDRL, TPHA)
- **Complications:**
 - Genital Disfigurement
- **Treatment:**
 - **Doxycycline/Azithromycin**/Erythromycin



Creative Commons: <https://commons.wikimedia.org/wiki/File:SOA-Donovanosis-female.jpg>

HEPATITIS C:

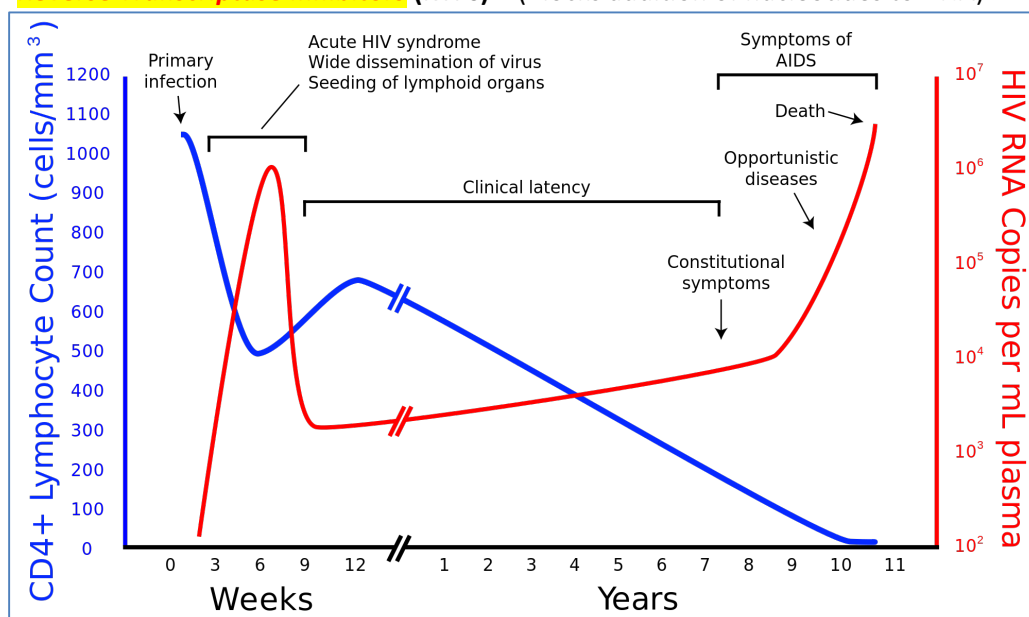
- **Aetiology:**
 - Hepatitis C Virus
- **Transmission:**
 - **Blood (Eg: IVDU/needle sharing):** As little as 0.0001 mL of blood can transmit the infection
 - **Body fluids (Eg: Sexual):** (Including Cervical Secretions and Semen)
 - **Vertical** (Uncommon)
- **Note: Epidemic Potential:**
 - **No Vaccines**
- **Pathogenesis:**
 - Viral Infection (Horizontal/Vertical) → Virus Replicates in the Liver
 - **Note: Virus is NOT directly Cytopathic; Damage is due to CD8-T-Cell Attack**
 - → Cellular (CD8) Immune Attack on Infected Hepatocytes
 - → **Chronic, Low-Grade Inflammation** → Eventually leads to Fibrosis → Cirrhosis
- **Morphology – Mostly Chronic:**
 - Chronic 'Peri-Portal' Inflammatory Infiltrates
 - Necrosis, Apoptosis & Fibrosis → Cirrhosis
 - (Hep C – Mild Fatty Change [Microvesicular Steatosis])
- **Clinical Features:**
 - **10% → Acute with Recovery** – (Mild Viral Illness + Jaundice)
 - May have Non-Specific Viral Symptoms (Nausea/Anorexia/Fatigue)
 - May have Jaundice
 - **90% → Chronic with Extrahepatic & Intrahepatic Manifestations:**
 - Asymptomatic for years (Usually Incidental Diagnosis)
 - May have Sporadic Mild Viral Illnesses + Jaundice
 - +/- Arthritis
 - +/- Glomerulonephritis
 - **END STAGE (CIRRHOSIS):**
 - 20-30% → **Cirrhosis** (within 10-30yrs)
 - 5% → **Hepatocellular Carcinoma** – (Hep C Directly inactivates P53)



- **Investigations:**
 - Usually discovered on Routine LFTs – (Mildly ↑ ALT/AST)
 - Hep C Serology – ((+) Anti-HCV)
 - Hep C PCR – ((+) HCV-RNA)
- **Treatment:**
 - **Post-Exposure/Acute (Eg: Needlestick):**
 - **IFN**
 - **Ribavirin**
 - Previously incurable
 - Now up to 95% 'curable' with 'Direct-Acting Antivirals' (DAA's):
 - Epclusa® (sofosbuvir + velpatasvir)
 - Maviret® (glecaprevir/pibrentasvir)
 - Harvoni® (sofosbuvir + ledipasvir)

HUMAN IMMUNODEFICIENCY VIRUS:

- **Aetiology:**
 - HIV
- **Transmission:**
 - **Blood** (IVDU, Transfusion)
 - **Body Fluids** (Sexual – Particularly Anal Sex)
 - **Vertical** (Cross-Placental & Breastmilk)
- **Pathogenesis:**
 - **Lymphotropic** – Preferentially infects CD4-T-Cells → Integrates into Genome → Uses host DNA-Replication for Reproduction
 - CD4-T-Cell Lysis → CD4-T-Cell Depletion (Including Memory T-Cells) → **Immunosuppression By:**
 - ↓IFN γ Production
 - ↓Antibody Production
 - ↓Antibody Isotype Switching
 - ↓Macrophage Activation
 - ↓CD8-T-Cell Activation
- **Clinical Features:**
 - **Symptoms:**
 - **1-2 months:**
 - **Acute infection** (Flu-like symptoms + Maculopapular Rash (ITP))
 - Following the acute infection, Antibody titres rise (**Detectable after 2.5mths**)
 - **2-4 Years:**
 - **Asymptomatic** Chronic Infection – (Equilibrium between T-Cells & Viral Mutation Rate)
 - **8 years:**
 - **Symptomatic** Chronic Infection – (Disequilibrium – HIV Quasispecies outnumber T-Cell Diversity → Body starts to lose the battle)
 - **10-12 years:** (If no intervention)
 - **AIDS** - Advanced infection – (T-Cell Depletion)
- **Diagnosis:**
 - Serology (Ab Detection)
 - Viral PCR (Ag Detection)
- **Complications:**
 - ↑Infections
 - ↑Cancer (Especially **Kaposi's Sarcoma**),
- **Treatment:**
 - **Fusion Inhibitors** – (Eg: **CCR5 Inhibitors**) - Prevent binding of HIV to Cell
 - **Reverse Transcriptase Inhibitors (RTI's)** – (Blocks addition of nucleotides to DNA)



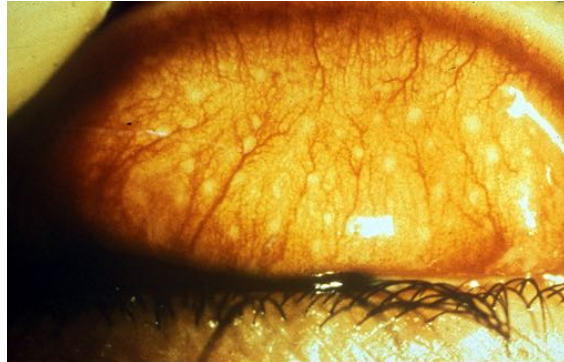
https://www.wikidoc.org/index.php/HIV_AIDS_natural_history_complications_and_prognosis

TRACHOMA:

- **What is it?**
 - The leading cause of the world's infectious blindness
 - Untreated, repeated trachoma can result in "*Entropion*", where the eyelids turn inward → eyelashes scratch the cornea
- **What causes it?**
 - **Causative Organism = Chlamydia Trachomatis**
 - Often acquired in the birth canal → Neonatal infections
 - **Chlamydia Trachomatis is also associated with:**
 - Chlamydia STI's
 - Lymphogranuloma Venereum
 - Neonatal Infections (Eyes & pneumonia)
- **How is it Spread?:**
 - Close Contact, especially with poor facial hygiene
 - Contact with Affected Individuals:
 - Eye Secretions
 - Nasal Secretions
 - Throat Secretions
 - Or contact with objects (shared Clothes/Towels/Flies) that have had similar contact with these secretions
- **Main Risk Groups & Factors:**
 - Children
 - Mothers
 - Carers of children
 - Other people in close contact with infected children
 - Poor Personal Hygiene
 - Overcrowding
- **Disease Process (Up to 30 years):**
 - Acquisition of C-Trachomatis
 - Chronic asymptomatic infection
 - Chronic inflammation of the conjunctiva
 - → Limbal follicles
 - → Pannus
 - → Herbert's Pits
 - Inflammation of Conjunctiva
 - → Scarring of the conjunctiva underneath the eyelid
 - → Contraction of eyelid scarring (Entropion)
 - → Eyelashes pulled inwards by contracting eyelid
 - Eyelashes touch the eye (Trichiasis)
 - Corneal Scarring
 - Loss of vision
- **Pathophysiology:**
 - Infection with C-Trachomatis → Chronic Inflammation of the Upper tarsal conjunctiva → Scarring of the Conjunctiva → Retraction of scarring → Pulls eyelid inwards → Eyelashes abrade the cornea → scarring of the cornea → Opacity & Blindness
- **Signs and Symptoms of Trachoma:**
 - **Conjunctivitis:**
 - Inflammation of the Conjunctiva
 - The result of "Active Trachoma"

○ **Trachomatous Inflammation, Follicular (TF):**

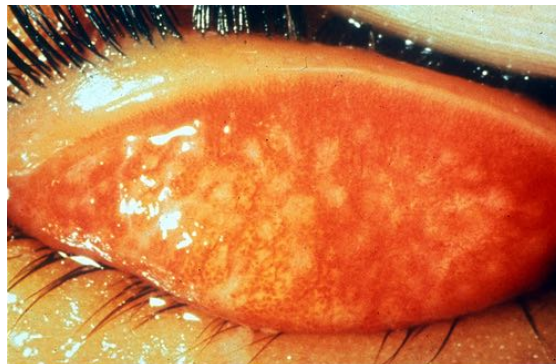
- **Conjunctival Follicles** = situated on the Undersurface of the Upper Eye-Lid
 - 5 or more is Diagnostic of Trachomatous Inflammation
- **Limbal Follicles** = Follicles can also appear at the Limbus (Junction between Cornea and Sclera)
 - A sign of chronic inflammation of the Corneal Conjunctiva
- **Follicles are White Dots** = Essentially Lymphoid germinal centres



<https://wikem.org/wiki/Trachoma>

○ **Trachomatous Inflammation, Intense (TI):**

- 'Papillary' (tiny projection-like) Hypertrophy (Red Dots) & Inflammatory Thickening of the upper Conjunctiva
- Red Dots = 'Papillae'
- Rough surface
- Obscured blood vessels



<https://wikem.org/wiki/Trachoma>

○ **Trachomatous Scarring (TS):**

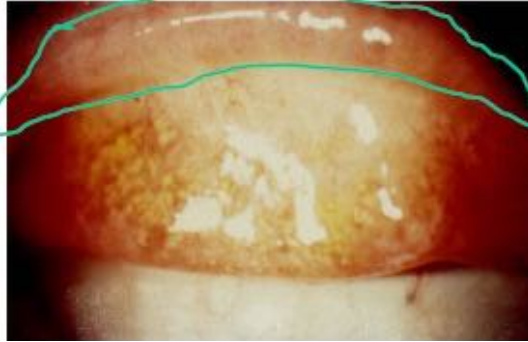
- (AKA: "Cicatricial" (Scarred) Trachoma)
- Scarring of upper-eyelid Conjunctiva



<https://wikem.org/wiki/Trachoma>

- **Entropion:**

- Contraction of scarring on the underside of the lid → Pulls the eyelid inwards
- Edge of the eyelid has a 'rolled' appearance



Unattributable

- **(Trichomatous) Trichiasis (TT):**

- Eyelashes touching the eye



<https://wikem.org/wiki/Trachoma>

- **Corneal Opacity (CO):**

- Causes blindness



<https://www.cehjournal.org/article/who-simplified-trachoma-grading-system/>

- **Treatment: "SAFE" Trachoma Management Strategy:**

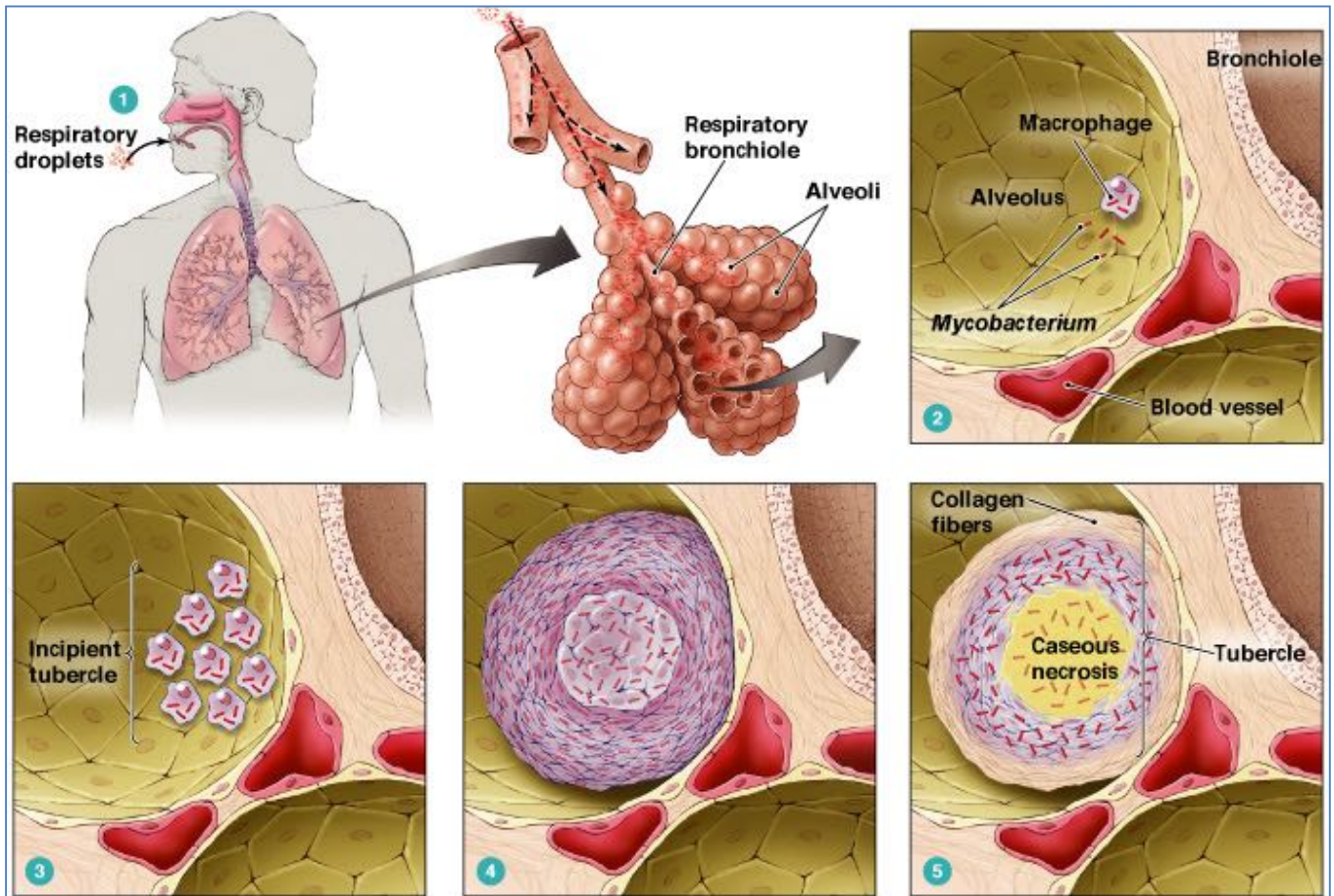
- Surgery
- Antibiotics (1-2 doses of Azithromycin)
- Face-Washing (Hygiene)
- Environmental control

- **Trachoma Screening:**

- Test person's visual acuity using a Snellen's chart
- General examination of the external eye
- **Look for trachoma in both eyes using the WHO Screening Criteria (See Below)**

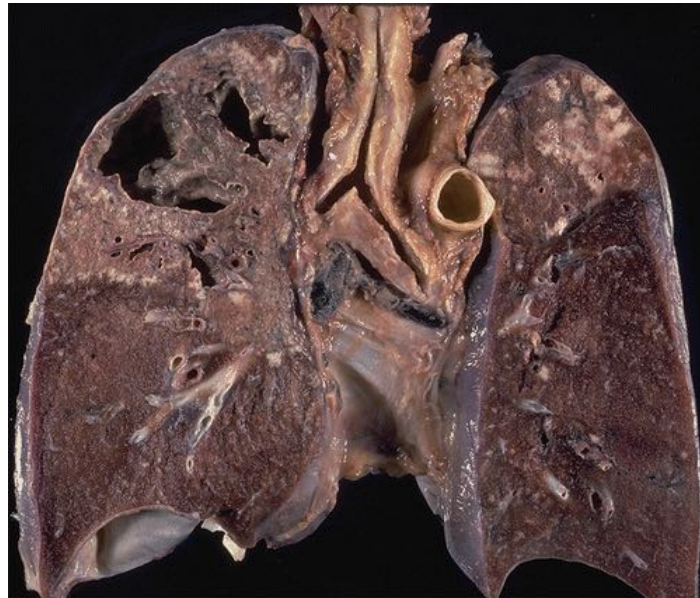
PULMONARY TUBERCULOSIS:

- **Aetiology:**
 - Infection with *Mycobacterium Tuberculosis* (An Acid-Fast Bacilli) (Droplet Transmission)
- **Pathogenesis:**
 - **Pulmonary Tuberculosis:**
 - M-Tuberculosis Inhaled → Reaches Alveoli
 - → **Invade & Replicate within Alveolar Macrophages**
 - **(3wks Later) T-Cell Sensitization → Chronic Hypersensitivity** reaction to TB Antigens
 - Th-Cells Secrete IFN γ → Activate Macrophages → Caseating Granulomatous Inflammation



<https://slideplayer.com/amp/3863668/>

- **Miliary Tuberculosis:**
 - M-tuberculosis overrun draining Lymph Nodes and enter the Circulation
 - Organisms are 'seeded' back into the lung → Forming Many lesions
 - Miliary lesions Coalesce & Erode the lung parenchyma → Pleural Effusion/Haemoptysis/Empyema
- **Morphology:**
 - **Typically Affects Upper Lung Lobes First**
 - **Caseating Granulomas** (Pulmonary or Miliary/Systemic)
 - Nodular, Cavitating, Fibrosing
 - T/B-Lymphocytes, Macrophages, Langerhan's Giant Cells & Fibroblasts
 - Caseating Necrosis (looks like soft, white cheese)
 - Rim = Fibroblastic + Lymphocytes
 - Centre = Multinucleated Langerhan's Giant Cells
 - **Ghon Focus:**
 - 1-1.5cm area of Gray-white inflammation with consolidation
 - Central Caseous Necrosis
 - **Ghon Complex:**
 - Ghon Focus + Nodal Involvement



<https://pubs.rsna.org/doi/full/10.1148/rg.2017160032>

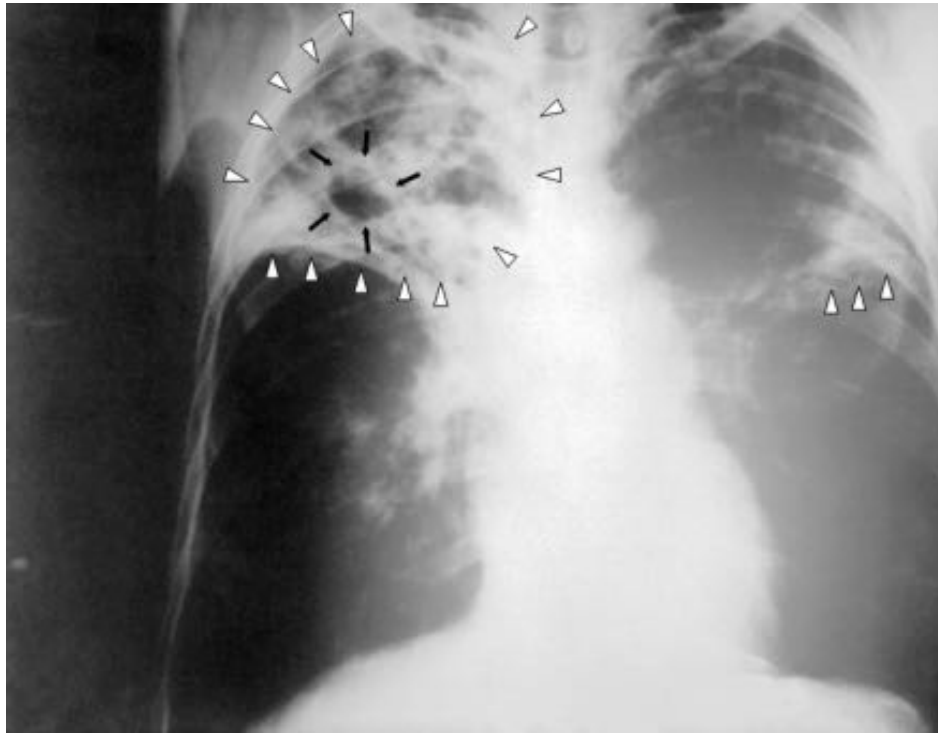
- **Clinical Features:**

- May be Asymptomatic/Latent
- **Classic Symptoms:**
 - Chronic Cough
 - Fever, Night Sweats
 - Weight Loss
 - Pleuritic Chest Pain
 - Cavitation & Erosion can → Pleural Effusion &/or Haemoptysis
 - Extrapulmonary Symptoms – Depend on the Organ Affected
- **Miliary Tuberculosis:**
 - M-tuberculosis overrun draining Lymph Nodes and enter the Circulation
 - Organisms are 'seeded' back into the lung → Forming Many lesions
 - Miliary lesions Coalesce & Erode the lung parenchyma
- **Diagnosis:**
 - **Mantoux Test (Tuberculin Test):**
 - Intradermal Hypersensitivity test to injected PPD (Purified Protein Derivative)
 - Only works after 2-4wks post infection; but once infected, will be positive for life
 - Signifies T-Cell Sensitivity to Mycobacterial Antigens



CDC: <https://phil.cdc.gov/details.aspx?pid=6806>

- **CXR:**
 - Upper Lobe Consolidation
 - "Ghon Focus" - ≈1.5cm area of gray-white inflammation with Caseous Necrosis
- **Serology:**
 - PCR Amplification (Much more sensitive)
- **Microscopy:**
 - Acid-Fast Sputum Smears
 - Culture & Sensitivity



<https://www.tbonline.info/posts/2016/3/31/pulmonary-tb/>

- **Treatment:**
 - Combined Antibiotics
 - Pyrazinamide
 - Ethambutol
 - Isoniazid
 - Rifampicin

INTESTINAL TUBERCULOSIS:

- **Aetiology:**
 - Reactivation of Primary Mycobacterium tuberculosis (Typically from Pulmonary TB)
 - Typically in Immunocompromised (HIV/Drugs)
- **Pathogenesis:**
 - Spread/Reactivation of Tuberculosis:
 - (i) hematogenous spread from the primary lung focus
 - (ii) ingestion of bacilli in sputum from active pulmonary focus;
 - (iii) direct spread from adjacent organs;
 - (iv) through lymph channels from infected nodes
- **Morphology:**
 - Mesenteric Thickening
 - Lymphadenopathy
 - Ulceration of Transverse Colon
 - Multiple Granulomas in Lymph Nodes or Below Ulcers
 - Fibrosis, Thickening and Strictureing of the bowel wall
- **Clinical Features:**
 - **Symptoms/Signs:**
 - **Fever + Night Sweats
 - **Weight Loss
 - *Ileocecal Area is most commonly affected → RIF Abdominal Pain, Palpable Masses
 - Generalised Peritonitis
 - Anaemia
 - Obstruction
 - **Diagnosis:**
 - Histology & Culture
 - CXR (50% have evidence of Pulmonary TB)
 - **Treatment:**
 - **Combination Antibiotics:**
 - Rifampicin
 - Isoniazid
 - Pyrazinamide
 - Ethambutol



Yale Rosen from USA, CC BY-SA 2.0 <<https://creativecommons.org/licenses/by-sa/2.0>>, via Wikimedia Commons

LEPROSY

- **Organism:**
 - *Mycobacterium leprae*
- **Pathogenesis:**
 - Chronic disease of skin and nerves
- **Presentation:**
 - Some skin lesions of leprosy can look like dermatophytosis
 - Decreased sensation and no sweating
 - Lesions can be:
 - Depigmented or Reddish/Copper-coloured
 - flat or raised
 - do not itch/hurt
 - Can appear anywhere
 - Becomes severely disfiguring if untreated
- **Differential Diagnoses:**
 - Birthmark
 - Vitiligo
 - Contact Dermatitis
 - Lichenoid Dermatitis
 - Tinea Versicolor
- **Diagnosis Of Leprosy:**
 - **Clinical**
 - Skin lesions
 - Thickening of cutaneous nerves
 - Loss of sensation
 - **Split Skin Smears**
 - Acid fast bacilli (AFB)
 - **Biopsy**



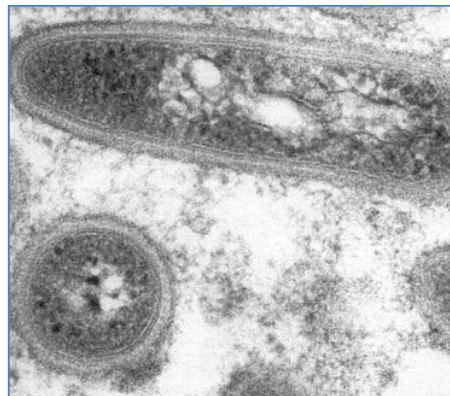
Centers for Disease Control (USA), CC0, via Wikimedia Commons



CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

WHIPPLES DISEASE

- **Aetiology:**
 - Chronic Infection with Bacterium: *Tropheryma Whipplei*
- **Pathogenesis:**
 - *Tropheryma Whipplei* is a Relative of Mycobacteria → ∴ Intracellular (in Macrophages)
 - Systemic Infection → Systemic Disease
- **Morphology:**
 - Endoscopy – Pale, Shaggy Duodenal Mucosa + Eroded, Red Friable Patches
 - **Biopsy – Characteristic 3-Layered Cell Wall of *T-whipplei* Within Foamy Macrophages
- **Symptoms/Signs:**
 - Initially – Arthritis & Arthralgia (but in Middle Aged)
 - Progression to – Weight Loss, Diarrhoea, Abdo Pain, Fever
 - Involvement of – Lymph nodes, Heart, Lung, Joints & Brain (Neuro Symptoms)
- **Investigations:**
 - Blood Tests – Features of Chronic Inflammation & Malabsorption
 - Endoscopy – Pale, Shaggy Duodenal Mucosa + Eroded, Red Friable Patches
 - **Biopsy – Characteristic 3-Layered Cell Wall of *T-whipplei* Within Macrophages
 - Immunohistochemistry – *T-whipplei* Antibodies
 - PCR
- **Treatment:**
 - Long-Course Antibiotics that Cross the BBB – Eg: Trimethoprim or Co-Trimoxazole
- **Prognosis:**
 - Fatal if untreated



Electron Micrograph showing the Tri-Laminar Cell Wall of *Tropheryma whipplei*

The characteristic feature of Whipple's disease; foamy macrophages are present in the lamina propria:



Nephron, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

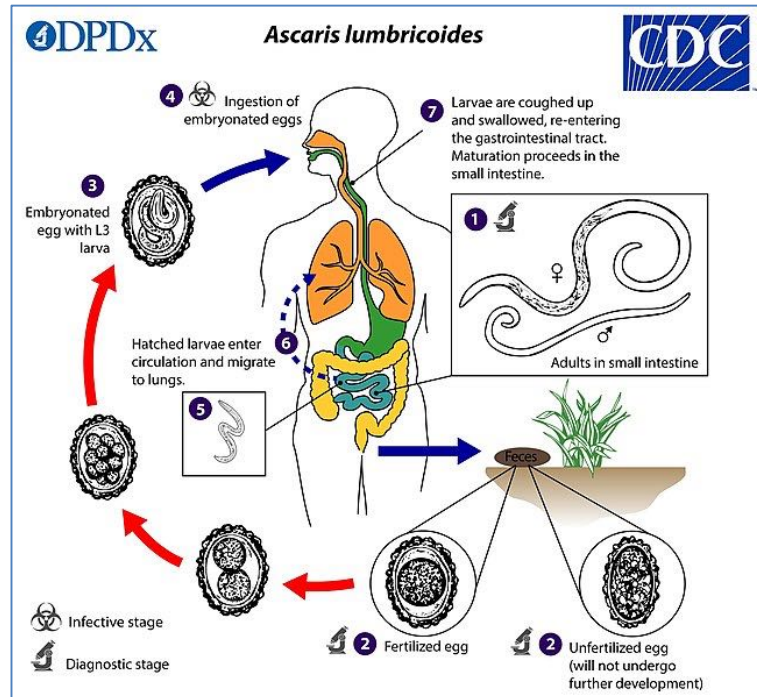
METAZOAN PARASITES:

- Soil Transmitted Helminths:

- Live in GIT
- Pass Progeny in the Faeces (Usually Eggs, Sometimes larvae)
- Use soil for Development from Early Stage (Egg/Larvae) → Infective Stage → Host
- Common in tropical Climates – Warmth & Humidity Critical

○ ROUNDWORM (“ASCARIS LUMBRICOIDES”):

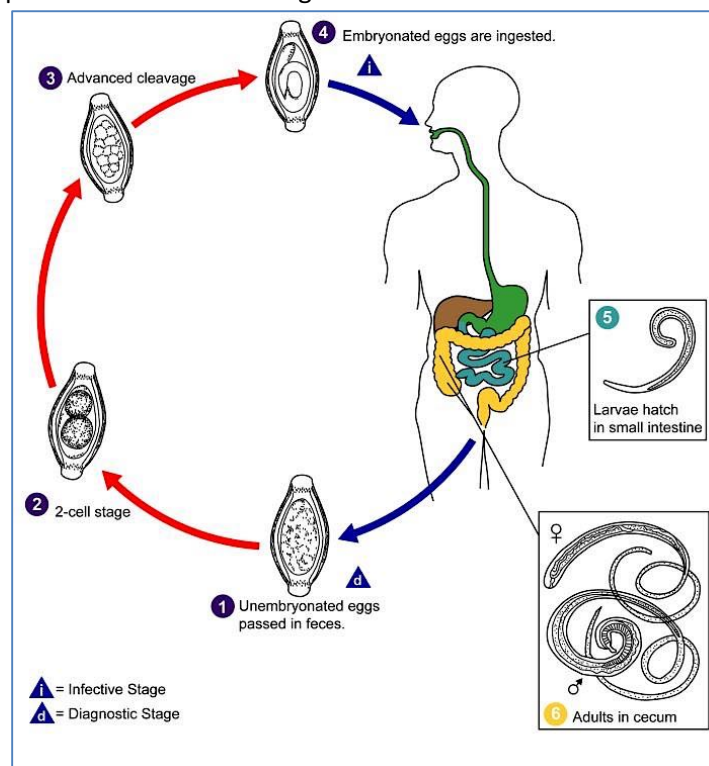
- Larvae grow in the Lungs
- Adults Live in Small Intestine
 - Feeds on Intestinal Contents
 - Strong Swimmer (has no ‘attachment organ’)



<https://www.cdc.gov/dpdx/ascariasis/index.html>

○ WHIPWORM:

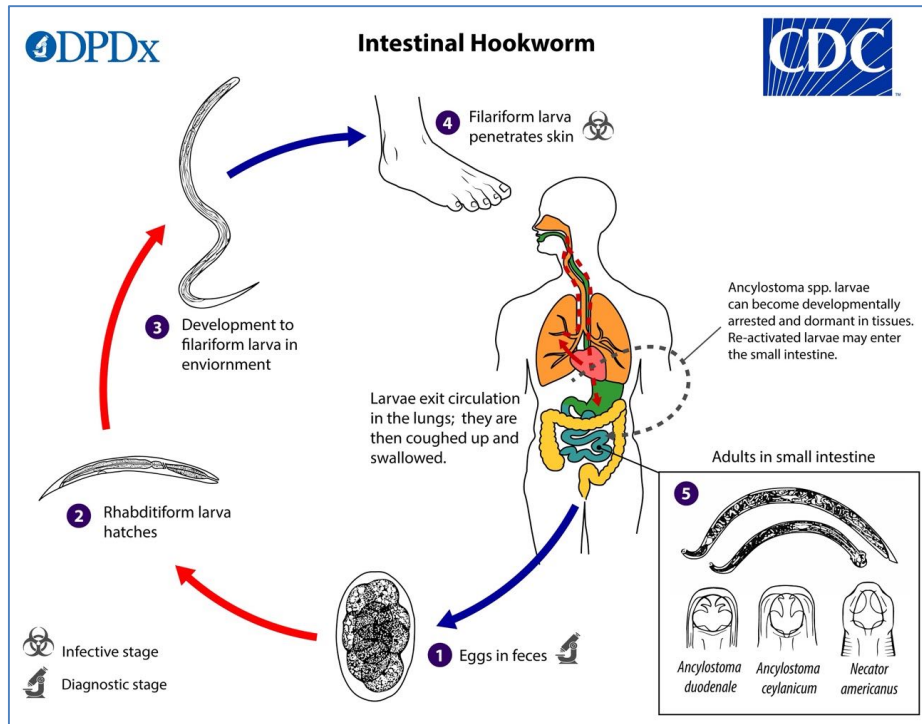
- Lives in Large Intestine
- Whip-like Tail Anchors to Large Intestine Wall



https://upload.wikimedia.org/wikipedia/commons/9/92/Trichuriasis_lifecycle.jpg

○ **HOOKWORM:**

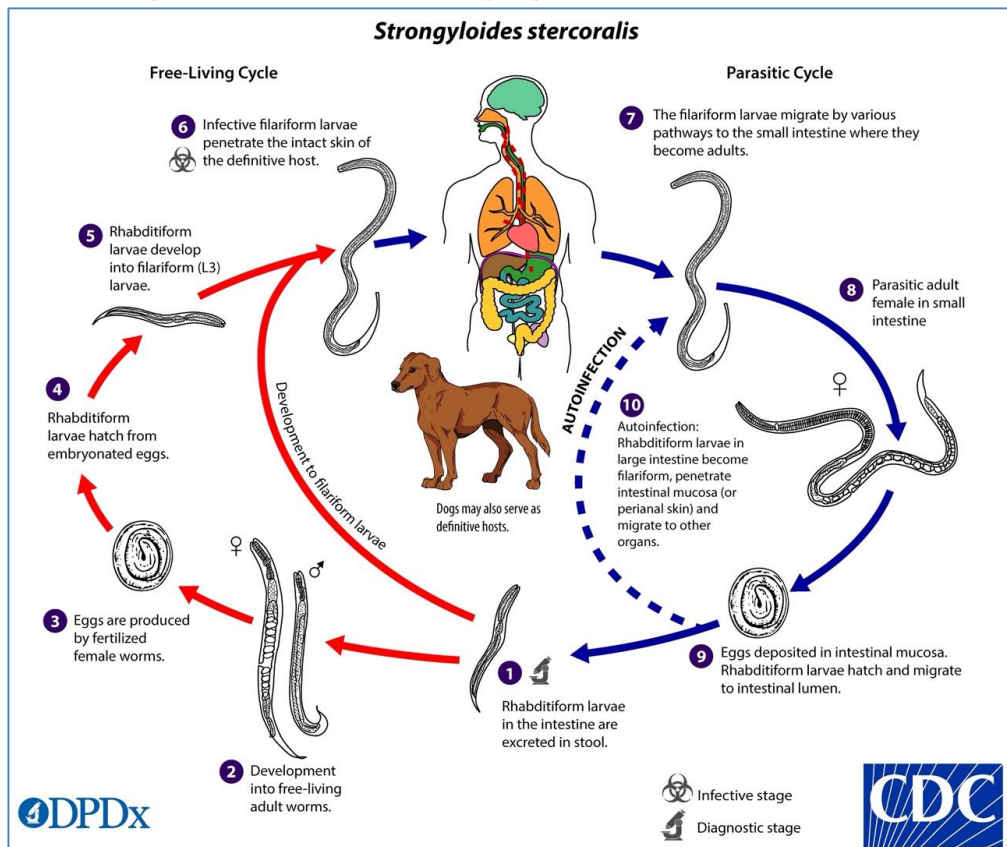
- Live in Small Intestine
- Uses Mouth to Attach to Intestine Wall → Feed on Blood
- Eggs → Soil → *Hatches in Soil* → Larvae Chase Heat → Burrow Through Skin → Circulation → Lungs → Trachea → Down Oesophagus → Stomach → Small Intestine



<https://www.cdc.gov/parasites/hookworm/biology.html>

○ **STRONGYLOIDES:**

- Lives in Small Intestine
- Eggs → Soil → *Hatches in Soil* → Larvae Chase Heat → Burrow Through Skin → Circulation → Lungs → Trachea → Down Oesophagus → Stomach → Small Intestine



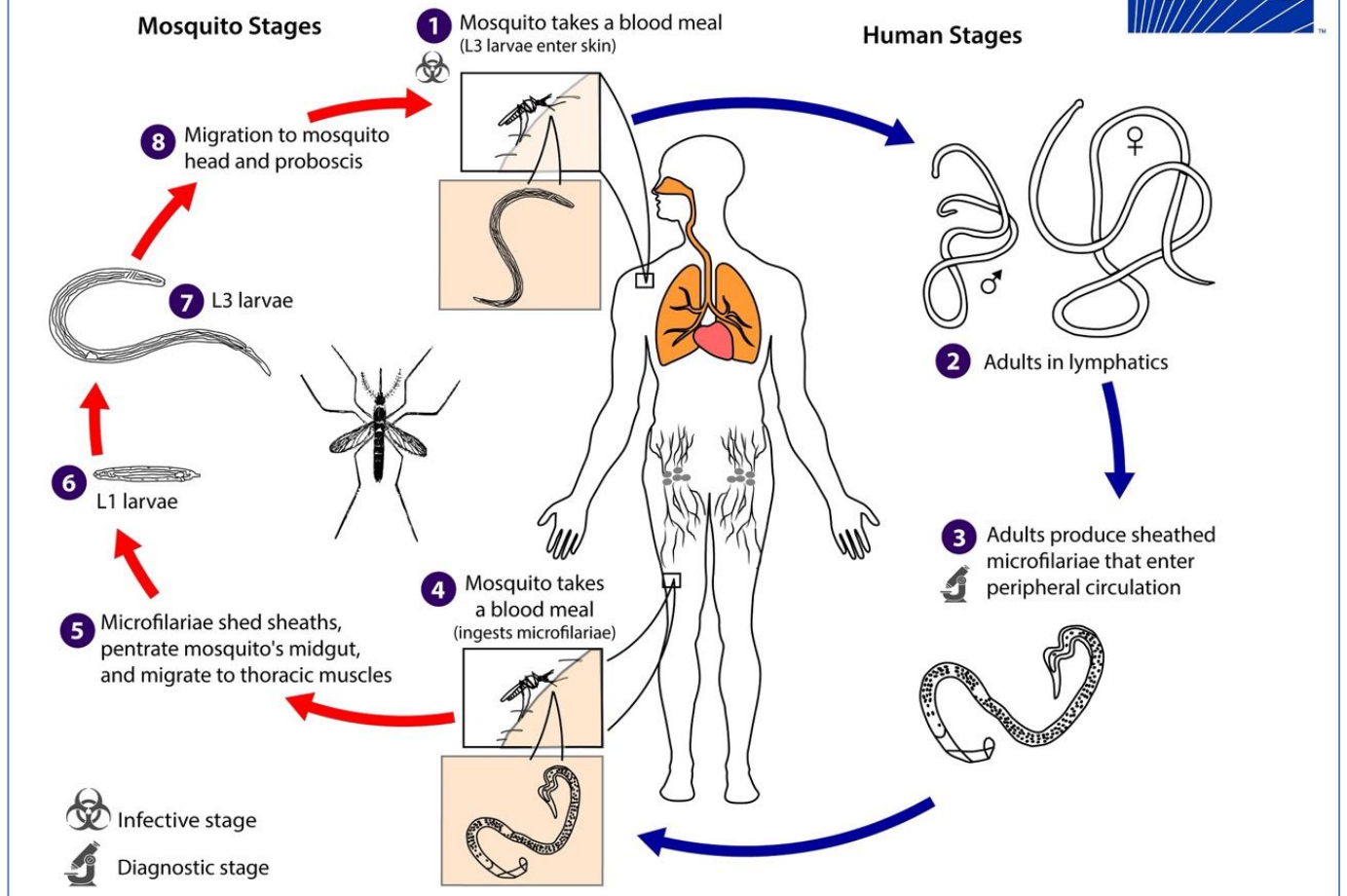
<https://www.cdc.gov/parasites/strongyloides/biology.html>

- **LYMPHATIC FILARIASIS:**

- **Vector:**
 - Mosquitos
- **Pathogen:**
 - Filarial Worms (Parasite)
 - Live in Lymphatics + Nodes
- **Life Cycle:**
 - Adults in Lymphatics → Release Baby Worms (Microfilaria)
 - Microfilaria → Sucked up By Mosquito → Develops inside mosquito
 - New Host Next Bite
- **Results in *Morbidity*, not *Mortality***
 - **Elephantiasis – Massive Oedema**
 - Extensive Lymphatic Damage
 - Suppresses Immune System → Recurrent Infections
 - Fevers
 - Genital Disease
 - Social Isolation/Stigmatisation/Depression
- **4 Aspects of Management:**
 - 1: Preventative Chemotherapy (Prevention)
 - 1x Dose every year for 5 years = good protection
 - 2: Hygiene
 - Care of Entry Lesions (wounds)
 - Wash affected limb with Soap + Water
 - Prevents Secondary Infections
 - 3: Elevation:
 - To Maximise Lymphatic Drainage
 - 4: Exercise:
 - To Maximise Lymphatic Drainage
- **Acute Attack of Filariasis:**
 - Caused by secondary bacterial infection
 - Increased swelling
 - Fever
 - Sore Glands
 - Headache
 - Nausea
 - **Treatment:**
 - 1: Cool leg with cold, clean water
 - 2: Take medicines for Fever + Drink More Water
 - 3: Keep Washing as per Usual
 - 4: Rest



<https://mectizan.org/diseases/lf-2/>



<https://www.cdc.gov/parasites/lymphaticfilariasis/biology.html>

MALARIA:

- **Pathogen = *Plasmodium***

- Eukaryotic Protozoan Parasite
- Widespread in Tropical & Subtropical regions
- **5 Species:**
 - *Plasmodium Falciparum* (**Most Serious**) (Not Persistent in Liver)(80% of Cases)
 - *Plasmodium Vivax* (**Less Serious**) (**Persistent in Liver**)
 - *Plasmodium Ovale* (**Less Serious**) (**Persistent in Liver**)
 - *Plasmodium Malariae* (**Less Serious**) (Not Persistent in Liver)
 - (*Plasmodium Knowlesi* – Mostly A Zoonosis)

- **Vector:**

- **Anopheles Mosquito:**
 - (A Night Biter)
 - Breeds in shaded, heavily vegetated permanent water

- **Two Patterns of Transmission:**

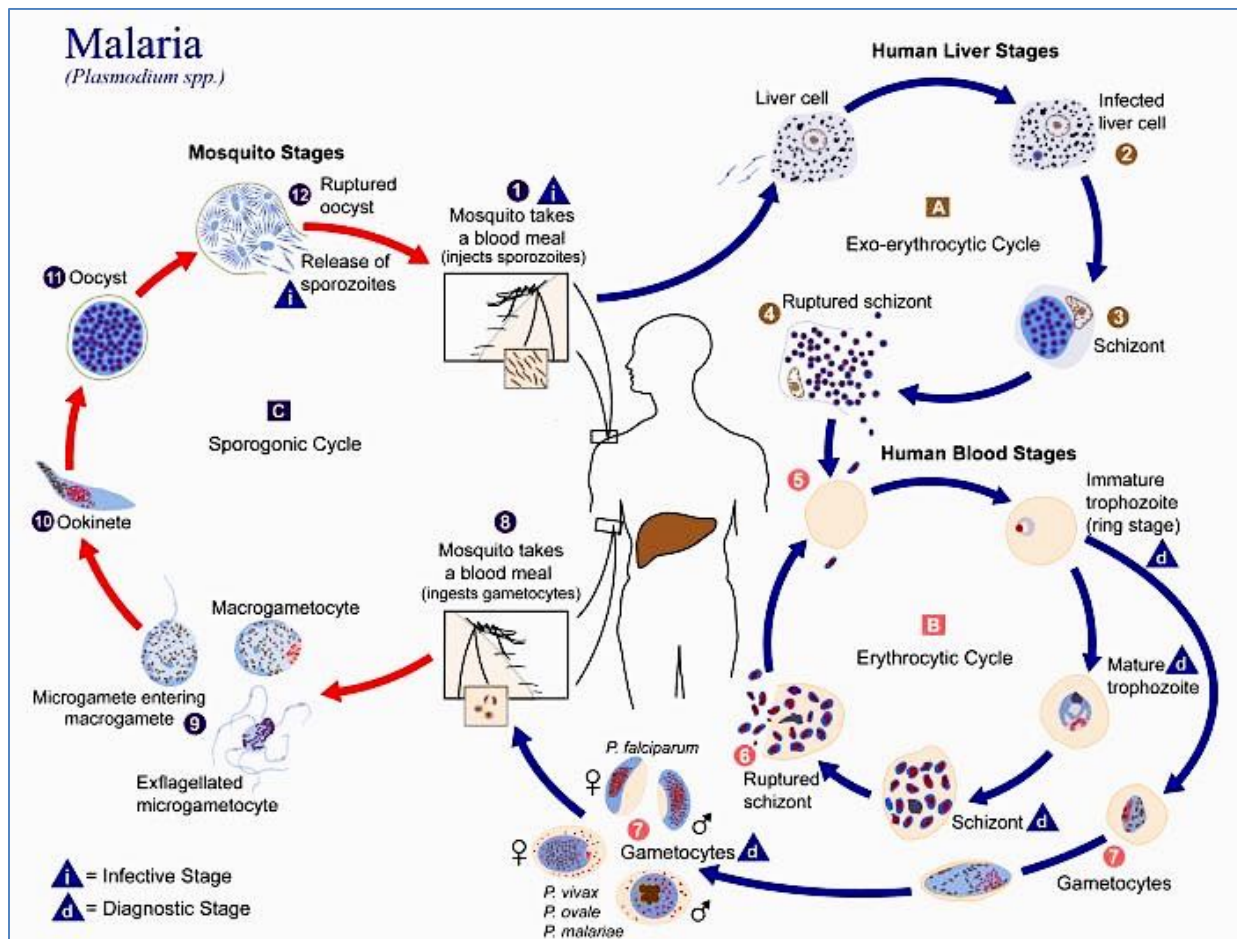
- **1: Stable Transmission:**
 - Constant Endemic Rates
- **2: Unstable Transmission:**
 - – Periodic Epidemic Outbreaks

- **Lifecycle:**

- By the bite of a female *Anopheles Mosquito*
- 1: Bites an Infected Person (Blood contains malaria *Gametocytes*)
- 2: *Gametocytes* develop in the *Anopheles Mosquito* → *Oocysts* in the Gut Wall
- 3: *Oocysts* rupture → Sporozoites Released → Migrate to Mosquito's Salivary Glands
- 4: *Sporozoites* are injected in the *Anopheles Mosquito's* Saliva → Into the Human Host
- 5: *Sporozoites* in Bloodstream → Infect Liver & Multiply → Thousands of *Merozoites*
- 6: *Merozoites* lyse Hepatocytes → Infect RBCs & Multiply
- 7: *Merozoites* → Form *Gametocytes* → Sucked up by *Anopheles Mosquito*

- **Incubation:**

- Between 2wks and several months



<https://www.cdc.gov/dpdx/malaria/index.html>

- **Pathogenesis:**
 - **RBC Invasion and Lysis** →
 - Release of Pyrogens → Fever
 - Extravascular haemolysis – (in spleen)
 - →Haemoglobinuria
 - →Anaemia
 - Headache
 - **RBC's Become 'Sticky' → Adhere to Endothelium → Capillaries Clogged → Tissue Hypoxia → Multiorgan Failure**
 - Cerebral Malaria
 - Pulmonary Oedema
 - Renal Failure
 - **Immune Complex Deposition (Type III Hypersensitivity):**
 - Glomerulonephritis
 - Arthritis
- **Symptoms:**
 - **(The Common Symptoms):**
 - **Episodic Fever** (6-8hrs) – Due to consecutive *Waves* of Merozoites Escaping from RBCs & Reinfesting Other RBCs
 - May be '*Tertian*' (Every 2nd day) or '*Quartan*' (Every 3rd day)
 - Vomiting/Headache/Diarrhoea
 - **"Complicated Malaria" (Acute) – Exclusively by *P-Falciparum* (80% of Cases; 90% of Deaths):**
 - Severe Headache/Nausea/Vomiting
 - Cerebral Ischaemia/Hallucinations
 - Severe Anaemia (Hb of 10-20)
 - Haemoglobinuria (+ Renal Failure)
 - Hepatomegaly/Splenomegaly
 - Hypoglycaemia/Acidosis
 - Seizures/Coma
 - Death (Fatality Rate ≈20% *with Treatment*; 100% *without treatment*)(Within hours/days)
 - **Treated with Artemisinins** – Target Gametocytes in the blood (kills active infection)
 - **"Uncomplicated Malaria" (Chronic) – With *P-Vivax* & *P-Ovale*:**
 - Headache
 - Fever, Chills, Sweating (Fever may be periodic)
 - Muscle Fatigue/Joint Pain
 - Dry Cough
 - Splenomegaly/Haemolytic Anaemia/Jaundice
 - Nausea/Vomiting
 - (Relapses can occur months/years after exposure – Due to *Latent Hypnozoites in Liver*)
 - **Treated with Primaquine** - Targets Hypnozoites in Liver (Preventing recurrence of P Vivax)
- **Diagnosis:**
 - Symptomatic Diagnosis – (Classical Symptoms + Endemic Area ≈ Malaria)
 - Old way - Microscopic Examination of Blood (Thick & Thin Films)(Still highly Effective)
 - New way – RDTs (Rapid Diagnostic Tests) – Antigen Tests Similar to Pregnancy Test
- **Possible Treatments:**
 - **Gametocidal Drugs:**
 - **Artemethers** – Target Gametocytes in the blood (The form that's infective to mosquitoes)
 - (Single Dose Primaquine is also effective)
 - **Drugs Targeting Hypnozoites – (in the Liver):**
 - **Primaquine** - Targets Hypnozoites in Liver (Which can lie dormant and cause recurrences)→Preventing recurrence of P Vivax
 - **Vaccine Development:**
 - Some currently in trial phases and offer partial protection:
 - Seem to be showing partial protection, but not total prevention
 - Short lived nature of natural immunity
 - Parasites' able to mutate

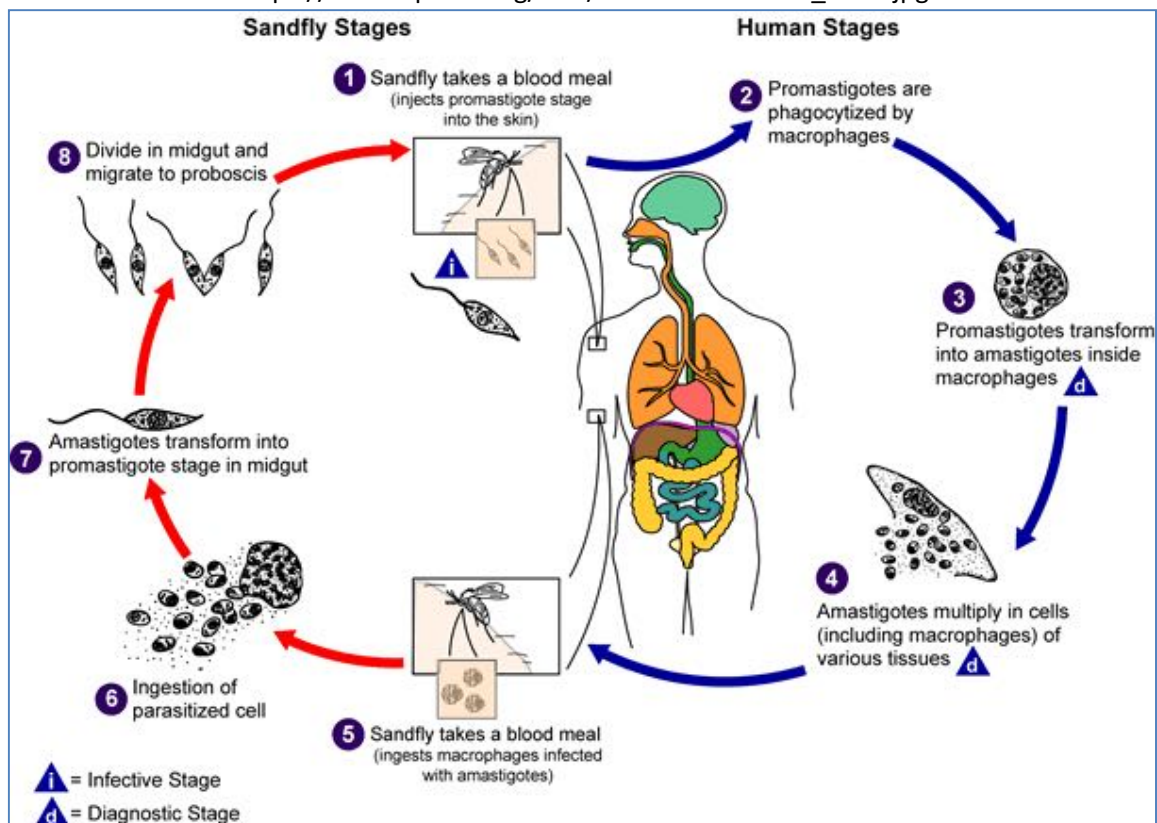
- **Genetic Protection Against Malaria:**
 - **Sickle Cell Trait (heterozygotes) is Protective from Malaria:**
 - In a Sickle Cell carrier, Infected Sickle RBCs rupture prematurely → Plasmodium is Unable to Reproduce
 - ↓O₂ → ↓Plasmodium Growth
 - ↑Macrophage Phagocytosis of the Infected Sickle Cells (Eliminates the parasites in the sickle cell population)
 - **Others:**
 - Lack of the Duffy Antigen (A RBC surface receptor which makes a RBC susceptible to P-Vivax)
 - G6-phosphate dehydrogenase deficiency
 - Thalassemia
- **Immunity to Malaria:**
 - **Immunes:**
 - After Repeated exposure over many years in an endemic area
 - →Malaria episodes are brief and rarely severe
 - **Non-immunes:**
 - Infants/children
 - Travellers from non-malarious areas
 - →Very symptomatic
 - →Susceptible to severe, life-threatening malaria
 - **Loss of Immunity to Malaria:**
 - Pregnant women
 - Previously immune residing outside of endemic areas
 - →also susceptible to severe, life-threatening malaria
- **Disease Prevention:**
 - Prophylactic Drugs
 - Mosquito Nets & Repellents
 - Indoor Residual Spraying (Insecticides) in houses
 - Vector Control (DDT Spraying, Poisoning Breeding Grounds)
 - Public Education about Disease Epidemiology

LEISHMANIASIS:

- **Vector:**
 - o Transmitted Via Ectoparasites (Namely Sandflies)
- **Pathogen:**
 - o Leishmania Parasites
 - o **2 Forms in Lifecycle:**
 - Amastigotes – In man (mostly Intracellular)
 - Promastigotes – In the Sandfly
- **Disease:**
 - o **Visceral Leishmaniosis – AKA- Kala Azar:**
 - Fever
 - Weight loss
 - Anaemia
 - Swelling of liver & spleen
 - Patient Turns Black
 - o **Cutaneous Leishmaniosis:**
 - Nodular & Ulcerated Skin Lesions
 - o **Mucocutaneous Leishmaniosis:**
 - Destructive Nasopharyngeal Lesions



https://en.wikipedia.org/wiki/File:Leishmaniasis_ulcer.jpg



<https://www.cdc.gov/parasites/leishmaniasis/biology.html>

ARBOVIRUSES:

- Aetiology:

○ Alphaviruses:

- **Ross River Virus** - Alphavirus - Mosquitoes - **Fever + Rash + Arthritis**
- Barmah Forest Virus - Alphavirus - Mosquitoes - Indistinguishable from RRV

○ Flaviviruses:

- **Dengue (4x Serotypes)** - Flavivirus - **Aedes Aegypti** - **Haemorrhagic Fevers**
- **Murray Valley Encephalitis** - Flavivirus - Mosquitoes - **Encephalitic Fevers**

- Pathogenesis:

- 1: Bite of an arthropod → Infection
- 2: Virus may replicate in the endothelium and lymphatics
- 3: Viraemia and infection of Target Organs → Fever and malaise (Often due to cytokines)
- 4: Adaptive Immunity to Viral Infections is Cell Mediated (Tc-Cells, NK-Cells)
- 5: Long-Term Immunity to Re-Infection is via Humoral Response (Antibodies & Complement)
 - Prevent Re-Infection by neutralising free viruses in blood & preventing Fusion with Cells

- Clinical Features:

○ 3x Typical Presentations:

▪ **ROSS RIVER VIRUS & BARMAH FOREST VIRUS:**

- (Fever)
- ***Rash** (Maculopapular) (On Trunk)
- ***Arthritis** (Symmetrical Polyarthritis)
- Lethargy
- (Barmah Forest – Indistinguishable from RRV)

▪ **DENGUE FEVER:**

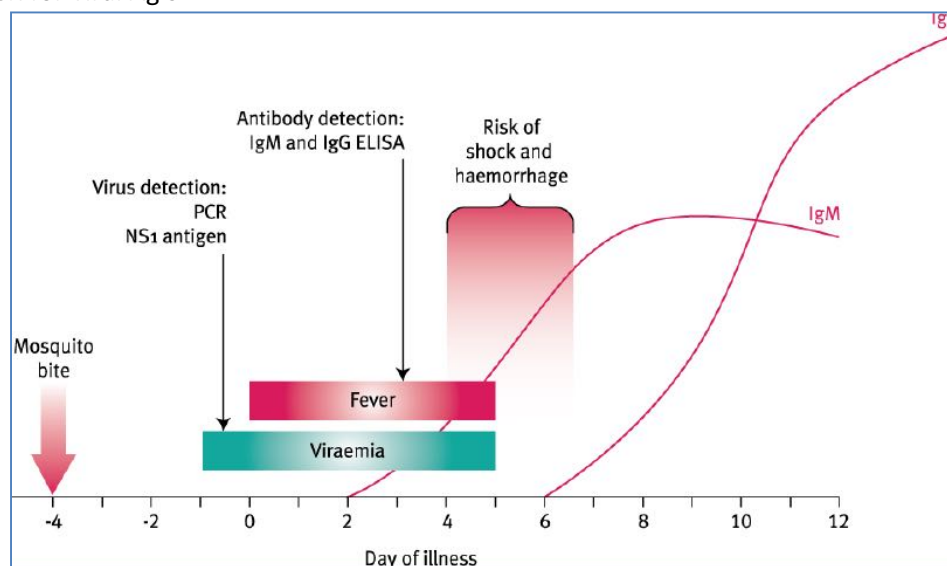
- (Fever)
- ***Rash** (Haemorrhagic/Petechial – due to DIC → Thrombocytopenia)
- ***Myalgia** (“Breakbone Fever” – Severe Muscle Pain)
- (+/- Vom, Diarr, Abdo Pain)
- If 2nd Infection with *Different Serotype* → **Dengue Haemorrhagic Fever/Shock (DHF)**
 - Severe Bleeding
 - Leaky Capillaries
 - Shock

▪ **MURRAY VALLEY ENCEPHALITIS:**

- (Fever)
- ***CNS Involvement** → Headache, Neck Rigidity, Nausea, Convulsions, ALOC
- ~20% Mortality; 50% of survivors have significant neurological disabilities

- Diagnosis:

- Serology for Ab's Test
- PCR for viral Ag's



- Treatment:

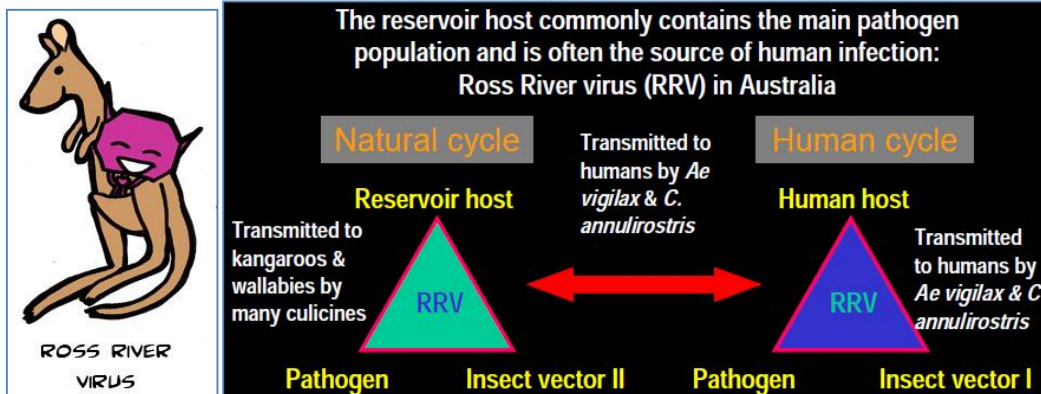
- Supportive Treatment

ROSS RIVER VIRUS (RRV)

- **Causative Organism:**
 - o Ross River Virus
 - o (RRV is an Alphavirus)
- **Vectors:**
 - o **Aedes Vigilax** – Breeds in stagnant Salt Water (Mudflats/Mangroves/Tidal Flats/Etc)
 - o **Culex Annulirostris** – Breeds in freshwater pools/ponds/wetlands/lakes/dams/etc)

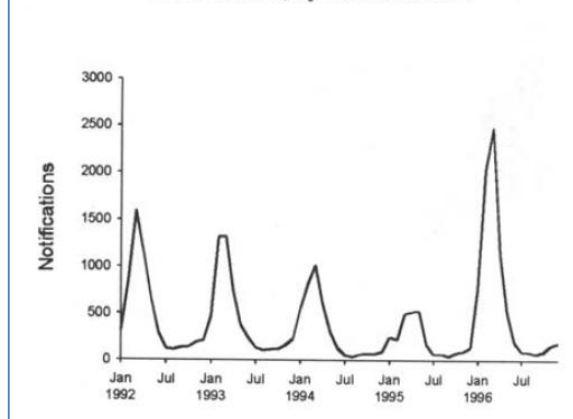


- **Reservoir Hosts:**
 - o Kangaroos & Wallabies



- **Symptoms:** (Similar to “Barmah Forest Disease”)
 - o **95% Polyarthrititis** in small, joints, fingers, hands, feet & wrist
 - o **30-50% suffer fever**
 - o **Maculopapular Rash** (Red/Raised – but not itchy)
 - o **Arthralgia** (sore joints)
 - o **Nausea, myalgia, anorexia & lethargy**
 - o Symptoms can last from 30 weeks to 2 years (Especially Arthritic Symptoms)
- **Infections are Epidemic/Seasonal:**
 - o Seasonal Rainfall (affects breeding environment of vector)
 - o Lunar influences on the tides → flooded marshlands (affects breeding environment of vector)
 - o Temperature & Humidity changes

Figure 28. Notifications of Ross River virus infection, 1992 to 1996, by month of onset

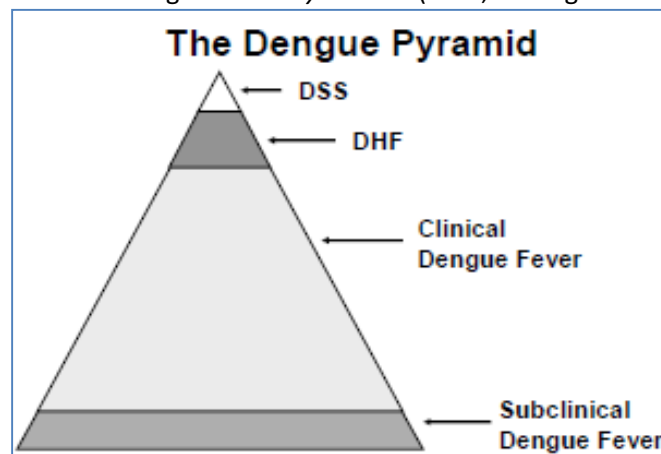


DENGUE VIRUS:

- **Causative Organism:**
 - **Dengue Virus (A Flavivirus)**
 - **4x Serotypes:**
 - (Ie: Different epitopes on the envelope → Specific for adaptive responses)
- **Vector:**
 - ***Aedes Aegypti***
 - Urban Environment (Pots/Gutters/Puddles/Around the house)
 - Infective Vector radius of ~200m from breeding ground



- **General:**
 - It is Extremely Common
 - Its Incidence is Increasing
- **Presentation:** - (Note: Most present *Before Immune Response*)
 - **Typical Presentation:**
 - **Fever & Malaise** (Death warmed up/"Breakbone Fever")
 - **Polyarthrititis** (Muscle & Joint Pain)
 - **Hemorrhagic Rash**
 - **Dengue Haemorrhagic Fever (DHF):**
 - Severe Bleeding
 - Leaky Capillaries
 - Shock
 - **Children may suffer from Dengue Haemorrhagic Shock Syndrome (DHSS)(DSS):**
 - A result of Immune Enhancement due to a *Second Infection* with a different Serotype
 - **(The Dengue Pyramid):**
 - Some infected will be Sub-Clinical
 - Most infected will be Clinically Obvious
 - Some will have Dengue Haemorrhagic Fever (DHF)
 - A few will have Dengue Shock Syndrome (Rare, but high mortality rate)



- **Pathophysiology:**

○ → **Dengue Haemorrhagic Fever (DHF):**

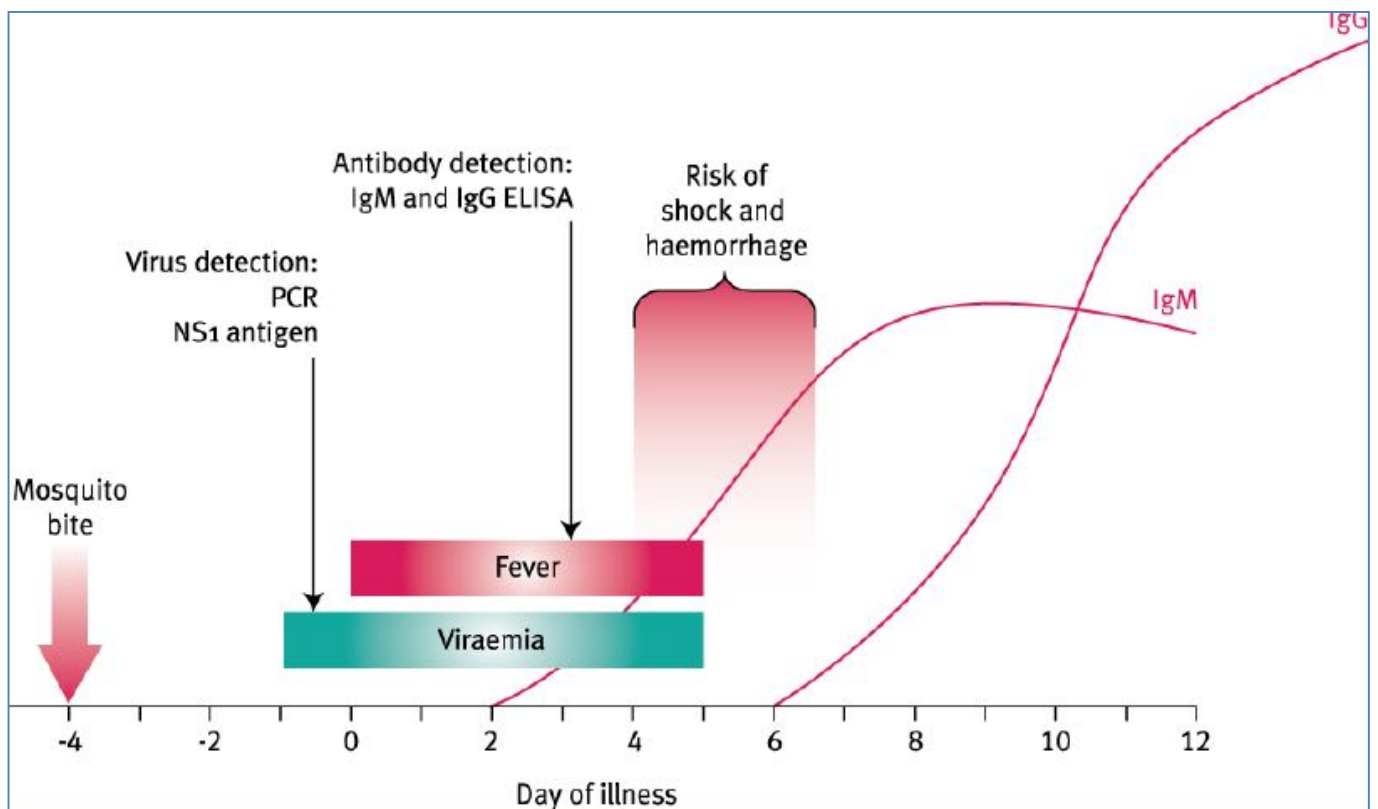
- **1: Primary Infection** → Production of Antibody to *Non-Neutralising Epitopes*
- **2: Secondary Infection** → Binding of Ab to *Non-Neutralising Epitopes* → ↑ Fc-Mediated Uptake of Dengue Virus by Macrophages
 - **3: → Activated Macrophages** → Massive Cytokine Production (Especially $\text{TNF}\alpha$, $\text{TNF}\beta$ and $\text{IFN}\gamma$)
 - **4: Cytokines + Complement** → ↑ Vessel Permeability → Vascular Leakage & Haemorrhage

○ → **Dengue Shock Syndrome (DSS):**

- **Due to Immune Enhancement** following a second infection with a different Dengue Serotype
- **Secondary Infection** → Binding of Ab to *Non-Neutralising Epitopes* → ↑ Fc-Mediated Uptake of Dengue Virus by Macrophages
 - → **Activated Macrophages** → Massive Cytokine Production (Especially $\text{TNF}\alpha$, $\text{TNF}\beta$ and $\text{IFN}\gamma$)
 - **Cytokines + Complement** → ↑ Vessel Permeability → Vascular Leakage & Haemorrhage → **If Severe → SHOCK!**

- **Disease Progression & Diagnostic Tests:**

- - (Note: Most present *Before Immune Response*)
 - Note: Early negative serology is irrelevant because there may not be antibodies yet
 - Note: Also difficult to distinguish between Antibodies against different serotypes
- *** - Dengue Non-Structural Protein 1 (NS1) (detected via ELISA)**
 - **RDT's Available**
- - High IgG Titre to Flaviviruses
- - Dengue Virus Antigens (detected via PCR)



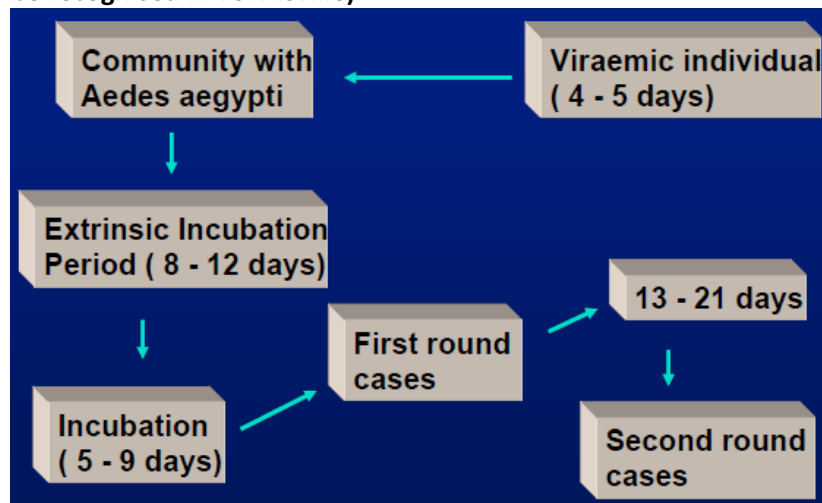
- **Epidemiology & Transmission:**

○ **Endemic and epidemic where vectors present:**

- **Vector = *Aedes Aegypti*** (Mosquito)
 - Urban Environment (Pots/Gutters/Puddles/Around the house)
 - Infective Vector radius of ~200m from breeding ground
- **Reservoir Host = Monkeys**

○ **Events Leading to Dengue Epidemic:**

- 1: Viraemic Individual
- 2: Community must have *Aedes Aegypti*
- 3: Extrinsic Incubation Period (Time from infection of vector, to when it can transmit it to others)
- 4: Intrinsic Incubation (Time from infection of human host, to onset of symptoms)
- 5: First Cases
- **(Note: Cycle takes 13-21 Days – Hence it may take several weeks for a Dengue Outbreak to be recognised in Townsville)**



- **Treatment:**

- **Fluid therapy** (usually very effective)
- **(Avoid Aspirin or Brufen – ie: Stuff that makes bleeding worse)**
- **NO Vaccines present**

YELLOW FEVER:

- **Causative Organism:**
 - o Yellow Fever Virus (A Flavivirus)
- **Vector:**
 - o ***Aedes Aegypti***
 - Urban Environment (Pots/Gutters/Puddles/Around the house)
 - Infective Vector radius of ~200m from breeding ground
- **Pathophysiology:**
 - o **Virus Infects Viral Organs (Especially the Liver):**
 - →Liver Necrosis →Jaundice
 - o The virus **also damages the kidney and heart**
- **Presentation:**
 - o Characterised by Jaundice
 - o **High case fatality rate**
- **Transmission**
 - o **Urban cycle:**
 - Requires man to man transmission
 - o **Sylvatic cycle:**
 - Involves other animals/environment (Especially monkeys)



Yellow Fever

Transmitted to humans from infected mosquitoes

Symptoms		Prevention	
 SUDDEN FEVER	 YELLOWNESS OF THE EYES/DARK URINE	 ENSURE YOU GET VACCINATED (SAFE & FREE)	 WEAR PROTECTIVE CLOTHING TO PREVENT MOSQUITO BITE
 BODY PAIN	 HEADACHE	 SLEEP UNDER A LONG-LASTING INSECTICIDE TREATED NET	 GET RID OF STAGNANT WATER
 BLEEDING (FROM THE NOSE, EAR MOUTH)		 KEEP WATER CONTAINERS COVERED	

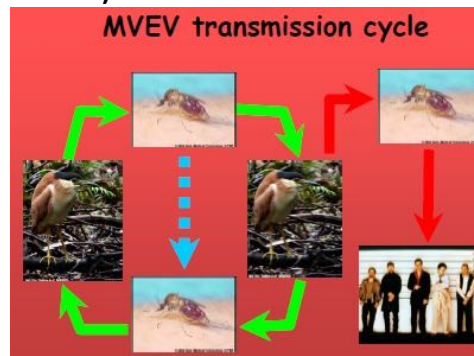
ANYONE WITH THE ABOVE SYMPTOMS SHOULD REPORT TO THE NEAREST HEALTH FACILITY
CONTACT NCDC ON THE TOLL-FREE NUMBER: 0800-970000-10



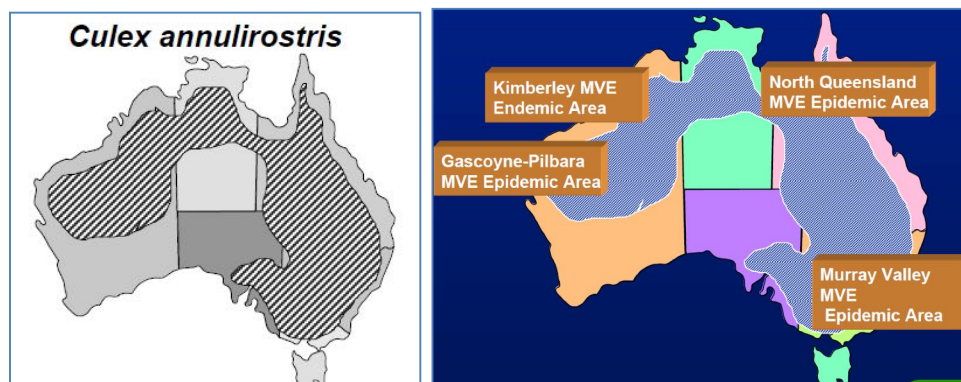
<https://ubth.org/yellow-fever-symptoms-prevention/>

MURRAY VALLEY ENCEPHALITIS:

- **Causative Organism:**
 - Murray Valley Virus
 - (A Flavivirus)
- **Vector:**
 - *Culex Annulirostris* – Breeds in freshwater pools/ponds/wetlands/lakes/dams/etc
 - (Reservoir Host = Water Birds)



- **Potentially Fatal CNS infection:**
 - **Virus crosses the Blood Brain Barrier during initial Viraemia**
 - → CD8-Tc-Cells invade the CNS → Attack infected *Glial Cells* → Damages brain
 - **Note: Knockout Mice with No Cell-Mediated Immunity (Tc-Cell Cytotoxins: Perforins/Granzymes), do not get Encephalitis associated with Infection. Why?**
 - Lack of CD8-Tc-Cell Cytotoxic Enzymes → No cytotoxicity of Tc-Cells → No cell-mediated damage of Virally-Infected Glial Cells in the brain → NO Encephalitis
- **Presentation:**
 - Fever
 - Headaches
 - Nausea & Vomiting
- **Severity of brain damage varies:**
 - Complete Recovery
 - Mild Residual Neurological Symptoms
 - Severe Neurological Damage
 - Death
- **Prognosis of Encephalitis:**
 - ~20% fatal
 - ~50% of survivors have significant neurological disabilities
- **Distribution in Australia:**
 - **Requires Culex Annulirostris** (Which breeds in freshwater/ponds/etc)
 - **Endemic** in Wet, Tropical Areas of the Northern Territory:
 - Eg: Kimberly
 - **(Epidemic in NQ & Murray Valley) – An Epidemic Requires:**
 - Very wet summer
 - Massive growth of the organism
 - Susceptible individuals & reservoir & amplificatory hosts & Migratory Birds
 - Continuous channels of water





MEDSTUDENTNOTES.COM

Best Wishes From Our Team!

On behalf of the MedStudentNotes team, we sincerely hope you found these study notes valuable. Our mission is to help aspiring medicos like yourself to become the best clinicians that you can possibly be. We wish you all success in your ongoing studies and career ahead!!

If you have any comments, suggestions, or feedback, please feel free to email us at admin@medstudentnotes.com

If you've found these notes valuable and you haven't yet got our [Bundle Deal](#), (which includes ALL of our subjects at an 80% discount), then NOW is the time! To upgrade, simply visit the link below for instructions:

>>> <https://www.medstudentnotes.com/pages/bundle-upgrade-process> <<<

Just like you, we have big plans for the future, so be sure to keep an eye on your email inbox for updates.

